UNIVERSITY OF GONDAR
COLLEGE OF MEDICINE AND HEALTH SCIENCES
INSTITUTE OF PUBLIC HEALTH
ETHIOPIAN FIELD EPIDEMIOLOGY AND LABORATORY TRAINING PROGRAM

Compiled Body of Works *in Field Epidemiology*

Field residency outputs, research and scientific dissemination

BY

Tesfahun Taddege GEREMEW

Ethiopian FELTP Fellow · Master of Public Health in Field Epidemiology

ACADEMIC MENTOR
Lemma Derseh

FIELD MENTOR
Meklit Mekonen

FIELD SUPERVISOR
Getie Zeru

JANUARY 2019 · GONDAR, ETHIOPIA

GET ONLINE VERSION →

UNIVERSITY OF GONDAR
COLLEGE OF MEDICINE AND HEALTH SCIENCES
INSTITUTE OF PUBLIC HEALTH
ETHIOPIAN FIELD EPIDEMIOLOGY AND LABORATORY TRAINING PROGRAM

Compiled Body of Works in Field Epidemiology

Field residency outputs in outbreak investigation, surveillance, health profiling, rapid assessment, research and dissemination

BY

Tesfahun Taddege GEREMEW

tesfahuntaddege.github.io

Ethiopian FELTP Fellow

Field residency outputs submitted to the Institute of Public Health, College of Medicine and Health Sciences, University of Gondar, in partial fulfilment of the requirements for the degree of Master of Public Health in Field Epidemiology.

ACADEMIC MENTOR
Lemma Derseh (MPH)

FIELD MENTOR
Meklit Mekonen (MPH)

FIELD SUPERVISOR
Getie Zeru (MPH)

JANUARY 2019 · GONDAR, ETHIOPIA

Suggested citation: Geremew TT. *Compiled Body of Works in Field Epidemiology.* Ethiopian Field Epidemiology and Laboratory Training Program, University of Gondar; 2019. Correspondence: ttaddege@gmail.com

Acknowledgement

First of all, I would like to express my sincere gratitude to my mentors and advisors Mr. Lemma Derseh (academic mentor), Ms. Meklit Mekonnen (field mentor), and Mr. Getie Zeru (field supervisor) for their continuous and valuable guidance and support throughout the development of this body of works in field epidemiology.

Beside my mentors and advisor, I gratefully acknowledge the public health emergency management case team staffs of West Gojjam zone health department for their valuable support during the residency period, the district health offices of Bahir Dar zuria, Yilmana Densa, Jabi-Tehinan and the TB prevention and control program of West Gojjam zone health department for their cooperation in providing the necessary data.

Finally, I would like to thank Centers for Disease Control and Prevention Ethiopia for the financial support during the residency period.

Contents

FRONT MATTER

Title page ii

Acknowledgement iii

List of Tables vii

List of Figures ix

Abbreviations xi

Executive Summary xiii

PART I · FIELD RESIDENCY OUTPUTS

1 Outbreak Investigation 1

Abstract 1

1.1 Introduction 2

1.2 Objectives 3

1.3 Methods 4

1.4 Results 7

Observation findings 13

1.5 Discussion 15

1.6 Conclusion 16

1.7 Recommendation 16

Public health actions 16

References 17

Annex 1.1: Information sheet and Informed consent form 18

Annex 1.2: Questionnaires for AWD outbreak investigation 19

2 Surveillance Data Analysis 21

Abstract 21

2.1 Introduction 22

2.2 Objectives 23

2.3 Methods 24

2.4 Results 26

2.5 Discussion 38

2.6 Conclusion 39

2.7 Recommendations 40

References 41

3 Surveillance System Evaluation 42

Abstracts 42

3.1 Introduction 43

3.2 Objectives 45

3.3 Methods 46

3.4 Result 47

3.5 Discussion 64

3.6 Conclusions 65
3.7 Recommendations 65
References 66
Annex 3.1: Malaria surveillance system evaluation data collection checklists 68

4 Community Health Profile 74
Abstract 74
4.1 Introduction 75
4.2 Objectives 76
4.3 Methods 77
4.4 Results 79
4.5 Discussion 104
4.6 Conclusion 106
4.7 Recommendation 106
References 107
Annex 4.1: Health profile assessment data extraction tool 108

5 Rapid Nutrition Assessment 115
Abstracts 115
5.1 Introduction 116
5.2 Objectives 118
5.3 Methods 119
5.4 Results 121
5.5 Discussion 128
5.6 Conclusions 129
5.7 Recommendations 129
References 130
Annex 5.1: Rapid nutrition assessment data extraction checklist 132

PART II · RESEARCH AND DISSEMINATION

6 Research Project 136
Abstract 136
6.1 Introduction 137
6.1.1 Background 137
6.1.2 Statement of the problem 137
6.1.3 Justification 138
6.1.4 Literature review 138
6.1.5 Conceptual framework 141
6.2 Objectives 142
6.3 Methods 143
6.4 Results 148
6.5 Discussion 158
6.6 Conclusion 160
6.7 Recommendation 160
References 161

7 Scientific Manuscripts 165
Abstract 165

7.1 Background 166
7.2 Methods 167
7.3 Results 171
7.4 Discussion 181
7.5 Conclusion 183
Declarations 183
References 185

8 Conference Abstracts 188
Abstract 188
Abstract 189
Abstract 190

9 Additional Outputs 191
9.1 Sample weekly bulletins 191
9.2 Sample poster for scientific presentation 192
9.3 Publication 193

List of Tables

TABLES

Table 1.1	Percentage distribution of cases and controls by background characteristics, Abune-Hara Holy water, Bahir Dar zuria district, northwest Ethiopia, May 24-June 16 2017	7
Table 1.2	Distributions of cases by clinical characteristics, Abune Hara holly water, Bahir Dar Zuriya district, Northwest Ethiopia, 2017 (n=141)	10
Table 1.3	Distribution of respondents (cases and controls) by exposure factors at Abune-Hara Holy water, Bahir Dar zuria district, Northwest Ethiopia, 2017.	12
Table 1.4	Factors associated with the AWD outbreak in Abune Hara Holy water, Bahir Dar Zuriya district, Northwest Ethiopia, 2017	13
Table 2.1	The trend in TB case detection rates by district in West Gojjam Zone, Northwest Ethiopia from 2012/3 to 2016/7.	26
Table 2.2	Distribution of all forms of TB cases by sex, age and TB category among total new and relapse TB cases from 2014/5-2016/7	26
Table 2.3	Community contributions to TB case notifications and treatment adherence support for TB patients (all forms) in districts of West Gojjam Zone, Northwest Ethiopia, 2016/7	35
Table 2.4	Percentage of HIV positive clients screened for TB and HIV positive clients confirmed with active TB, West Gojjam zone, Northwest Ethiopia, 2014/5 - 2016/7.	36
Table 2.5	Percentage distribution of drug resistant TB cases by background characteristics in West Gojjam, Northwest Ethiopia, 2012/3-2016/7	36
Table 3.1	Proportion of malaria cases detected by the surveillance system in Jabi-Tehinan district, northwest Ethiopia, 2009 EFY.	47
Table 3.2	Percentage of households with at least one ITN, ITN utilization and IRS, Jabi-Tehinan district, Northwest Ethiopia, 2009 EFY	48
Table 3.3	Proportion of malaria species among all ages by RDT and microscopy, Jabi-Tehinan district, Northwest Ethiopia, 2009 EFY	51
Table 3.4	Meta regression to explore the potential sources of heterogeneity in API across reports of health facilities, northwest Ethiopia, 2009 EFY.	58
Table 3.5	Meta regression of malaria positivity rate to explore the potential sources of heterogeneity across reports, Yilmana Densa district, northwest Ethiopia, EFY 2009.	58
Table 3.6	Population under surveillance in Jabi-Tehinan district, Northwest Ethiopia, 2009 EFY	59
Table 3.7	Number of health facilities that had necessary materials for malaria surveillance in Jabi-Tehinan District, Northwest Ethiopia, 2009 EFY.	60
Table 3.8	Timeliness and completeness of the last five epidemiological weeks' report in Jabi-Tehinan district, Northwest Ethiopia, 2009 EFY	63
Table 4.1	Estimated number of and population and households by clusters in Yilmana Densa district, Northwest Ethiopia, EFY 2008.	80
Table 4.2	The HF to population ratios compared to the standard of HF to population ratio in Yilmana Densa district, Northwest Ethiopia, EFY 2008.	81
Table 4.3	Number of health staffs by sex and health professional to population ratio in Yilmana Densa district, Northwest Ethiopia, EFY 2008.	83
Table 4.4	Proportion of pregnant women who received antenatal care, tested for HIV and births attended by skilled health personnel in Yilmana Densa district, EFY 2008.	86
Table 4.5	Proportion of still births among deliveries attended at health facilities by clusters, Yilmana Densa district, EFY 2008	87
Table 4.6	Proportion of SAC and PAC disaggregated by age among total Comprehensive abortion care service users, Yilmana Densa, Northwest Ethiopia, EFY 2008 (n=186).	87
Table 4.7	Proportion of live births received BCG, DPT1-HepB1-Hib1, DPT3-HepB3-Hib3 and measles vaccines among surviving infants, Penta3 and Measles dropout rates by cluster, Yilmana Densa, EFY 2008	89
Table 4.8	Percentage of malnourished children under-five years based on weight-for-age anthropometric indice, Yilmana Densa, Northwest Ethiopia, EFY 2008.	90
Table 4.9	Percentage of low birth weight newborns among live births weighted by cluster, Yilmana Densa district, EFY 2008.	90
Table 4.10	Percentage of under-five age children treated for Pneumonia, diarrhea and neonatal sepsis at health facilities, Yilmana Densa district, Northwest Ethiopia, EFY 2008.	91
Table 4.11	The ten top leading causes of morbidity for all ages in the outpatient departments of health facilities in Yilmana Densa district, Northwest Ethiopia, EFY 2008	92
Table 4.12	The ten top leading causes of morbidity for under-five children in the outpatient departments of health facilities in Yilmana Densa district, Northwest Ethiopia, EFY 2008.	92
Table 4.13	TB case detection rate by cluster in Yilmana Densa district, northwest Ethiopia, EFY 2018.	93
Table 4.14	TB treatment outcomes by TB category, Yilmana Densa district, Northwest Ethiopia, EFY 2008.	94
Table 4.15	Percentage distribution of HIV screening for TB patients and HIV positive rate among TB Patients by cluster, Yilmana Densa district, northwest Ethiopia, EFY 2008.	95
Table 4.16	Percentage distribution of TB screening for HIV positive clients and TB positive rate among HIV patients by cluster, Yilmana Densa district, northwest Ethiopia, EFY 2008	95
Table 4.17	Number of individuals who have been tested for HIV and HIV test positivity rate in Yilmana Densa district, Northwest Ethiopia, EFY 2008.	96
Table 4.18	Percentage of HIV positive PLW among those who were tested for HIV, Yilmana Densa district, Northwest Ethiopia, EFY 2008.	98
Table 4.19	Malaria positivity rate disaggregated by age, sex and malaria species among tests done for malaria diagnosis, Yilmana Densa district, EFY 2008 (N=13,620)	99
Table 4.20	Proportion of rural kebeles that have been declared ODF, Yilmana Densa, EFY 2008.	100
Table 4.21	Distribution of OPD attendance per capita and admission rate by cluster, Yilmana Densa district, Northwest Ethiopia, EFY 2008	102
Table 5.1	Percentage of facilities with basic equipment items available and trained personnel for Nutrition screening and management, Yilmana Densa district, Northwest Ethiopia, 2010 EFY.	122
Table 5.2	Percentage of children with severe acute malnutrion and underweight based on weight-for-age and MUAC measurements, Yilmana Densa, northwest Ethiopia 2018.	122
Table 5.3	Factors associated with the inconsistency of underweight prevalence among children across health facilities in Yilmana Densa district northwest Ethiopia, 2010 EFY.	126

Table 5.4	Factors associated with inconsistency of prevalence of acute malnutrion among under-five children across health facilities in Yilmana Densa district northwest Ethiopia, 2010 EFY.	126
Table 5.5	Treatment outcome for management of severe acute malnutrition in Yilmana Densa district, Northwest Ethiopia, 2010 EFY.	127
Table 6.1	Independent variables and categorization	145
Table 6.2	Percentage of children age 12-35 months who received the first dose Measles vaccine at any time before the survey, according to background characteristics, Ethiopia, 2016.	148
Table 6.3	Percentage of children age 12-35 months who received the first dose Measles vaccine at any time before the survey, according to community level background characteristics, Ethiopia DHS 2016.	150
Table 6.4	The most likely clusters from a purely spatial scan statistic (discrete Poisson model) of children received MCV1 in Ethiopia, 2016.	152
Table 6.5	Multilevel logistic regression analysis of individual and community level factors with childhood MCV1 among 12--23-month old children in Ethiopia, 2016.	154
Table 6.6	Measures of variation in childhood MCV1 in Ethiopia, 2016.	157
Table 7.1	Independent variables and categorization	168
Table 7.2	Percentage of children age 12-35 months who received MCV1 at any time before the survey (weighted) by background characteristics, Ethiopia, 2016.	171
Table 7.3	The most likely clusters from a purely spatial scan statistic (discrete Poisson model) of children received MCV1 in Ethiopia, 2016.	175
Table 7.4	Multilevel logistic regression analysis of individual and community level factors associated with childhood MCV1 in Ethiopia, 2016.	177
Table 7.5	Model fit statistics and measures of variation (random intercept models) in childhood MCV1 in Ethiopia, 2016.	180

List of Figures

FIGURES

Figure 1.1	Map of Bahir Dar Zuriya district by kebele, Northwest Ethiopia, 2017	4
Figure 1.2	Distribution of cases and Controls by Zone, where the Cases came from, Abune-hara Holly Water, Northwest Ethiopia from May 24 -- June 17, 2017(Cases=141, Controls =282).	8
Figure 1.3	District Addresses for AWD patients at Abune Hara Holly, Bahir Dar Zuriya District, Northwest Ethiopia from May 24-June 17, 2017(n=139)	9
Figure 1.4	Epi-curve of Acute Watery Diarrhea (AWD) by date of onset in Abune Hara Holy Water, Bahir Dar Zuriya District, Northwest Ethiopia, 2017.	10
Figure 1.5	Photo on June 2, 2017: A living camp (room 21) for the holy water attendants and a Holy water attendant cocking food inside the living room with a floor unpleasant to look at, Abune-Hara Holly water, Bahir Dar Zuriya District, Northwest Ethiopia)	14
Figure 2.1	TB case notification disaggregated by age and sex (male in blue; female in green), West Gojjam zone, 2016/7.	27
Figure 2.2	Percentage of new and relapse TB cases in under-fifteen year children, 2016/7.	27
Figure 2.3	TB CDR by district, West Gojjam zone, Northwest Ethiopia (20113/4-2016/	28
Figure 2.4	Trend of TB incidence rate per 100,000 population in West Gojjam Zone, Northwest Ethiopia, 2012/3 - 2016/7.	29
Figure 2.5	Estimated TB incidence rate by district, West Gojjam Zone, Northwest Ethiopia, 2012/3 - 2016/7	30
Figure 2.6	Trend in TB CDR, TCR and TSR, West Gojjam Zone, Northwest Ethiopia, 2012/3 - 2016/7	31
Figure 2.7	Trend in TB TSR by district, West Gojjam Zone, Northwest Ethiopia, 2012/3 - 2016/7	32
Figure 2.8	Trend in Unsuccessful treatment outcome (absolute numbers) West Gojjam Zone (2012/13-2016/17)	33
Figure 2.9	Trend in unsuccessful treatment outcomes, West Gojjam Zone, 2012/3-2016/7	33
Figure 2.10	Trend in TB death Rate, West Gojjam Zone, 2013/4-2016/7	34
Figure 2.11	Trend in TB retreatment cases, West Gojjam Zone, 2014/5-2016/7	34
Figure 2.12	Trend in proportion of HIV positive clients screened for TB and confirmed with active TB in West Gojjam zone, Northwest Ethiopia, 2014/5-2016/7.	36
Figure 3.1	Trends in number of malaria cases by Ethiopian Epidemiological Weeks, Jabi-Tehinan District, Northwest Ethiopia, 2005 EFY- 2009 EFY.	50
Figure 3.2	Trend in percentage of malaria species, Jabi-Tehinan district, Northwest, Ethiopia, 2009 EFY	51
Figure 3.3	Trend in malaria positivity rate, Jabi-Tehinan district, Northwest Ethiopia, 2009 EFY.	52
Figure 3.4	Age distribution of malaria in Jabi-Tehinan district, Northwest Ethiopia, 2009 EFY.	52
Figure 3.5	Risk map of kebeles by API, Jabi-Tehinan district, northwest Ethiopia, 2009 EFY.	53
Figure 3.6	Pooled API estimate (ES) per 1000 population in Jabi-Tehinan district, northwest Ethiopia, 2009 EFY.	54
Figure 3.7	Subgroup Malaria API estimate (ES) per 1000 population in Jabi-Tehinan district, northwest Ethiopia, 2009 EFY.	55
Figure 3.8	Subgroup malaria positivity rate estimates (ES) among microscopies and RDTs tested for malaria diagnosis in Jabi-Tehinan district, northwest Ethiopia, 2009 EFY.	56
Figure 3.9	Subgroup malaria positivity pate estimate (ES) by health facility type in Jabi-Tehinan district, northwest Ethiopia, 2009 EFY.	57
Figure 3.10	Flow of Malaria Surveillance data at Jabi-Tehinan district, West Gojjam Zone, Northwest Ethiopia, 2009 EFY.	61
Figure 4.1	Map of Yilmana Densa district by cluster and kebele, Northwest Ethiopia, 2008 EFY.	77
Figure 4.2	Population pyramid, Yilmana Densa district, Northwest Ethiopia, EFY 2008	80
Figure 4.3	Organogram of the district health system, Yilmana Densa district, EFY 2008	82
Figure 4.4	Comparison of baseline, performance and target of CAR by cluster, Yilmana Densa district, Northwest Ethiopia, EFY 2008	84
Figure 4.5	Proportion of modern contraceptive acceptors disaggregated by methods among the total new and repeat acceptors, Yilmana Densa district, Northwest Ethiopia, FEY 2008	84
Figure 4.6	Trend in Contraceptive acceptance rate, Yilmana Densa district, EFY 2000-2008	85
Figure 4.7	Percentage of LAFP coverage by cluster, Yilmana Densa district, EFY 2008.	85
Figure 4.8	Trend in antenatal coverage, percentage of births attended by skilled health personnel and postnatal coverage, Yilmana Densa district, Northwest Ethiopia, EFY 2000-2008	86
Figure 4.9	Percentage of infants aged less than 12 months who had received basic vaccinations, Yilmana Densa district, Northwest Ethiopia, EFY 2008	88
Figure 4.10	Trend in proportion of BCG, DPT1-HepB1-Hib1, DPT3-HepB3-Hib3 and measles vaccination coverage, Yilmana Densa district, Northwest Ethiopia, EFY 2000-2008.	89
Figure 4.11	Proportion of bacteriologically confirmed new PTB, smear negative new PTB, new extra-pulmonary TB and re-treatments among all forms of TB detected in EFY 2008 (N=312).	94
Figure 4.12	Number of HIV positive clients and those who received HIV care, Yilmana Densa district, Northwest Ethiopia, EFY 2008.	96
Figure 4.13	Percentage of pregnant women tested for HIV and knew their result during pregnancy by cluster, Yilmana Densa district, EFY 2008.	97
Figure 4.14	Malaria Positivity Rate by Cluster, Yilmana Densa district, EFY 2008	99
Figure 4.15	Percentage of latrine construction and utilization in the communities of Yilmana Densa district, Northwest Ethiopia, EFY 2008.	101
Figure 4.16	Distribution of outpatient attendants by age, sex and type of visit, Yilmana Densa district, Northwest Ethiopia, EFY 2008 (N=581,481).	102
Figure 4.17	Trend in percentage of total budget allocation to the health sector, Yilmana Densa district, Northwest Ethiopia, EFY 2005-2009	103
Figure 5.1	Availability of services for undernutrition among health facilities in Yilmana Densa district, Northwest Ethiopia, 2010 EFY.	121
Figure 5.2	Pooled estimates for underweight prevalence by type of health facility among children aged 6-59 months in Yilmana Densa district, Northwest Ethiopia, EFY 2010.	124
Figure 5.3	Pooled estimates for global acute malnutrition prevalence by type of health facility among children aged 6-59 months in Yilmana Densa district, northwest Ethiopia, EFY 2010.	125

Figure 6.1 Conceptual Framework for determinants of first dose Measles vaccination among children in Ethiopia[36] 141
Figure 6.2 Regional states of federal democratic republic of Ethiopia[39]*.* 143
Figure 6.3 The global spatial autocorrelation based on feature locations and attribute values (MCV1) across the study areas in Ethiopia, 2016. 151
Figure 6.4 The spatial clustering of areas with low childhood MCV1 coverage in Ethiopia, 2016. 153
Figure 7.1 The global spatial autocorrelation based on feature locations and attribute values (MCV1) across the study areas in Ethiopia, 2016. 174
Figure 7.2 The spatial clustering of childhood MCV1 in Ethiopia, 2016. 176

Abbreviations

AFI	Acute Febrile Illness
ANC1	Antenatal Care atleast one Visit
ANC4	Antenatal Care atleast four visit
API	Annual Paracite Incidence/Index
AR	Attack Rate
AWD	Acute Watery Diarrhea
CAC	Comprehensive Abortion Care
CAPI	Computer assisted personal Interviewing
CAR	Contraceptive Acceptance Rate
CDC	Centers for Disease Control and Prevention
CDR	Case Detection Rate
CFR	Case Fatality Rate
CI	Confidence Interval
CSA	Central Statistics Agency
CTC	Case Treatment Center
DHS	Demographic and Health Survey
DOR	Dropout Rate
DOTS	Directly Observed Therapy Short course
DPT1-HepB1-Hib1	First dose Diphtheria, Tetanus toxoid, Pertussis, Hepatitis B and Haemophilus influenza Type b vaccines
DR-TB	Drug Resistance Tuberculosis
EAs	Enumeration Areas
EDHS	Ethiopian Demographic Health Survey
EFY	Ethiopian Fiscal Year
eHMIS	Electronic Health Management Information System
EMIS	Ethiopia Malaria Indicator Survey
EPI	Expanded Programme on Immunization
EPTB	Extra-Pulmonary Tuberculosis
ERIAs	Enhanced Routine Immunization activities
F100	Therapeutic milk used in Transition Phase and Phase 2 of treatment
F75	Therapeutic milk used only in Phase 1 of treatment for SAM
FMOH	Federal Ministry of Health
GAM	Global Acute Malnutrition
HC	Health Center
HDA	Health Development Army
HEW	Health Extension Worker
HMIS	Health Management Information System
HP	Health Post
HSDP	Health Sector Development Program
HSTP	Health Sector Transformation Plan
ICCM	Integrated Community Case Management
ICT	Information & Communication Technology
IDSR	Integrated Diseases Survielance And Response
IMNCI	Integrated Management Of Newborns and Childhood Illness
IPC	Inpatient Therapeutic Care
IPH	Institute of Public Health
IQR	Interquartile Range
IR	Incidence Rate
IRS	Indoor Residual Spraying
ITN	Insecticide-Treated Net
IUCD	Intra-Uterine Contraceptive Device
LLITN	Long-Lasting Insecticide-Treated Nets
LLR	Log Likelihood Ratio
MAM	Moderate Acute Malnutrition
MCV	Measles-Containing Vaccine
MCV1	First dose of Measles-Containing Vaccine
MDR	Multi Drug Resistance
MPR	Malaria Positivity Rate
MUAC	Mid-upper arm circumference
MUW	Moderately Underweight
NTP	National TB Programme
OCP	Oral Contraceptive Pills
OR	Odd Ratio
OTP	Outpatient Therapeutic Programme
PAC	Post Abortion Care
PHC	Population and Housing Census
PHEM	Public Health Emergency Management
PITC	Provider Initiated Testing and Counseling
PLWH	People Living with HIV/AIDS
PTB	Pulmonary Tuberculosis
RDT	Rapid Diagnostic Test
RED	Reaching Every District
RR	Rifampicin Resistance

RRT	Rapid Response Team
RUTF	Ready-to-Use Therapeutic foods
SAC	Safe Abortion Care
SAM	Sever Acute Malnutrition
SARA	Service Availability and Readiness Assessment
SD	Standard Deviation
SIAs	Supplemental Immunization Activities
SS+ PTB	Smear Positive Pulmonary Tuberculosis
SS- PTB	Smear Negative Pulmonary Tuberculosis
TB	Tuberculosis
TCR	Treatment Cure rate
TFP	Therapeutic Feeding Program
TSR	Treatment Success rate
WHO	World Health Organization

Executive Summary

The Field Epidemiology and laboratory Training Program (FELTP) is an in-service training program in field epidemiology. The program is predominantly field-based with residents spending 75% of their time in the field. Each resident is expected have opportunities to participate in field investigations, program evaluations, analysis of data sets, surveillance and control activities, scientific report writing, oral presentations, and other public health activities at various levels of the health system. Hence, this compiled body works consists all outputs of field residency and expectations by the program at the end of the two residency periods.

The first chapter is an outbreak investigation of acute watery diarrhea at Abune-Hara Holy water, Bahir Dar zuria district northwest Ethiopia. It was a case-control study in which 141 cases and 282 controls were included. The second chapter was survielance data analysis of tuberculosis in west Gojjam zone, northwest Ethiopia. It was a five year tuberculosis trend analysis using the tuberculosis prevention and control program data. The third chapter is evaluation of malaria survielance system report. The evaluation was conducted in Jabi-Tehinan district, one of the highly malarious districts in west Gojjam zone, northwest Ethiopia. It highlights the existing status of malaria survielance system in the district.

The fourth chapter was description of community health profile which was conducted at Yilmana Densa district. It provides a summary of health and health related information for Yilmana Densa district community. The fifth chapter is rapid assessment of nutrition status among children in Yilmana Densa district. It highlights the childhood nutrition status in the district using the secondary data from growth and monitoring program. In addition, it assessed the nutrition specific service availability and readiness in the health facilities of the district. Chapter six is a research project which entitled “Spatial clustering of measles vaccination and associated factors among children in Ethiopia”. It is a spatial and multilevel analysis using the 2016 demographic and health survey. The last three chapters are scientific manuscript for peer-reviewed journal (already submitted to BMC public health), abstracts presented at different scientific conferences and other expected outputs such as bulletin preparation, poster for scientific presentation and publication. The last three chapters include only some sample outputs, some other outputs could not include in this compiled body of works.

PART I

Field Residency Outputs

Outbreak investigation, surveillance analysis and evaluation, community health profile and rapid assessment.

CHAPTER 1

Outbreak Investigation

Acute watery diarrhea outbreak investigation and response at Abune-Hara Holy water in Bahir Dar zuriya district, northwest Ethiopia, 2017.

Tesfahun Taddege GEREMEW[1], Lemma Derseh, Getie Zeru and Meklit Mekonnen

1. Correspondence to : ttaddege@gmail.com↩

Abstract

Introduction: Acute watery diarrhea (AWD) has become a public health problem in Ethiopia. Since the beginning of 2017 to 7 July 2017, a total of 38,230 cases with 792 deaths (CFR 2.1%) were reported. However, the AWD outbreak at Abune-Hara Holy water in Bahir-Dar zuria district was not investigated. Hence, the purpose of this study was to identify the potential risk factors of AWD outbreak for guiding responses activities.

Methods: A case-control study was conducted at Abune-Hara Holy water in Bahir-Dar Zuria district, Northwest Ethiopia. We defended cases as any person aged 5 years or more at the Holy water presenting with watery like rise water diarrhea (>3 lose stool per day), with or without vomiting from May 24 to June 17, 2017. Cases were identified through passive and active case search and controls were selected randomly from family members or neighbors of the cases. A total of 141 cases and 282 controls were interviewed. The data was collected through computer assisted personal interviewing (CAPI) using android device loaded with Epi-Info^{TM} templet. We fitted binary logistic regression model to identify associated exposures.

Result: A total of 141 epidemiologically linked cases with three deaths (CFR 2.1%) were identified. The median age of cases was 35 years [IQR=24]. Close contact history [AOR=5.81, 95%CI=2.85-11.85], water interruption than any other before [AOR=4.30, 95%CI =2.09-8.85], living in overcrowded room [AOR=3.85, 95%CI=1.48-10.05], hand washing after toilet [AOR=0.15, 95%CI=0.08-0.28] and before eating [AOR=0.10, 95%CI =0.02-0.41], storing water in narrow mouthed plastic container [AOR=0.42, 95% CI =0.23-0.78] and eating food immiadetely after cocked [AOR=0.21, 95%CI=0.12-0.38] were statistically significant predictors for AWD illness.

Conclusion: Close contact history, living in overcrowded room, Water supply interruption were the possible risk factors and water storage in narrow-mouthed container, hand washing before eating and after toilet, and eating food immiadetely after cocked were the possible protective factors. Hence, result suggests a need to give more attention for adequate water supply, water storage in narrow-mouthed container, feeding practices, building additional rooms and increasing the awareness of hygienic practices.

Key Words: AWD, Outbreak, Abune-Hara Holy Water, Northwest Ethiopia

1.1 Introduction

Acute Watery Diarrhea is an acute diarrheal illness which has remained as one of the important public health problems in Africa causing substantial morbidity and mortality[1]. It is the typical feature cholera, a diarrheal disease caused by infection of the intestine with the bacterium Vibrio cholera, either type O1 or O139 and both children and adults can be infected[2, 3]. Cholera is usually transmitted through faecally contaminated water or food. New outbreaks can occur sporadically in any part of the world where water supply, sanitation, food safety, and hygiene are inadequate. The greatest risk occurs in over-populated communities and refugee settings characterized by poor sanitation, unsafe drinking-water, and increased person-to person transmission[3]. Flooding during rainy seasons can favor the emergence or the upsurge of acute watery diarrhea (AWD) outbreak[1]. Crowded areas where people practice improper hygiene and sanitation practices pose higher risk of transmission (e.g. Holy water sites visited by thousands of religious pilgrims)[4]

From late 2014 to end of 2015, an outbreak of AWD was occurred in Kenya causing close to 7000 cases of morbidity and over 100 deaths. The outbreak occurred in light of the *El Niño floods* which created favorable conditions for the spread of the illness. After spreading in Kenya for over a year, the disease moved to Ethiopia. Recently in Ethiopia, AWD was first reported in districts of Oromia and Ethiopian Somali region close to the Ethio-Kenyan border in February 2016. Then after, it continued spreading east and north reaching many parts of the country including the capital city and the northern parts of the country. Despite an ongoing effort to contain the outbreak, it continued spreading to large areas of the country affecting a large number of people; its course remained protracted and kept many people at risk[1]. Since the beginning of 2017 to 7 July 2017 (week 27), a total of 38,230 cases including 792 deaths (case fatality rate 2.1%) have been reported from seven regions, namely: Somali, Oromia, Amhara, Afar, SNNP, Tigray, and Benishangul Gumuz. Eighty-nine percent of these cases and 95% of the deaths were reported in Somali Region alone[5]

Bahir Dar Zuriya district experienced three outbreaks of acute watery diarrhea since 2016. The first on June 2016 at Andasa Holy water in which 650 cases and 14 deaths (CFR=2.2%) were recoded and five samples were confirmed with Vibrio cholerae O1 (positivity rate 100%) with rapid diagnostic test (RDT). The second, on February 2017 at Andasa Holy water which causes for 214 cases and 6 deaths (CFR = 2.8%) and the third one occurred on May 2017 at another Holy water called Abune-Hara Holy water.

However, the possible exposure factors are not still investigated in the areas. Hence, this study aimed to investigate the potential risk factors so that appropriate control measures can be taken. Primarily, the findings of this investigation may help for rapid response and control of the outbreak by guiding response activities in the district. Second, the investigation may generate valuable evidences that can help for local planners and health authorities in evidence-based decision making so as to prevention of furthing occurrence AWD outbreak in the future.

1.2 Objectives

GENERAL OBJECTIVE

★ To investigate the acute watery diarrhea outbreak at Abune Hara holy water in Bahir Dar zuriya district, northwest Ethiopia, 2017.

SPECIFIC OBJECTIVES

1. To describe the epidemiological characteristics of the AWD outbreak.
2. To identify the potential risk factors associated with the AWD outbreak

1.3 Methods

Study design and setting

An unmatched case-control study was conducted from May 24 to June 17 2017 at Abune-Hara Holy water in Bahir Dar Zuria district, northwest Ethiopia. Abune-Hara Holy water, one of the famous Holy waters in Amhara region, is found nearly 15 km distance due North from Bahir Dar town, the capital of Amhara region (Figure 1.1). Many religious pilgrims of Orthodox Christianity religion followers visited by the Holy water every day. During the outbreak, many people (specific list couldn't be discovered) were attending the Holy water.

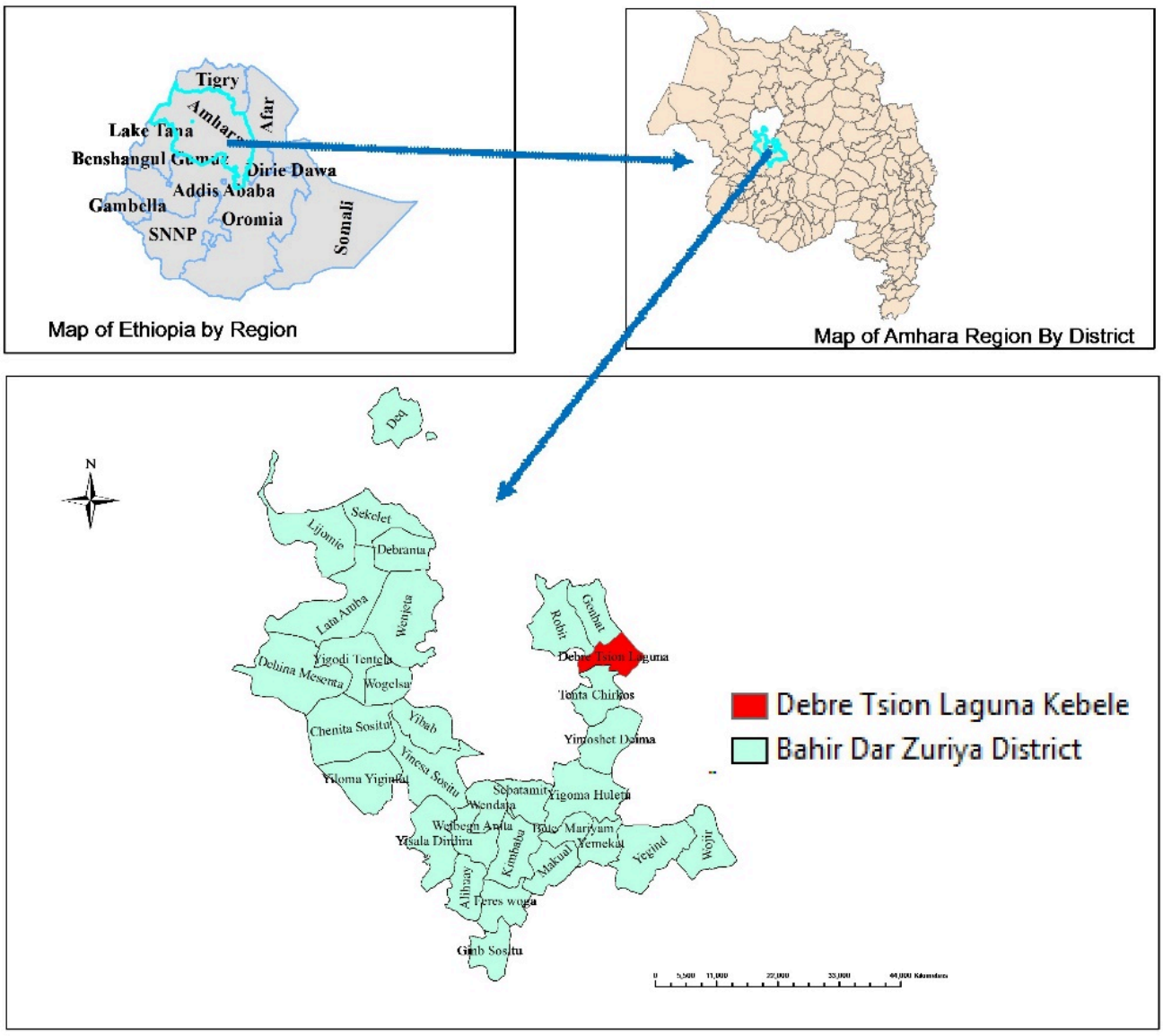


Figure 1.1. Map of Bahir Dar Zuriya district by kebele, Northwest Ethiopia, 2017

Sample size determination and sampling procedure

Sample size determination

StatCalc, an epidemiological calculator of Epi Info™ 7.2 software, was used to compute the sample size with an assumption of 95% level of confidence, 90% power, odd ratio of 1.99, ratio of controls to cases 2:1, and assuming that 27.2% of the controls experienced interruption of water supply than any other before[6]. A total of 471 sample (157 cases and 314 controls) was planned and during the actual data collection, a total of 423 participants (141 cases and 282 controls) were included in the study.

Sampling procedure

Cases were identified using the WHO case definition of suspected cases of cholera in an area where there is a cholera epidemic. Hence, a working case definition, “a patient aged 5 years or more which develops acute watery diarrhea, with or without vomiting”, was used to identify AWD cases[3]. All the cases occurred during the study period were included in the study. Controls were recruited randomly among family members or/and neighbors of the cases who did not report clinical symptoms consistent with AWD and stayed in the Abune Hara holly water attendants shelter since last one week. Despite reporting AWD with or without vomiting, cases and controls were identified irrespective of other health related status.

Data collection

After designing structured questionnaires in English and creating form in Epi Info^{TM} 7.2, the form was loaded onto android device for computer assisted personal interviewing (CAPI), which was used during interviews to ask questions and also record responses. Data were obtained through interviewing participants and reviewing of line list records using a structured questionnaire adapted from the line list, case based report format and checklist. The data from cases were collected using both active and passive surveillance systems from patients who visited cases treatment centers (CTCs) at Abune-Hara Holy water and Addis-Alem hospital.

Study variables

Demographic characteristics such as age, sex, residence, level of education, occupation and hygienic & sanitation practices such as latrine availability, toilet use, hand washing at critical times, source of water, water treatment, adequate water supply and travel history in the last one weak preceding the interview and travel place .

Data quality control

To assure the quality of data, the data was collected by two field epidemiology and laboratory training program residents through computer assisted personal interviewing (CAPI) using smart phone loaded with electronic questionnaires.

Data management and statistical analysis

Data was edited and checked for consistency with Epi-Info^{TM} 7.2 and analyzed Using SPSS Version 22. Descriptive measures were computed to describe the study participants. Binary logistic regression model was fitted to identify factors associated with AWD. Simple binary logistic regression analysis was carried out and independent variables with p-value

less than 0.25 were included in multivariable logistic regression analysis to control possible confounders[7]. Finally, adjusted odd ratio (AOR) with 95% confidence level (CI) were used to declare statistical significant association between the outcome variable and the predictors.

Ethical Consideration

A Legal Letter was obtained from the University of Gondar, Department of Epidemiology and Biostatistics. The protocol was approved by West Gojjam zone health department, Public Health Emergency Management (PHEM) case team. Informed consent was obtained from all participants after explaining the study procedures and purposes. All data were kept confidential through the use of codes.

1.4 Results

Background characteristics of cases and controls

A total 423 samples, 141 cases and 282 controls, were included in this investigation and 54% were males and nearly one forth (75%) had no education, eighty three percent were came from rural area, and 88% were holly water attendants. Among 141 epidemiologically linked AWD cases were recorded from May 24 to June 17, 2017 in Abune-Hara Holly water attendants' camp with three deaths (Case fatality rate 2.1%), more than half (57.4%) of the cases were males and 51 percent were in the 15-44 years age group with the median age of 35 years (IQR 26-50). The majority of the cases had no education (72%) and were farmers (72%). Three forth (75%) of the cases were come from rural area of different parts of the country (Figure 1). Most of the cases (87%) and Controls (90%) were wholly water attendants and the rest were Care givers for the holly water attendants (Table 1.1).

Table 1.1. Percentage distribution of cases and controls by background characteristics, Abune-Hara Holy water, Bahir Dar zuria district, northwest Ethiopia, May 24-June 16 2017

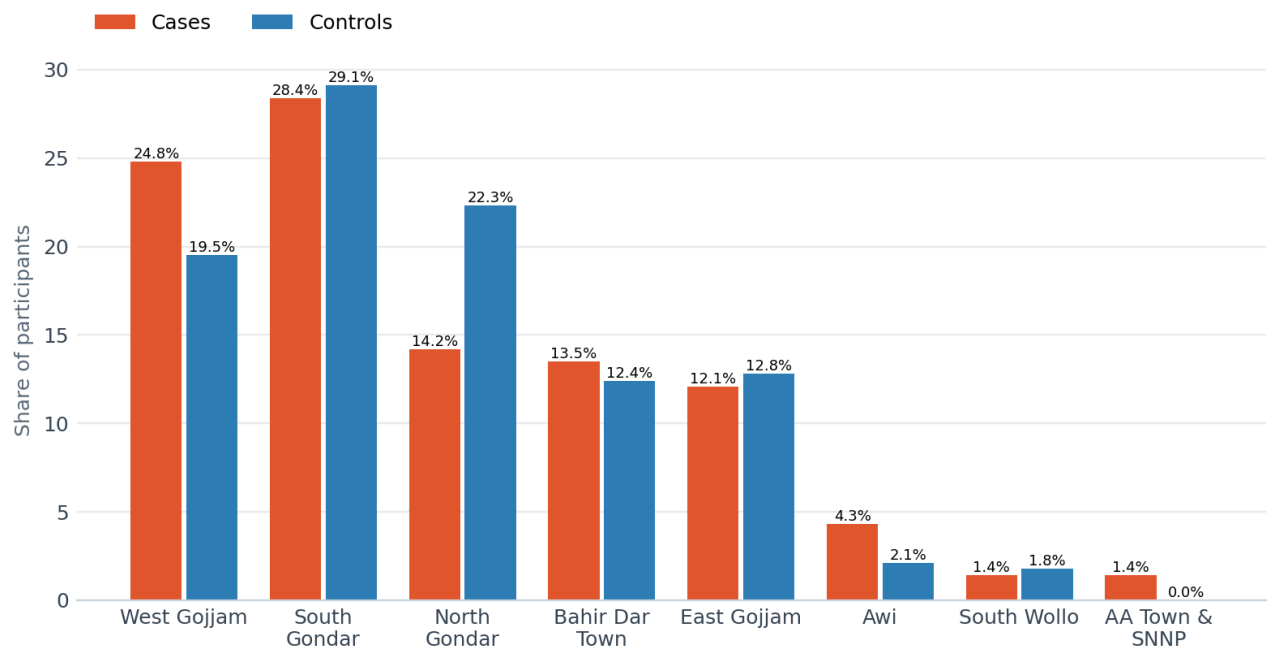
Background characteristics	Cases (n=141)	Controls (n=282)	Total (N= 423)
Sex			
Female	42.6	47.2	45.6
Male	57.4	52.8	54.4
Age group (years) [Median = 35, IQR= 24]			
5 - 14	4.9	6.0	5.2
15 - 44	51.1	56.4	54.6
45+	44.0	37.6	39.7
Educational Status			
No Education	71.6	73.8	73.0
Primary	15.6	11.0	12.5
Secondary	9.2	10.6	10.2
Higher	1.4	3.2	2.6
NA	2.1	1.4	1.7
Occupation			
Farmer	71.6	73.8	73.0
Merchant	12.8	6.7	8.7
Student	12.1	14.9	13.9
Others*	3.6	4.6	4.3
Residence where participants came from			
Rural	75.2	86.5	82.7
Urban	24.8	13.5	17.3
Region where participants came from			
Amhara	98.6	100.0	99.5
Others	1.4	0.0	1.4

Background characteristics	Cases (n=141)	Controls (n=282)	Total (N= 423)
Reason to came to the holly water			
Holy water attendant	87.2	90.1	88.2
Holy water attendant care giver	12.8	9.9	11.8

NA = Children for whom educational status and occupation is not applicable due to their age.

*Gov't worker, Jobless and NA**,** ** Addis Ababa town and SNNP

Nearly all (99%) of the cases came from different areas of Amhara region, mainly from South Gondar (28%), West Gojjam (25%), North Gondar (14%), Bahir Dar Town (14%) and East Gojjam (12%) (Figure 1.2 and 1.3).



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 1.2. Distribution of cases and Controls by Zone, where the Cases came from, Abune-hara Holly Water, Northwest Ethiopia from May 24 – June 17, 2017(Cases=141, Controls =282).

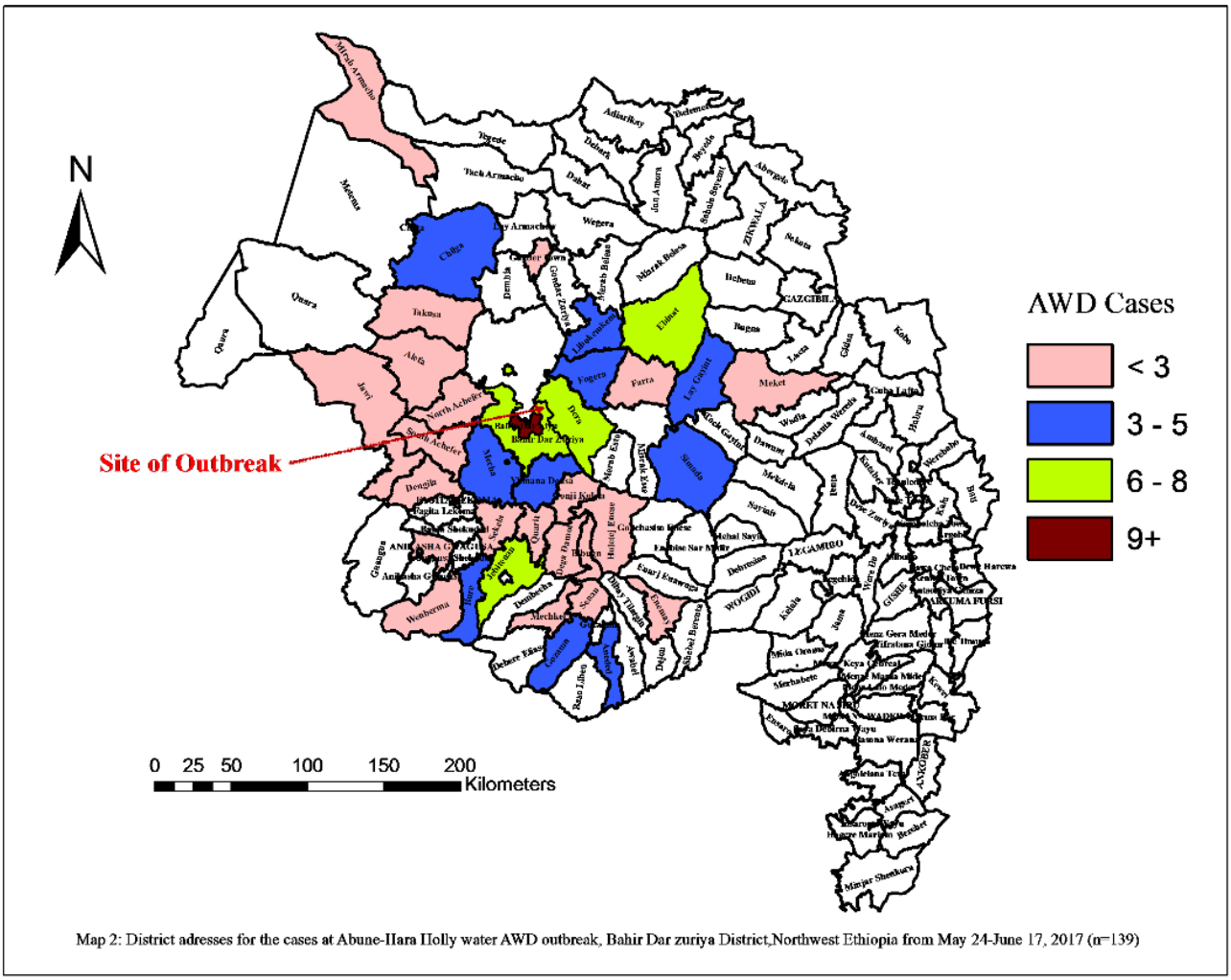


Figure 1.3. District Addresses for AWD patients at Abune Hara Holly, Bahir Dar Zuriya District, Northwest Ethiopia from May 24-June 17, 2017(n=139)

The onset date of the first case (index case) was on May 24th 2017 in Abune Hara holly water attendants' camp and the outbreak reached peak on May 27th (after four days). As the first case came to Addisalem Hospital on May 25th it was notified immediately to the next higher level (district health office, zone health department and region health bureau. The index case denied any travel history to AWD affected area. The outbreak was control after three weak (Figure 1.4).

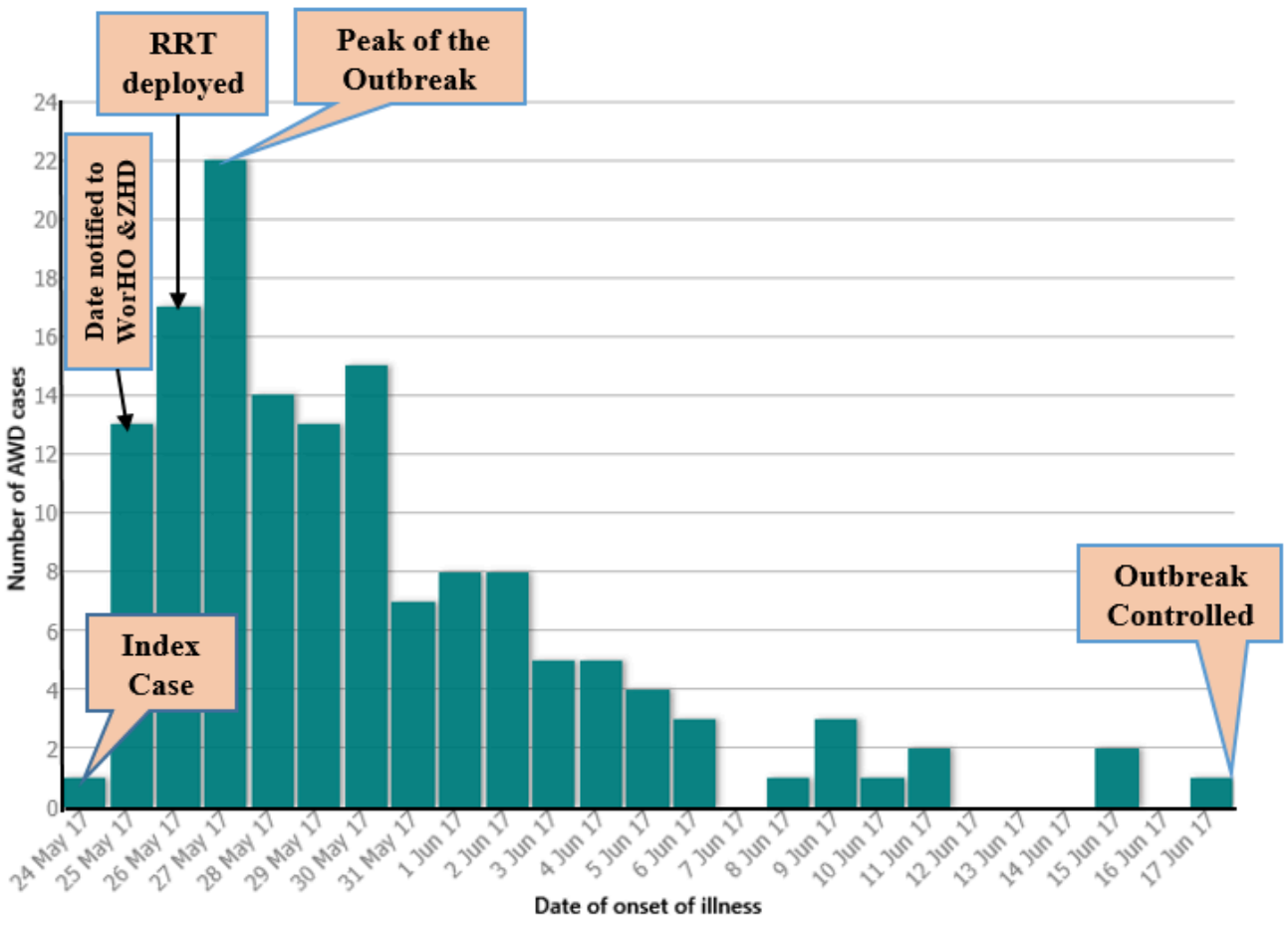


Figure 1.4. Epi-curve of Acute Watery Diarrhea (AWD) by date of onset in Abune Hara Holy Water, Bahir Dar Zuriya District, Northwest Ethiopia, 2017.

Clinical Features of the cases

From May 24 to June 17 2017, AWD outbreak caused over 141 clinically recognized cases and 3 deaths among Abune-Hara Holly water attendants and holly water attendant care givers living in the camp of the holly water. All cases had symptoms of Watery like rise water diarrhea with a frequency more than three loose stool per day. Ninety- five percent of the cases had vomiting and only 3.5 percent of the cases were reported as they had fever. More than four fifth (85%) of the cases were treated as inpatient, 64% were severely dehydrated and more than half (58%) of the cases were treated at the Case Treatment Center established at Addis Alem Hospital in Bahir Dar town (Table 1.2).

Table 1.2. Distributions of cases by clinical characteristics, Abune Hara holly water, Bahir Dar Zuriya district, Northwest Ethiopia, 2017 (n=141)

Clinical features, diagnosis and outcome	Frequency	Percent
Symptoms		
Watery like rise water diarrhea	141	100.0
> 3 lose stool per day	141	100.0
Vomiting	134	95.0
Abdominal cramps	61	43.3

Clinical features, diagnosis and outcome		Frequency	Percent
Fever		5	3.5
CTC where the Cases were admitted			
	Abune-Hara CTC	59	41.8
	Addis-Alem Hospital CTC	82	58.2
Service type			
	Inpatient (IP)	120	85.1
	Outpatient (OP)	21	14.9
Dehydration status			
	No dehydration	10	7.1
	Some dehydration	41	29.1
	Severe dehydration	90	63.8
Sample collection and Lab. result			
Specimen taken for RDT/Culture		0	0.0
Final classification of the cases			
	Probable	141	100.0
	Confirmed	0	0.0
Final outcome			
	Alive	138	97.9
	Died	3	2.1
	Total Cases	141	100

Exposure factors

In this study, all the participants (both cases & controls) did not access to latrine and they were defecated at open field. Sixty (43%) of the cases and 22 (8%) of the controls had close contact history in the last one weak**.** Despite the source of water for all participants were protected hand dug well, only 5 percent of the cases and 12 percent of the controls reported as receiving adequate water (at least 20 liters per day per capita). More than half (54%) of the controls and 28% of the cases stored water at home in narrow mouthed plastic container. However none of the participants were treated water before drinking. Thirteen percent of the case and nineteen percent of the controls eat/drink from the lounge of the holly water before a weak of the illness or interview date, 40% of the cases and three fourth (76%) of the controls ate their food immediately after cocked and only 11% of the cases and 11% of controls ate fresh fruit in the last week (Table 1.3).

Table 1.3. Distribution of respondents (cases and controls) by exposure factors at Abune-Hara Holy water, Bahir Dar zuria district, Northwest Ethiopia, 2017.

Exposure variables	Cases (n=141)	Controls (n=282)	Total (N=423)
Access to latrine	0.0	0.0	0.0
Open field defecation	100.0	0.0	100.0
Close contact history with AWD patients	42.6	7.8	19.4
Washing hands after defecation/toilet	17.0	64.9	48.9
Washing hands before food preparation	30.5	35.1	33.6
Washing hands before eating	76.6	98.9	91.5
Drank from the Holy water	95.7	94.7	95.0
Any travel history to an area affected with AWD in the last one week	0.0	0.0	0.0
Water source : protected hand dug well	100.0	100.0	100.0
Adequate water supply	5.0	12.4	9.9
Stored water at home in good condition (Narrow mouthed plastic container)	28.4	53.5	45.2
Treat water at home before drinking	0.0	0.0	0.0
Interruption of water supply (than any other before)	83.7	67.0	72.6
Eating/drinking in the lounge in a week before your illness	12.8	18.8	16.8
Lived in overcrowded room	94.3	78.4	83.7
Eat food immediately after cocked	39.7	75.5	63.6
Eat fresh fruit/vegetables in the last one weak	11.3	10.6	10.9

Multivariable Analysis

All the exposure variables were entered into simple binary logistic regression and then variables with p-value less than 0.25 were entered into multivariable logistic regressions to predict the association with AWD and exposures of 141 case-patients and 282 unmatched controls were compared.

The study revealed that hand washing after toilet (AOR=0.15, 95%CI=0.08-0.28), hand washing immediately before eating (AOR=0.10, 95%CI =0.02-0.41), storing water in narrow mouthed plastic container (AOR=0.42, 95% CI =0.23-0.78) and eating food immiadetely after cocked (AOR = 0.21, 95% CI =0.12-0.38) were statistically significant protective factors of the AWD illness. Whereas, close contact history with the cases (AOR=5.81, 95%CI=2.85-11.85), water interruption than any other before (AOR=4.30, 95%CI =2.09-8.85) and living in an overcrowded area (AOR=3.85, 95% CI=1.48-10.05) had shown statistically significant association with being a case. Multivariable logistic regression analysis revealed that AWD patients were more likely to have close contact history (AOR=5.81, 95%CI=2.85-11.85), experience water interruption than any other before (AOR=4.30, 95%CI =2.09-8.85) and live in an overcrowded area (AOR=3.85, 95% CI=1.48-10.05). (Table 1.4).

Table 1.4. Factors associated with the AWD outbreak in Abune Hara Holy water, Bahir Dar Zuriya district, Northwest Ethiopia, 2017

Exposure variables	Cases	Controls	Crude OR (95% CI)	Adjusted OR (95% CI)
	N (%)	N (%)		
Close Contact History with AWD patient	60 (42.6)	22 (7.8)	8.75 (5.06-15.15)	**5.81 (2.85-11.85)**
Washing Hands after defecation/toilet.	24 (17.0)	183 (64.9)	0.11 (0.07-0.18)	**0.15 (0.08-0.28)**
Washing hands before food preparation	43 (30.5)	99 (35.1)	0.81 (0.53-1.25)	0.75 (0.41-1.39)
Washing hands before eating	108 (76.6)	279 (98.9)	0.04 (0.01-0.12)	**0.10 (0.02-0.41)***
Drank from the Holly water	135 (95.7)	267 (94.7)	1.26 (0.48-3.33)	1.84 (0.43-7.826)
Adequate water supply	7 (5.0)	35 (12.4)	0.37 (0.16-0.85)	0.56 (0.19-1.64)
Use good water storage conditions in the house (Narrow mouthed plastic container)	40 (28.4)	151 (53.5)	0.34 (0.22-0.53)	**0.42 (0.23-0.78)***
Interruption of water supply than any other before	118 (83.7)	189 (67.0)	2.52 (1.51-4.21)	4**.30 (2.09-8.85)**
Eating/drinking in the lounge a week before your illness	18 (12.8)	53 (18.8)	0.63 (0.35-1.13)	1.09 (0.49-2.42)
Lived in overcrowded room	133 (94.3)	221 (78.4)	4.59 (2.13-9.89)	**3.85 (1.48-10.05)***
Eat food immediately after cooked	56 (39.7)	213 (75.5)	0.21 (0.14-0.33)	**0.21 (0.12-0.38)**
Eat fresh fruit/Vegetable in the last one weak	16 (11.3)	30 (10.6)	1.08 (0.56-2.05)	1.83 (0.65-5.17)

*P < 0.05, **P < 0.001

Observation findings

We visited and observed shelters of the community at the Holy water site. Despite the sources of drinking water were protected hand dug well, there was no latrine access, separate kitchen, and the floor of the living room was unpleasant to look at (*see photo June 2 2017 (Figure 1.5)*). The participants dried spoilage foods by sun light which in turn exposes to house flays to contaminate the food. We also observed that houses where cases came from were not disinfected. Since it was a religious place, no any health information was disseminated by the local health extension workers and other responsible health professionals routinely. The environment of living camps was contaminated due to open defecation.



Figure 1.5. Photo on June 2, 2017: A living camp (room 21) for the holy water attendants and a Holy water attendant cocking food inside the living room with a floor unpleasant to look at, Abune-Hara Holly water, Bahir Dar Zuriya District, Northwest Ethiopia)

We also observed the case management and infection prevention practices at the case treatment center (CTC). The CTC established at Abune-Hara Holy water was not standardized-i.e. separate zones for case screening, observation, admission and recovery were not available. It was not adequately staffed and patient charts (proper history, progress note, medication order, discharge summary, death summary) were not properly filled. Patient's vital sign were not followed closely and regularly.

1.5 Discussion

A total of 141 epidemiologically linked AWD cases were identified. Three deaths were occurred among the 141 cases admitted at the case treatment centers (CFR =2.1%). The case fatality rate was consistent with the case fatality rate reported nationally (2%) from beginning of 2017 to the epidemiological week 26, 2017[5], and low as compared to the AWD case fatality rate reported by another study in Afar (4.4%)[6]. However, it was high as compared to the WHO recommendation, case fatality rate < 1% if treatment is straightforward (basically rehydration), and if applied appropriately[3]. The reason to this high case fatality rate could be explained by late detection of the cases, poor case management, underlying factors such as malnutrition or low accessibility of health care facilities or case treatment centers.

The majority of the cases were males (57%). This could be due to poor housing cleanness and overcrowding as compared to females. Three-fourth of the cases were came from rural area, suggesting poor hygiene practices compared to those who came from urban area.

The Epi-curve depicted in figure 2 revealed one peak with sharp rise and fall, which could suggest a short incubation period of the case, point source for the outbreak and rapid response activity of the district epidemic task force.

Cases were less likely to practice hand washing after toilet (AOR=0.15, 95%CI: 0.08-0.28) and immediately before eating (AOR=0.10, 95%CI: 0.02-0.41) compared to controls. This indicated that individuals practicing hand washing at critical times (in this case before eating and after toilet) were less likely to develop the illness. Similarly, cases were less likely to store drinking water in narrow mouthed plastic container (AOR=0.42, 95%CI: 0.23-0.78) and to eat food immiadetely after cocked (AOR = 0.21, 95%CI: 0.12-0.38) compared to their counter parts. This indicated relatively safe storage of drinking water and eat food immiadetely after cocked were protective factors for AWD illness. It is obvious that it is due to the low chance of drinking water and food contamination. Generally, the study revealed that hand washing after toilet, hand washing immediately before eating, storing water in narrow mouthed plastic container and eating food immiadetely after cocked were protectives of AWD illness. These findings were biologically plausible and consistent with the findings of other studies conducted in Afar region[6], and Oromia region[8]. It is also supported by another evidence from the Ethiopian weekly epidemiological record[9].

Individuals having close contact history with the AWD case were more likely to develop the illness (AOR=5.81, 95%CI=2.85-11.85) compared to individuals without close contact history with AWD patients. This finding is consistent with another similar study conducted in Afar region[6]. In addition, individuals living in overcrowded rooms were more likely to have the case compared to their counter parts (AOR=3.85, 95% CI=1.48-10.05). This could be explained by the relatively high chance of close contact and contamination in overcrowded rooms. Cases were exposed more likely to water interruption than any other before compared to controls (AOR=4.30, 95%CI =2.09-8.85).This could be justified as individuals with lack of adequate water supply could have poor hygiene practices.

Lack of laboratory confirmation of cases was the major limitation of this study. Since, the setting was open and the population was not well defended, the population at risk was not known. So, the overall and age specific attack rates could not be calculated and this was another limitation of the study.

The individual's perceived expectation of diarrhea following Holy water drinking was the major challenge for early detection of cases and timely response activities. In addition, the refusal of the religious leaders to switch off the holly water temporarily for the entrance of new comers was other challenge to control the outbreak.

1.6 Conclusion

The overall case fatality rate of AWD was high. Close contact with a case, living in overcrowded room, water interruption than any other before were positively associated (risk) factors with AWD illness. Hand washing after toilet, hand washing before eating, storing water in narrow mouthed plastic container and eating food immiadetely after cocked were found to be negatively associated (protective) factors of the outbreak.

1.7 Recommendation

It is better if the district health office, in collaboration with other concerned bodies work more on:

- Promoting personal hygiene and sanitation practices
- Accessing adequate water supply to the Holy water attendants and religious pilgrims
- Improving the housing characteristics (to decrease overcrowededness of the shelters) at Holy water site

Public health actions

- A consultative meeting with the religious leaders of the holy water and other concerned Orthodox Church representatives was conducted.
- We established case treatment center at the Holy water site in addition to the treatment center at Addis Alem Hospital at Bahir Dar town, disseminated health information, distributed printed materials such as brushers and leaflets prepared by regional health and water and energy bureau.
- Supervising and supporting the case management team at the CTC to monitor and strengthen the case management protocols according to the standard and adherence to infection prevention and control.
- Active case searching was conducted in the Holy water attendants
- Daily feedback was provided to the district health office.
- The key findings were communicated to the public health emergency management case teams of West Gojjam zone health department, and Bahir Dar zuria district health office. In addition, the findings were presented at the 2nd annual scientific conference of Ethiopian field epidemiology and laboratory training program (EFELTP).

References

1. Haileamlak, A., Why is the Acute Watery Diarrhea in Ethiopia Attaining Extended Course? Ethiop J Health Sci, 2016. 26(5): p. 408. .
2. Ethiopian Health and Nutrition Research Institute, Federal Democratic Republic of Ethiopia. Guideline on chollera outbreak management Ethiopia. May 2011, Addis Ababa, Ethiopia.
3. WHO, Globlal Task Force on Chollera Outbreak. Chollera outbreak ; Assessing the outbreak response and improving prepardeness. Geneva, WHO/CDS/CPE/ZFK/2004.4. 2004.
4. UNICEF-Ethiopa (2016), AWD awareness Poster: available at: www.facebook.com/UNICEFETH/photos/a.439016859511842.1073741826.429787893768072/1085721648174690/?type=3&theater.
5. Wolrld Health Organization, Regional Office for Africa, Health Emergencies Programme. Weekly bulletin on outbreaks and other emergencies. Week 27 : 1 – 7 July 2017.
6. Beyene B, et al., Epidemiology of Acute Watery Diarrhea Outbreak and Challenges of Control—Afar, Ethiopia, 2009. International Invention Journal of Medicine and Medical Sciences, 2014. 1(10): p. 162-169.
7. Hosmer, D. and L. S, Applied Logistic Regresion. New York: John Wiley. 2000.
8. Bartels, S.A., et al., Investigation of a cholera outbreak in Ethiopia's Oromiya Region. Disaster Med Public Health Prep, 2010. 4(4): p. 312-7.
9. Outbreak news. Acute watery diarrhoeal syndrome, Ethiopia. Wkly Epidemiol Rec, 2006. 81(41): p. 385.

Annex 1.1: Information sheet and Informed consent form

Hello, I am _____ working for west Gojjam zone health department PHEM case team. We are investigating the acute watery diarrhea outbreak to identify the possible risk factors. We assumed that the cases may be resulted from poor hygiene and sanitation practices but we do not know what hiyegen and sanitation problems are. It could be through a contaminated food or drink or any one of a number of sources. Hence, we will ask you information on socio-demographic and exposure factors to learn the kinds of exposures that each group has and to compare the likelihood of specific exposures in cases and control groups. This will give us information about the types of exposures that may lead to infection.

Primarily, our study will include individuals who were diagnosed with AWD at Abune Hara holly water. We will also include a comparison group that includes individuals that who did not report clinical symptoms consistent with AWD and stayed in the Abune Hara holly water attendants' camp for the last one week

> Do you understand what I explained and may I continue?
>
> If yes, continue interviewing.

Annex 1.2: Questionnaires for AWD outbreak investigation

Participant ID: _____ Date of interview: _____

NO	QUESTIONS	CHOICE	SKIP
Section I Socio-demographic characteristics (Circle or fill the answers)			
1	Age (in years)	_____	
2	Sex	1 Male 2 Female	
3	Educational status	1 No education 2 Primary 3 Secondary 4 Higher	
4	Occupation	1 Farmer 2 Student 3 Merchant 4 Gov't worker 5 Other (Specify)	
5	Residence	1 Urban 2 Rural	
6	Region	1 Amhara 2 Other (Specify) _____	
7	Zone	_____	
8	District	_____	
9	Reason to came to the holly water:	1 Holly water attendant 2 Holly water attendant giver 3 Other (Specify) _____	
10	Subject:	1 Control 2 Case	Skip to Q 29
Section II: History of illness related to the outbreak (Circle or fill the answers)			
11	Date of onset of illness	_____	
12	Symptoms (Multiple answers are possible)	1 Acute diarrhea 2 Vomiting 3 Abdominal cramps 4 Fever 5 Rectal pain 6 Other (Specify)	
13	Type of diarrhea	1 Bloody 2 Watery like rise water 3 Mucous/Pus	
14	Frequency of diarrhea	1 < 3 loose stool/day 2 > 3 loose stool/day	
15	Location when symptom started	_____	
16	Date seen at HF/CTC	_____	
17	Name of the HF/CTC	_____	
18	Patient service	1 IP 2 OP	
19	Dehydration status	1 No dehydration 2 Some Dehydration 3 Severe Dehydration	
20	Treatment given	1 No 2 Yes	Skip to Q 21
21	Type of treatment given	1 ORS 2 IV Fluid 3 Antibiotics	
22	Patient outcome	0 Died 1 Alive	
23	Date notified to the next higher level	_____	
24	Specimen taken	1 No 2 Yes	Skip to Q 28
25	Type of specimen taken	1 Stool 2 Blood 3 Serum 4 Other (Specify)…	
26	Date specimen received to lab	_____	
27	Type of diagnostic test	1 RDT 2 Culture 3 Other	
28	Diagnostic test result	0 Negative 1 Positive	
29	Final Classification of the case	1 Suspected 2 Probable 3 Confirmed 4 Discarded	
Section III: Exposure factors (circle or fill the answers)			

NO	QUESTIONS	CHOICE	SKIP
30	Close contact history with AWD case in the last one weak	0 No 1 Yes	
31	Wash hands after toilet	0 No 1 Yes	
32	Wash hands before food preparation	0 No 1 Yes	
33	Wash hands immediately before eating	0 No 1 Yes	
34	Travel history to an area affected with AWD.	1 No 2 Yes	Skip to Q 45
35	Where did you go?	_____	
36	Ate/drank in the lounge before illness	0 No 1 Yes	
37	Lived in overcrowded area	0 No 1 Yes	
38	Eat food immediately after cocked	0 No 1 Yes	
39	Ate fresh fruit/vegetable in the last one weak	0 No 1 Yes	
40	Source of drinking water	0 No 1 Yes	
41	Interruption of water supply than any other before	0 No 1 Yes	
42	How many liters of water do you get per individual per day	_____	
43	Store water in good conditions in the house (narrow mouthed plastic container)	0 No 1 Yes	
44	Treat water at home before drinking	1 No 2 Yes	Skip to Q 45
45	What did you use to treat water	1 Boiling 2 Filtration 3 Chlorination 4 Others (specify) …	
46	Drank from the holly water	0 No 1 Yes	
47	Access to latrine	1 No 2 Yes	Skip to Q 48
48	Utilize the latrine	0 No 1 Yes	
49	Where did you defecate?	1 Open field 2 Other specify ….	

Thank you once again!

CHAPTER 2

Surveillance Data Analysis

A five year tuberculosis surveillance data analysis, West Gojjam zone, northwest Ethiopia, 2012/3-2016/7.

Tesfahun Taddege GEREMEW[2], Lemma Derseh, Getie Zeru and Meklit Mekonen

2. Correspondence to : ttaddege@gmail.com↩

Abstract

Background: Tuberculosis is a global concern for both developing and developed countries. Currently it becomes more complex due to increasing levels of drug resistance and HIV co-infection. Ethiopia is the one among the tuberculosis high burden countries and countries that contribute for high number of missed cases of tuberculosis. Analysis of TB surveillance data is essential for TB Programme evaluation, guiding planners and decision makers on TB prevention and control strategies. Hence, this analysis aimed to describe the trend in TB case notification and treatment outcomes in West Gojjam zone.

Methods: A five years secondary data analysis was carried out on 15,682 all forms of TB patients registered from July 2012/3 to June 2016/7 in West Gojjam Zone, Northwest Ethiopia. The data were extracted from the eHMIS database in excel spreadsheet. Descriptive measures were computed using SPSS Version 22 and ArcGIS 10.3 softwares.

Result: Over the five years, TB case detection rate rose from 45% in 2012/3 to 55.5% in 2014/5 and then declined to 54.8% in 2016/7. The proportion of TB in under-fifteen children among all forms accounted 10%. The most recent report revealed 114 of TB incidence rate (per 100,000 populations), 95% of treatment success rate and 3.1% of TB death rate. A total of 35 drug resistance TB cases were notified during the five years period.

Conclusion: The overall TB CDR in West Gojjam Zone was low. The unfavorable TB treatment outcomes have been increasing and a total of 35 Drug resistance TB cases were notified in five years period. The study result suggests a need to give more attention for TB case detection and unfavorable treatment outcomes.

Keywords: Tuberculosis, Case detection, Treatment outcomes, Northwest Ethiopia.

2.1 Introduction

Tuberculosis (TB) is an infectious disease caused by the bacillus Mycobacterium tuberculosis. It typically affects the lungs (pulmonary TB), but can also affect other sites (extra-pulmonary TB). The disease is spread via droplet infection when people with pulmonary TB expel the bacilli while coughing, sneezing, talking, etc.[1, 2].

Tuberculosis (TB) is a global concern for both developing and developed countries. Currently it becomes more complex due to increasing levels of drug resistance and HIV co-infection [3]. Since 1998, the concept of a “high burden country” (HBC) became familiar and widely used in the context of TB. Starting from 2015, three new HBC lists (HBCs for TB, TB/HIV and MDR-TB) have been defined to use for the period 2016–2020 [4].

For the first 5 years of the sustainable development goals (SDGs) and End TB Strategy (2016–2020), World Health Organization (WHO) has defined three lists of high burden countries (HBCs): for TB, TB/HIV and multidrug-resistant TB (MDR-TB)[5].

Ethiopia is among the high burden countries for drug susceptible TB, MDR-TB and TB-HIV co-infections as well as among countries that contribute for high number of missed cases of tuberculosis[6, 7]. TB remains one of the leading causes of mortality. According to the 2014 WHO report, the prevalence and incidence of all forms of TB are 211 and 224 per 100,000 of the population, respectively. Among TB patients with known HIV status, about 11% were HIV co-infected. According to the recent national TB drug resistance surveillance report, 2.3% of new TB cases and 17.8% of previously treated TB cases were estimated to have MDR[8, 9].

According to the 2016 Ethiopia Service Availability and Readiness Assessment (SARA) summary report, more than two-third (63 percent) of facilities excluding health posts and 36 percent of health posts offer TB services, more than half (52 percent) of the facilities excluding health posts have the capacity to diagnose TB by sputum smear microscopy examination[9]. Sixty-seven percent of Health facilities excluding health posts and 74 percent for health post have at least 1 trained staff for the diagnosis & treatment of TB[9]. Among facilities excluding health posts, 80 percent of them have a system for diagnosing HIV among TB clients[9].

2.2 Objectives

GENERAL OBJECTIVE

★ To assess the TB diseases burden and treatment outcomes in West Gojjam Zone, Northwest Ethiopia, 2012/3-2016/7.

SPECIFIC OBJECTIVES

1. To estimate TB disease burden in West Gojjam zone, Northwest Ethiopia
2. To describe the TB treatment outcomes in West Gojjam zone, Northwest Ethiopia.

2.3 Methods

Study design and setting

An institution-based cross-sectional study was conducted in West Gojjam zone, Northwest Ethiopia from June 15 to 30, 2017. West Gojjam zone is one of the eleven zones of Amhara National Regional State. Administratively, it is subdivided into thirteen districts and five town administrations. In 2016/7, the catchment population of the zone was 2,560,131 and it has 7 hospitals (1 private and 6 government), 103 health centers (83 TB diagnostic HCs) and 374 health posts (255 HPs started DOTs) and 154 private clinics (6 Diagnostic and DOTS).

Study subject

The zonal health department was the study subject

Operational definitions

According to the standard definitions of the national tuberculosis and leprosy control program guideline[2], the following clinical case and treatment outcome definitions were used:

- Tuberculosis case detection rate (all forms) is the number of new and relapse tuberculosis cases notified in a given year, divided by the estimated of the number of incident tuberculosis cases for the same year, expressed as a percentage.
- Cured: A patient with bacteriologically confirmed pulmonary TB at the beginning of treatment who was smear or culture-negative in the last month of treatment
- Treatment completed: A patient with TB who completed treatment without evidence of failure, but with no record of sputum smear or culture results, in the last month of treatment.
- Successful treatment outcome: If PTB patients were cured (i.e., negative smear microscopy at the end of treatment and on at least one previous follow-up test) or completed treatment with resolution of symptoms.
- Unsuccessful treatment outcomes: Unfavorable TB treatment outcomes which include failure (remaining smear-positive at the fifth month or later during treatment), Lost to follow-up (LTFU) (patients who interrupted their treatment for two consecutive months or more after registration), or death (patient who died from any cause during TB treatment).
- Rifampicin resistance TB (RR-TB): resistance to Rifampicin detected using phenotypic or genotypic methods, with or without resistance to other anti-TB drugs.
- Multidrug resistance (MDR-TB): resistance to at least both Isoniazid and Rifampicin.

Data source and extraction

The data were extracted from the electronic health management information system (eHMIS) database and TB prevention and control program annual reports through using semi-structured data extraction tool. The tuberculosis survielance data reported from 2012/3-2016/7 were included in this analysis.

Data management and statistical analysis

The data was entered into excel spreadsheet and analyzed using Microsoft-excel 2013, SPSS Version 22 and ArcGIS 10.3 softwares. We computed descriptive statistical measures and absolute number, percentage and rate were used to present the findings using words, tables, graphs and map.

Ethical considerations

A legal letter was obtained from the University of Gondar, department of biostatistics and epidemiology. Permission was obtained from West Gojjam zone health department, HMIS and TB control and prevention program officers prior to the data extraction.

2.4 Results

TB case detection rate (all forms)

A total of 15,682 patients with all forms of new and relapse TB were notified in the five years period. The absolute number of all forms of new and relapse TB declined from 3,474 in 2012/3 to 2,953 in 2016/7. The overall TB case detection rate (CDR) was 45% in 2012/3 and 55% in 2016/7. In the last year, the highest CDR (75%) was in Debub Achefer district and the lowest (34%) in Wonberma district (Table 2.1).

Table 2.1. The trend in TB case detection rates by district in West Gojjam Zone, Northwest Ethiopia from 2012/3 to 2016/7.

District	2012/3	2013/4	2014/5	2015/6	2016/7
Bahir Dar Zuriya	40.0	45.6	46.5	59.3	42.7
Bure Zuria	37.0	20.7	33.6	28.8	43.7
Burie Metema	95.0	96.5	84.7	74.9	55.6
Debub Achefer	37.0	55.5	73.5	81.4	75.1
Dega Damot	37.0	31.1	37.4	25.6	36.1
Dembecha	45.0	69.8	81.6	63.6	62.7
Finote-Selam Town	37.0	53.4	86.8	79.1	74.5
Gonji Kollella	40.0	32.1	33.5	46.3	43.9
Jabi-Tehinan	40.0	72.8	82.7	66.7	70.8
Mecha	55.0	56.0	54.7	49.1	54.1
Quarit	64.0	57.0	55.3	63.2	52.6
Sekela	36.0	36.4	37.6	38.8	50.9
Semen Achefer	40.0	37.3	46.5	35.7	57.1
Wonberma	37.0	43.5	40.6	38.1	34.4
Yilmana Densa	55.0	51.8	49.1	52.1	50.6
Overall case detection rate (CDR)	45.0	50.6	55.5	52.5	54.8
Number of all forms of new & relapse cases*	3,474	3,113	3,130	3,012	2,953

*includes extra-pulmonary TB (EPTB), smear negative PTB, new smear positive TB and relapse TB

The male to female sex ratio of all forms of new and relapse TB cases notified from 2014/5 to 2016/7 was 1.2, 1.1 and 1.2, respectively. Among all forms of new and relapse TB cases detected in each year, the percentage of extra-pulmonary TB (EPTB) cases increased from 52 % in 2014/5 to 58% in 2016/7, whereas, the percentage of relapse cases remained nearly the same, 3.7% (Table 2.2).

Overall, children (aged <15 years) accounted for 10% of the new and relapse cases that were notified in 2016/7 with the highest percentage (14.4) in Wonberma district and lowest (2.3) in Sekela district. Figure 2.2 showed the shade of the districts correspond to the magnitude of the percentage; the darker the shade, the higher the value (Figure 2.2).

Table 2.2. Distribution of all forms of TB cases by sex, age and TB category among total new and relapse TB cases from 2014/5-2016/7

Background Characteristics	2014/5	2015/6	2016/7
Sex			

Background Characteristics	2014/5	2015/6	2016/7
Male	54.8	52.8	53.6
Female	45.2	47.2	46.4
Age (in years)			
< 5	1.3	1.0	1.5
05-14	8.3	9.1	8.6
15-24	24.1	23.0	25.0
25-34	23.8	25.2	22.7
35- 44	17.4	17.7	17.0
45-54	13.5	11.8	12.6
55-64	6.5	7.3	6.4
65+	3.8	3.4	4.0
Unknown/NA	1.2	1.7	2.2
New TB cases (all forms)	96.2	96.3	96.3
Pulmonary TB (PTB)	44.3	39.1	38.1
Smear positive PTB	20.2	19.6	18.4
Smear negative PTB	24.2	19.5	19.7
Extra-pulmonary TB (EPTB)	51.9	57.2	58.2
Relapse cases	3.8	3.7	3.7
Smear positive PTB relapses	2.6	2.1	1.6
Other relapses*	1.2	1.6	2.1
Number of all forms of new and relapse cases	3,130	3,012	2,953

*include cases that have been re-registered as treatment after failure, lost to follow up or as other previously treated (whose outcome after the most recent course of treatment is unknown or undocumented).

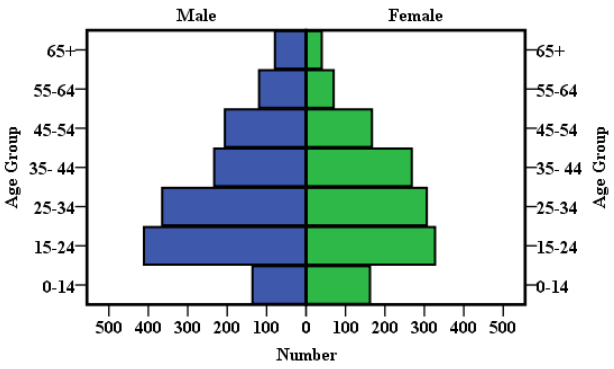


Figure 2.1. TB case notification disaggregated by age and sex (male in blue; female in green), West Gojjam zone, 2016/7.

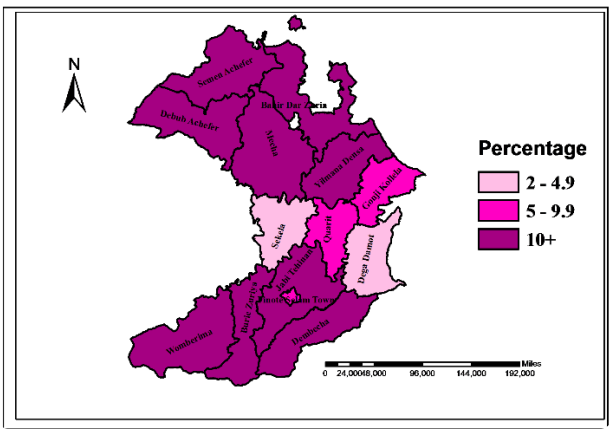


Figure 2.2. Percentage of new and relapse TB cases in under-fifteen year children, 2016/7.

Figure 2.3 revealed the geographical distribution of Tuberculosis case detection rate (all forms) by district from 2013/4 to 2016/7.The shade of the district corresponds to the magnitude of the indicator. The darker the shade, the higher the value. The district with the highest value in the zone is Finote- Selam town, with a value of 75 percent. The district with the lowest value in the zone is Wonberma, with a value of 34 percent.

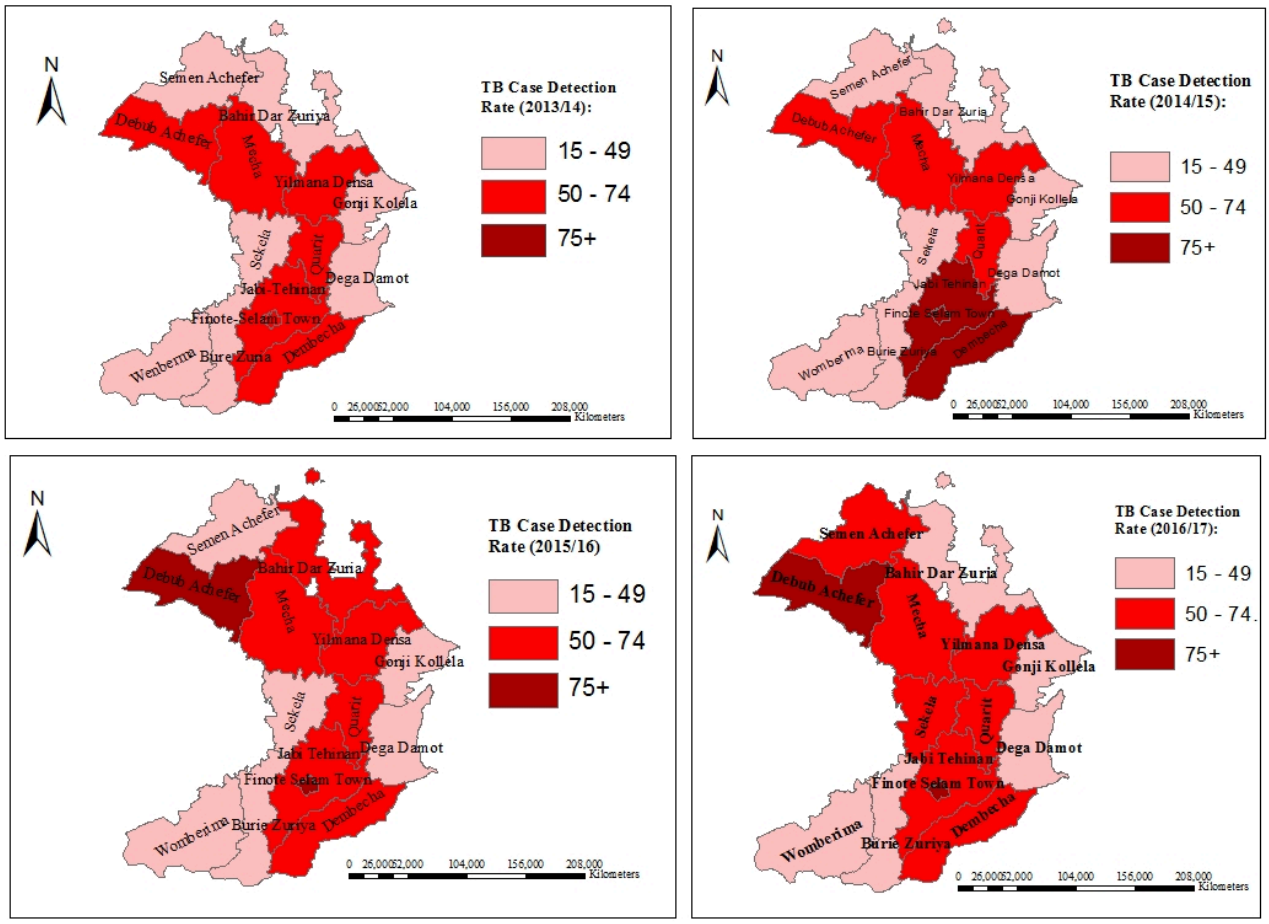
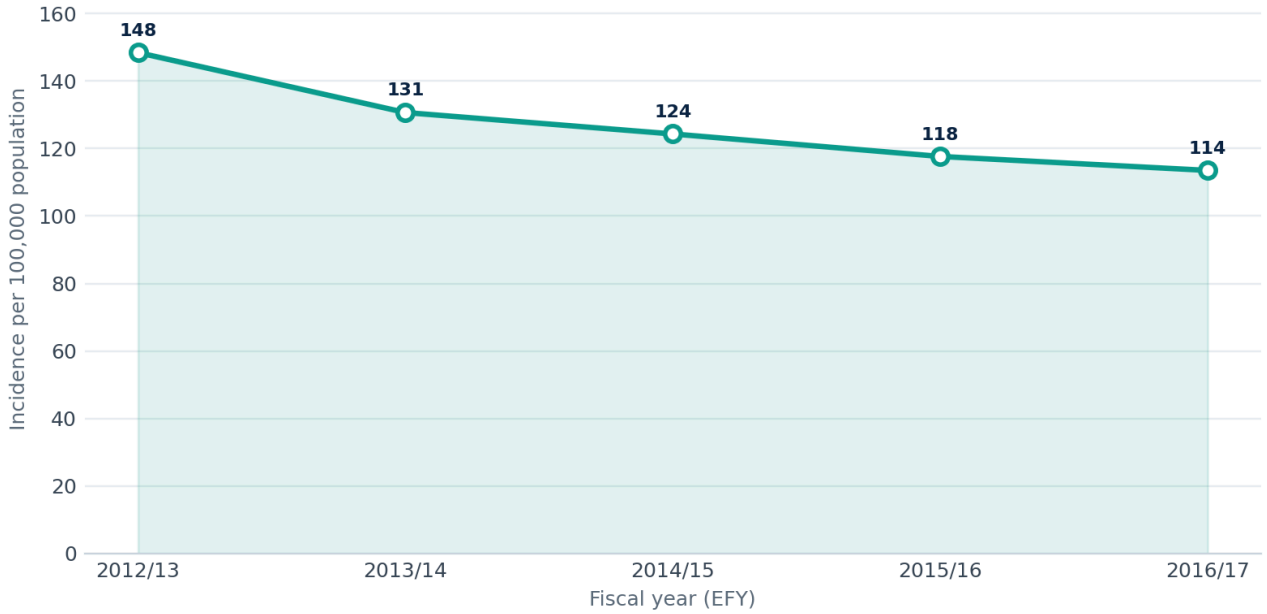


Figure 2.3. TB CDR by district, West Gojjam zone, Northwest Ethiopia (20113/4-2016/

Incidence of Tuberculosis

Over five years period, the annual TB incidence rate (per 100,000 population) declined from 148 in 2012/3 to 114 in 2016/7 (Figure 2.4). The annual number of incident TB cases relative to population size (the incidence rate) varied among the districts in 2015/6, from 25 to 99 per 100 000 population in one-third of the districts to 100–199 in two-third of the districts (Figure 2.5).



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 2.4. Trend of TB incidence rate per 100,000 population in West Gojjam Zone, Northwest Ethiopia, 2012/3 - 2016/7.

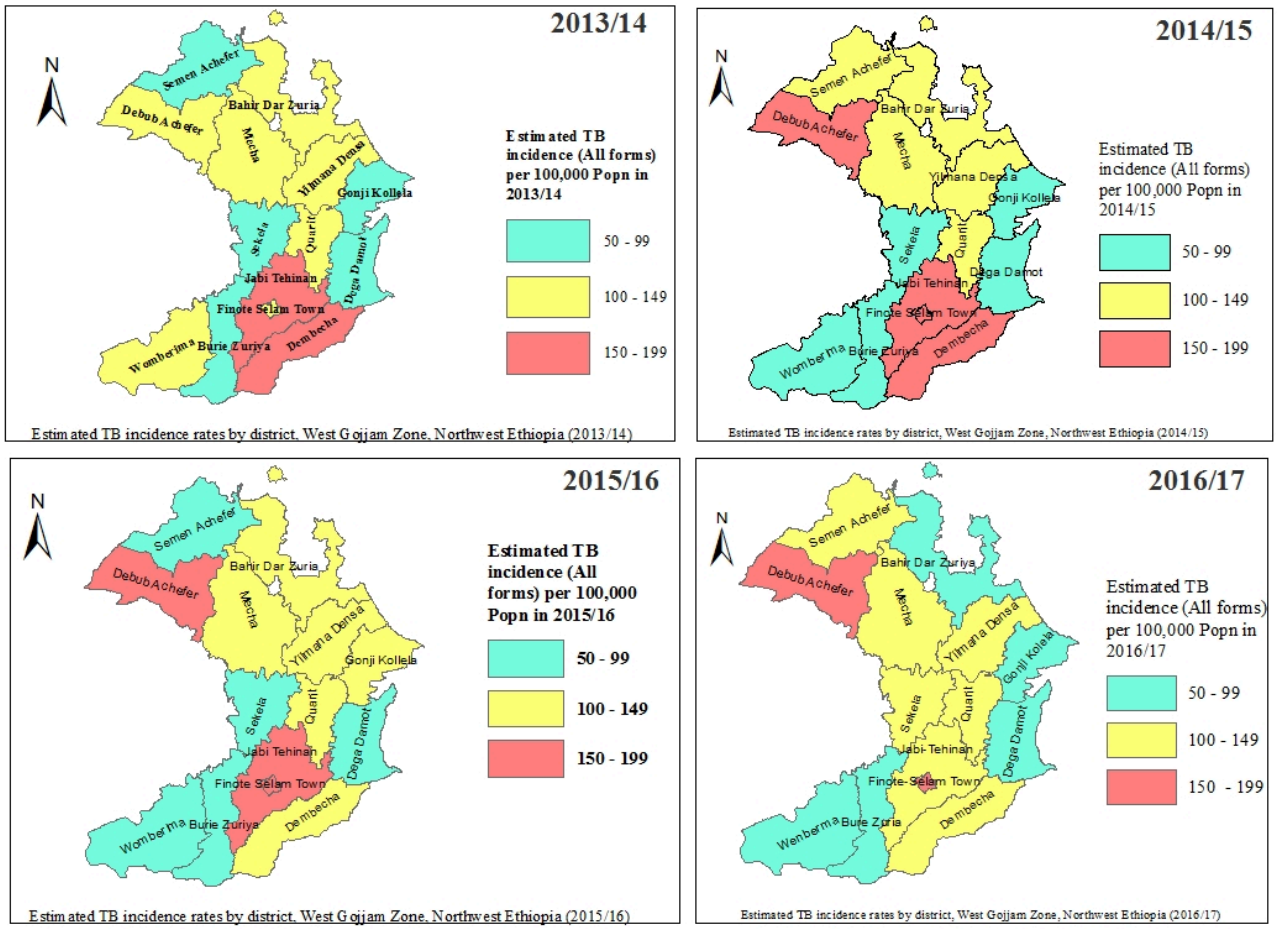
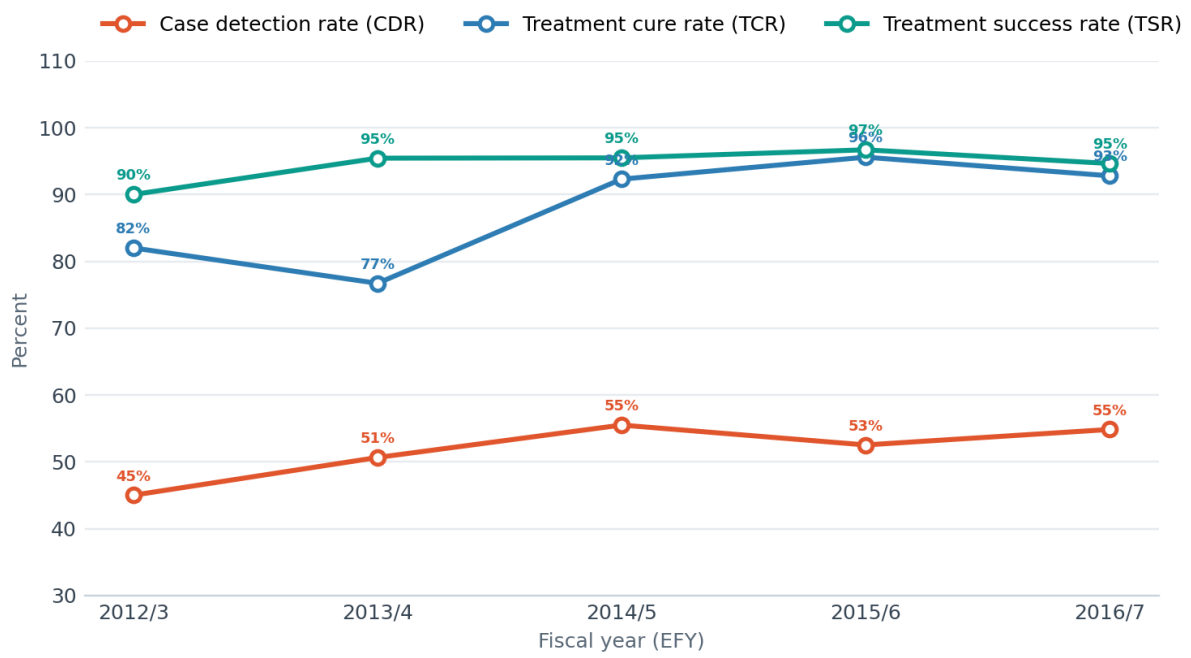


Figure 2.5. Estimated TB incidence rate by district, West Gojjam Zone, Northwest Ethiopia, 2012/3 - 2016/7

TB treatment outcomes

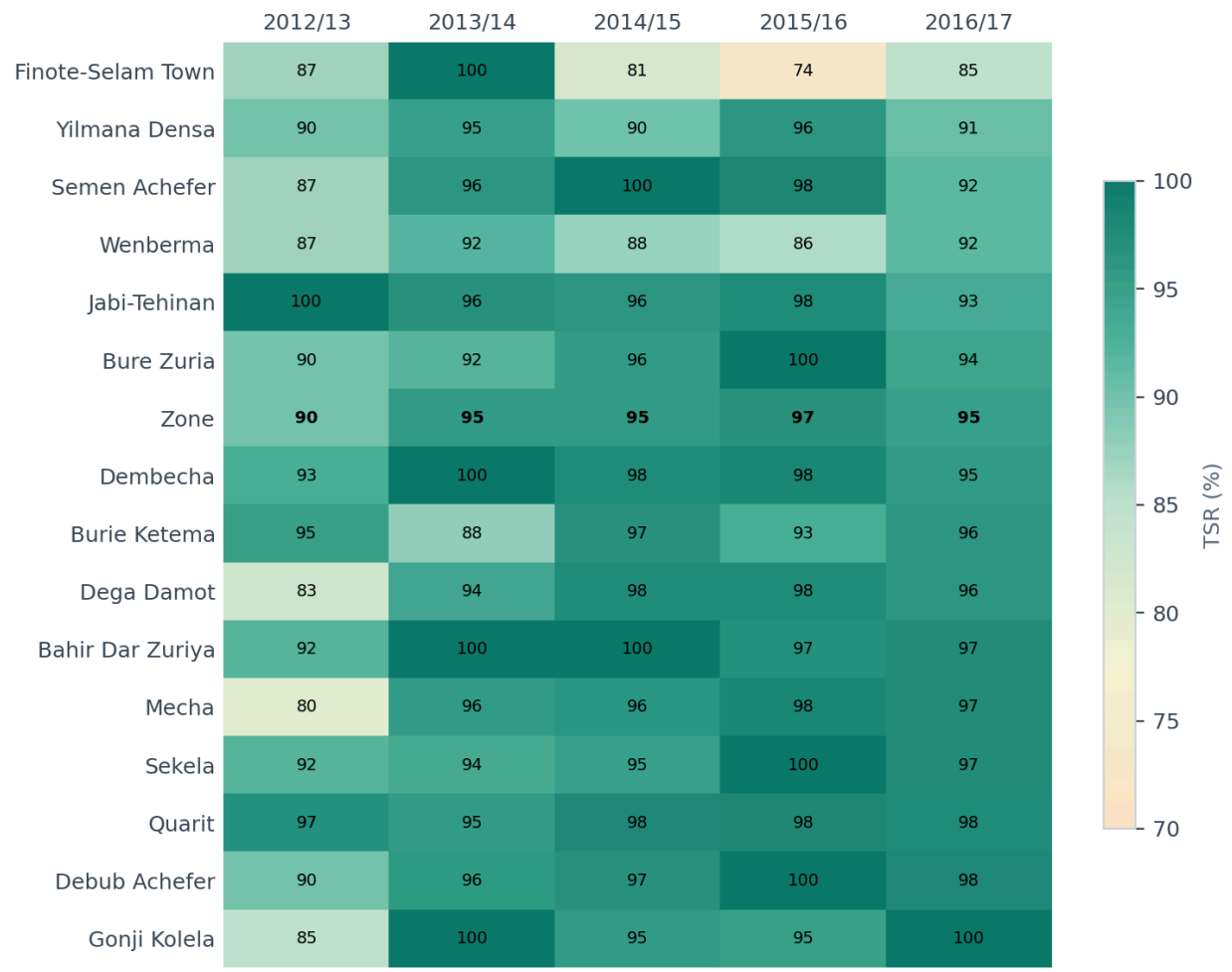
Successful treatment outcomes

TB treatment cure rate (TCR) was 82% in 2012/3 and 93% in 2016/7. The TB treatment successful rate (TSR) was increased from 90% in 2012/3 to 95% in 2016/7 (Figure 2.6). As figure 2.7 shows the trend of TB TSR by district, relatively higher decrement was observed in Finote-Selam town from 100% in 2014/4 to 85% in 2015/6.



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 2.6. Trend in TB CDR, TCR and TSR, West Gojjam Zone, Northwest Ethiopia, 2012/3 - 2016/7

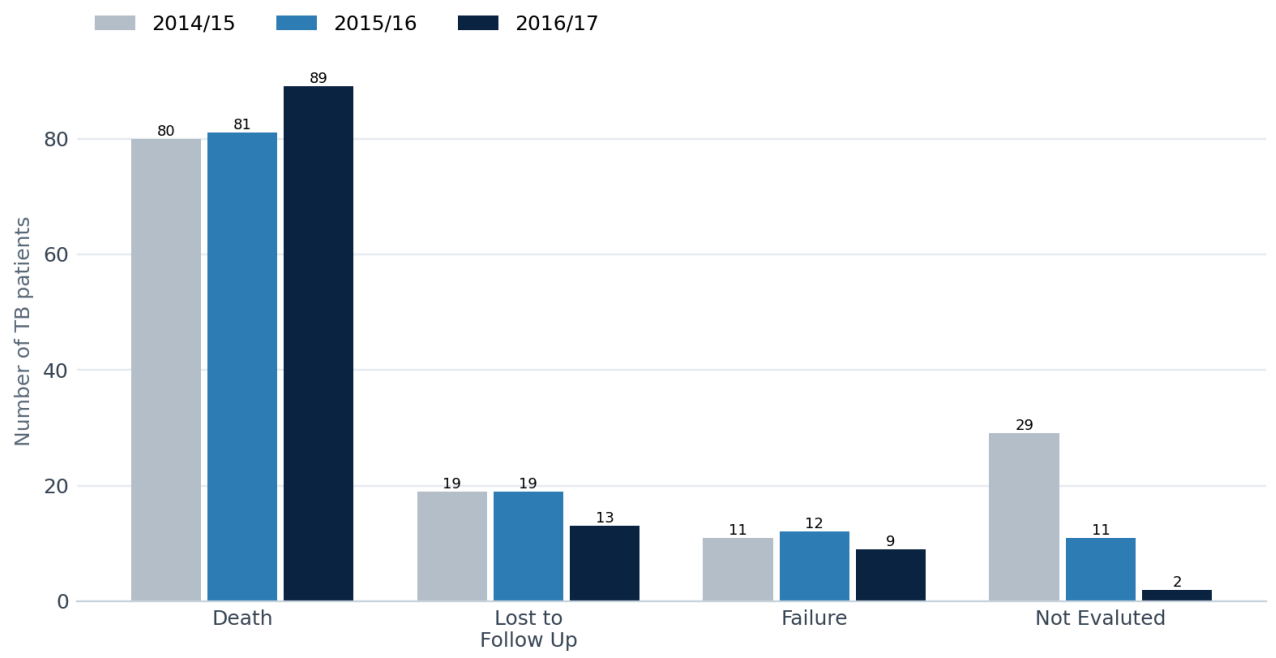


Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 2.7. Trend in TB TSR by district, West Gojjam Zone, Northwest Ethiopia, 2012/3 - 2016/7

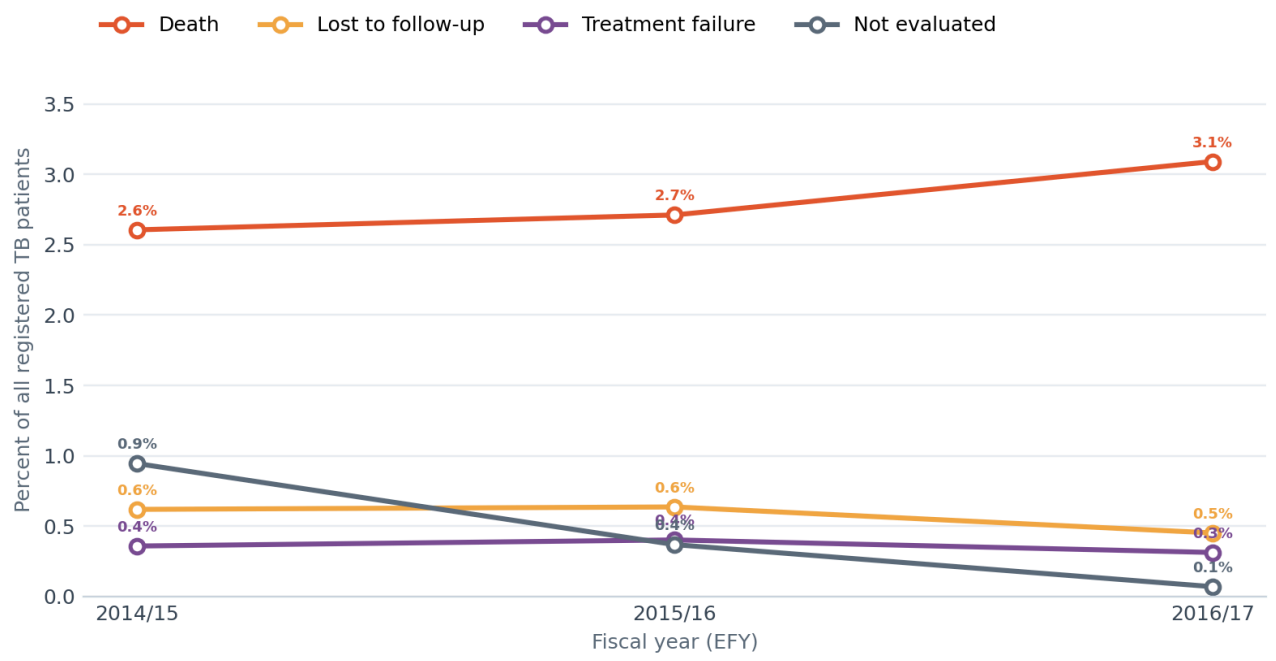
Unsuccessful treatment outcomes

The number of tuberculosis death is rising slowly, in both absolute numbers and death rates (Fig. 2.8 and Fig. 2.9). In zone, the total number of TB death was 80 (2.6%) in 2014/5, 81 (2.7%) in 2015/5 and 89 (3.1%) in 2016/7. The TB treatment failure rate was 3.6 percent in 2014/5 and nearly the same until 2016/7 (3.1 Percent).



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

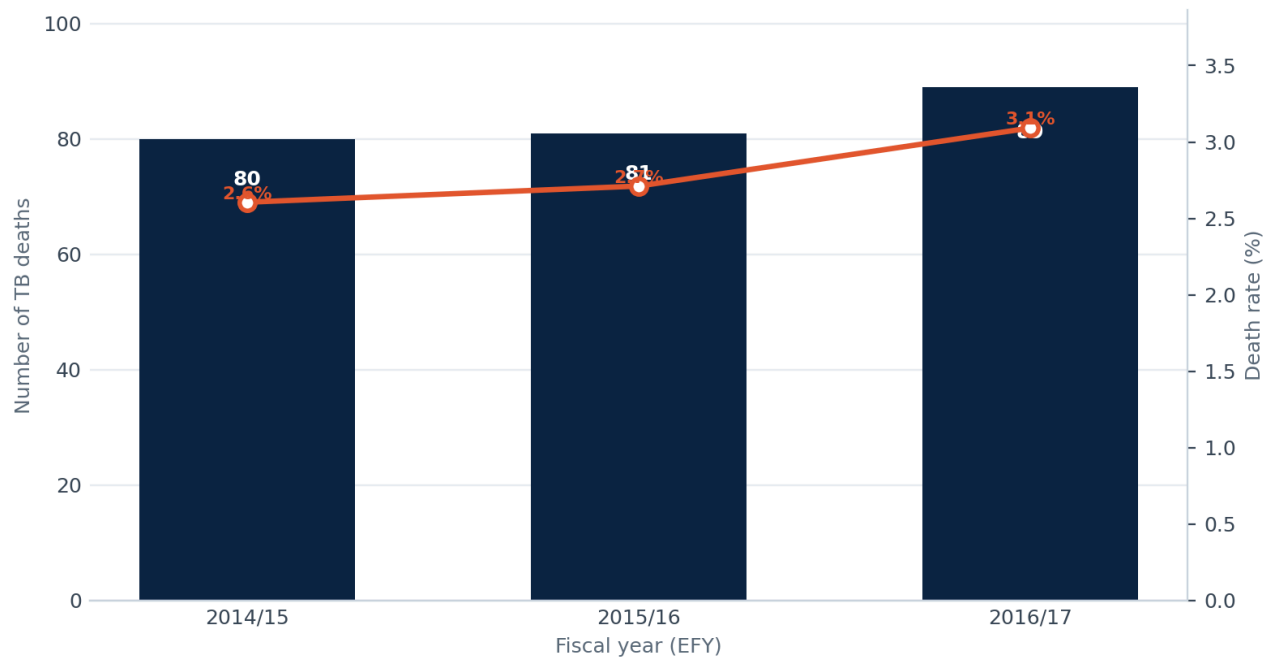
Figure 2.8. Trend in Unsuccessful treatment outcome (absolute numbers) West Gojjam Zone (2012/13-2016/17)



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

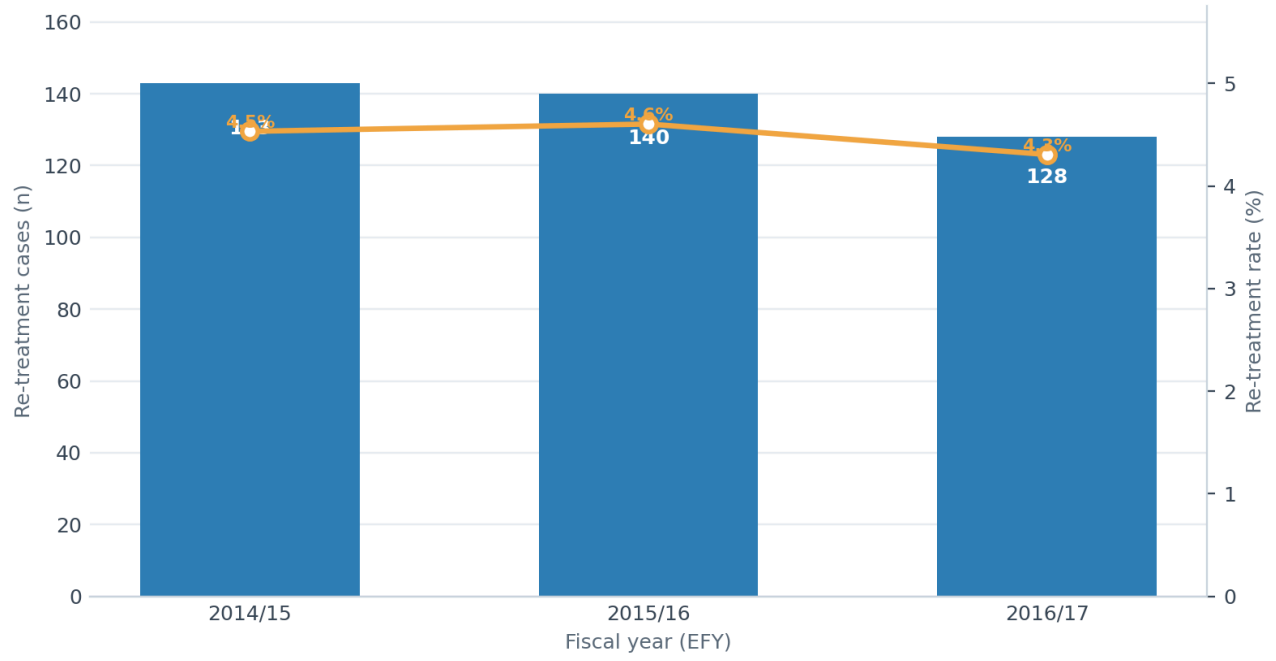
Figure 2.9. Trend in unsuccessful treatment outcomes, West Gojjam Zone, 2012/3-2016/7

The most recent report showed 3.1% of TB deaths rate and 4.3% of TB retreatment rate in 2016/7 (Figures 2.10 and 2.11).



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 2.10. Trend in TB death Rate, West Gojjam Zone, 2013/4-2016/7



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 2.11. Trend in TB retreatment cases, West Gojjam Zone, 2014/5-2016/7

Community contributions to TB notifications and treatment support

In 2016/7, the percentage of notified TB patients by community referrals accounted for 11%, with lowest notification (2.7%) in Gonji-Kollella district and highest notification (18.2%) in Debub Achefer district. Overall, the community contributions to TB case notifications and treatment adherence support for TB patients were 11% and 24%, respectively (Table 2.3).

Contribution of private health facilities to TB case notifications

Of the total 2,953 all forms of TB cases notified in 2016/7, 28% is attributed to private health facilities with the highest (55%) contribution of private health facilities to the district case notification in Gonji Kollella district and lowest (13%) in Dembecha(Table 2.3)

Table 2.3. Community contributions to TB case notifications and treatment adherence support for TB patients (all forms) in districts of West Gojjam Zone, Northwest Ethiopia, 2016/7

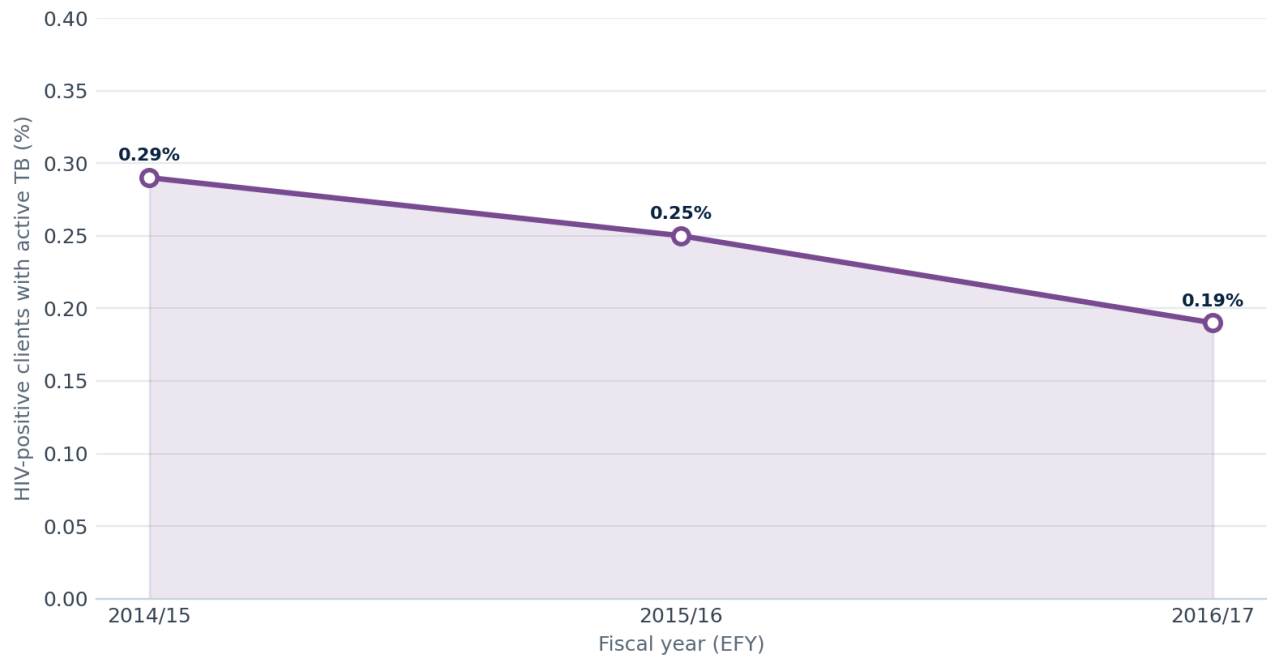
District	Number TB cases (all forms) notified	TB notifications from community referrals		TB patients given treatment adherence support by community health workers		TB cases (all forms) diagnosed through private health facilities	
		Number	Percent	Number	Percent	Number	Percent
Bahir Dar Zuriya	195	21	10.8	59	30.3	49	25.1
Bure Zuria	115	10	8.7	31	27.0	44	38.3
Burie Town	65	6	9.2	10	15.4	21	32.3
Debub Achefer	275	50	18.2	61	22.2	95	34.5
Dega Damot	184	26	14.1	30	16.3	28	15.2
Dembecha	203	20	9.9	63	31.0	26	12.8
Finote-Selam	68	2	2.9	1	1.5	22	32.4
Gonji Kollella	112	3	2.7	43	38.4	61	54.5
Jabi-Tehinan	320	41	12.8	63	19.7	57	17.8
Mecha	427	54	12.6	119	27.9	102	23.9
Quarit	146	9	6.2	46	31.5	77	52.7
Sekela	171	31	18.1	20	11.7	71	41.5
Semen Achefer	187	15	8.0	60	32.1	68	36.4
Wonberma	90	8	8.9	23	25.6	18	20.0
Yilmana Densa	283	24	8.5	81	28.6	84	29.7
Total	2,953	320	10.8	710	24.0	836	28.3

TB screening for HIV positive Clients

TB screening for HIV positive clients is one of the HMIS indicators related to TB/HIV co-infection. Among the total number of 80,069 HIV positive clients who received HIV care, about 78,927 (99%) clients were screened for tuberculosis in the last three years and 189 (0.24%) of the were diagnosed with active TB. The percentage of HIV positive clients screened for TB and confirmed with active TB was declined from 0.29 percent in 2014/5 to 0.19 percent in 2016/7 (Table 2.4 and figure 2.12).

Table 2.4. Percentage of HIV positive clients screened for TB and HIV positive clients confirmed with active TB, West Gojjam zone, Northwest Ethiopia, 2014/5 - 2016/7.

Years	Number HIV positive clients who received HIV care	HIV positive clients screened for TB		HIV positive clients screened for TB and confirmed with active TB	
		Number	Percent	Number	Percent
2014/5	22,666	21,703	95.8	64	0.29
2015/6	28,365	28,187	99.4	71	0.25
2016/7	29,038	29,037	100.0	54	0.19
Total	80,069	78,927	98.6	189	0.24



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 2.12. Trend in proportion of HIV positive clients screened for TB and confirmed with active TB in West Gojjam zone, Northwest Ethiopia, 2014/5-2016/7.

Drug Resistance Tuberculosis

A total of 35 drug resistance tuberculosis (DR TB) cases were notified from 2012/3 to 2016/7. Nearly two-third of the patients were males and under-five children accounted for 6%. The majority of the cases (80%) were smear positive pulmonary TB (Table 2.5).

Table 2.5. Percentage distribution of drug resistant TB cases by background characteristics in West Gojjam, Northwest Ethiopia, 2012/3-2016/7

Background characteristics	Yearly notification					Overall notification
	2012/3	2013/4	2014/5	2015/6	2016/7	
Sex						
Female	71.4	42.9	10.0	28.6	25.0	34.3

Background characteristics	Yearly notification					Overall notification
Male	28.6	57.1	90.0	71.4	75.0	65.7
Age Group						
< 15	0.0	0.0	10.0	0.0	25.0	5.7
15 - 24	28.6	42.9	40.0	14.3	0.0	28.6
25 - 34	28.6	14.3	40.0	42.9	25.0	31.4
35 - 44	14.3	42.9	10.0	28.6	25.0	22.9
45 - 54	14.3	0.0	0.0	0.0	0.0	2.9
55 - 64	0.0	0.0	0.0	14.3	25.0	5.7
65+	14.3	0.0	0.0	0.0	0.0	2.9
District						
Bahir Dar Zuriya	0.0	14.3	0.0	0.0	0.0	2.9
Burie Town	28.6	14.3	10.0	14.3	0.0	14.3
Debub Achefer	14.3	0.0	0.0	0.0	0.0	2.9
Dega Damot	0.0	0.0	0.0	14.3	25.0	5.7
Dembecha	14.3	14.3	30.0	14.3	0.0	17.1
Finote-Selam Town	14.3	0.0	0.0	0.0	25.0	5.7
Gonji Kollella	0.0	0.0	20.0	0.0	0.0	5.7
Jabi-Tehinan	28.6	0.0	0.0	14.3	25.0	11.4
Mecha	0.0	14.3	10.0	0.0	0.0	5.7
Sekela	0.0	14.3	0.0	14.3	0.0	5.7
Semen Achefer	0.0	0.0	10.0	0.0	0.0	2.9
Wonberma	0.0	0.0	0.0	0.0	25.0	2.9
Yilmana Densa	0.0	28.6	20.0	28.6	0.0	17.1
TB category						
EPTB	14.3	0.0	10.0	28.6	25.0	14.3
Smear negative PTB	0.0	0.0	10.0	14.3	0.0	5.7
Smear positive PTB	85.7	100.0	80.0	57.1	75.0	80.0
HIV Status						
Non-reactive	100.0	100.0	80.0	14.3	50.0	71.4
Reactive	0.0	0.0	20.0	71.4	50.0	25.7
Unknown	0.0	0.0	0.0	14.3	0.0	2.9
Resistance type						
RR-TB	57.1	42.9	0.0	0.0	0.0	20.0
MDR-TB	42.9	57.1	100.0	100.0	100.0	80.0
Treatment Outcome						
Completed	14.3	0.0	10.0	0.0	0.0	5.7
Cured	42.9	71.4	10.0	0.0	0.0	25.7
Died	28.6	0.0	0.0	0.0	0.0	5.7
LTFP	14.3	28.6	10.0	0.0	0.0	11.4
Transferred out	0.0	0.0	20.0	0.0	0.0	5.7
On treatment	0.0	0.0	50.0	100.0	100.0	45.7
Number of DR-TB cases	7	7	10	7	4	35

2.5 Discussion

After the end of the eras of the MDGs and Stop TB Strategy, WHO's global TB programme initiated the development of a post-2015 global TB strategy called End TB Strategy. Early diagnosis TB cases and treatment of all people with active TB are among the components of the Pillar 1 of the End TB Strategy [1, 10].

In this analysis, we found that tuberculosis cases detection rate of West Gojjam zone has increased from 45% in 2012/3 to 55 % in 2016/7. Despite this increment, it is still below the national target for the year 2016/7, 81%[6]. However, it is comparable to the 2015 national TB case detection rate (54%)[6]. This may indicate poor tuberculosis surveillance systems (i.e. the number of cases may be missed by surveillance systems).

Although it is quite difficult to estimate TB incidence because of it would require long-term studies among cohorts of people, which would involve high costs and challenging logistics, notifications of TB cases provide a good proxy indication of TB incidence in areas that have high-performance surveillance systems (e.g. with little underreporting of diagnosed cases), and in which the quality of and access to health care means that few cases are not diagnosed. Hence, the TB incidence rate was 114 cases per 100,000 populations per year in 2016/7. This indicates low TB incidence rate (per 100 000 population per year) compared to the 2016/7 national estimated target,191 cases [6]. This finding was also lower than the global TB report for Ethiopia in TB incidence estimates (per 100,000 population per year) , 192 (95%CI:142-250) in 2015[10], 182 (95% CI: 128-245) in 2016[5], and 164(95%CI: 115-221) in 2017[1]. However, this low TB incidence rate in West Gojjam zone may not indicate have high-performance surveillance system rather it may be due to poor performance surveillance system (high number of missed cases, underreporting of diagnosed cases) and low community engagement in TB case detection. The trend of TB incidence rate has shown a steady decline at an average rate of 8.7% per year since 2012/3, while the national TB incidence fell at an average rate of about 3.6% per year between 2000 and 2013[6].

Of the 2,953 new and relapse TB patients notified in 2016/7, 58% were extrapulmonary TB (EPTB) followed by smear negative PTB (20%) and smear positive PTB (bacteriologically confirmed) (18%). This finding was supported by other studies reported elsewhere in the country[11-14]. This finding was inconsistent to the 2017 global estimate of EPTB, 14% [1] and a local finding from Gambella region in 2014[15]. The higher proportion of EPTB might be due to the over diagnosis of EPTB using clinical data, imaging and pathologic evidence. Reasons for a low proportion of cases being bacteriologically confirmed should be assessed at zonal level. The microbiological detection of TB is critical for infection control because it allows patients to be correctly diagnosed and started on the most effective treatment regimen as early as possible. Most clinical features of TB and abnormalities on chest radiography or histology results generally associated with TB have low specificity, which may lead to false diagnoses of TB, and hence to people being enrolled in TB treatment unnecessarily.

Notification rate of TB was highest among young people and adults (in the 15–34 year age category) and then declined with age. This may be as a result of higher HIV prevalence in young people and adults compared to children and elders. Children (aged <15 years) accounted for 10% of the new and relapse cases that were notified in 2016/7. This finding is higher than the global proportion of children (aged <15 years) among the new and relapse cases that were notified in 2016, 6.9%[5]. The proportion of children (age <15 years) varies across districts ranged from 2.3% in Sekela district to 14.4% in Wonberma district. This variation in the percentage of childhood cases among districts in may reflect real differences in epidemiology, differences in access to or use of reliable health care services, or differences in reporting practices[16].

The treatment success rate for the 597 new and relapse cases who were treated in the 2015/6 cohort was 95%, which was comparable to the national target for TB treatment success rate, (95%)[6]. This finding was relatively higher than the WHO minimum recommendation for successful outcomes (85 %), the global treatment success rate for new and relapse cases who were treated in the 2015 cohort, 83%[5], and 2016 cohort, 82%[1]. It is also relatively high as compared to the findings of local studies in the Northern Ethiopia, 81% in Felege-Hiwot referral hospital[8], 87% in Debre Tabor[13], and 89% in Tigray[11]. This relatively improved TB treatment success rate could be as a result of an improved access to TB control services, particularly community-TB care and the expansion of TB treatment centers.

In the latest TB treatment outcome data, the unsuccessful treatment outcomes (default rate (0.5) and death rate (3.1%)) found in this study were lower than the findings of studies conducted in northwest Ethiopia (1.7% of default rate and 8.5% of death rate)[17] and Northern Ethiopia, 3.2 default rate and 3.9 death rate[11]. However, the trend of TB death rate has steadily increased over time. This may be due to poor quality and effectiveness of the treatment service, higher HIV prevallenece.

The role of community engagement in contributing to TB prevention, diagnosis and treatment, especially where people with TB have poor access to formal health services, is well-recognized and it can increase TB case detection and notifications and also help to improve treatment outcomes [7]. Fostering such community participation is an explicit component of the *Stop TB Strategy* and a “strong coalition with communities” is one of the four principles underpinning the post-2015 global TB strategy[7]. This analysis showed that about 68% of health posts were providing DOTS services (treatment follow up), which was high as compared to the finding of the 2016 Ethiopia service availability and readiness assessment (ESARA), 36%[9]. However, it was below the national target to reach 80% of the health posts[6]. In addition, this analysis revealed that the community contributions to TB case notifications and treatment adherence support for TB patients (all forms) were 11% and 24%, respectively in 2016/7. The proportion of total TB notifications (all forms) from community referrals (11%) was consistent with national level community contributions to TB case notifications, 11%[7].

Both TB and HIV programmes need to routinize HIV testing of people with diagnosed and presumptive TB as well as TB screening among HIV-positive people to ensure early access to life-saving interventions[7]. This study showed the percentage TB cases among HIV positive clients screened for TB declined from 0.29 to 0.19 since 2014/5 to 2026/7. The decrease in this indicator may suggest that a lower proportion of PLWH were being screened for TB, a turnover in trained staff, decreased supervision visits, shortage of screening tools.

2.6 Conclusion

The overall TB case detection rate and incidence rate in West Gojjam zone were low. The TB treatment success rate was encouraging, despite increased trend of TB death rate. TB among HIV patients has been declined. Community-based TB care was low.

2.7 Recommendations

To improve the TB case notification and treatment outcome, it is better If West Gojjam zone department TB prevention and control program work more on:

- Improving the TB case notification across all districts
- Maintaining the successful treatment outcomes
- Engaging community to TB prevention, diagnosis and treatment.

References

1. *World Health Organization. (2018). Global tuberculosis report 2018. World Health Organization. www.who.int/iris/handle/10665/274453. License: CC BY-NC-SA 3.0 IGO.*
2. *FMOH (2013). Guidelines for clinical and programmatic management of TB, TB/HIV and leprosy. 5th edition.*
3. Zenebe, Y., et al., *Profile of tuberculosis and its response to anti-TB drugs among tuberculosis patients treated under the TB control programme at Felege-Hiwot Referral Hospital, Ethiopia.* BMC Public Health, 2016. **16**: p. 688.
4. WHO, *Use of high burden country lists for TB by WHO in the post-2015 era: Summary.*
5. *World Health Organization (2017). Global tuberculosis report 2017. Geneva: World Health Organization.*
6. *FMOH (2015). Health Sector Transformation Plan (HSTP) 2015-2020 (EFY 2008-2012). Addis Ababa, Ethiopia.*
7. *World Health Organization (2014). Global tuberculosis report 2014. World Health Organization. www.who.int/iris/handle/10665/137094.*
8. *Zenebe, Y., et al., Profile of tuberculosis and its response to anti-TB drugs among tuberculosis patients treated under the TB control programme at Felege-Hiwot Referral Hospital, Ethiopia. BMC Public Health, 2016. 16: p. 688.*
9. *Ethiopia. Service Availability and Readiness Assessment. 2016 Summary Report. Ethiopian Public Health Institute. Addis Ababa, Ethiopia. Federal Ministry of Health. Addis Ababa, Ethiopia. World Health Organization (WHO). Addis Ababa, Ethiopia.*
10. *World Health Organization (2016). Global tuberculosis report 2016. World Health Organization. www.who.int/iris/handle/10665/250441.*
11. *Berhe, G., F. Enquselassie, and A. Aseffa, Treatment outcome of smear-positive pulmonary tuberculosis patients in Tigray Region, Northern Ethiopia. BMC Public Health, 2012. 12: p. 537.*
12. *Hailu, D., W.E. Abegaz, and M. Belay, Childhood tuberculosis and its treatment outcomes in Addis Ababa: a 5-years retrospective study. BMC Pediatr, 2014. 14: p. 61.*
13. *Addisu Melese, Balew Zeleke, and B. Ewnete, Treatment Outcome and Associated Factors among Tuberculosis Patients in Debre Tabor, Northwestern Ethiopia: A Retrospective Study. Hindawi Publishing Corporation, 2016.*
14. *Tessema B, et al., Treatment outcome of tuberculosis patients at Gondar University Teaching Hospital, Northwest Ethiopia. A five - year retrospective study. BMC Public Health, 2009. 9:37.*
15. *Sisay, S., et al., Directly Observed Treatment Short-course for tuberculosis control program in Gambella Regional State, Ethiopia: ten years' experience. BMC Res Notes, 2014. 7: p. 44.*
16. *Vijay S, et al., Treatment Outcome and Mortality at One and Half Year Follow-Up of HIV Infected TB Patients Under TB Control Programme in a District of South India. PLoS ONE 2011. 6(7).*
17. *Zelalem Addis, et al., Treatment Outcome of Tuberculosis Patients in Azezo Health Center, NorthWest Ethiopia. International Journal of Biomedical And Advance Research(IJBAR), 2013. 04(03).*

CHAPTER 3

Surveillance System Evaluation

Evaluation of malaria surveillance system in Jabi-Tehinan district, northwest Ethiopia, 2010 EFY

Tesfahun Taddege GEREMEW, Lemma Derseh, Giete Zeru and Meklit Mekonnen

Abstracts

Background: Globally approximately 216 million malaria cases with 445,000 deaths were reported in 2016. Ethiopia is one of malaria endemic sub-Saharan Africa countries. Periodic evaluation of malaria surveillance system is crucial to ensure that malaria is monitored effectively and efficiently. However, the malaria surveillance system was not evaluated in Jabi-Tehinan district, one of the high malarious districts in Northwestern Ethiopia. Hence, this study was aimed to evaluate the malaria surveillance systems in Jabi-Tehinan district, Northwest Ethiopia.

Methods: A malaria surveillance system evaluation was conducted from April 15 to 25 2018 in Jabi-Tehinan district, Northwest Ethiopia. The 2009 EFY malaria data from the malaria surveillance was collected using semi-structured questionnaires adapted from CDC (2001) guidelines for evaluating public health surveillance systems. Randomly selected five health centers and ten health posts including the district health office were included in this evaluation. Descriptive measures and narrative summary were used to describe the system. Heterogeneity in annulal paracite index and malaria positivity rate among reports of health facilities was assessed by I2 statistic and random effect model was used in the presence of heterogeneity.

Results: Almost all households in the district had received atleast one insecticide-treated net (ITN), of which, 53 percent were using the net. Sixty percent of the households were received indoor residual spraying (IRS) in the past 12 months. The malaria surveillance system detected only 58% of confirmed malaria cases. The usefulness of the surveillance system was limited and the positive predictive value was 58%. Completeness and timeliness were 98% and 83 %, respectively. The annual paracite incidence was 52 cases per 1000 population per year (95%CI; 45-69). The surveillance data analysis, interpretation and using information for decision making was limited.

Conclusions: The malaria surveillance system in Jabi-Tehinan district was weak. The usefulness of the malaria surveillance system was limited. Hence, it is better if the Jabi-Tehinan district health office with other concerned bodies work more on strengthening the malaria surveillance system through improving insecticide-treated net utilization, case detection, surveillance data analysis, interpretation and utilization. It is also good to conduct further studies to identify factors associated with malaria case detection and insecticide-treated net utilization.

Key words: Malaria, Surveillance system, Attributes, Evaluation, Northwest Ethiopia

3.1 Introduction

Globally, approximately 216 million cases of malaria with 445,000 deaths from malaria were reported in 2016. The WHO African Region accounted for 90% of all malaria cases and 91% of all malaria deaths. Of the 91countries reporting indigenous malaria cases, 15 countries accounted for 80% the global malaria cases and deaths; all of these countries are in sub-Saharan Africa, except for India[1]. The incidence rate of malaria is estimated to have decreased by 18%, from 76 to 63 cases per 1000 population at risk, between 2010 and 2016 in the world. The WHO South-East Asia Region recorded the largest decline (48%) followed by the WHO Region of the Americas (22%) and the WHO African Region (20%). Despite these reductions, between 2014 and 2016, substantial increases in case incidence occurred in the WHO region of the Americas, and marginally in the WHO South-East Asia, Western Pacific and African Regions[1].

Ethiopia is one of the malaria endemic sub-Saharan Africa countries, in which 75% of the landmass is malarious and 60% of the population is at risk of malaria. Malaria is a major public health problem in Ethiopia despite relatively low malaria prevalence compared to most other malaria-endemic countries in Africa. It is seasonal in most parts of the country, and the massive scale-up of malaria control interventions, including case diagnosis and treatment, distribution of long-lasting insecticidal nets (LLINs), and indoor residual spraying (IRS) of households with insecticides have preferentially targeted these areas[2]. The 2015 Ethiopia National Malaria Indicator Survey (EMIS) results indicated that the malaria prevalence was 1.2 percent by rapid diagnostic test (RDT) and 0.5 percent by microscopy. Two-thirds (64 percent) of households in malarious areas own at least one insecticide-treated net (ITN) and 29 percent of all households in malarious areas were received IRS[2].

The third Pillar of the global technical strategy for malaria 2016–2030 is to transform malaria surveillance into a core intervention. This involves the establishment of passive case detection systems that perform optimally and are complemented, in elimination settings, with active case detection, to ensure that potentially all malaria cases are detected and treated. The information from surveillance systems is then used to optimize the deployment and impact of interventions to accelerate elimination, and eventually to confirm the interruption of transmission. Cases and deaths from the surveillance system reported by countries are often from the public health sector, predominantly through passive case detection. In elimination settings, data on cases may also be recorded during active case detection. Often, data from the private sector remain sparse, and in countries with moderate to high transmission, it is possible that a substantial proportion of patients who do not seek care remain undocumented by the surveillance system[1].

The rationale of the evaluation

Effective surveillance of malaria is essential for identifying which areas or population groups are most affected, and for targeting resources to communities most in need. The surveillance also alerts the health sector and other responsible bodies to epidemics, enabling control measures to be intensified when necessary[3]. In addition, malaria surveillance system that monitors the disease effectively and efficiently is crucial in the rapid detection of outbreaks and in timely response activities.

Therefore, a surveillance system should be evaluated periodically to assess its performance in interms of its targets. A periodic evaluation a surveillance activity is useful for obtaining an objective, more in-depth evaluation of the design of the system, protocol and procedures, implementation and analysis[4]. Moreover, an evaluation of the malaria surveillance system may improve the information provided to ensure that malaria is monitored effectively and efficiently.

However, there was no evaluation of malaria surveillance system in Jabi-Tehinan district, one of the districts in the high malaria transmission strata in Northwest Ethiopia[2]. Hence, we conducted this malaria surveillance system evaluation in Jabi-Tehinan district to describe the performance of the malaria surveillance system. The finding from this evaluation may help for generating information for evidence-based decision making in the surveillance system. In addition, it may also help for generating hypothesis for further studies on the malaria surveillance system in the district.

3.2 Objectives

GENERAL OBJECTIVE

★ To evaluate the malaria surveillance system in Jabi-Tehinan district, Northwest Ethiopia, 2009 EFY.

SPECIFIC OBJECTIVES

1. To describe the malaria surveillance system in Jabi-Tehinan district.
2. To estimate annual parasite incidence and malaria positivity rate
3. To identify factors associated with the malaria positivity rate and incidence in Jabi-Tehinan district.

3.3 Methods

Study design and setting

A facility-based cross sectional study was conducted in Jabi-Tehinan district, one of the highly malarious districts in West Gojjam zone, from April 15 to 25 2018. Administratively, Jabi-Tehinan district is sub-divided into 11 clusters and 42 kebeles (3 urban and 39 rural kebeles). The catchment population of the district was 214,636 and all the kebeles of the district are malarious [5]. The district has 11 Health centers and 41 Health posts and 12 private health facilities.

Study population and selection

Government health institutions were the study subjects. The district was selected on the basis of relatively high burden of malaria in West Gojjam zone. Five randomly selected health centers, ten randomly selected health posts, and the district health office were included in this evaluation.

Data Extraction and management

The data was extracted through document review and face to face interview using checklists adapted from the CDC (2001) updated guidelines for evaluating public health surveillance system [6]. The data was entered into Excel spreadsheet.

Statistical methods and analysis

The data were analyzed with MS-Excel 2013, STATA Version-14 [7] and ArcGIS 10.3 softwares. Descriptive measures and narrative summary were used. Heterogeneity among reports in annual parasite incidence (API) and malaria positivity rate (MPR) was assessed using the Higgins's I^2 test statistic [8, 9]. In the presence of heterogeneity, we used a random effects model to calculate the pooled estimate of malaria incidence and positivity rate [10]. Subgroup analysis and meta-regression were carried out to explore the potential sources of heterogeneity in malaria incidence and positivity rate across health facility reports [11, 12] using a statistical software of Stata V-14. The results of the pooled analysis are presented in a Forest plot.

Ethical consideration

Legal letters were obtained from University of Gondar, West Gojjam zone health department and Jabi-Tehinan district health office. Permission was obtained from the health facilities.

3.4 Result

Description of malaria surveillance system

Effective surveillance of malaria cases and deaths is essential for identifying which areas or population groups are most affected by malaria, and for targeting resources to communities most in need. Such surveillance also alerts ministries of health to epidemics, enabling control measures to be intensified when necessary[13, 14]. Malaria is among the major public health problems of the 20 priority diseases under surveillance in Ethiopia and 23 priority diseases under surveillance in the Amhara national regional state[15-18].

However, surveillance systems do not detect all malaria cases for several reasons. First, not all malaria patients seek care or, if they do, they may not seek care at health facilities that are covered by a country's surveillance system. Second, not all patients seeking care receive a diagnostic test. Finally, recording and reporting within the surveillance system are not always complete[3].

The two common indicators related to malaria surveillance systems are the proportion of expected health facility reports received at the higher level and proportion of malaria cases detected by surveillance systems[1].

Health facility reports received at the district level

The completeness of health facility reporting is a good indicator of a surveillance system's performance because achieving a high reporting rate requires health facilities to adhere to several processes. At the district level, the health facility reporting rate was 98% in the last five epidemiological weeks of 2009 EFY (Table 3.1). However, this indicator could not be calculated for the health centers, because the number of reports received from health posts was not found.

Malaria cases detected by surveillance systems

The malaria surveillance system detected 58% of the estimated malaria cases to be existed in the district. Among 10 health posts, the proportion of estimated cases detected by surveillance systems was lowest in Agomamit (1.4%) and highest in Berkegn (95%), whereas, the proportion of estimated cases reported by surveillance systems was lowest in Maksegnin (15%) and highest in Awunt (95%) among five health centers (Table 3.1)

Table 3.1. Proportion of malaria cases detected by the surveillance system in Jabi-Tehinan district, northwest Ethiopia, 2009 EFY.

Health facilities	Estimated number of malaria cases*	Malaria cases detected by surveillance systems	
		Number	Percent
Health posts			
Agomamit	702	10	1.4
Guay	700	116	16.6
Awunt	694	94	13.5
Jimat	610	430	70.5
Fetegem	360	286	79.5
Hodansh	829	557	67.2
Jiga Yelmdar	472	148	31.3

Health facilities		Estimated number of malaria cases*	Malaria cases detected by surveillance systems	
Berkegn	401	382	95.3	
Arbawash	791	85	10.7	
Azemer	410	173	42.2	
Health centers				
Agomamit	1,951	386	19.8	
Awunt	2,588	2,469	95.4	
Fetegem	1,646	901	54.8	
Jiga	3,219	1,511	46.9	
Maksegnin	2,213	327	14.8	
District health office		21,464	12,442	58.0

* The burden of malaria was estimated from the annual paracite incidence new stratification where Jabi-Tehinan district is in the high transmission strata (>100 cases/1000 population).

Malaria prevention and control

The two major malaria prevention services implemented in Ethiopia are targeted indoor residual spraying (IRS) and distribution of ITNs, specifically, long-lasting insecticide-treated nets (LLINs) for universal coverage[2].

Ownership and utilization of ITN

Almost all households of Jabi-Tehinan district had received at least one ITN. Among a total of 3,230 households visited in the 49th Ethiopian epidemiological week (Ginbot 21-27, 2009 EFY), only 46% were using the ITN at the time of visit. However, there were no documented evidence that can show the proportion of households with sufficient nets, prioritized children and pregnant women to sleep under ITN. Nearly three-fourth (72%) of households are protected with IRS in the past 12 months (Table 3.2). In addition, other vector control activities, mainly larval control through environmental management (draining, filling, compacting and clearing) were practiced in the district. However, no chemical larviciding was practiced.

Table 3.2. Percentage of households with at least one ITN, ITN utilization and IRS, Jabi-Tehinan district, Northwest Ethiopia, 2009 EFY

Health Institution	Total Number of Households	Percentage of Households received at least one ITN	Percentage of HHs with IRS	ITN Utilization*	
				Number of HHs Visited	Percentage of HHs using ITN
Health Posts					
Agomamit	302	100.0	81.4	50	57.3
Guay	301	100.0	74.7	42	44.4
Awunt	298	100.0	75.0	56	50.0
Jimat	175	100.0	67.3	80	30.8
Fetegem	155	100.0	74.2	66	42.9
Hodansh	356	100.0	51.3	10	26.7
Jiga Yelmdar	203	100.0	76.8	15	53.8
Berkegn	115	100.0	64.1	45	30.0
Arbawash	340	100.0	85.3	78	67.0

Health Institution	Total Number of Households	Percentage of Households received at least one ITN	Percentage of HHs with IRS	ITN Utilization*	
Azemer	176	100.0	80.2	54	56.0
Health Centers					
Agomamit HC	839	100.0	83.4	180	66.1
Awunt HC	742	100.0	74.4	211	44.2
Fetegem HC	708	100.0	68.8	215	31.5
Jiga HC	1384	100.0	82.5	321	62.8
Maksegnin HC	952	100.0	86.7	102	68.2
District health office	9229	100.0	72.0	3,230	46.0

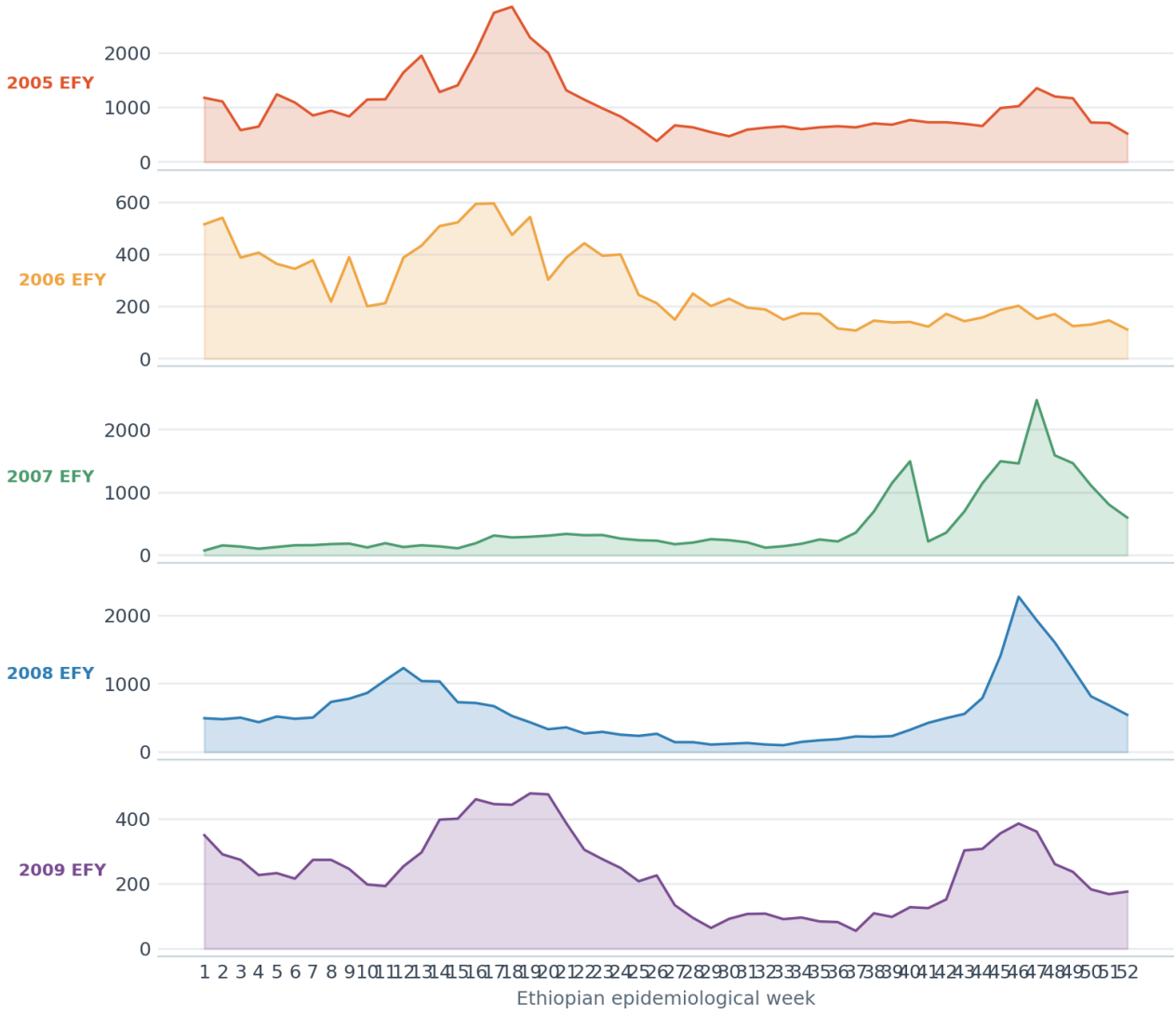
*The ITN utilization was calculated from the Ethiopian 49th epidemiological week (Ginbot 21-27, 2009 EFY) PHEM weekly report

Diagnostic testing

World health organization (WHO) and the national malaria guide line recommends that every suspected malaria case be confirmed by microscopy or an RDT before treatment. In the malaria surveillance system, once malaria is suspected on clinical grounds, screening for malaria and diagnosis was made by light microscopy or Rapid diagnostic tests (RDTs) for the presence of parasites and speciation according to WHO guidelines[19].

So, rapid diagnostic tests (RDTs) had been widely used at health post level and microscopy at the health center level in Jabi-Tehinan district. A confirmed malaria case was defined as anyone with Plasmodium species parasites on microscopy or RDT, regardless of parasite density.

The weekly trend of confirmed malaria cases in the last five years showed two outbreaks of malaria in 2007 EFY and 2008 EFY in Jabi Tehinan District. Among the total 137, 717 outpatient and inpatient cases, malaria accounted for 9,651 (7%) (Figure 3.1).



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

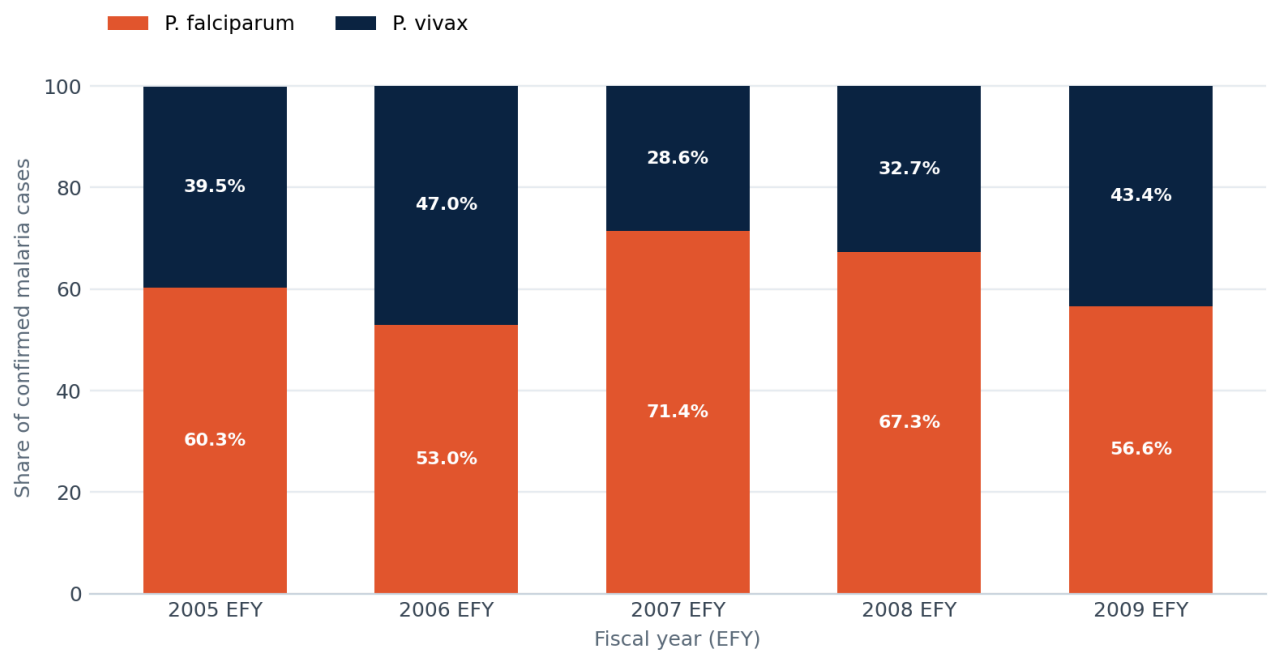
Figure 3.1. Trends in number of malaria cases by Ethiopian Epidemiological Weeks, Jabi-Tehinan District, Northwest Ethiopia, 2005 EFY- 2009 EFY.

The proportion of plasmodium species by diagnostic test

Plasmodium falciparum species accounted for 68 percent of the RDT confirmed malaria cases and 50 percent of the microscopy confirmed cases. Overall, 57 percent of the malaria cases confirmed with RDT or microscopy were falciparum (Table 3.3). During the malaria outbreaks, in 2007 EFY and 2008 EFY, plasmodium falciparum species accounted 71 percent and 67 percent respectively. The trend in the percentage of malaria species among all ages by RDT and microscopy is shown in figure 3.2.

Table 3.3. Proportion of malaria species among all ages by RDT and microscopy, Jabi-Tehinan district, Northwest Ethiopia, 2009 EFY

Malaria Species	RDT		Microscopy		Total	
	Number	Percent	Number	Percent	Number	Percent
P. Falciparum	2,479	67.9	2,986	49.8	5,465	56.6
P.Vivax	1,174	32.1	3,012	50.2	4,186	43.4
Total	3,653	100.0	5,998	100.0	9,651	100.0

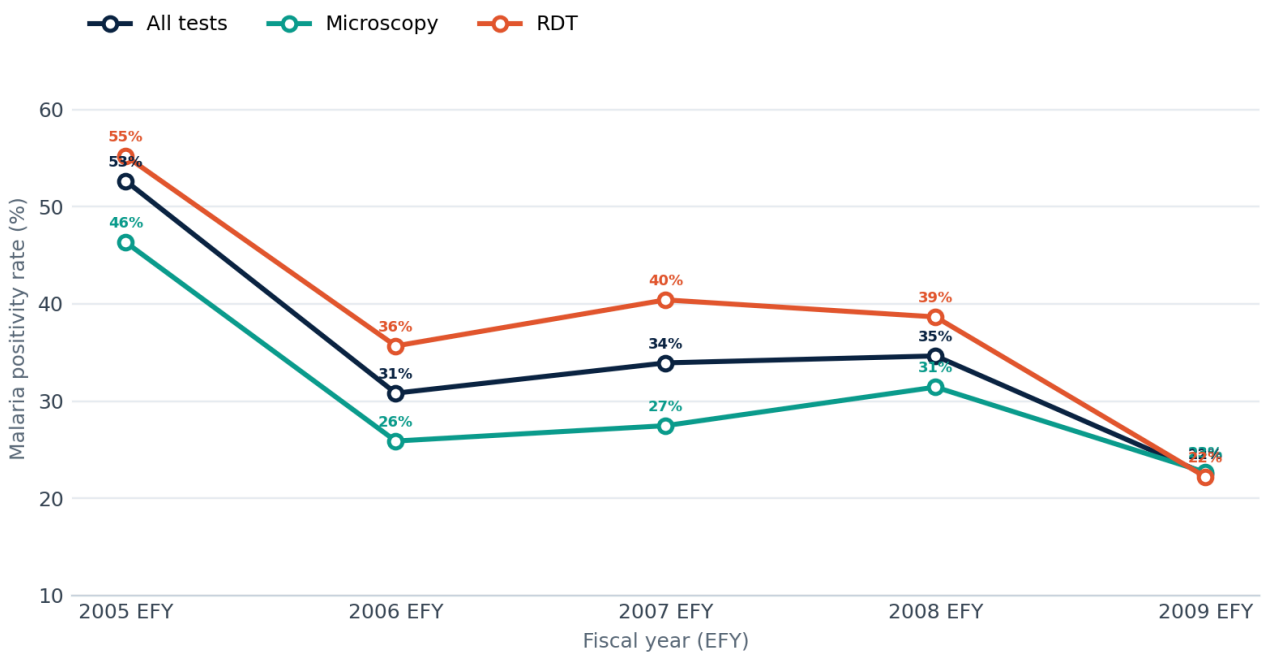


Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 3.2. Trend in percentage of malaria species, Jabi-Tehinan district, Northwest, Ethiopia, 2009 EFY

As shown in figure 5, the RDT positivity rate (55%) was above the slide (microscopy) positivity rate (46%) from 2005 EFY to 2008 EFY. In 2009 EFY, the RDT positivity rate, slide positivity rate and overall malaria positivity rate were approximately the same (22%) (Figure 3.3).

Under-five years old children accounted for nearly one-fifth (22%) of the total malaria cases and nearly half (48%) of the cases were accounted by patients aged 15 years and above (Figure 3.4).



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 3.3. Trend in malaria positivity rate, Jabi-Tehinan district, Northwest Ethiopia, 2009 EFY.

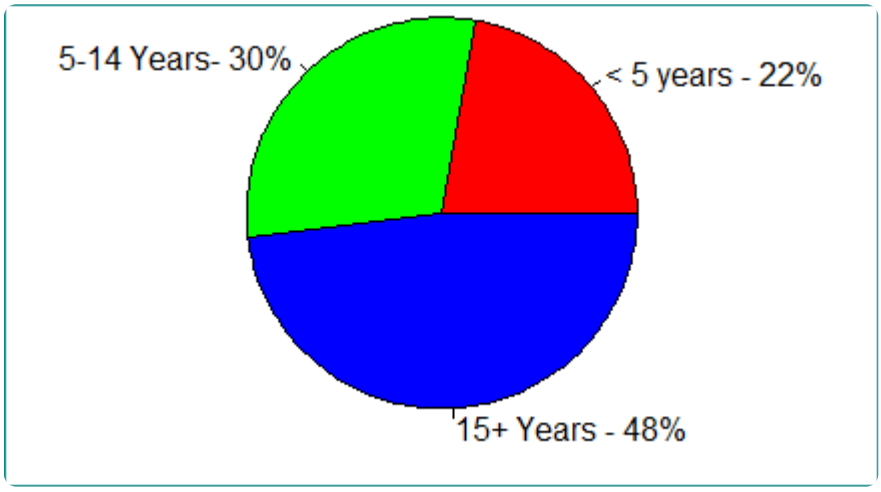


Figure 3.4. Age distribution of malaria in Jabi-Tehinan district, Northwest Ethiopia, 2009 EFY.

Annual Paracite Index (API)

According to the epidemiological malaria stratification, five (12%) kebeles in the district were in the high malaria transmission strata, while seven kebeles in the low malaria transmission strata. There was no malaria free (API=0) kebele in the district (Figure 3.5).

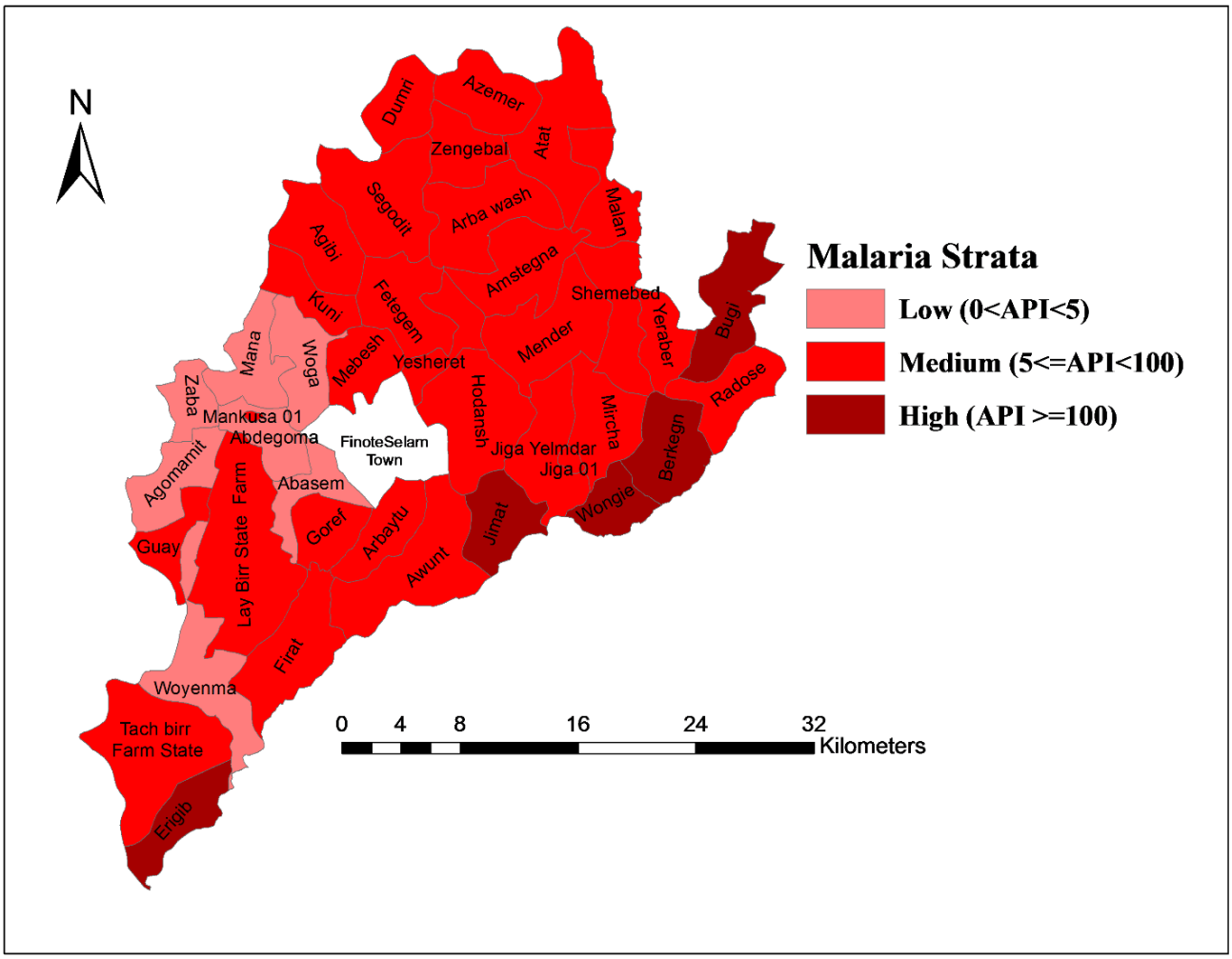


Figure 3.5. Risk map of kebeles by API, Jabi-Tehinan district, northwest Ethiopia, 2009 EFY.

The pooled analysis for API per 1000 population shows variations in API, ranged from 1.42 (Agomamit health post) to 143.1 (Awunt health center) with a high statistical heterogeneity across the reports of health facilities (I^2=99.8%, P<0.0001) (Figure 3.6).

In the presence of such high heterogeneity, calculation of a combined estimate from individual reports may be inappropriate, and, at least, potential reasons for such heterogeneity need to be explored by using subgroup analysis or/and meta-regression.

The overall pooled estimate of API in Jabi-Tehinan district from the random effect model was 52.2 (95%CI: 35.5-67.5) (Figure 3.6).

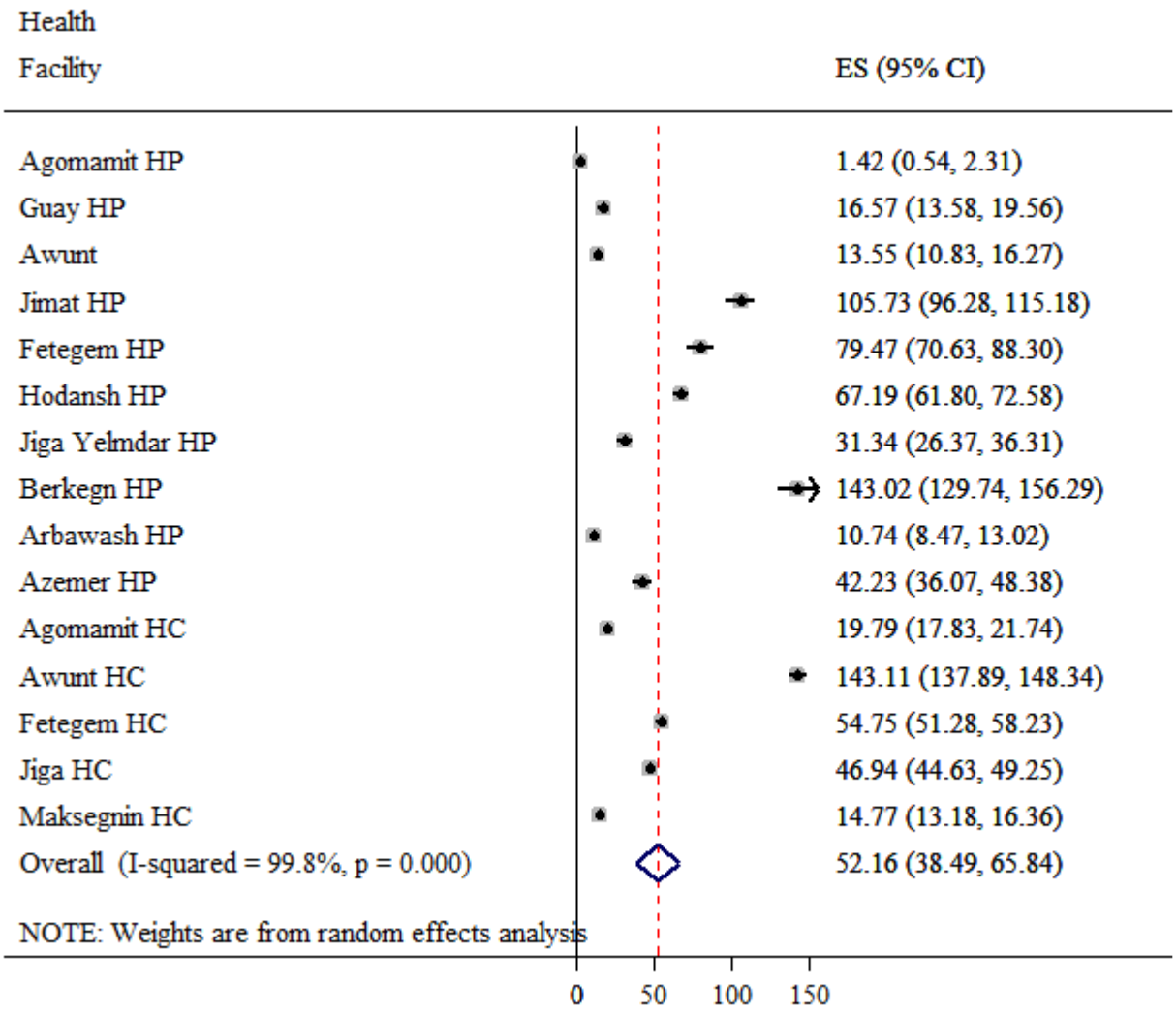


Figure 3.6. Pooled API estimate (ES) per 1000 population in Jabi-Tehinan district, northwest Ethiopia, 2009 EFY.

Based on the subgroup analyses by health facility type, the pooled estimate of API was 50.1% (95% CI 35.37–64.91) at health post level and 55.8% (95% CI 27.92–83.62) at health center level (Figure 3.7).

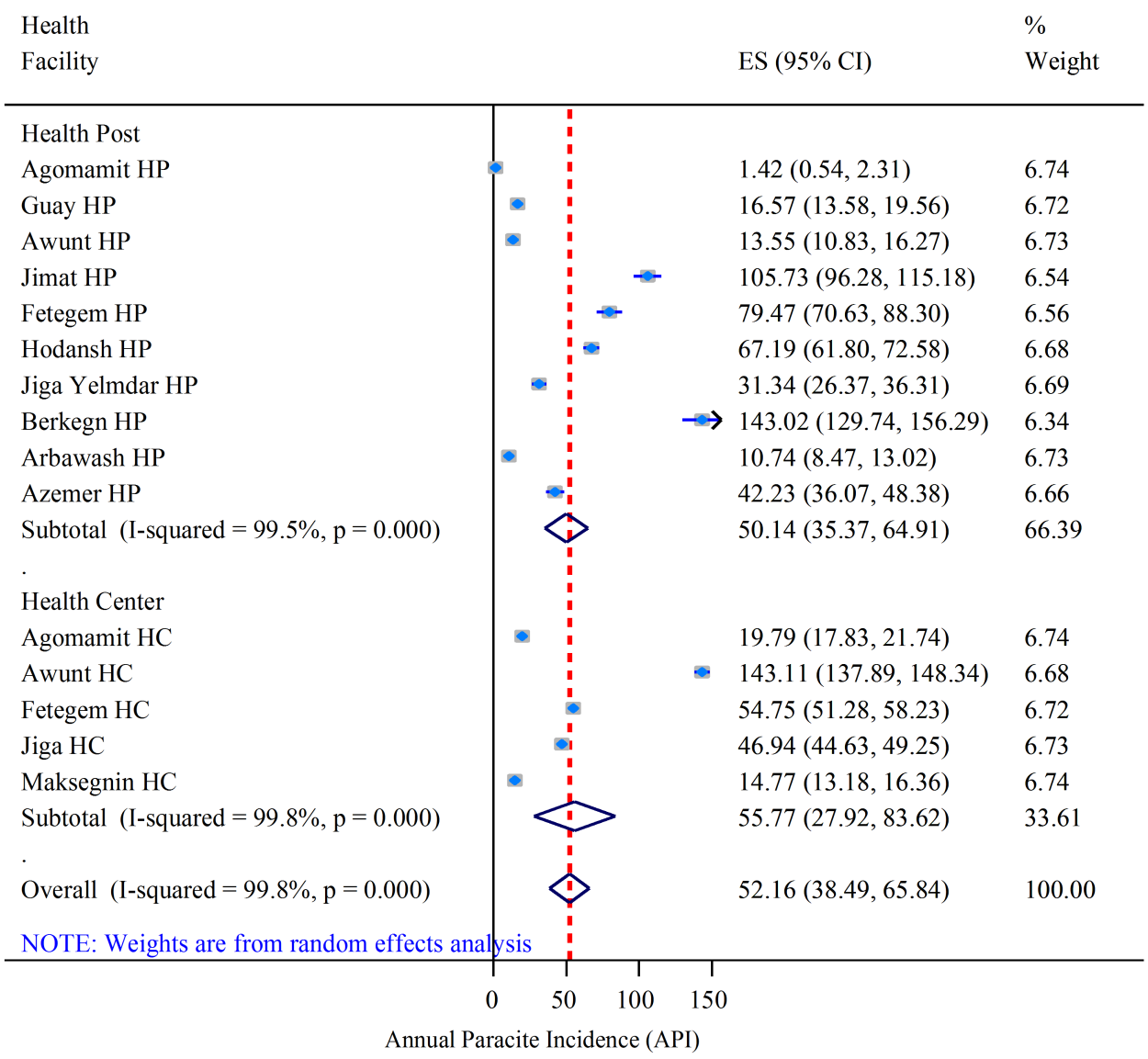


Figure 3.7. Subgroup Malaria API estimate (ES) per 1000 population in Jabi-Tehinan district, northwest Ethiopia, 2009 EFY.

Malaria Positivity Rate (MPR)

The pooled estimate of malaria positivity rate from fixed effect model revealed high heterogeneity across reports of health facilities (I^2=99.7%, P<0.0001). So, random effect model was used and the overall pooled estimate of malaria positivity rate was 27.8% (95%CI: 20.2-35.4) with a high heterogeneity across reports (I^2=99.7%, P<0.000). The lowest malaria positivity rate was 4.1% at Maksegnin health center and the highest was 47.9% at Berkegn health post.

In the subgroup analyses by cluster, the lowest pooled estimate of malaria positivity rate was 13.91% (95% CI: 1.99–25.83) at Maksegnin cluster, and the highest was 40.26% (95% CI: 32.32–48.20) at Fetegem cluster (Figure 3.8).

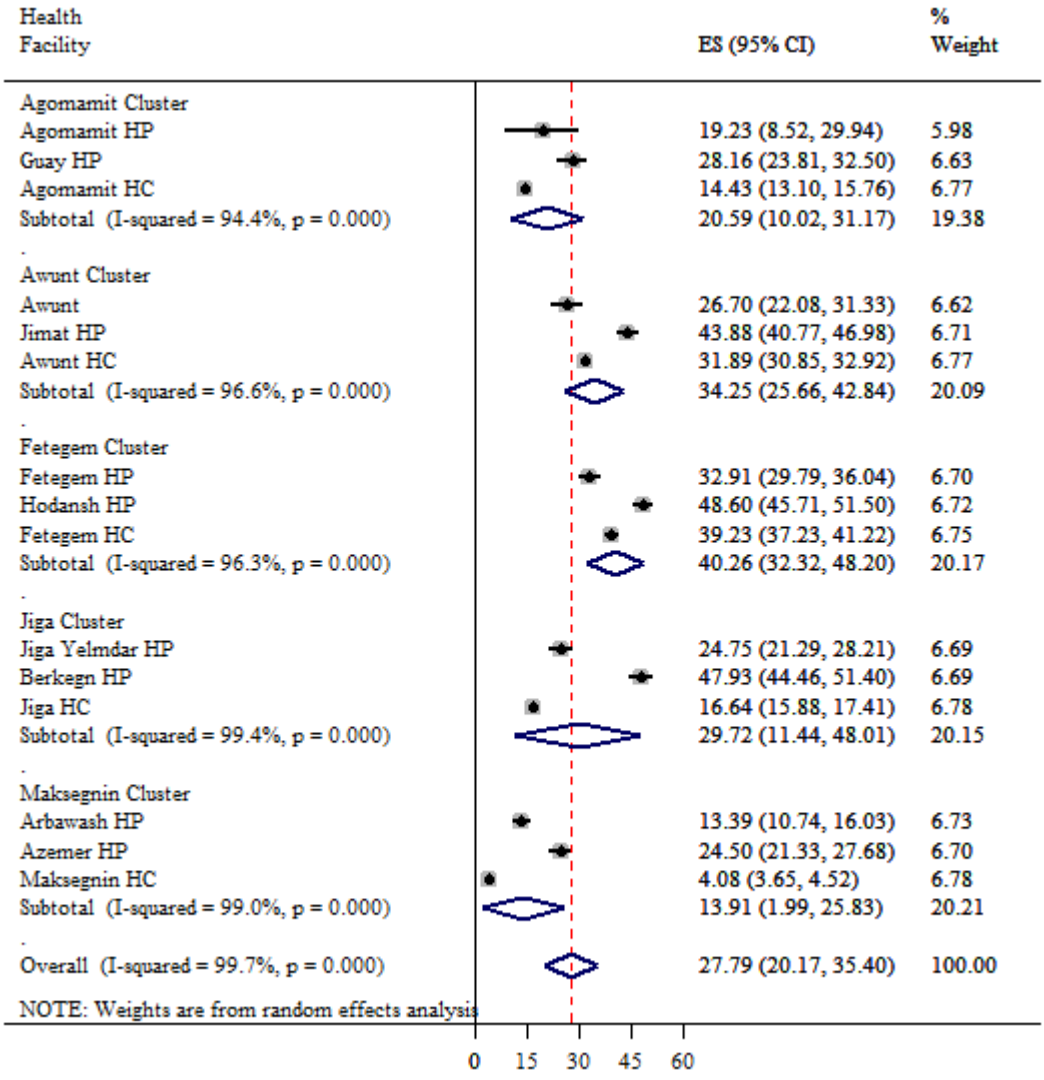


Figure 3.8. Subgroup malaria positivity rate estimates (ES) among microscopies and RDTs tested for malaria diagnosis in Jabi-Tehinan district, northwest Ethiopia, 2009 EFY.

In the subgroup analyses by type of health facility, the pooled estimates of malaria positivity rate were 31.17% (95% CI: 22.86–39.48) from the health post level and 21.24% (95% CI: 9.42–33.05) from the health centers (Figure 3.9).

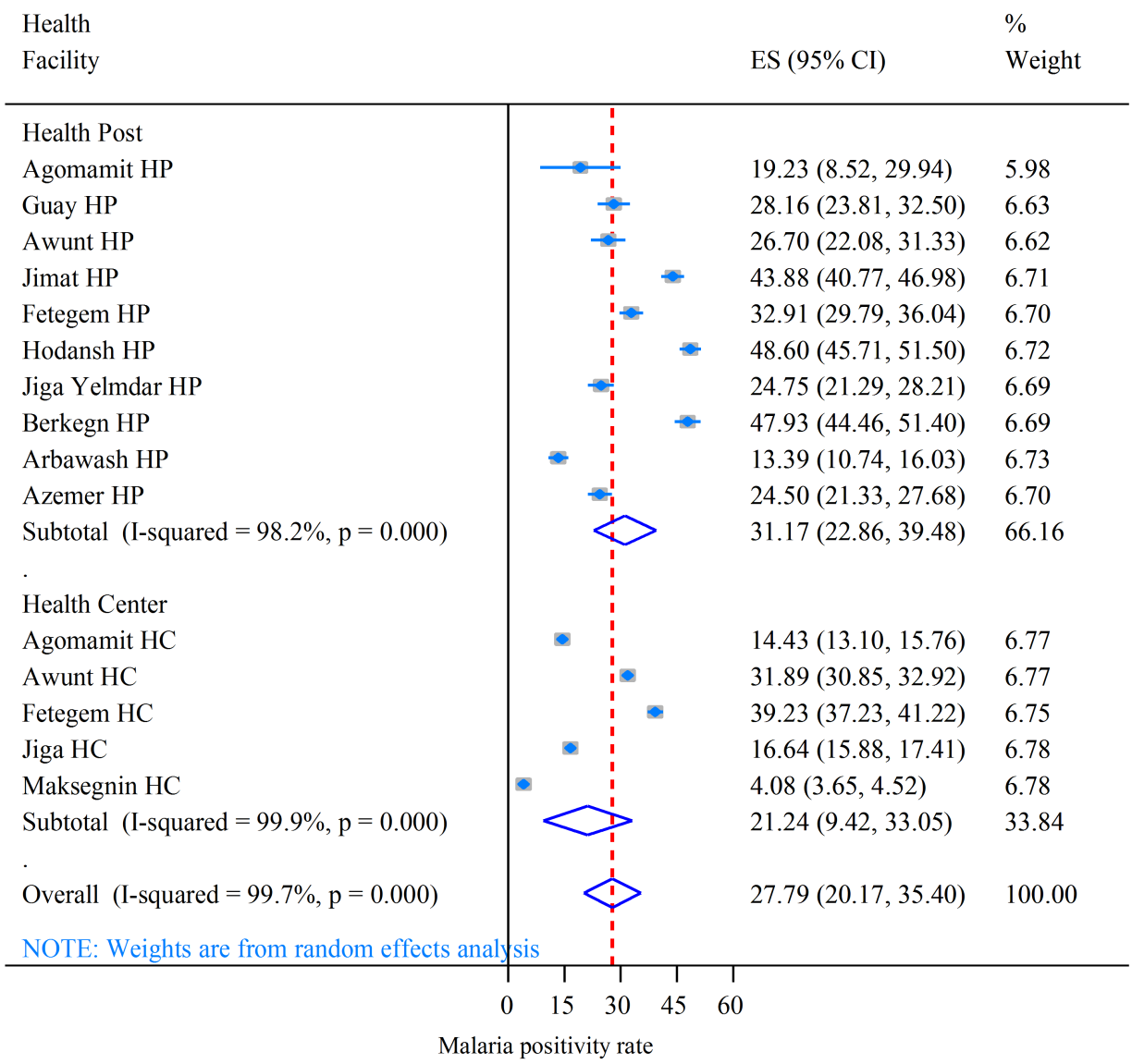


Figure 3.9. Subgroup malaria positivity pate estimate (ES) by health facility type in Jabi-Tehinan district, northwest Ethiopia, 2009 EFY.

Meta regression

In addition to subgroup analysis, Meta-regression was carried out to explore the potential sources of inconsistency in the annual paracite incidence (API) and malaria positivity rate (MPR) by considering type of health facility, cluster, ITN utilization and IRS as a potential sources of variation. Hence, the ITN use and IRS were significant factors associated with API and MPR heterogeneities among reports of health facilities (Table 3.4 and 3.5).

Table 3.4. Meta regression to explore the potential sources of heterogeneity in API across reports of health facilities, northwest Ethiopia, 2009 EFY.

Variables	Coefficients	P-value
Clusters		
Agomamit (constant)	12.59	Reference
Awunt	74.74	0.06
Fetegem	54.50	0.11
Jiga	60.63	0.07
Maksegnin	9.95	0.77
Type of health facility		
Health post (constant)	50.88	Reference
Health center	4.95	0.851
ITN use and IRS		
Percentage of ITN use	-2.14	**<0.01**
Percentage of households protected with IRS	-2.80	**0.01**

Table 3.5. Meta regression of malaria positivity rate to explore the potential sources of heterogeneity across reports, Yilmana Densa district, northwest Ethiopia, EFY 2009.

Variables	Coefficients	P-value
Clusters		
Agomamit (Constant)	20.60	Reference
Awunt	13.60	0.12
Fetegem	19.65	0.06
Jiga	9.05	0.31
Maksegnin	-6.69	0.45
Type of health facility		
Health Post (Constant)	31.18	Reference
Health Center	-9.95	0.154
ITN use and IRS		
Percentage of ITN use	-0.90	**<0.001**
Percentage of HHs with IRS	-1.33	**<0.001**

The purpose of the malaria surveillance system

The malaria surveillance system in the district was using the case definition as recommended by the national and subnational public health emergency management guidelines. The malaria standard case definition: Any person with fever or fever with headache, rigor, back pain, chills, sweats, myalgia, nausea, and vomiting diagnosed clinically as malaria (suspected) or suspected case confirmed by microscopy or RDT for plasmodium parasites (confirmed) and malaria community case definition: Any person with fever or fever with headache, back pain, chills, rigor, sweating, muscle pain, nausea and vomiting or Suspected case confirmed by RDT.

Population under surveillance

The total population under-surveillance were 214,636, of which 13% were estimated to be under-five children and 3.37% pregnant women (Table 3.6).

Table 3.6. Population under surveillance in Jabi-Tehinan district, Northwest Ethiopia, 2009 EFY

Health Institution	Total Population	Under-five years population	Pregnant women
Health Posts			
Agomamit	7,022	951	237
Guay	6,999	948	236
Awunt	6,938	939	234
Jimat	4,067	551	137
Fetegem	3,599	487	121
Hodansh	8,290	1,122	279
Jiga Yelmdar	4,722	639	159
Berkegn	2,671	362	90
Arbawash	7,911	1,071	267
Azemer	4,097	555	138
Health Centers			
Agomamit HC	1,9506	2,641	657
Awunt HC	1,7252	2,336	581
Fetegem HC	1,6456	2,228	555
Jiga HC	3,2188	4,358	1,085
Maksegnin HC	2,2134	2,997	746
District health office	214,636	29,062	7,233

Necessary materials for malaria survielance system

(For case diagnosis, recording and report)

Health facilities reported malaria surveillance data collected from Monday to Sunday on weekly basis through hard copy or telephone to the district health office. At district level, the routine reports of the health facilities submitted to the zone health department public health emergency management case team. The necessary registers and report formats were available in all health facilities of the district. Only two (40%) of the health centers, one (10%) of the health posts and the district health office had rumor log-book and none of the health facilities had emergency response and preparedness plan. RRT was established in four (80%) health centers, three (30%) health posts and in the district health office. However, none of those established RRTs were functional (Table 3.7).

Table 3.7. Number of health facilities that had necessary materials for malaria surveillance in Jabi-Tehinan District, Northwest Ethiopia, 2009 EFY.

Materials	Health Centers (N=5)	Health Posts (N=10)	District Health Office
HFs that had the national/regional PHEM guideline	3	4	1
HFs that had the national malaria guide line	4	0	1
HFs that had standard case definition	5	8	1
HFs that had community case definition	2	7	1
HFs that posted the case definitions	2	4	1
HFs that had rumor log book	2	1	1
HFs that had OPD/IPD registers	5	5	NA
HFs that had epidemic report form	3	3	1
HFs that had Case based reporting format	2	1	0
HFs that had functional computer	3	0	1
HFs that had Microscope	5	NA	NA
HFs that had RDT	5	10	NA
HFs that had emergency response & preparedness plan	0	0	0
HFs that had established rapid response team (RRT)	4	3	1
HFs that had functional rapid response team (RRT)	0	0	0

HFs= Health Facilities, NA=Non-applicable

Data analysis and interpretation

The PHEM officer at district level has trained on basic surveillance data analysis but the focal persons and health extensions had no any training about survielance data analysis. At the district level, the data analysis was done by cluster level disaggregation. At all levels, there were no any plan for emergency response and preparedness. Epidemic management committee or rapid response team (RRT) were established from both at the health center and district health office level from different disciplines, but there was no any evidence that can show the functionality of the committees. At district level, about 75 thousand Ethiopian birrs were allocated for any emergency condition but no specific malaria budget. All health facilities lack trained persons in epidemic investigation and management and active rapid response committee. All the health facility had and used a weekly malaria monitoring chart.

Training, supervision, and feedback

At all levels, there were on plan for supervision. However, each health facility was supervised only twice by the supervisors from the district health office. But there was no any documented evidence that can show feedback provision from district health office to health facilities or from the health centers to health posts on malaria the surveillance system. The health workers at health facility level were not trained on survielance system except the district PHEM officer.

Laboratory function

All health the five health centers and the eleven health posts are malaria diagnostic. The laboratories in health centers were equipped with a microscope and the health posts diagnose malaria using RDT.

Attributes of malaria surveillance system

Malaria surveillance is integrated with all other priority diseases which include laboratory support and passive surveillance which becomes active during outbreaks. We assessed the surveillance system attributes such as usefulness, simplicity, flexibility, data quality, sensitivity, acceptability, timeliness, stability, and representativeness.

Usefulness

At the district level, the data from malaria surveillance system was analyzed and interpreted by cluster but not kebele. However, there was no evidence for taking actions as a result of this data analysis and interpretation. At the health center & health post level, the malaria surveillance data analysis and interpretation were limited and no evidence for using the data to make decisions and take actions. This indicates that the malaria surveillance system was less useful in the district.

Simplicity

All respondents agreed that the case definition of malaria for the identification of suspected cases is easy to understand and apply to all levels of health professionals. Though forms are easy to complete, Data are sent with hard copies to all levels, with delays at each level and this makes it difficult to get a timely surveillance data. In addition, manual data entry of reports from multiple registers at health facility level & multiple reporting sources at the district level and using multiple forms for reporting makes the malaria surveillance system more complex.

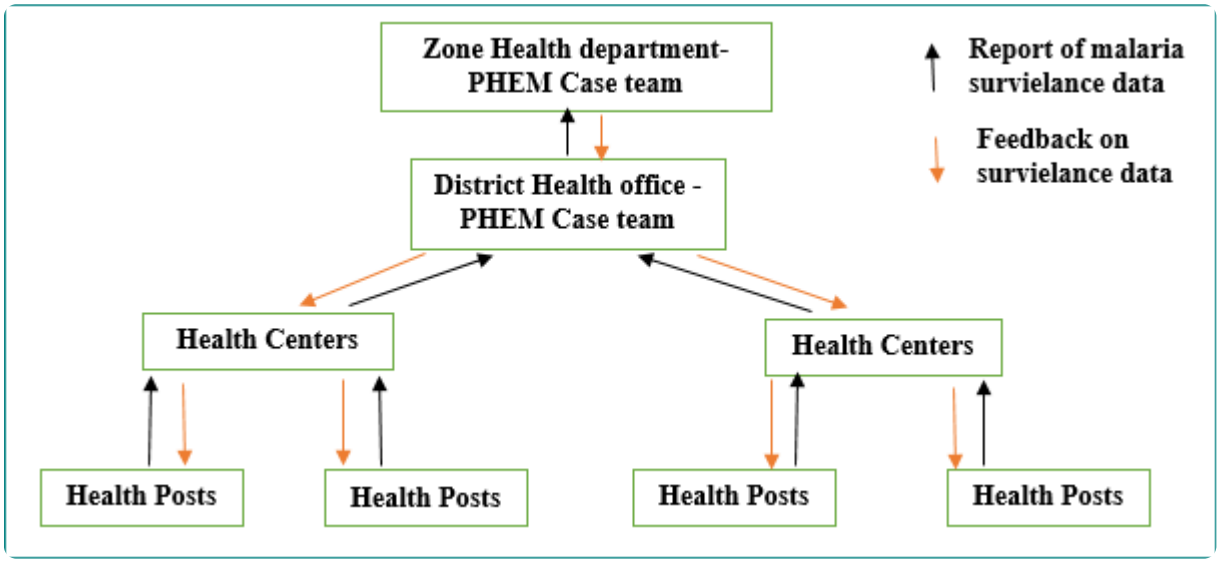


Figure 3.10. Flow of Malaria Surveillance data at Jabi-Tehinan district, West Gojjam Zone, Northwest Ethiopia, 2009 EFY.

Flexibility

At district and health facility level, the eHMIS reporting format is less flexible for accommodating changes when there is a need for additional information or modes of operation, however, the PHEM weekly report format can be modified to include other variables to report other newly occurring health events without much difficulty. Generally, the malaria surveillance system is flexible and able to adapt and accommodate changes when there is a need for additional information or reporting on malaria forms. However, the system is considerably donor-dependent and cessation of funding and support is a concern for its sustainability.

The sensitivity of the surveillance system

We defined sensitivity as the ability of the surveillance system to truly detect cases of malaria in the district (sensitivity = cases detected/cases existing×100). However, there is no local survey conducted to establish the exact burden of malaria in Jabi-Tehinan district. Therefore, we used the national estimate for the district. According to the new stratification of malaria risk in Ethiopia by annual parasite incidence, Jabi-Tehinan district is in the high transmission stratum (API>100/per 1000 population)[2]. So, taking the minimum expected estimate for confirmed malaria cases per 1000 population/parasite prevalence, we calculated the sensitivity of malaria surveillance system in case of case detection using the formula

$$\text{Sensitivity} = \frac{\text{Confirmed malaria cases detected per year}}{\text{Expected malaria cases in the same year}} \times 100$$

where, the expected malaria cases per year are 21,464 [2] and the confirmed malaria cases detected in the same year is 12,442. Hence, the sensitivity of the malaria surveillance system to detect case was 58 percent. The active case detection is another problem which is not done by the system and reduces its sensitivity. In addition, the PHEM weekly report did not include private health facility reports this also may reduce the sensitivity.

Predictive Value Positive

We defined predictive value positive (PPV) as the proportion of true positive slides or rapid diagnostic tests among all slides and rapid diagnostic tests performed. Therefore, we calculated the PPV using the formula:

$$\text{PPV} = \frac{\text{Number of RDT or slide confirmed malaria cases}}{\text{Number of RDT or slide tests performed for malaria diagnosis}} \times 100$$

where, the number of RDT/slide tests performed for malaria diagnosis and number of RDT or slide confirmed malaria cases were 42, 919 and 9, 651 respectively. Hence, the PPV of the malaria surveillance system was 22.5 percent. The PHEM weekly report did not include private health facility reports, this may reduce the PPV of the malaria survielance system.

Data quality

Data quality reflects the completeness and validity of the data recorded in the public health surveillance system. Examining the percentage of "unknown" or "blank" responses to items on surveillance forms is a straightforward and easy measure of data quality. However, a full assessment of the completeness and validity of the system's data might require a special study. On the other hand, completeness can be measured as the percentage of health facilities reported weekly to health facilities that are expected to report weekly [15-18]. At the district level, the PHEM weekly report completeness was followed with a follow-up chart and it was 98% (Table 3.8).

Timeliness

Timeliness of health post reports at health center level was difficult to measure since the date of receiving the report was not recorded and there was no any means of the following timeliness and completeness. At the district level, the PHEM weekly report timeliness was followed with a follow-up chart and it was 84% (Table 3.8).

Table 3.8. Timeliness and completeness of the last five epidemiological weeks' report in Jabi-Tehinan district, Northwest Ethiopia, 2009 EFY

Health Centers	Ethiopian week numbers*					Total timely reports
	48	49	50	51	52	
1. Mankusa	1	1	1	1	1	5
2. Agomamit	1	1	1	1	1	5
3. Kuni	1	0	1	1	2	3
4. Asewa	1	1	1	2	1	4
5. Yeraber	1	1	1	1	2	4
6. Jiga	2	1	1	1	1	4
7. Fetegem	1	1	2	1	1	4
8. Woyna	1	1	2	1	1	4
9. Maksegnin	1	2	1	2	1	3
10. Malan	1	1	1	1	1	5
11. Awunt	1	1	1	1	1	5
HCs Expected to report	11	11	11	11	11	55
HCs Reported	11	10	11	11	11	54
Timeliness (%)	90.9	81.8	81.8	81.8	81.8	83.6
Completeness (%)	100.0	90.9	100.0	100.0	100.0	98.2

*Reported timely=1, Reported late=2, No report at all=0

Representativeness: All the malaria surveillance data were reported from all government health facilities in the district, in which the primary health service coverage is 96%. Hence, the data from this surveillance system is representative.

3.5 Discussion

The most commonly used methods to prevent mosquito bites are sleeping under an ITN and spraying the inside walls of a dwelling with an insecticide – an intervention known as IRS. Long-lasting insecticidal nets (LLINs) are the main type of ITNs distributed in the district and are estimated to have an effective lifespan of 3 years[20]. Almost all households at the malarious area of Jabi-Tehinan district had at least one ITN, of which nearly half (53%) were using the net. The ownership of ITN was high as compared with the report of EMIS 2015, 64%[2], study findings in Dabat HDSS site, northwest Ethiopia, 16%[21] and sub Saharan Africa in 2016, 80%[1]. The utilization of ITN was 53% and it was low as compared with recent study findings in Northwest Ethiopia particularly, in Dabat district, 69%[21] and Kola Diba town, 92%[22]. However, there was no documented record that can show the proportion of households with sufficient nets, prioritized children and pregnant women sleeping under ITN. In addition, other vector control activities, mainly larval control through environmental management (draining, filling, compacting and clearing) were practiced in the district.

The malaria surveillance system detected only 58% of cases in the district which is weak as compared the national surveillance with 66% of cases detected by the surveillance system[1]. This may be due to the unknown exact burden of malaria, a large proportion of patients seek treatment in the private sector, and these cases are not captured by existing surveillance systems. Also, patients attending public health facilities may not receive a diagnostic test[3]. On the other hand, the system detects more as compared with the 2015 global malaria surveillance system with 19% cases detected by the surveillance system[3].

Malaria accounted for seven percent of all outpatient and inpatient cases in the health facilities. This was in line with a study on in southern Ethiopia, 6.1% [23]. However, the finding was low as compared to study findings in Benna Tsemay district of south Ethiopia, 28% [24], a rural health center in Northwest Ethiopia, 36%[25], in West Armachiho district, Northwest Ethiopia, 18%[26]. On the other hand, the study finding was higher than the malaria prevalence reported by a study done in Woreta health center, Northwest Ethiopia, 5.6% [27].

The malaria prevalence among all age groups was 22.2 percent and 22.7 percent by RDT and microscopy, respectively. This was high as compared with the national malaria prevalence which was 1.2 percent and 0.5 percent by RDT and microscopy, respectively[2]. Overall, malaria prevalence in the district is high, 22.5 percent , compared with national malaria prevalence[2], findings in Jimma town, southwestern Ethiopia, 5%[22], 19%[28], Northwest Ethiopia, 5% [27] and Southern in Ethiopia, 6%[22]. On the other hand, the prevalence was low as compared with study findings in Northwest Ethiopia, 33% [29], 37%[25] and Southern Ethiopia, 28%[24].

The annual paracite incidence was high in Jimat and Berkegn health posts, 105 case per 1000 population per year and 143 cases per 1000 population per year, respectively. Aunt health center also is in the high malaria strata with 143 cases per 1000 population per year. The overall malaria incidence was in the moderate strata, 52 cases per 1000 population per year.

A public health surveillance system is useful if it contributes to the prevention and control of adverse health-related events as a result of evidence based decision making[6]. However, the analysis and interpretation of the data from the malaria surveillance system and using information for decision making was limited to Jabi-Tehinan district. This indicates the less usefulness of malaria surveillance system in the district. Usefulness might be affected by the other attributes of a public health surveillance system. For example, increased sensitivity might afford a greater opportunity for identifying outbreaks and understanding the natural course of an adverse health-related event in the population under surveillance, improved timeliness allows control and prevention activities to be initiated earlier, increased predictive value positive enables public health officials to more accurately focus resources for control and prevention measures, a representative surveillance

system will better characterize the epidemiologic characteristics of a health-related event in a defined population. In addition, Public health surveillance systems that are simple, flexible, acceptable, and stable will likely be more complete and useful for public health action[6].

The report completeness and timeliness of the surveillance system were 98 percent and 83% respectively. This was good as compared to the WHO recommendation, 80% and above [15-18]. However, a full assessment of the completeness and validity of the system's data might require a special study. Quality of data is influenced by the performance of the screening and diagnostic tests (i.e., the case definition) for the health-related event, the clarity of hardcopy or electronic surveillance forms, the quality of training and supervision of persons who complete these surveillance forms, and the care exercised in data management. A review of these facets of a public health surveillance system provides an indirect measure of data quality.

Acceptability is a largely subjective attribute that encompasses the willingness of persons on whom the public health surveillance system depends to provide accurate, consistent, complete, and timely data. The sensitivity of malaria surveillance system was low. However, a public health surveillance system that does not have high sensitivity can still be useful in monitoring trends as long as the sensitivity remains reasonably constant over time. Questions concerning sensitivity in surveillance systems most commonly arise when changes in the occurrence of a health-related event are noted. Changes in sensitivity can be precipitated by some circumstances (e.g., heightened awareness of a health-related event, the introduction of new diagnostic tests, and changes in the method of conducting surveillance).

The positive predictive value of the surveillance system was 22.5% which was nearly in agreement with the national slide positivity rate according to world malaria report 2017, 27%[1]. However, it is high as compared to the 2015 malaria indicator survey report, 1.2 percent RDT positivity rate and 0.5 percent microscopy positivity rate[2]

3.6 Conclusions

The malaria surveillance system in Jabi-Tehinan district was weak. The usefulness of the malaria surveillance system was limited. The overall malaria incidence was moderate and the ITN utilization was low. Surveillance data analysis, interpretation and utilization of information for evidence-based decision making was limited. ITN use and IRS were factors negatively associated with the malaria incidence and prevalence.

3.7 Recommendations

To improve the performance of malaria survielance system, It is better if the Jabi-Tehinan district health office work more on strengthening the malaria surveillance system through improving insecticide-treated net utilization, case detection, surveillance data analysis, interpretation and data and utilization. It is also good to conduct further studies to identify factors associated with malaria case detection and insecticide-treated net utilization.

References

1. WHO, Globlal Task Force on Chollera Outbreak. Chollera outbreak ; Assessing the outbreak response and improving prepardeness. Geneva, WHO/CDS/CPE/ZFK/2004.4. 2004.
2. Morbidity and Mortality Weekly Report (MMWR), Global routine vaccination coverage. 2013.
3. Ethiopian Health and Nutrition Research Institute, Federal Democratic Republic of Ethiopia. Guideline on chollera outbreak management Ethiopia. May 2011, Addis Ababa, Ethiopia.
4. Bartels, S.A., et al., Investigation of a cholera outbreak in Ethiopia's Oromiya Region. Disaster Med Public Health Prep, 2010. 4(4): p. 312-7.
5. Tarantola, D., et al., Is immunization coverage in Africa slipping? An evaluation of regional progress to 2013. Ann Vaccines Immun, 2014. **1**.
6. Dickson Shey Nsagha, et al., Assessing the risk factors of cholera epidemic in the Buea Health District of Cameroon. BMC Public Health, 2011.
7. Central Statistical Agency (CSA) (Ethiopia) and ICF International, Macro ORC. Ethiopia Demographic and Health Survey 2011. Central Statistical Agency and ICF International: Addis Ababa, Ethiopia and Calverton, Maryland, USA; 2011.
8. Introduction of Rotavirus vaccine in Ethiopia. Training Manual for Health Worker. 2013.
9. Mohamud, A.N., et al., Immunization coverage of 12–23 months old children and associated factors in Jigjiga District, Somali National Regional State, Ethiopia. BMC Public Health, 2014. **14**.
10. Animaw, W., et al., Expanded program of immunization coverage and associated factors among children age 12–23 months in Arba Minch town and Zuria District, Southern Ethiopia, 2013. BMC Public Health, 2014. **14**.
11. Sedgwick, P., Meta-analyses: what is heterogeneity? BMJ, 2015. **350**: p. h1435.
12. Nordmann, A.J., B. Kasenda, and M. Briel, Meta-analyses: what they can and cannot do. Swiss Med Wkly, 2012. **142**: p. w13518.
13. Vijay S, et al., Treatment Outcome and Mortality at One and Half Year Follow-Up of HIV Infected TB Patients Under TB Control Programme in a District of South India. PLoS ONE 2011. 6(7).
14. Zenebe, Y., et al., Profile of tuberculosis and its response to anti-TB drugs among tuberculosis patients treated under the TB control programme at Felege-Hiwot Referral Hospital, Ethiopia. BMC Public Health, 2016. 16: p. 688.
15. Machingaidze S, Wiysonge CS, Hussey GD. Strengthening the Expanded Program on Immunization in Africa: Looking beyond 2015. PLoS Med. 2013;10(3).:e1001405. doi:10.1371/journal.pmed.1001405: 2-3.
16. Negussie, A., et al., Factors associated with incomplete childhood immunization in Arbegona district, southern Ethiopia: a case – control study. BMC Public Health, 2016. **16**(1): p. 27.
17. Ethiopian Health and Nutrition Research Institute, Public Health Emergency Management Centre: Basic Level Public Health Emergency Management Training Module Participant Modules October 2012, Addis Ababa, Ethiopia.
18. Ayele, D.G., T.T. Zewotir, and H.G. Mwambi, Prevalence and risk factors of malaria in Ethiopia. Malaria journal, 2012. **11**: p. 195-195.
19. Nshimirimana, D., R. Mihigo, and C.J. Clements, Routine immunization services in Africa: back to basics. J Vaccines Immun, 2013. **1**.
20. Berhan, Y. and A. Berhan, Skilled health personnel attended delivery as a proxy indicator for maternal and perinatal mortality: a systematic review. Ethiopian journal of health sciences, 2014. **24 Suppl**(0 Suppl): p. 69-80.
21. Muchie, K.F., et al., Rural households at risk of malaria did not own sufficient insecticide treated nets at Dabat HDSS site: evidence from a cross sectional re-census. BMC Public Health, 2017. **17**(1): p. 888.
22. Alemu, M.B., et al., Utilization of insecticide treated bed net and associated factors among households of Kola Diba town, North Gondar, Amhara region, Ethiopia. BMC Res Notes, 2018. **11**(1): p. 575.
23. Debo, G.W. and D.H. Kassa, Prevalence of malaria and associated factors in Benna Tsemay district of pastoralist community, Southern Ethiopia. Trop Dis Travel Med Vaccines, 2016. **2**: p. 16.
24. Belete, E.M. and A.B. Roro, Malaria Prevalence and Its Associated Risk Factors among Patients Attending Chichu and Wonago Health Centres, South Ethiopia. J Res Health Sci, 2016. **16**(4): p. 185-189.

25. Birhanie, M., et al., Malaria, Typhoid Fever, and Their Coinfection among Febrile Patients at a Rural Health Center in Northwest Ethiopia: A Cross-Sectional Study. Adv Med, 2014. **2014**: p. 531074.

26. Aschale, Y., et al., Prevalence of malaria and associated risk factors among asymptomatic migrant laborers in West Armachiho District, Northwest Ethiopia. Res Rep Trop Med, 2018. **9**: p. 95-101.

27. Derbie, A. and M. Alemu, Five Years Malaria Trend Analysis in Woreta Health Center, Northwest Ethiopia. Ethiop J Health Sci, 2017. **27**(5): p. 465-472.

28. Degefa, T., et al., Malaria incidence and assessment of entomological indices among resettled communities in Ethiopia: a longitudinal study. Malar J, 2015. **14**: p. 24.

29. Alelign, A., Z. Tekeste, and B. Petros, Prevalence of malaria in Woreta town, Amhara region, Northwest Ethiopia over eight years. BMC Public Health, 2018. **18**(1): p. 990.

Annex 3.1: Malaria surveillance system evaluation data collection checklists

> Date of interview: _____, Region: _____, Zone: _____

Part I: Background information:

- District name _____, Health facility name :_____, Type of Health facility:_____
- Catchment population:
  - By sex: 1. Male: _____, 2. Female: _____
  - By residence: 1. Rural: _____, 2. Urban: _____
  - Under-five children:_____
  - Pregnant women:_____
  - Population at risk of malaria:_____
- Number of kebeles:
  - 1. Rural_____, 2. Urban _____
  - Number of malarous kebeles: _____
- Health facilities:
  - Government: 1. Hospital:_____, 2. HCs____, 3. HPs:_____
  - Private: 1. Hospital_____, 2. All other private clinics ____

Part II: Communication and reporting system assessment

- Which communication material did you have?

  a.

  b. E-mail

  c. Wired-phone

  d. Hard copy

  e. Mobile

  f. Other: _____

  g.

- Did you have address of health bureau PHEM officers? 1. Yes 2. No
- How frequently you communicate with the health office PHEM officers on emergencies and other daily activities?

a.

b. Daily

c. Weekly

d. Monthly

e. Other (Specify) ……

- Did you have address of HP HEWs? 1. Yes 2. No
- How frequently you communicate with the HP HEWs on emergencies and other daily activities?

a.

b. Daily

c. Weakly

d. Monthly

e. Other (specify) …..

- When are you expected to send weekly report to the next higher level? (Saturday/ Sunday/ Monday/ Other…..)
- When are you expected to receive weekly report from HPs? (Sat/Sun/ Mon/ Other…..)

Part III: Malaria prevention and control

- Total number of households _____
- Number of households received at least one ITN:_____
- Number of households protected with IRS in the past 12 months:_____
- ITN utilization:

A. Number of households visited (at a randomly selected epidemiological week)____

B. Number of households using ITN at the time of visit:_____

- Other vector control activities _____
- Total number of tests performed for malaria diagnosis disaggregated by test type:

A.

B. RDT_____

C. Microscopy/Slides _____

- Number of RDT confirmed malaria cases disaggregated by species

A.

B. Plasmodium falciparum _____

C. Plasmodium Vivax _____

- Number of microscopy confirmed malaria cases, disaggregated by species

A.

B. Plasmodium falciparum _____

C. Plasmodium Vivax _____

- Total number of patients with microscopy or RDT confirmed malaria disaggregated by age:

A. < 5 years _____, B. 5-14 years_____, C. 15 and above years _____

- Total number of patients with microscopy or RDT confirmed malaria disaggregated by sex:

A.

B. Male _____

C. Female_____

Part V: Epidemic responses and preparedness assessment

- Is the established rapid response team (RRT)? (check the document) _____
- Is the established RRT functional? (check for activities done by the RRT) _____
- Did the RRT have regularly scheduled meeting time? (Yes/No) _____
- It the ant budget allocated for emergency response (Yes _____ ETB/ No)

Part IV: Necessary materials for malaria survielance system

24. Check for the of availability of necessary guidelines, report forms, registers and other materials used for surveillance (If available, “Yes”, otherwise “No”)

MATERIALS	YES	NO	MATERIALS	YES	NO
The national/regional PHEM guideline	☐	☐	Functional computer	☐	☐
The national malaria guide line	☐	☐	Microscope	☐	☐
Standard case definition	☐	☐	RDT	☐	☐
Community case definition	☐	☐	Emergency response and preparedness plan	☐	☐

MATERIALS	YES	NO	MATERIALS	YES	NO
Was the case definition posted?	☐	☐	Established rapid response team	☐	☐
Rumor log book	☐	☐	Functional rapid response team	☐	☐
OPD/IPD registers	☐	☐			
Epidemic report form	☐	☐			
Case based reporting format	☐	☐			

Section IV: Supervision, feedback and survielance training assessment

25. Are you trained on survielance system particularly on malaria survielance?

26. It there a supervision plan?

27. The frequency of supervision plan _____

28. How many times did you supervise the next lower level HFs?

29. Had you faced any challenge on supervision?

30. Did you send feedback of your supervision to the health facilities?

31. Did you have regular supervision checklist**?**

32. How did you supervise?

33. Had you reviewed about surveillance practice by higher level supervision?

34. Had you faced any challenge on supervision and feedback

35. Were you supervised by higher level officers in 2009 EFY?

36. How many times were you supervised by higher level officer?

Part IV: Malaria survielance data analysis, result presentation and information use

37. Basic computer skill (Yes/No):

    A. Microsoft word _____

    B. Microsoft excel _____

    C. Microsoft PowerPoint _____

    D. Other statistical software(s) (specify)___

38. Survielance data/information management system

A. Paper based,

B. Electronic

C. Both paper based and electronic

39. Was the malaria survielance data analyzed in a meaningful way?

40. Are the technical reports sound and well prepared?

41. Was the analysis result presented to decision makers and other concerned bodies?

42. Were the reports disseminated on time?

43. Have program managers/ decision makers used the data?

Part-III: Attributes of malaria surveillance system

	ATTRIBUTES		JUSTIFICATION (IF NECESSARY)
1	**Usefulness**		
	Was the data from malaria surveillance system analyzed?	1 Yes 2 No	
	Was the analyzed data from malaria surveillance system interpreted?	1 Yes 2 No	
	Was the information used for decision-making and take action? (prove or see any evidence)	1 Yes 2 No	
2	**Simplicity**		
	Is the case definition for malaria is easy to understand?	1 Yes 2 No	
	It the malaria report form easy to complete?	1 Yes 2 No	
	Malaria survielance system is time taking	1 Yes 2 No	
	Malaria data is easy to manage and analyses	1 Yes 2.No	
	Sending report to the next higher level is easy	1 Yes 2 No	
	Malaria cases were investigated through	1 HF visit by patients 2 Home visit 3 Telephone contact 4 Other (specify) _	
3	**Flexibility**		
	Is the system easy to add new variables?	1 Yes 2 No	
	Is the surveillance system easy to integrate with other systems?	1 Yes 2 No	
	Any difficulty for flexibility?	_____	
	Sensitivity		
	Number of malaria cases expected to be existed?	_____	
	Total number confirmed malaria cases detected by the survielance system	_____	
4	**Predictive Value Positive**		
	Number of RDT or slide tests performed for malaria diagnosis	_____	
	Number of RDT or slide confirmed malaria cases	_____	
5	**Completeness and timeliness**		
	Number of health facilities expected to report (use atleast five Epi weeks data)	_____	
	Number of health facilities reported (use atleast five Epi weeks data)	_____	
	Number health facilities reported on time (use atleast five Epi weeks data)	_____	
6	**Acceptability**		

	ATTRIBUTES		JUSTIFICATION (IF NECESSARY)
	Do you think all the reporting agents accept and well engaged to the surveillance activities?		
7	**Representativeness**		
	Was the survielance data from all areas of the community (all health facilities in the district)		

CHAPTER 4

Community Health Profile

A health profile description of Yilmana Densa district of West Gojjam zone, northwest Ethiopia, EFY 2008

Tesfahun Taddege GEREMEW[3], Lemma Derseh, Getie Zeru and Meklit Mekonnen

3. Correspondence to : ttaddege@gmail.com↩

Abstract

Background: Community health profile is a summary of health and health related information for a specific community and provides quick and easy access to the most commonly requested health indicators. Hence, this assessment was aimed to describe the health status of the community in the assessment area.

Methods: A descriptive cross-sectional study was conducted in Yilmana Densa district of northwest Ethiopia from Miyazya 05 -20 EFY 2009.The primary and secondary data was collected though. The secondary data were extracted from eHMIS database and annual reports of the health and other related sectors using data extraction tool. Descriptive measures were computed.

Result: Yilmana Densa is the second most populous district next to Mecha district in west Gojjam Zone, located in the Northwest part of Ethiopia. Administratively, it is subdivided into 11 clusters which are fatherly subdivided into 36 kebeles (3 urban & 33 rural), the smallest administrative unites. The estimated catchment population in EFY 2008 was 265,615 of which 51 percent were estimated to be females and 13.54 percent under-five children. The district has a total of 69 health facilities serving the community in the district with one primary hospital, 11 health centers , 36 health posts (33 rural & 3 Urban) and a total of 14 private clinics. The primary health care coverage is 66.4 percent. The functional 1 to 5 networks were 55%, antenatal care atleast one visit and births attended by skilled personnel were 76% and 31 respectively. The percentage of infants who received all basic vaccination was 64% and the percentage of severely underweight under five children was 1.2%. TB case detection 52%.

Conclusion: The health facilities in the district lack basic infrastructures and not adequately staffed with health professionals. The potential health coverage and major health service indicators' coverage were low. The health budget allocation was below the recommended percentage.

4.1 Introduction

Community health profile is a summary of health and health related information for a specific community [1-4]. It measures the health of a community across a wide array of indicators[2] and provides quick and easy access to the most commonly requested health indicators[3]. Despite heightened investments in health systems of Ethiopia, few health sectors have full and up-to-date information on the health profile that covers the health and health related sectors. Fewer still have accurate, up-to-date information required to assess and monitor the "utilization" of health information[5].

Bringing together the existing information into one place to provide a snapshot of health information across each local authority, health profile is a useful health information summary to support local authorities, planners, and decision makers to improve health of people in their communities[1]. It updates the health and health related indicators of a community to include the most recent information from health management information system, which health professionals currently use[6].

In the current revised health management information system (HMIS) implementation, although routine reporting contribute to understand health status, local health status assessment are required to further complement the available routine data. It is impossible to obtain all health and health related data exclusively through HMIS, conducting health profile assessment is crucial to capture selected set of data and triangulate various sources in order to improve health information synthesis and utilization.

Improving community health through effective planning and decision-making requires good information about the health status of the community[4, 7] and enabling updated health estimates will help for local planners and decision makers for evidence-based decision making[3].

Hence, the finding in this community health profile assessment can be used to monitor and evaluate the performances of the health and health related programs, identify health priorities, allocate resources, and guide the development partners for new projects or programs. This report may also reveal current gaps in the district collected data. For example, notably low rate of specific service utilization may indicate the need for improvements to that specific service.

4.2 Objectives

GENERAL OBJECTIVE

★ To assess the community health status of Yilmana Densa district, West Gojjam zone, Northwest Ethiopia, EFY 2008.

SPECIFIC OBJECTIVES

1. To describe the heath and health related basic infrastructures and human resources
2. To explain health service coverages in the district
3. To describe the health budget allocation for the health sector of the district.

4.3 Methods

Study area and settings

This health profile assessment was conducted in Yilmana Densa district, the second most populous district, in West Gojjam zone administration of Northwest Ethiopia from Miyazya 05 - 20 EFY 2009. Administratively, the district is subdivided into 11 clusters (subgroups of kebeles) and 36 Kebeles (smallest administrative units). A descriptive cross sectional study was conducted from Miyazya 4 -20 EFY 2009 by visiting different sectors (Health office, Finance & economic development office, district administration office, culture and truism development office, water, agriculture and rural development offices) in the district which have relation on the outcome of health activity.

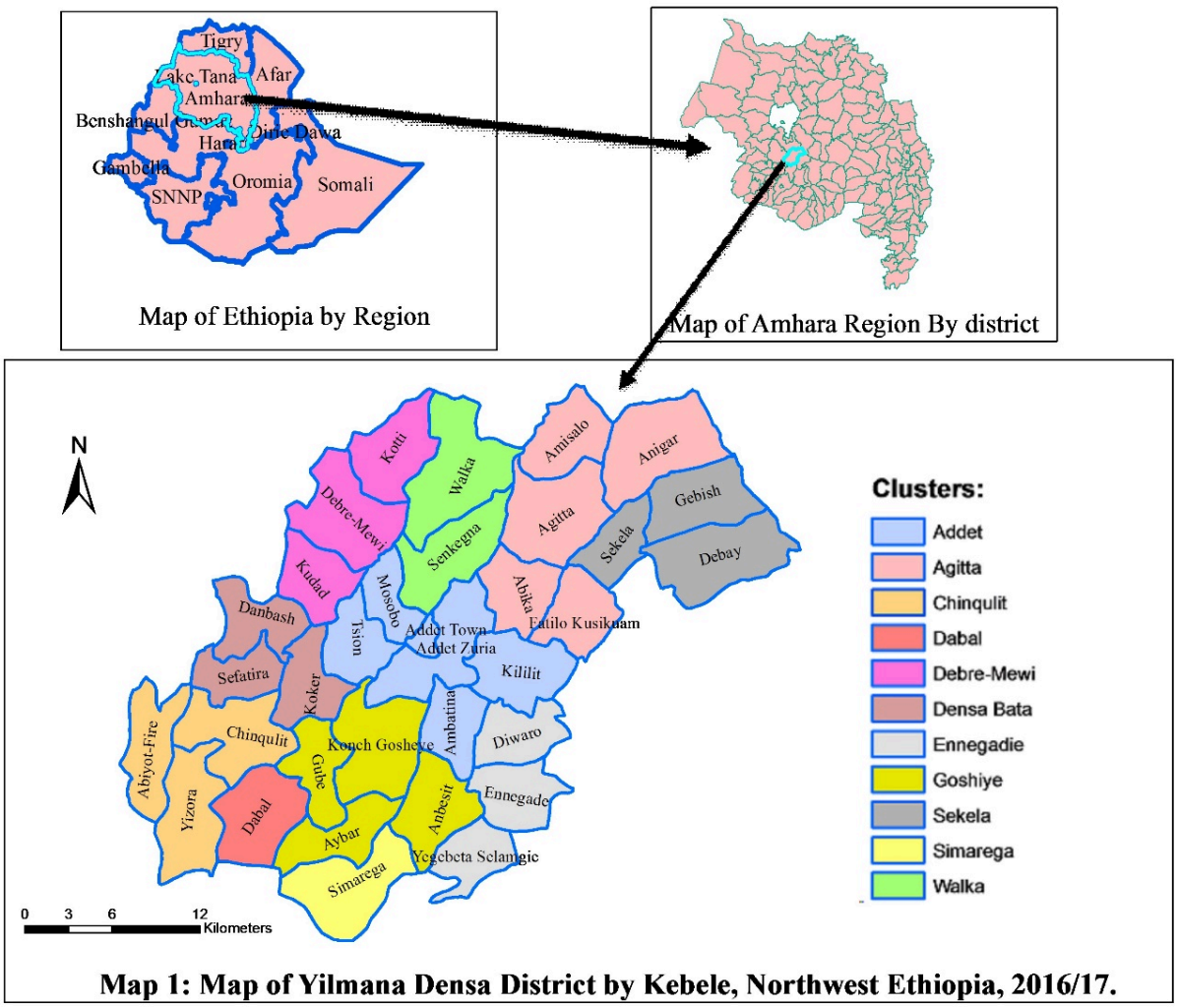


Figure 4.1. Map of Yilmana Densa district by cluster and kebele, Northwest Ethiopia, 2008 EFY.

Data sources and extraction

Data was collected through key informants interview (primary data) and document review (secondary data). The secondary data were extracted from the eHMIS database and annual reports of the district health office using data extraction tool. Additionally, health related data were collected from other sectors (Education office, Finance and

economic development office, district administration office, culture and truism development office, water and energy office and agriculture & rural development office).The analysis for this health profile was based on to the EFY 2008 data, but the performances of some common health indicators previous to EFY 2008 were reviewed for comparison purposes.

Data management and analysis

The aggregated count data were entered into excel spreadsheet and further recoding was done. The data was analyzed using MS-Excel 2013 and SPSS 25. Descriptive measures were computed and absolute numbers, percentage and rate were used to present the data in words, tables, charts and graphs.

Ethical consideration

Legal letters were obtained from University of Gondar, department of epidemiology and biostatistics and West Gojjam zone health department. Permission was obtained from Yilmana Densa district health office.

4.4 Results

Background characteristics of the study area

Historical background

Yilmana Densa (named after "*Yilmana*" and "*Densa"*, the two brothers who were the leaders of the area) is the second most populous district in West Gojjam zone, Amhara national regional state of Ethiopia. Its Major town is Addet (named after Madam"Ade" who lived in the area, which means "Mother or Mother Land" in "Tigrigna") but no clear evidence when it was established. Since 2013, the district has been divided into two administrations, Addet town administration and Yilmana Densa woreda administration, but most of the sectoral offices (Like Health, education, Agriculture, etc.) are not still separated into those two administrations. So, this health profile description incudes both the town and the district administration profiles.

Geography and climate

Yilmana Densa district is located in the Northwestern part of Ethiopia, at a distance of 41km East from Bahir Dar, the capital of Amhara regional state, lying between 11º 29' 59.99''N latitude and 37º 19' 60.00" E longitude. The district is stretched in an area of 99180 hectares with an altitude of 2044 meters above sea level. It is bordered Quarit district to South, Sekela district to southwest, Mecha district to west, Bahir Dar Zuria district to North, Gonji-Kollella district to East and Abay River which separates from south Gondar zone. The climatic variations broadly categorized into three: Kolla (12%), Woyna Degas (64%) and Dega (24%)**.** The annual temperature ranged from 8.8^{o}c to 25.2^{o}c with 1270 mm of annual average rainfall.

Organizational administrative and demography

Administratively, Yilmana Densa district is subdivided into 11 clusters (subgroups of kebeles): Addet, Agitta, Debre-Mewi, Walka, Densa Bata, Chinqulit, Gosheye, Daval, Simarega, Ennegadie and Sekela. Furtherly, each cluster is subdivided into Kebeles, the smallest administrative unit. Overall, there are 33 rural and 3 urban kebeles in the district.

In 2008 EFY, the total population was estimated to 265,615, of which 51.4% were females. The sex ratio was 95 males per 100 females which was higher in urban areas (98 Males per 100 females) than in rural areas (95 Males per 100 females).

The pyramidal age structure reflects the large number of children under age 15. Children under age 15 accounted for nearly 37 percent of the total population, a feature of populations with high fertility levels, while only about 3.6 percent of the population is over age 65. The estimated young and old dependency ratio is 62% and 4% respectively. Overall, the estimated dependency ratio in the district is 66.5%. This population distribution is almost similar to that observed in the 2011 EDHS (Figure 4.2).

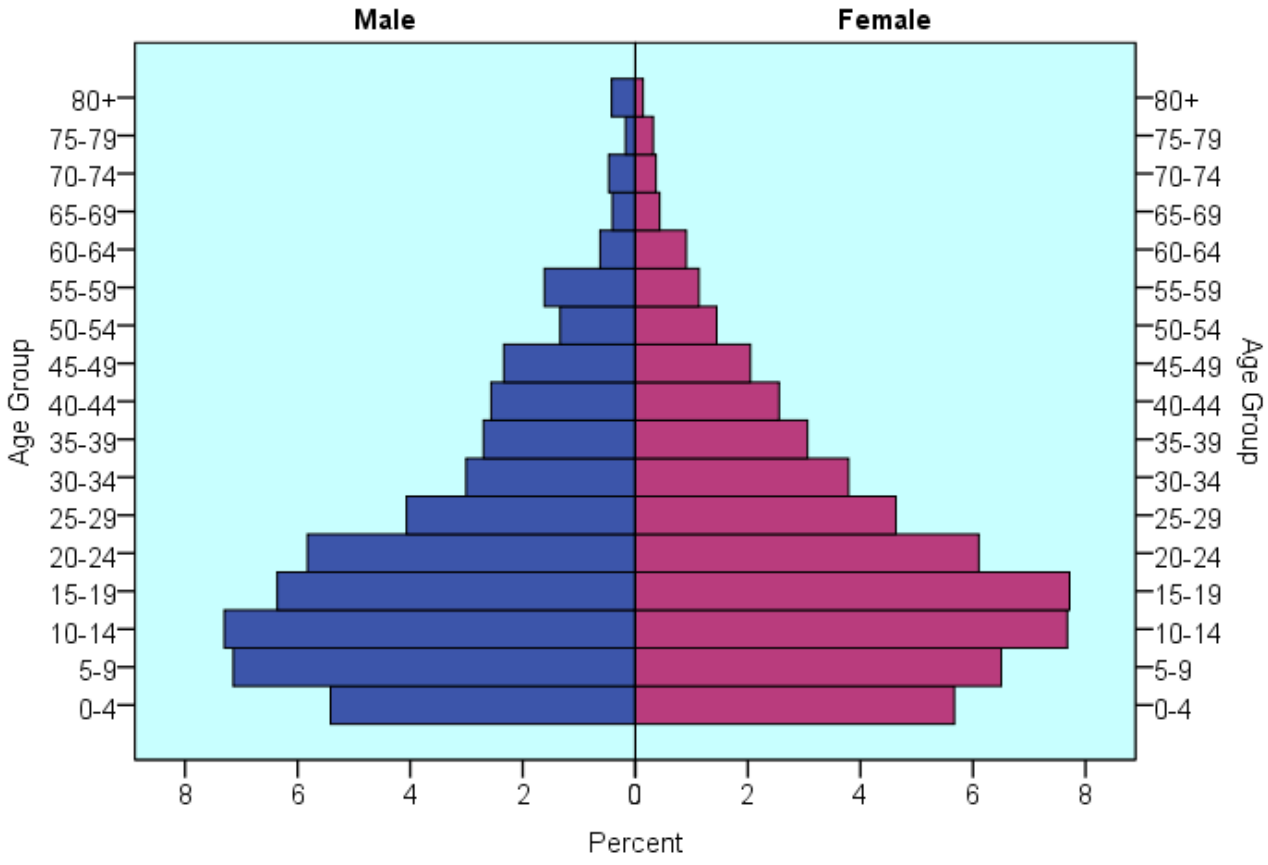


Figure 4.2. Population pyramid, Yilmana Densa district, Northwest Ethiopia, EFY 2008

There were about 8,261 (3.11 percent) surviving infants, 35,964 (13.54 percent) estimated under-five children, 8, 951(3.11 percent) estimated pregnant women, 62,632 (20.23%) estimated women under 15-49 years age. The average estimated family size per household was 4.3 in a total 61,771 estimated households (Table 4.1).

Table 4.1. Estimated number of and population and households by clusters in Yilmana Densa district, Northwest Ethiopia, EFY 2008.

			Population (estimated)							
	Clusters	Total number of Households	Male	Female	Total	< 1 year Age	Children < 2 years	Children < 5 years	Women 15-49 years	Pregnant Women
1	Addet	13,259	27,710	29,306	57,016	1,773	2,879	7,720	13,444	1,921
2	Agitta	8,288	17,321	18,319	35,640	1,108	1,800	4,826	8,404	1,201
3	Chinqulit	5,900	12,330	13,040	25,370	789	1,281	3,435	5,982	855
4	Daval	1,781	3,722	3,936	7,658	238	387	1,037	1,806	258
5	Debre Mawi	5,434	11,356	12,010	23,366	727	1,180	3,164	5,510	787
6	Densa Bata	4,127	8,624	9,121	17,744	552	896	2,403	4,184	598
7	Ennegadie	4,477	9,357	9,896	19,253	599	972	2,607	4,540	649
8	Goshiye	7,813	16,327	17,268	33,595	1,045	1,697	4,549	7,922	1,132
9	Sekela	4,182	8,739	9,243	17,982	559	908	2,435	4,240	606
10	Simarega	2,557	5,343	5,651	10,995	342	555	1,489	2,593	371
11	Walka	3,953	8,260	8,736	16,996	529	858	2,301	4,008	573
	Total	61,771	129,089	136,526	265,615	8,261	13,414	35,964	62,632	8,951

Health and health related infrastructures

About 24 (73%) rural kebeles were connected with rural road access and 8 (22%) kebeles (5 rural and 3 urban) were accessed to electric power supply. Almost all the district areas were accessible to mobile network.

In the education sector, the district had one kindergarten (KG), 73 primary schools, four secondary schools including one preparatory school, and two higher institutions (colleges) including the technical and vocational education training (TVET) college. Fifty one (64%) schools were accessed to safe water supply and the all schools have common latrine.

In the health sector, the district has one primary hospital, 11 health centers, 36 health posts (33 rural and 3 Urban) and 14 private clinics. The primary health care coverage was 66.4 percent. Only 5 health centers (45.5%) were accessible for electric power supply and 4 health centers (36.4%) were accessible to safe water supply.

The health center health post density reflects the total number of health centers and health posts relative to population size (per 25000 & 5000 population respectively). The health center to population ratio was nearly 1 to 24,000, whereas the health post to population ratio was 1 to 7,300 (Table 4.2).

Table 4.2. The HF to population ratios compared to the standard of HF to population ratio in Yilmana Densa district, Northwest Ethiopia, EFY 2008.

Type of HF	Number of HFs	HF to population ratio	Standard of HF to population ratio
Primary hospital	1	1:265,615	1:100,000
Health center	11	1:24,147	1:25,000
Health posts	36	1:7,379	1:5,000
All private clinics	14	1:18,973	1:10,000
Drug stores	7	1:37,945	-
Total Health Facilities	69	-	-

Health development armies (HDAs)

There were a total of 9,570 one-to-five networks established until the end of EFY 2008, of which 57% of them were functional (perform activities that are expected from each 1 to 5 network).

Health system organization and human resource development

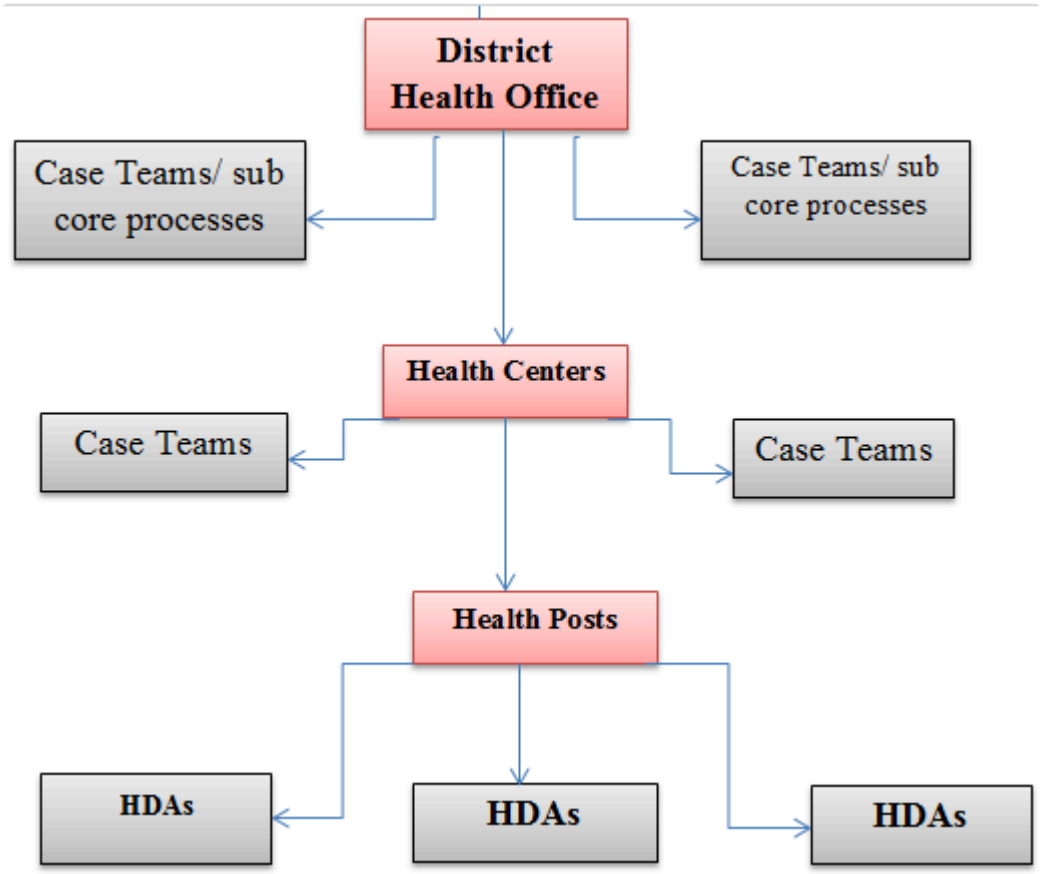


Figure 4.3. Organogram of the district health system, Yilmana Densa district, EFY 2008

There were a total of 464 heath staffs in the district of which 248 were males. The midwifery professional to population ratio is about 1 to 8, 569 while the medical laboratory professional to population ratio was 1 to 15,625 (Table 4.3).

Table 4.3. Number of health staffs by sex and health professional to population ratio in Yilmana Densa district, Northwest Ethiopia, EFY 2008.

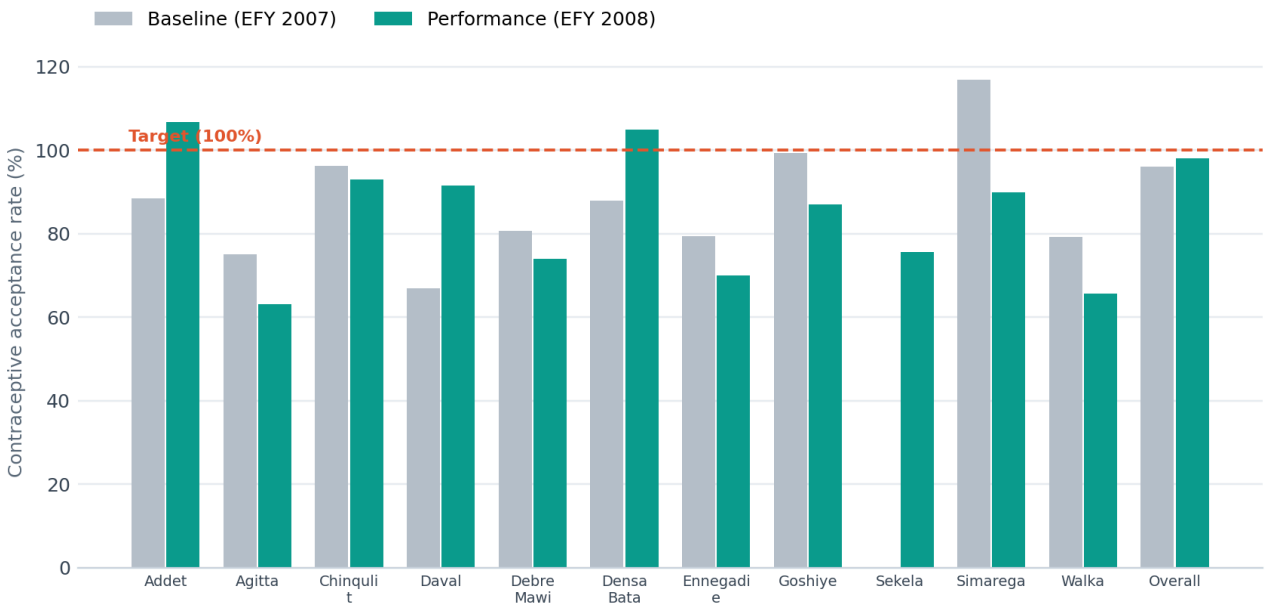
Professions	Sex			Health professional to population ratio	Standard of health professional to population ratio
	Female	Male	Both sexes		
General practitioners	1	5	6	1:44,270	1:10,000
Health officer	8	25	33	1:8,049	1:10,000
Nurses (diploma & degree)	33	49	82	1:3,240	1:5,000
Midwifery professionals (degree & diploma)	16	15	31	1:8,569	1:5,000
Medical lab. professionals (diploma & degree)	5	12	17	1:15,625	1:8,000
Pharmacy professionals (degree & diploma)	7	18	25	1:10,625	1:8,000
Surgeon (emergency & general)	1	2	3		
Environmental health professional	1	1	2		
Health extension worker	83	0	83	1:3,200	1:5,000
Health Information Technician (HIT)	4	3	7		
All supportive health workers	87	82	169		
Others*	1	5	6		
Total health staffs	**248**	**216**	**464**		

*Frontlines, Anesthetics, Radiographer & Psychiatry Professional

Maternal health services

Contraceptive acceptance rate (CAR)

The overall CAR in was 98% with the lowest rate at Agitta cluster (63%) and the highest rate at Densa Bata cluster (105%). An increment was observed in three clusters: Addet, Daval and Densa Bata, with the highest increment in Daval cluster, 24.6% (Figure 4.4). Nearly three-fifth (59%) were injectable among accepted methods (Figure 4.5)



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 4.4. Comparison of baseline, performance and target of CAR by cluster, Yilmana Densa district, Northwest Ethiopia, EFY 2008

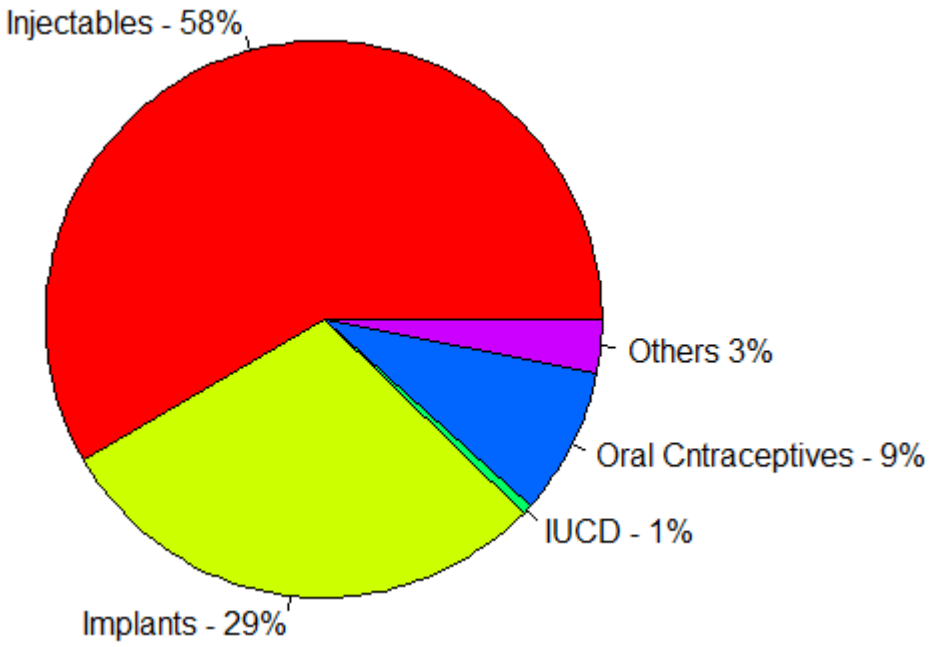
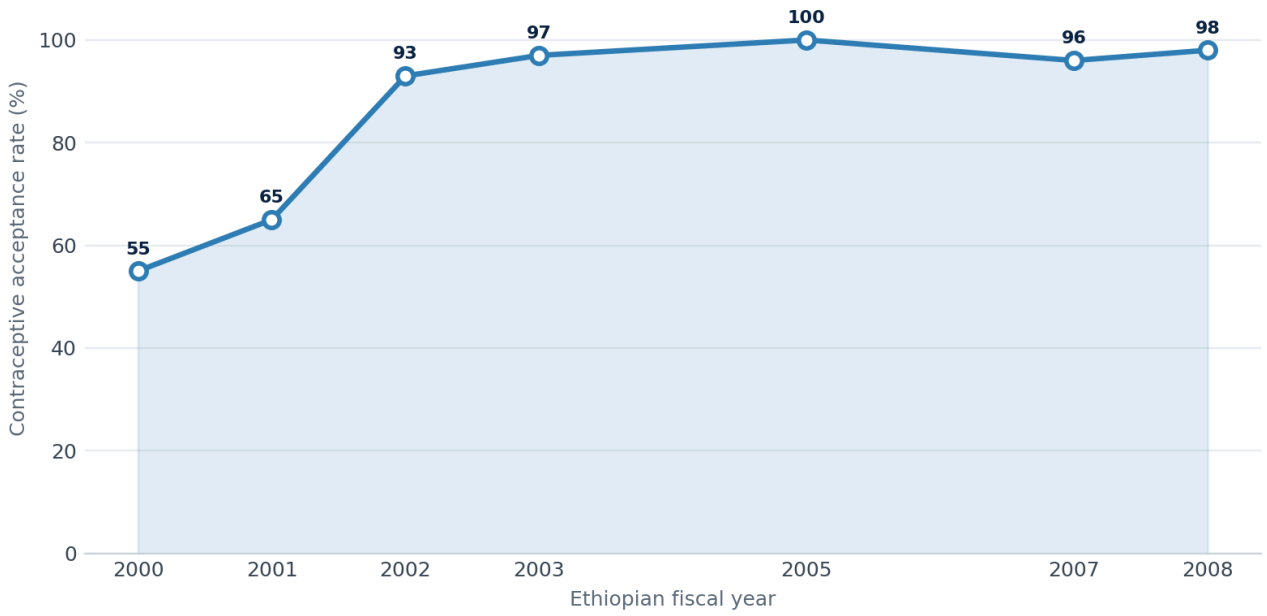


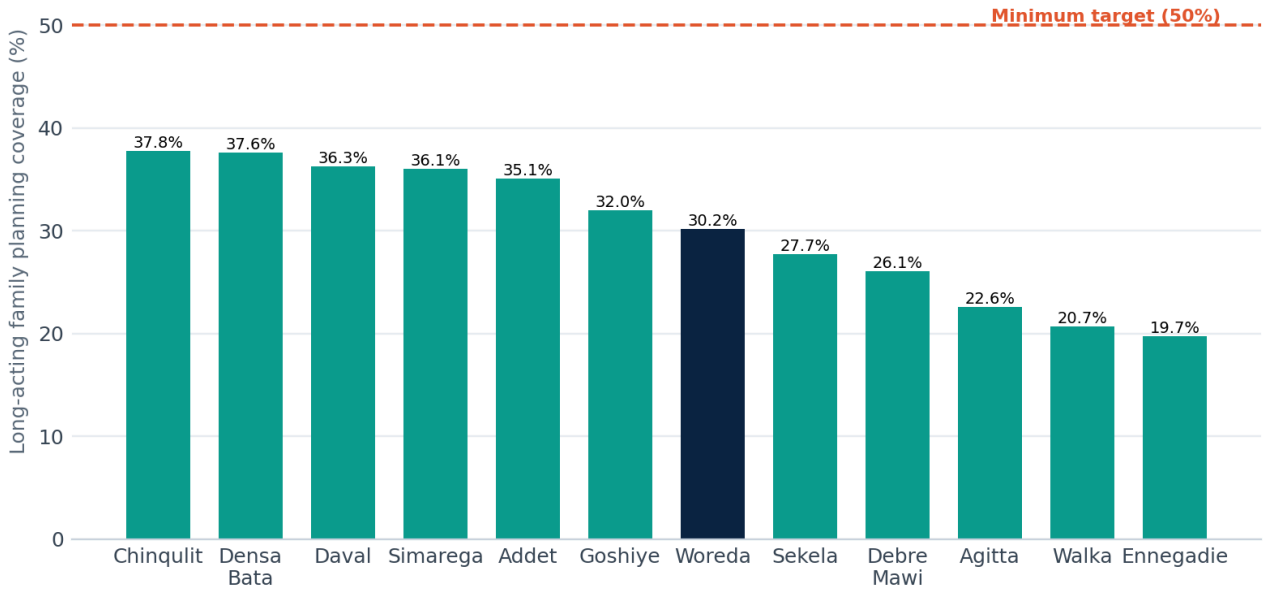
Figure 4.5. Proportion of modern contraceptive acceptors disaggregated by methods among the total new and repeat acceptors, Yilmana Densa district, Northwest Ethiopia, FEY 2008

CAR increased from 55% in EFY 2000 to 98% in EFY 2008 (Figure 4.6). The overall long acting family planning (LAFP) coverage was 30 % with relatively higher coverage in Chinqulit cluster (38%) and lower in Ennegadie cluster (18%) (Figure 4.7)



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 4.6. Trend in Contraceptive acceptance rate, Yilmana Densa district, EFY 2000-2008

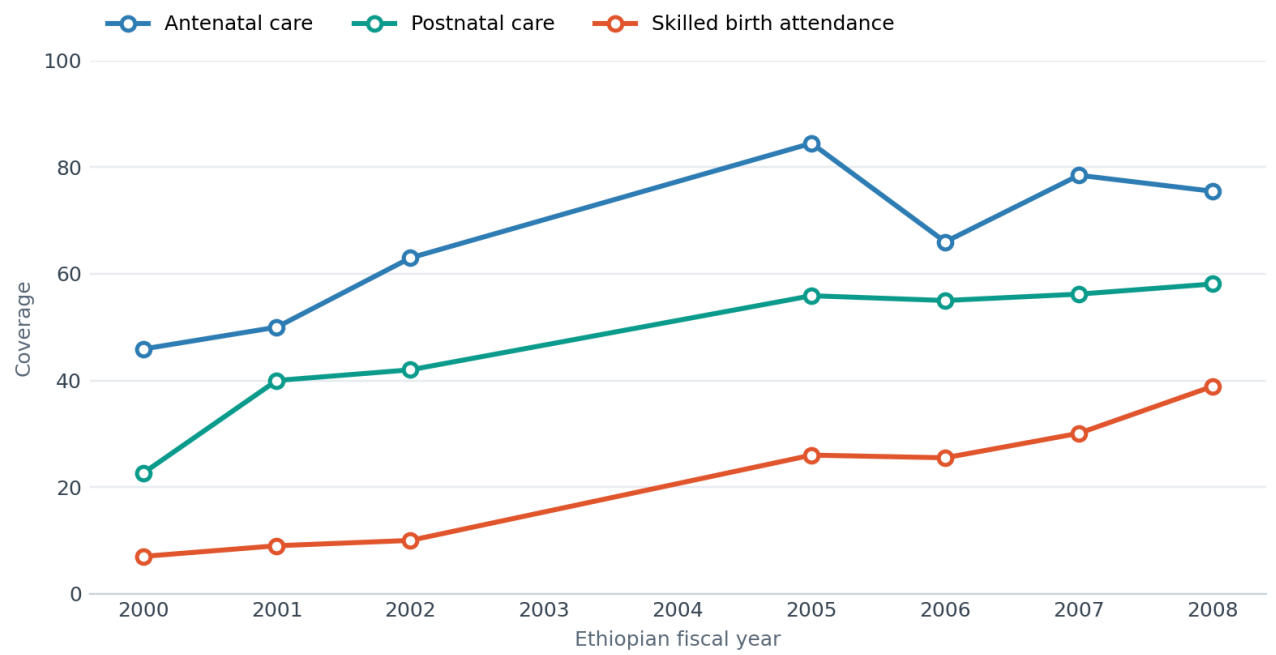


Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 4.7. Percentage of LAFP coverage by cluster, Yilmana Densa district, EFY 2008.

Antenatal care, birth by skilled personnel and postnatal care

The most recent report revealed that the coverage of first visit antenatal care was 76%, whereas, the coverages of skilled birth attendants and postnatal care were 39% and 58%, respectively (Figure 4.8). Among a total of 8,951 estimated pregnant women, only 50% were tested for HIV (Table 4.4).



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 4.8. Trend in antenatal coverage, percentage of births attended by skilled health personnel and postnatal coverage, Yilmana Densa district, Northwest Ethiopia, EFY 2000-2008

Table 4.4. Proportion of pregnant women who received antenatal care, tested for HIV and births attended by skilled health personnel in Yilmana Densa district, EFY 2008.

Health centers		Number of eligible PW	Proportion of PW who received ANC1	Proportion of PW who received ANC4	Percentage of pregnant and lactating women tested for HIV	Proportion of births attended by skilled health personnel
1	Addet	1,921	90.6	62.5	47.0	53.2
2	Agitta	1,201	59.7	37.5	45.1	29.6
3	Chinqulit	855	80.9	66.9	57.9	33.7
4	Daval	258	84.9	60.8	48.4	52.7
5	Debre Mawi	787	70.2	58.5	57.3	35.1
6	Densa Bata	598	101.0	72.2	63.5	56.4
7	Ennegadie	649	68.1	59.0	61.8	37.1
8	Goshiye	1,132	63.2	40.5	35.9	24.7
9	Sekela	606	77.2	46.5	52.3	36.5
10	Simarega	371	63.7	41.0	30.2	34.3
11	Walka	573	59.2	46.4	53.1	33.5
Ov	erall 8,951	75.5	53.9	49.6	38.9	

ANC1: First visit antenatal care, ANC4: Fourth visit anetenatal care, PW: Pregnant women

Regarding to maternal vaccination for tetanus toxoid (TT), 49.2% of pregnant women and 3% of non-pregnant women were received atleast two doses of TT (TT^{2+}).

Still birth rate

Among a total of 3,751 deliveries attended at health facilities, 37 (1%) were still births with the highest still birth rate at Debre Mawi health center (3%) and zero still birth rate at Ennegadie health center (Table 4.5).

Table 4.5. Proportion of still births among deliveries attended at health facilities by clusters, Yilmana Densa district, EFY 2008

Cluster	Number deliveries at health facility	Number of live births	Number of still births	Still birth rate
1. Addet	1050	1046	6	0.57
2. Agitta	378	377	4	1.06
3. Chinqulit	334	335	2	0.60
4. Daval	147	145	2	1.36
5. Debre Mawi	327	320	10	3.06
6. Densa Bata	356	357	2	0.56
7. Ennegadie	250	250	0	0.0
8. Goshiye	331	331	3	0.91
9. Sekela	232	230	3	1.29
10. Simarega	138	136	2	1.45
11. Walka	205	203	3	1.46
Total	**3,751**	**3,733**	**37**	**0.99**

Comprehensive abortion care

A total of 186 women (2.1 percent of the estimated pregnant women) received comprehensive abortion care (CAC) and more than three-fifth of them (64%) received safe abortion care (SAC). Nearly 4% were below 18 years old age (Table 4.6).

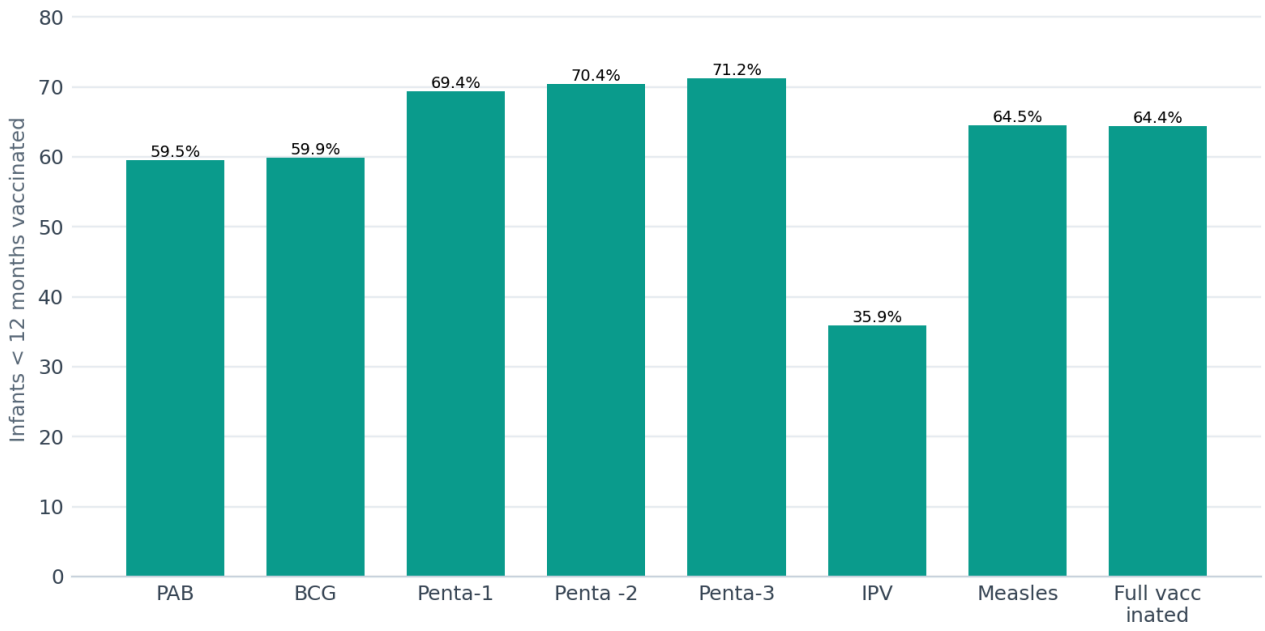
Table 4.6. Proportion of SAC and PAC disaggregated by age among total Comprehensive abortion care service users, Yilmana Densa, Northwest Ethiopia, EFY 2008 (n=186).

Age (in years)	SAC	PAC	CAC
< 18	3.2	0.5	3.8
18+	60.8	35.5	96.2
Overall	64.0	36.0	100.0

Child health and nutrition

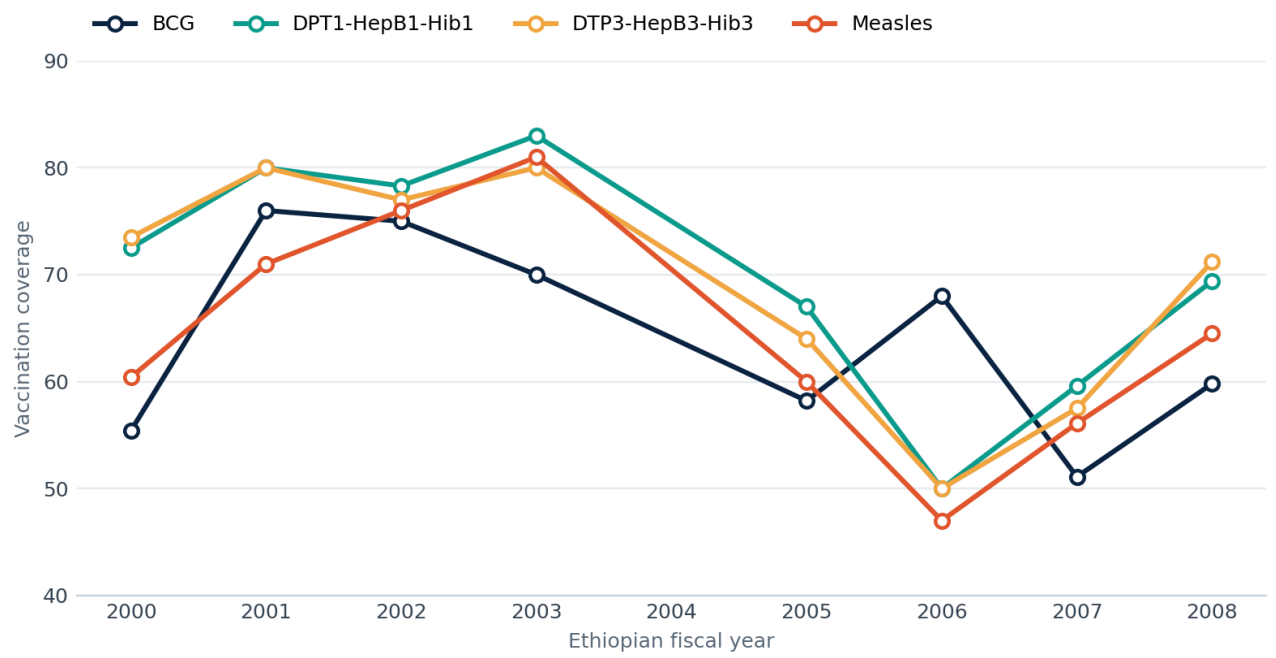
Childhood vaccination

Nearly three-fifth (60%) of children aged less than 12 months received BCG vaccine and 65% received measles vaccination through routine immunization program. Overall, about 64% of children aged below 12 months received all basic vaccinations (Figure 4.9).



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 4.9. Percentage of infants aged less than 12 months who had received basic vaccinations, Yilmana Densa district, Northwest Ethiopia, EFY 2008



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 4.10. Trend in proportion of BCG, DPT1-HepB1-Hib1, DPT3-HepB3-Hib3 and measles vaccination coverage, Yilmana Densa district, Northwest Ethiopia, EFY 2000-2008.

The percentage of children aged less than 12 months who received each vaccine varied across districts. The coverage for all basic vaccination was highest in Daval cluster (119 %) and lowest in Goshiye (44 %) (Table 4.7).

Table 4.7. Proportion of live births received BCG, DPT1-HepB1-Hib1, DPT3-HepB3-Hib3 and measles vaccines among surviving infants, Penta3 and Measles dropout rates by cluster, Yilmana Densa, EFY 2008

Note: Figures in parentheses are negative dropouts, in which Coverage rates increases for subsequent doses.
[1]Dropout rate

DOR: Dropout rate
Note: Figures in parentheses are negative dropouts, in which coverage rates increases for subsequent doses.

HF	Number of live births	Number of surviving Infants	Proportion of live births received BCG	Proportion of infants received DPT1-HepB1-Hib1	Proportion of infants received DPT3-HepB3-Hib3	Proportion of infants received Measles	Proportion of infants Fully Vaccinated	Penta1/ Penta3 DOR	Penta1/ Measles DOR
Addet	1,921	1,773	67.4	77.5	76.2	66.9	66.9	1.7	13.7
Agitta	1,201	1,108	43.8	57.6	61.2	55.7	55.7	(6.3)	3.3
Chinqulit	855	789	60.9	75.3	80.1	75.8	75.8	(6.4)	(0.7)
Daval	258	238	107.3	114.2	126.4	123.9	119.7	(10.7)	(8.5)
Debre Mawi	787	727	52.8	61.5	63.4	62.5	62.5	(3.1)	(1.6)
Densa Bata	598	552	79.4	83.5	76.7	69.6	69.6	8.2	16.7
Ennegadie	649	599	50.9	57.5	58.5	50.1	49.8	(1.7)	12.8
Goshiye	1,132	1,045	39.3	45.1	49.5	42.8	43.7	(9.8)	5.1
Sekela	606	559	60.6	66.9	61.0	57.8	57.8	8.8	13.6
Simarega	371	342	73.4	82.5	98.6	71.1	71.1	(19.5)	13.8
Walka	573	529	75.1	90.1	93.3	90.2	90.2	(3.6)	(0.2)
District	**8,951**	**8,261**	**59.8**	**69.4**	**71.2**	**64.5**	**64.4**	**(2.6)**	**7.1**

Nutritional status of children

To describe the nutritional status of children an anthropometric indicator, weight-for-age measurement was used, which is a composite index of weight-for-height and height-for-age. This indice is expressed as standard deviation (SD) units from the median for the reference group. Children who fall below minus two standard deviations (-2SD) from the median of the reference population are regarded as moderately malnourished, while those who fall below minus three standard deviations (-3SD) from the reference population median are considered severely malnourished.

Among the total estimated number under-five children (35,964) in the district, about 11,434 (32%) were measured for weight, of which 16 percent were underweight (Table 4.8)

Table 4.8. Percentage of malnourished children under-five years based on weight-for-age anthropometric indice, Yilmana Densa, Northwest Ethiopia, EFY 2008.

Nutrition status	Number	Percent
Number of children under-five years measured for weight	11,434	-
Moderately underweight		
Children weighted below -2SD	1201	10.5
Severely underweight		
Children weighted below -3SD	583	5.1
Total Underweight	1784	15.6

Low birth weight

Among the total of 3,599 live births weighted in health facilities, 52 (1.4%) of newborns were low in birth weight with the highest percentage reported from Agitta cluster, 4.3%, followed by Addet cluster, 3.1% (Table 4.9).

Table 4.9. Percentage of low birth weight newborns among live births weighted by cluster, Yilmana Densa district, EFY 2008.

Cluster	Number of live births weighed	Number of newborns with birth weight less than 2,500 gm.	Percentage of newborns with low birth weight
1. Addet	1027	32	3.12
2. Agitta	324	14	4.32
3. Chinqulit	332	0	0.00
4. Daval	145	0	0.00
5. Debre Mawi	320	6	1.88
6. Densa Bata	355	0	0.00
7. Ennegadie	248	0	0.00
8. Goshiye	292	0	0.00
9. Sekela	228	0	0.00
10. Simarega	125	0	0.00
11. Walka	203	0	0.00
Total	**3,599**	**52**	**1.44**

Vitamin A supplementation (VAS) and De-worming

For sustainable reduction of malnutrition, Vitamin A supplementation and De-Warming are among the major targets to be achieved.

Hence, out of the eligible population of 6-59 Months children, nearly all children (100%) were supplemented with Vitamin A in two rounds and among the children aged 2-5 years, eligible population for deworming, 98% were dewormed.

Childhood pneumonia, diarrhea and neonatal sepsis

Among the common childhood illnesses, we assessed under-five children treated for pneumonia and diarrhea as well as newborns treated for neonatal sepsis in the health facilities of the district. The percentages of pneumonia, diarrhea and neonatal sepsis treatment were computed among the estimated number of children with pneumonia, diarrhea and neonatal sepsis based on estimated prevalence in Ethiopia-i.e. 6.4% with 3 episodes for pneumonia, 13.7% with 3 episodes for diarrhea and 7.6% for neonatal sepsis.

Hence, among the estimated cases for each illness to be existed in the district, 28 percent of under-five children were treated for pneumonia, 20 percent treated for diarrhea and 6.8 percent treated for neonatal sepsis. The highest percentage of pneumonia treatment (57%) was reported in Daval Cluster, while the lowest percentage (15%) was reported by Sekela Cluster (Table 4.10).

Table 4.10. Percentage of under-five age children treated for Pneumonia, diarrhea and neonatal sepsis at health facilities, Yilmana Densa district, Northwest Ethiopia, EFY 2008.

Cluster	Children treated for pneumonia*		Children treated for diarrhea*		Neonates treated for sepsis*	
	Number	Percent	Number	Percent	Number	Percent
1. Addet	408	27.5	676	21.3	1	0.7
2. Agitta	198	21.4	352	17.7	2	2.2
3. Chinqulit	211	32.0	429	30.4	12	18.5
4. Daval	114	57.3	70	16.4	0	0.0
5. Debre Mawi	151	24.9	219	16.8	0	0.0
6. Densa Bata	172	37.3	283	28.7	21	46.2
7. Ennegadie	75	15.0	149	13.9	8	16.2
8. Goshiye	140	16.0	225	12.0	1	1.2
9. Sekela	61	13.0	149	14.9	0	0.0
10. Simarega	88	30.8	10	1.6	0	0.0
11. Walka	116	26.3	205	21.7	1	2.3
Total	**1968**	**28.5**	**2940**	**19.9**	**46**	**6.8**

* Among estimated number of children with pneumonia, diarrhea and neonatal sepsis based on estimated prevalence in Ethiopia, EFY 2008

Prevention and control of communicable diseases

Causes of morbidity

Ten top leading causes of morbidity

Among the population visited outpatient department in all health facilities, the ten top leading causes of morbidity accounted 27% in all ages and 91% in under five years children. Acute febrile illness (AFI) was the first leading cause of morbidity in the total population who visited outpatient department, accounting 13 percent and while non-bloody diarrhea in under-five children morbidity, accounting 91% (Tables 4.11& 10.12).

Table 4.11. The ten top leading causes of morbidity for all ages in the outpatient departments of health facilities in Yilmana Densa district, Northwest Ethiopia, EFY 2008

Rank	Diagnosis	Number	Percent
1	Acute Febrile illnesses (AFIs)	5992	12.8
2	Acute upper respiratory tract infections (AURTIs)	4482	9.5
3	Pneumonia	4097	8.7
4	Parasitic diseases	3659	7.8
5	Diarrhea non-bloody	3439	7.3
6	Trauma	3399	7.2
7	Infection of skin & subcutaneous tissue	3003	6.4
8	Dyspepsia	2610	5.6
9	Helmentiasis	2568	5.5
10	Urinary tract infections	1731	3.7
	Total of leading causes of morbidities in all ages	12,626	26.9
	All causes of morbidities in all ages	46,979	100.0

Table 4.12. The ten top leading causes of morbidity for under-five children in the outpatient departments of health facilities in Yilmana Densa district, Northwest Ethiopia, EFY 2008.

Rank	Diagnosis	Number	Percent
1	Diarrhea (Non-bloody)	2664	26.7
2	Pneumonia	1968	19.8
3	Acute febrile illnesses (AFI)	1237	12.4
4	Acute upper respiratory tract infections (AURTIs)	786	7.9
5	Infection of skin and subcutaneous tissue	673	6.8
6	Helmentiasis	524	5.3
7	Other or unspecified infectious & parasitic diseases	388	3.9
8	Diarrhea with blood (dysentery)	378	3.8
9	Disease of eye and adnexa	263	2.6
10	Trauma (Injury, fracture, etc.)	161	1.6
	Total of leading causes of morbidities in children	9,042	90.7
	All causes of morbidities in children	9,964	100.0

Tuberculosis

Tuberculosis case notification

In this report, tuberculosis case detection rate (TB CDR) is calculated as the number of newly detected all forms of TB cases (including relapses) divided by the total number of TB cases estimated to occur in the area during a given time period.

The overall TB CDR was 52.4% (95%CI: 48.4-56.4), with the lowest CDR at Walka cluster (21%) and highest CDR at Addet cluster (90%) (Table 4.13).

Table 4.13. TB case detection rate by cluster in Yilmana Densa district, northwest Ethiopia, EFY 2018.

Cluster	Number of estimated all forms of TB cases	The number of newly detected all forms of TB cases (including relapses)	TB CDR [95% CI]
Addet	128	115	90.0 [84.8, 95.2]
Agitta	80	57	71.4 [61.5, 81.3]
Chinqulit	57	13	22.9 [12.0, 33.8]
Daval	17	07	40.8 [17.4, 64.2]
Debre Mawi	52	30	57.3 [43.9, 70.7]
Densa Bata	40	19	47.8 [32.3, 63.2]
Ennegadie	43	13	30.1 [16.4, 43.8]
Goshiye	75	24	31.9 [21.4, 42.5]
Sekela	40	15	37.2 [22.2, 52.2]
Simarega	25	10	40.6 [21.4, 59.9]
Walka	38	08	21.0 [8.1, 33.9]
Overall	595	312	52.4 [48.4 ,56.4]

Among the total notified 312 TB cases, the majority (60 %) were extra-pulmonary tuberculosis (EPTB) and smear positive (bacteriologically confirmed) pulmonary tuberculosis (PTB) cases accounted 17 percent (Figure 4.11).

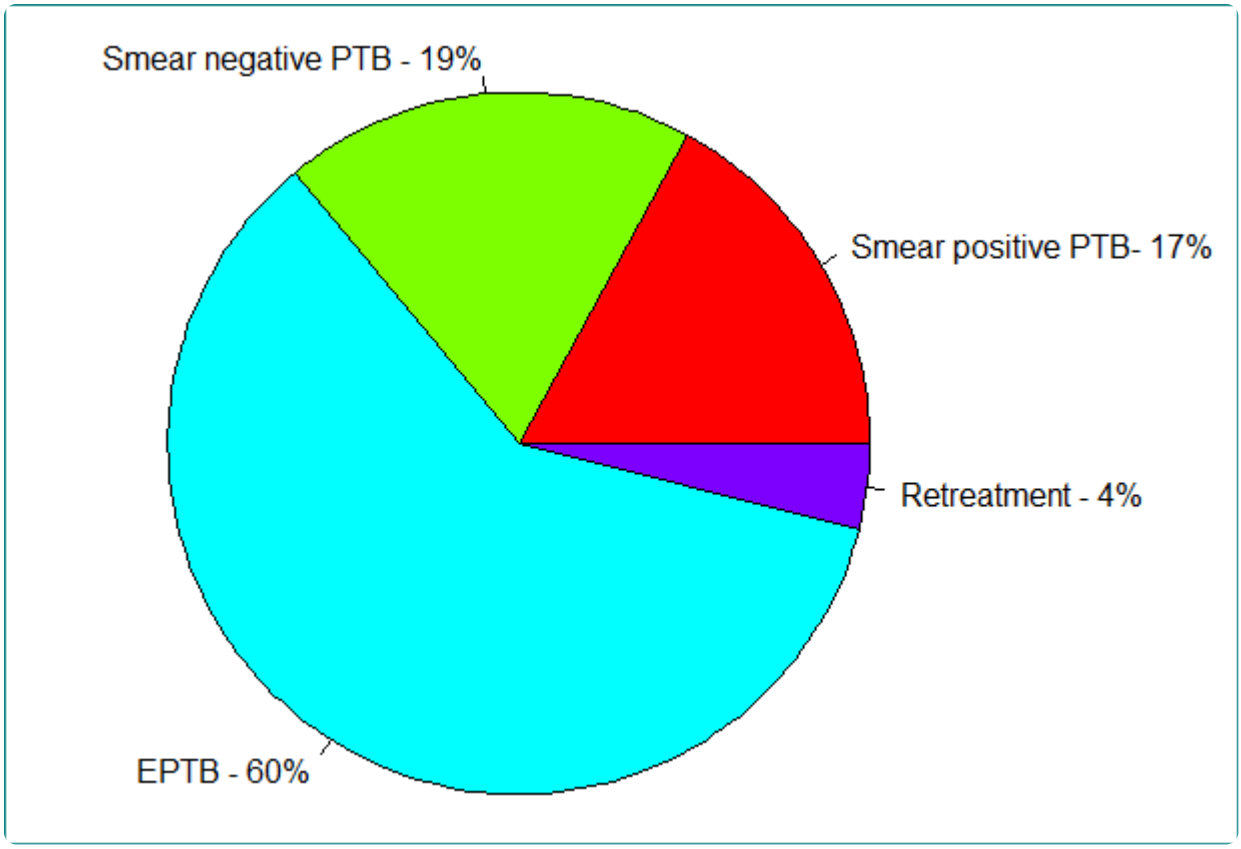


Figure 4.11. Proportion of bacteriologically confirmed new PTB, smear negative new PTB, new extra-pulmonary TB and re-treatments among all forms of TB detected in EFY 2008 (N=312).

TB treatment outcomes

Table 4.14. TB treatment outcomes by TB category, Yilmana Densa district, Northwest Ethiopia, EFY 2008.

TB category	Treatment outcomes (rate)					
	Cured	Treatment completed	Lost to follow up	Deaths	Failure	Not evaluated
PTB^{+} (N=53)[a]	96.2	0.0	0.0	0.0	3.8	0.0
PTB^{-} (N=75)[b]	na	93.3	1.3	4.0	0.0	1.3
EPTB (N=151)[c]	na	98.0	0.0	1.3	0.0	0.7
Overall (N=279)	96.2	78.1	0.4	1.8	0.7	0.7

N=Number of cohort cases enrolled to treatment

[a]PTB^{+}=smear positive pulmonary tuberculosis

[b]PTB^{-} = smear negative pulmonary tuberculosis

[c]EPTB =extrapulmonary tuberculosis

TB/ HIV

Among the total of 312 notified TB cases, only 242 (78%) were screened for HIV and 20 (8.3%) of them were HIV positive. Whereas, all HIV positive clients (4, 996) were screened for TB and five (0.1%) of them were diagnosed with active TB (Table 4.15).

Table 4.15. Percentage distribution of HIV screening for TB patients and HIV positive rate among TB Patients by cluster, Yilmana Densa district, northwest Ethiopia, EFY 2008.

	Health centers/ Private clinics	Number of TB cases detected (All forms)	TB patients screened for HIV		TB patients diagnosed with HIV positive	
			Number	Percent	Number	Percent
1	Addet	115	61	53.0	14	23.0
2	Agitta	57	54	94.7	0	0.0
3	Chinqulit	13	14	107.7	0	0.0
4	Daval	7	10	142.9	0	0.0
5	Debre Mawi	30	29	96.7	0	0.0
6	Densa Bata	19	11	57.9	0	0.0
7	Ennegadie	13	13	100.0	0	0.0
8	Goshiye	24	18	75.0	5	27.8
9	Sekela	15	15	100.0	0	0.0
10	Simarega	10	9	90.0	0	0.0
11	Walka	8	7	87.5	1	14.3
12	All Privates	1	1	100.0	0	0.0
	District 31	2 242	77.6 20		8.3	

Table 4.16. Percentage distribution of TB screening for HIV positive clients and TB positive rate among HIV patients by cluster, Yilmana Densa district, northwest Ethiopia, EFY 2008

Characteristics	Number HIV positive clients who received HIV care	HIV positive clients screened for TB		HIV positive clients with active TB	
		Number	Percent	Number	Percent
Sex					
Male	1,825	1,825	100.0	3	0.2
Female	3,171	3,171	100.0	2	0.1
HC/private Clinics					
Addet	4,850	4850	100.0	5	0.1
Agitta	145	145	100.0	0	0.0
Chinqulit	0	0	-	0	-
Daval	0	0	-	0	-
Debre Mawi	0	0	-	0	-
Densa Bata	0	0	-	0	-
Ennegadie	1	1	100.0	0	0.0
Gosheye	0	0	-	0	-
Sekela	0	0	-	0	-
Simarega	0	0	-	0	-
Walka	0	0	-	0	-

Characteristics	Number HIV positive clients who received HIV care	HIV positive clients screened for TB		HIV positive clients with active TB	
All Privates	0	0		0	-
Woreda	4,996	4,996	100.0	5	0.1

HIV/AIDS

HIV counseling and testing (HCT)

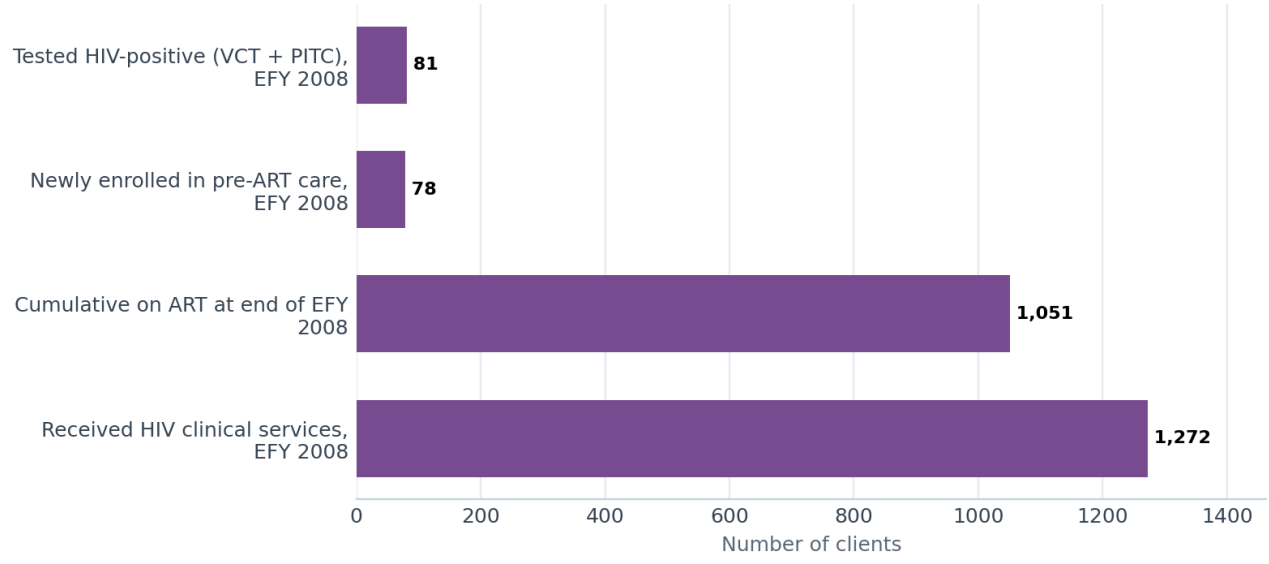
A total of 16,515 clients were tested for HIV both at VCT and PITC service units, of which 81 clients (0.5 percent) were positives (Table 4.17).

Table 4.17. Number of individuals who have been tested for HIV and HIV test positivity rate in Yilmana Densa district, Northwest Ethiopia, EFY 2008.

HIV testing and counseling	Number of individuals ever tested for HIV	Number of individuals tested positive for HIV	HIV test positive rate
VCT	6,104	39	0.64
PITC	10,409	42	0.40
Total	16,513	81	0.49

HIV care

Among 81 HIV positive clients, only 78 (96.3%) were enrolled into pre-ART care. At the end of EFY 2008, a total 1,051 HIV positive clients were on ART and about 1,272 HIV positive clients were received HIV clinical care (both Pre-ART & ART care) (Figure 4.12).



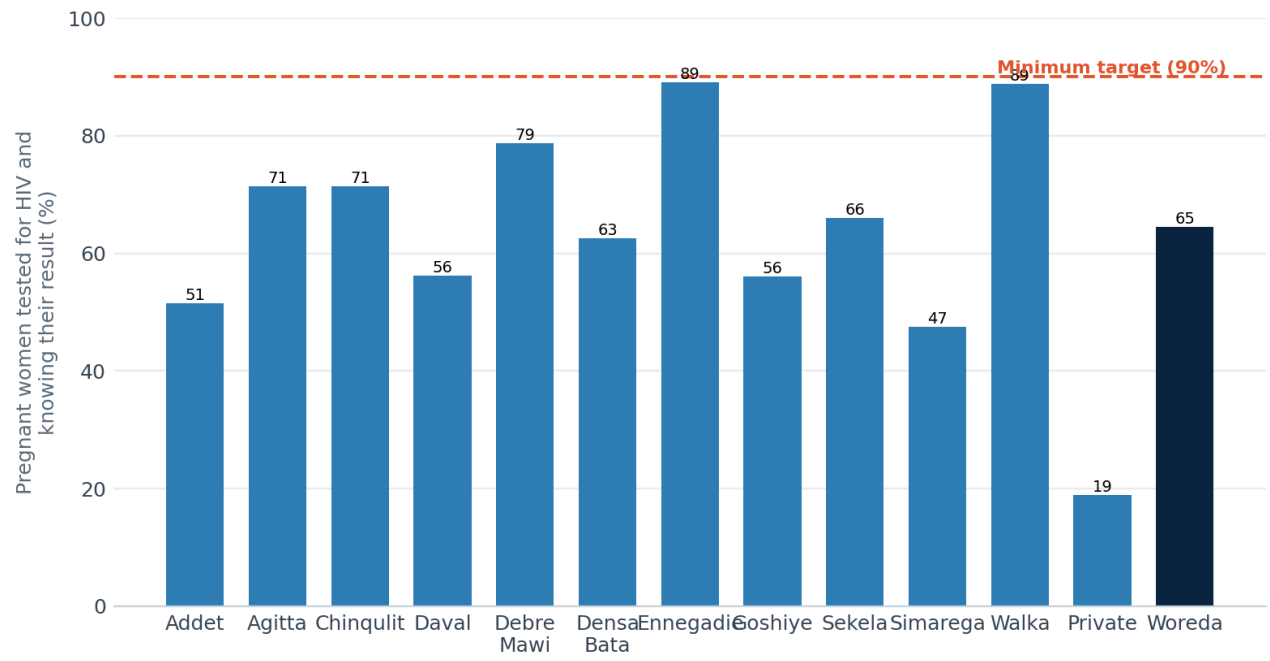
Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 4.12. Number of HIV positive clients and those who received HIV care, Yilmana Densa district, Northwest Ethiopia, EFY 2008.

Prevention of mother-to-child transmission (PMTCT)

To assess coverage of HIV testing services for pregnant and lactating women, this health profile collected data on pregnant and lactating women whether they had ever been tested for HIV during antenatal care, labor and delivery and postnatal care service. If they were tested and knew their result, the HIV positive pregnant and lactating women were checked whether they had ever been enrolled into ART care based on option B+.

Nearly two-third (65%) of pregnant women who had ever received antenatal care were tested for HIV during their pregnancy with the highest rate (89%) at Ennegadie health center (Figure 4.13).



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 4.13. Percentage of pregnant women tested for HIV and knew their result during pregnancy by cluster, Yilmana Densa district, EFY 2008.

About 4,444 pregnant and lactating women (PLW) were tested for HIV during pregnancy, labor and delivery and postnatal care, of which 40 (0.9 %) were HIV positive with the highest HIV positivity rate at Addet health center, 3.7% (Table 4.18).

Table 4.18. Percentage of HIV positive PLW among those who were tested for HIV, Yilmana Densa district, Northwest Ethiopia, EFY 2008.

	Health Center/ Privates clinics	Number of PLW who were tested for HIV and knew their results	HIV positive PLW among tested for HIV	
			Number	Percent
1	Addet	904	33	3.7
2	Agitta	542	2	0.4
3	Chinqulit	495	0	0.0
4	Daval	125	0	0.0
5	Debre Mawi	451	0	0.0
6	Densa Bata	380	0	0.0
7	Ennegadie	401	3	0.7
8	Goshiye	406	2	0.5
9	Sekela	317	0	0.0
10	Simarega	112	0	0.0
11	Walka	304	0	0.0
12	All Private clinics	7	0	0.0
	Woreda	4,444	40	0.9

Among the 40 HIV positive PLW, 8 (20 %) were newly diagnosed HIV positives and 32 (80%) were women who get pregnant while on ART. Among the eight newly HIV positive PLW, seven (88 percent) were enrolled into ART care.

Malaria

About 13,620 patients suspected for malaria were tested with microscopy or rapid diagnostic test (RDT) in the district health facilities, of which, 2,794 (20.5%) were diagnosed with confirmed malaria. Among the 2,794 patients diagnosed with confirmed malaria, the majority (87%) were adults aged 15 and above years and 6% were under-five children. The proportions of plasmodium falciparum and plasmodium vivax were equivalent (Table 4.19). The overall malaria positivity rate was 21% with the highest positivity rate (66%) reported from Simarega cluster (Figure 4.14).

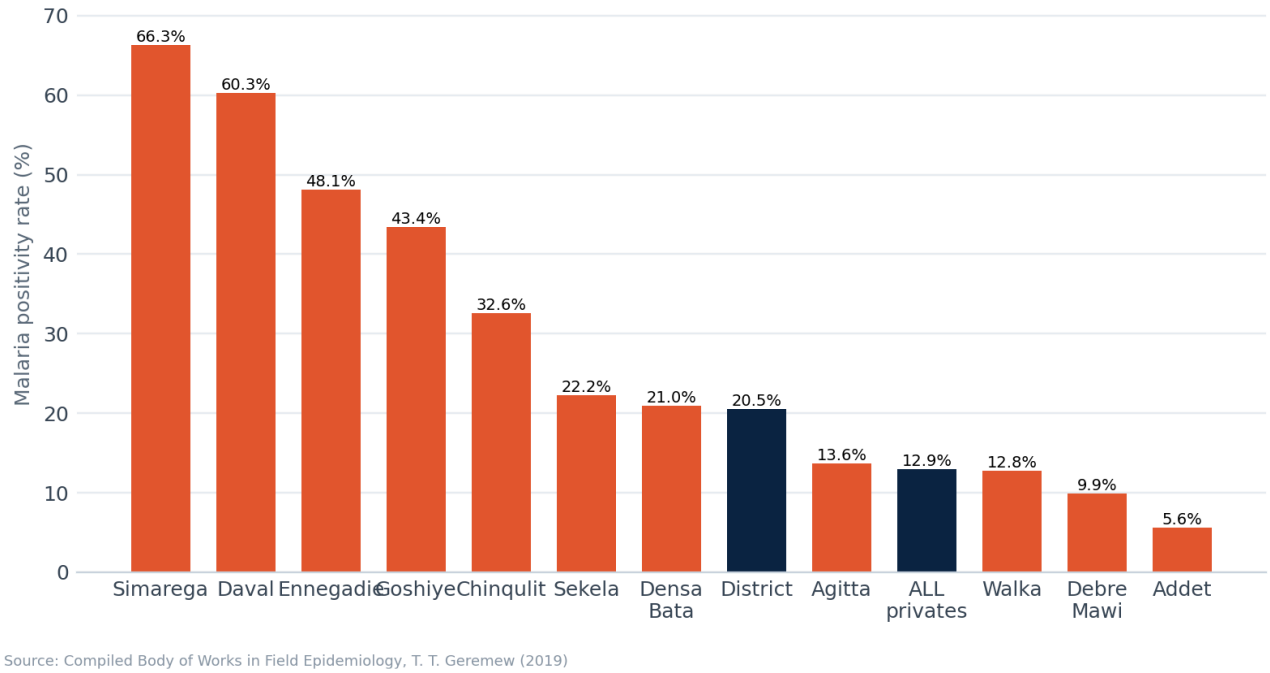
Table 4.19. Malaria positivity rate disaggregated by age, sex and malaria species among tests done for malaria diagnosis, Yilmana Densa district, EFY 2008 (N=13,620)

Characteristics Fr	equency Percen	t
Age (in years)		
<5	160	1.2
5-14	201	1.5
15+	2,433	17.8
Sex		
Female	847	6.2
Male	1,947	14.3
Malaria species		
Plasmodium falciparum*	1,392	10.2
Plasmodium vivax	1,403	10.3
Total 2,	794 20.5	

N = Total number of slides or RDT done for malaria diagnosis

*includes mixed malaria

Note: The source of this data is the eHMIS database, which is different from the PHEM weakly report data, when the total malaria cases was 2,974



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

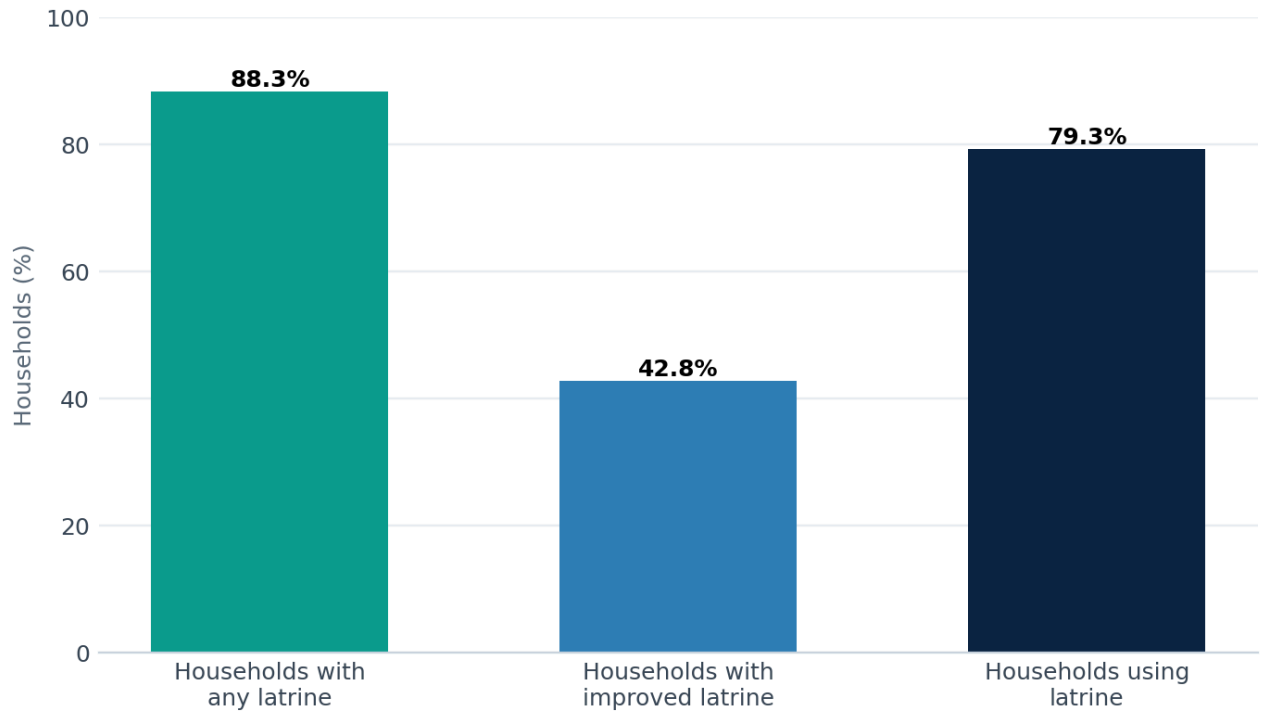
Figure 4.14. Malaria Positivity Rate by Cluster, Yilmana Densa district, EFY 2008

Environmental health and sanitation

Among a total of 33 rural kebeles, nearly 79 percent kebeles were declared open defecation free (ODF). However, all kebeles in Ennegadie cluster were not declared ODF yet (Table 4.20). The latrine coverage of the district was 88 percent, of which 42% were improved and 64% were unimproved latrine (Figure 4.15).

Table 4.20. Proportion of rural kebeles that have been declared ODF, Yilmana Densa, EFY 2008.

	Cluster	Number of rural kebeles	Kebeles that have been declared ODF	
			Number	Percent
1	Addet	5	4	80.0
2	Agitta	5	3	60.0
3	Chinqulit	3	3	100.0
4	Daval	1	1	100.0
5	Debre Mawi	3	3	100.0
6	Densa Bata	3	3	100.0
7	Ennegadie	3	0	0.0
8	Goshiye	4	4	100.0
9	Sekela	3	3	100.0
10	Simarega	1	1	100.0
11	Walka	2	1	50.0
	Total	33	26	78.8



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 4.15. Percentage of latrine construction and utilization in the communities of Yilmana Densa district, Northwest Ethiopia, EFY 2008.

Health service utilization

Outpatient department (OPD) attendance per capita and admission rate

The OPD attendance per capita was 2.2 with the highest visit in Daval and Debre-Mawi clusters (2.7) and lowest in Simarega cluster (1.2). The overall admission rate in the district health facilities was 0.1 per 1000 population per year (Table 4.21).

Among the total OPD attendants, 57% were males and 54% were children aged 5 to 14 years (Figure 4.16).

Table 4.21. Distribution of OPD attendance per capita and admission rate by cluster, Yilmana Densa district, Northwest Ethiopia, EFY 2008

Cluster	Population	OPD attendance		Admission	
		Number of OPD Visits*	OPD attendance per capita	Number of inpatient admission	Admission rate (per 1000 popn) |
Addet	57,016	123,258	2.2	23	0.2
Agitta	35,640	70,292	2.0	0	0.0
Chinqulit	25,370	44,340	1.7	4	0.1
Daval	7,658	20,873	2.7	7	0.3
Debre Mawi	23,366	62,925	2.7	0	0.0
Densa Bata	17,744	40,590	2.3	4	0.1
Ennegadie	19,253	36,744	1.9	0	0.0
Goshiye	33,595	59,341	1.8	0	0.0
Sekela	17,982	32,446	1.8	0	0.0
Simarega	10,995	12,828	1.2	0	0.0
Walka	16,996	33,054	1.9	5	0.2
Total	265,615	581,481	2.2	43	0.1

*Includes new and repeat visits

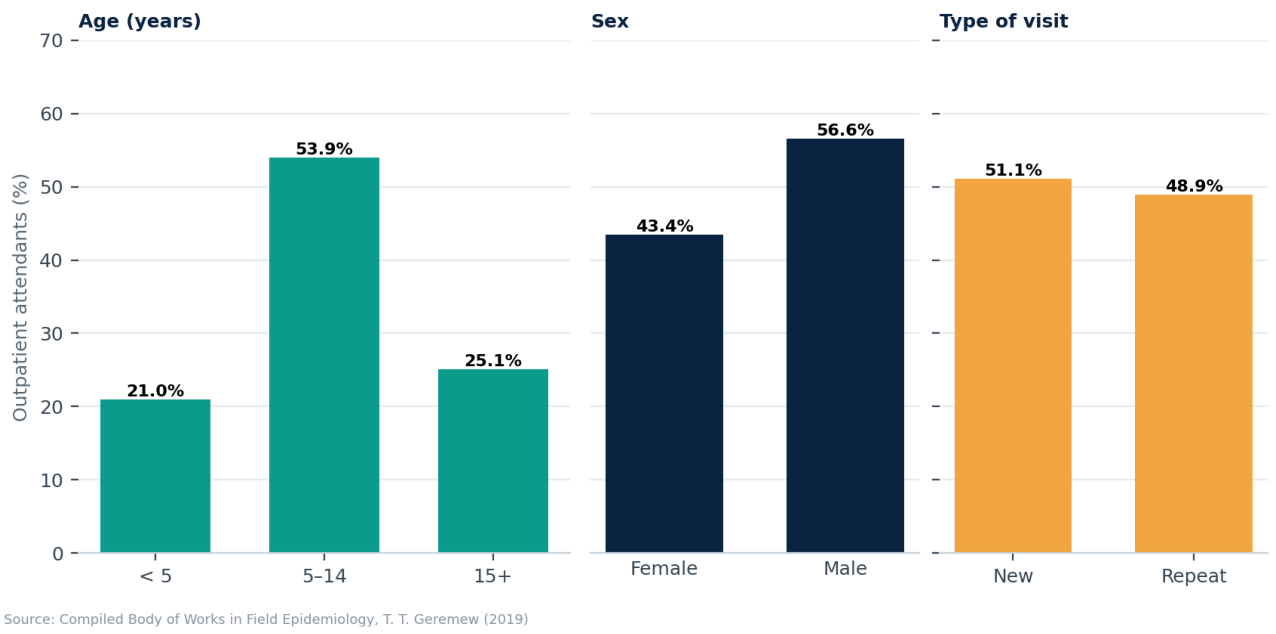


Figure 4.16. Distribution of outpatient attendants by age, sex and type of visit, Yilmana Densa district, Northwest Ethiopia, EFY 2008 (N=581,481).

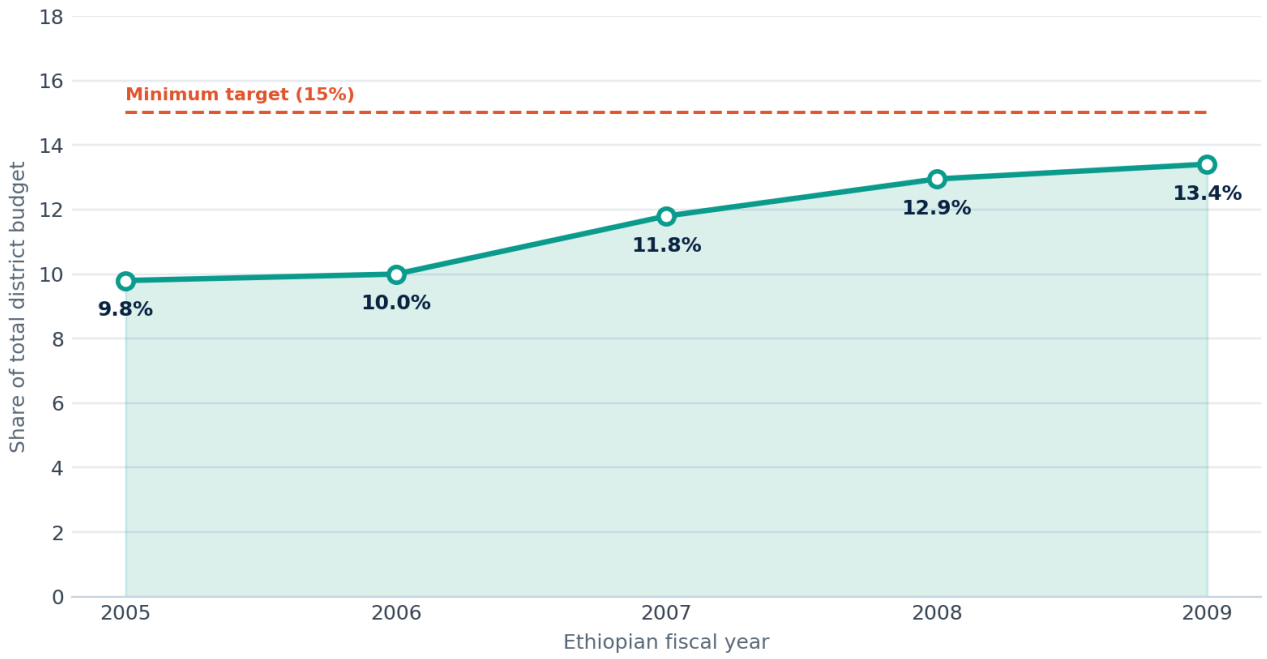
Health information system (HIS) implementation

Out of 11 health centers, only three (27%) were using eHMIS and electronic medical recording (eMR) at medical record unit (MRU). Eleven (30%) health posts were implementing community health information system CHIS with its limitation.

Health budget allocation and expenditure

For the 2008 budget year, nearly 119.4 million Ethiopian birr (ETB) were allocated for the district administration from the government, of which nearly 15.5 million ETB (12.9%) were allocated for the health sector. The percentage share of the health budget allocation increased from 9.8% in EFY 2005 to 13.4% in EFY 2009 (Figure 4.17).

In addition, the district health sector was donated nearly 2.9 million ETB from different development partners, which accounted nearly 16% of the total health sector budget expenditure. The popular development partners contributed to the district health sector were WHO, GAVI, Global fund and UNICEF. They supported mainly in the areas of health system strengthening, improving health service delivery, maternal newborn health and nutrition, prevention and control of communicable and non-communicable diseases including diseases survielance.



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 4.17. Trend in percentage of total budget allocation to the health sector, Yilmana Densa district, Northwest Ethiopia, EFY 2005-2009

Health Care Financing (HCF)

Health care financing is one of the reforms on implementation in the district health sector. Hence, all the health centers of the district were implementing HCF.

4.5 Discussion

The potential health service coverage and the coverage of a key health service indicators were low. The health facilities of the district were not adequately staffed and the percentage share of health budget was also low.

The potential health service coverage was 68 percent. This was low as compared to the national potential health service coverage (98%) and WHO recommendation. The primary hospital to population ratio is 1 to 265,615 populations which is less than national standard (1:100,000) and the health post to population ratio is 1 to 7,379 which is less than the national standard (1:5000).

Regarding maternal health service coverages, the report revealed that the modern contraceptive acceptance rate was 98% which was higher than the national prevalence, 35%[8], and it was comparable with national target, 95%[9]. This high contraceptive acceptance coverage may be as a result of improved quality of care, good client-care provider relationship, report bias including double reporting and low population projection. Among the total modern contraceptive methods used, LAFP methods accounted 30%. This finding was consistent with a study finding in Oromia region, 30.3%[10]. However, it was relatively higher than the national and regional percentage, 10% and 15%, respectively[8]. About three-fifth (60%) women had ANC atleast one visit which was comparable with the national ANC1 coverage 62%[8]. It was low as compared to the yearly target of the district (90%) and national(95%)[9].

Percent of deliveries attended by skilled health personnel is an indicator to provide information on women's use of delivery care services[11]. It helps program management at district, national and international levels by indicating whether safe motherhood programs are on target with making professional assistance at delivery available and used. In addition, the proportion of births attended by skilled health personnel is a measure of the health system's functioning and potential to provided adequate coverage for deliveries[11]. In this report the percent of births attended by skilled health personnel was 39% which was low as compared with the national reproductive health strategy target for 2015, 60%[12], the EFY 2007 national health and health related indicators report, 68% [13], the national target, 90%[9]. However, the finding was relatively higher than national coverage, 28%[8], and findings of studies in Awi zone, 19% [14], Dodota district of Oromia
region,18%[15]. The possible reasons for such low performance may be due to lack of health service accessibility, low quality of health service, unknown exact denominator as a result of population projection, poor community awareness on about antenatal care services, culture influence and geographical barrier.

Regarding child health, this assessment revealed that about 64% children aged less than 12 months received all basic vaccinations (Fully vaccination). It was higher than the recent national survey findings, 46% in Amhara region and 39 % at national level[8]. However, it is lower than the national target for all basic childhood vaccination , 95%[9] and a study finding in Oromiya region (73.3%)[16]. This low coverage may be due to inappropriate target setting as a result of increased population projection, service inaccessibility, low quality of care, low awareness of the community on child immunization and under reporting.

Nutritional status can be assessed with different nutrition assessment methods. In this report the nutritional status of under-five children was assessed using an anthropometric measurement, weight-for-age. According to WHO criteria, the prevalence cut-off values for public health significance for underweight is <10% for low prevalence, 10-19% for medium prevalence, 20-29% for high prevalence and ≥ 30% for very high prevalence[17, 18]. The finding of in this health profile assessment revealed that 15.6% under-five children were underweight among those who weighted through routine growth and monitoring program. This indicate a medium public health significance undernutrition in the district. Despite this medium public health significance, the prevalence of underweight found in this report was relatively low as compared to

the national prevalence of childhood underweight, 24%[8] and the national target for the year EFY 2008 based on the second national nutrition program strategy,17%[19]. It was also low compared to another study done in Takusa district, Northwest Ethiopia, 19.5%[20]. This possible explanation for this difference may be due to the weakness (low sensitivity) of growth and monitoring program (GMP), lack of documentation and underreporting as a result of poor knowledge or negligence of health workers.

Among the total of 46,979 patients of all ages who visited the outpatient departments in the district health facilities, the ten top causes of morbidity accounted for 27%. The first leading cause of morbidity was acute febrile illness, by accounting 13% of the total patients followed by acute upper respiratory tract infections by accounting 10%. This may indicate poor personal and environmental hygiene sanitation practices in the community. Among the patients of under-five children treated in the outpatient department, the ten top leading causes of morbidity constituted 91%. By constituting the 27%, non-bloody diarrhea was ranked first in the top leading causes of under-five children morbidities followed by pneumonia, accounted 20%. This finding was supported by the national health and health related indicators report for EFY 2007[13].

All forms TB detection rate was 52 percent which is low as compared to the national level TB case detection rate (67.3 Percent)[13] and TB treatment success rate was 96 percent, higher than the national level TB treatment success rate (92%) [13], WHO target, 85%, other studies done in University of Gondar Teaching hospital, 60.1% [21] and Tigray region, 89% [22] but still needs further effort to achieve the HSTP target, 95%[9]

Twelve percent of households had no toilet facility at all, relatively high as compared to the EDHS 2016, when 32 percent of all households in rural areas did not have a toilet facility[8]

Health service utilization reflects the interaction between demand and supply of outpatient care. The use of outpatient services is inversely related to certain barriers that may be physical (distance), economic (cost to patient), cultural (low awareness and health care seeking behavior) or technical (poor quality of health care). One of the indicators to measure health service utilization is OPD attendance per capita. In this report, the OPD attendance per capita was 2.2 which is higher than the recent national health and health related indicators report in 2007 EFY (0.48)[13], the national target (2 times)[9]. This finding is also in alignment with the WHO recommendation for OPD attendance per capita (2-3 times)[9].

Limitations

Not all issues important to the health of Yilmana Densa district communities are included in this report. Locally collected data may provide additional information about the health of community living in health service areas. Data presented in this report may be most useful when combined with individual data, stories about patients and community members, and local surveillance or survey data when available.

Lack of justification of low or high performances in health service coverages was another limitations of this health profile.

4.6 Conclusion

The potential health service coverage was low and the health facilities in the district were not adequately staffed with the recommended health professionals. The coverages in the most key maternal and child health indicators were low. The health budget allocation was low.

4.7 Recommendation

To the district health office

- It is better to work more one improving maternal and child health services, especially, on antenatal care and institutional deliver. In addition, it is advisable to improve tuberculosis case detection and health information system.

To the district administration office

- It is good if the district administration improves the percentage share of the health budget allocation

Priority areas identified

Based on the descriptive measure evidences in this health profile, the following top five priority areas of intervention have been identified using priority setting criterias:

1. Maternal and child health
2. Low potential health coverages
3. Health professional to population ratio.
4. Low maternal health service child health service coverages
5. Low access health related infrastructures
6. Low performance of Communicable diseases prevention and control Programme.
7. Low health Budget allocation.

Public health actions/Information dissemination

After compiling this health profile, further public health actions were not taken as response. However, the key findings of this report were presented to the district health office staff and the final draft of this profile was submitted to the district health office.

1.

References

1. Public Health England: (2014). Public Health England's Health Profiles 2014 Published. www.gov.uk/government/news/public-health-englands-profile-2014-published, accessed on May 14-2017.
2. Clinton, Eaton, and Ingham. Community Health Profile 2012. Healthy! Capital Counties. www.healthycapitalcounties.org/community-health-profile.html.
3. UCLA Center for Health Policy Research. Health Profile. healthpolicy.ucla.edu/health-profiles/Pages/default.aspx.
4. Urban Indian Health Institute, Seattle Indian Health Board. (2016). Community Health Profle: National Aggregate of Urban Indian Health Program Service Areas. Seattle, WA: Urban Indian Health Institute. www.uihi.org/wpfb-file/uihi_chp_2016_web_20170531-pdf/.
5. Ethiopia. Service Availability and Readiness Assessment. 2016 Summary Report. Ethiopian Public Health Institute. Addis Ababa, Ethiopia. Federal Ministry of Health. Addis Ababa, Ethiopia. World Health Organization (WHO). Addis Ababa, Ethiopia.
6. Sinclair, D.D., Health Profiles Project Director at PHE. 2014.
7. U.S. Centers for Disease Control and Prevention. Community Health Assessment for Population Health Improvement: Resource of Most Frequently Recommended Health Outcomes and Determinants, Atlanta, GA: Ofce of Surveillance, Epidemiology, and Laboratory Services, 2013.
8. Central Statistical Agency (CSA) [Ethiopia] and ICF. 2016. Ethiopia Demographic and Health Survey 2016. Addis Ababa, Ethiopia, and Rockville, Maryland, USA: CSA and ICF.
9. FMOH (2015). Health Sector Transformation Plan (HSTP) 2015-2020 (EFY 2008-2012). Addis Ababa, Ethiopia.
10. Fakadu H, et al., Prevallence and determinants Factors of long acting Contraceptive Utilization among maried of reproductive age in Adaba Town, West Arsi zone, Oromiya, Ethiopia. J Women's health care, 2017. 6:360.
11. MEASURE Evaluation (2018).Family plannin and reproductive health indicators data base. Percent of deliveries attended by skilled health personnel www.measureevaluation.org/prh/rh_indicators/womens-health/sm/percent-of-deliveries-attended-by-skilled-health.
12. FMOH(2006).The national reproductive health strategy 2006 -2015. Addis Ababa, Ethiopia.
13. FMOH. Health and health related indicators, EFY 2007. Addis Ababa, Ethiopia.
14. Mulunesh, A. and M. Wubgzier, The prevalence of Skelled birth attendant utilization and correlates in North-West Ethiopia. BioMed Research Institutional, 2015.
15. Fikre, A.A. and M. Demissie, Prevalence of institutional delivery and associated factors in Dodota Woreda (district), Oromia regional state, Ethiopia. Reproductive health, 2012. **9**: p. 33-33.
16. Animaw Worku, et al., Expanded program of immunization coverage and associated factors among children age 12 – 23 months in Arba Minch town and Zuria District, Southern Ethiopia, 2013. BMC Public Health 2014 14:464. 2013.
17. WHO. Physical status: the use and interpretation of anthropometry. Report of a WHO Expert Committee. Technical Report Series No. 854. Geneva, World Health Organization, whqlibdoc.who.int/trs/WHO_TRS_854.pdf.
18. World Health Organization. Nutrition Landscape Information System country profile indicators interpretation guide. Geneva: World Health Organization; 2010.
19. Federal Democratic Republic of Ethiopia (2016). National Nutrition Program II (NNP II) 2016-2020. Addis Ababa, Ethiopia.
20. Nigatu, G., et al., Prevalence and associated factors of underweight among children 6-59 months of age in Takusa district, Northwest Ethiopia. Int J Equity Health, 2018. 17(1): p. 106.
21. Minaleshewa, B., et al., Treatment outcomes of TB and associated factors in Ethiopian University. Hindawi Publishing Corporation, 2016. p. 9.
22. Berhe, G., F. Enquselassie, and A. Aseffa, Treatment outcome of smear-positive pulmonary tuberculosis patients in Tigray Region, Northern Ethiopia. BMC Public Health, 2012. 12: p. 537.

Annex 4.1: Health profile assessment data extraction tool

Region: _____, Zone: _____, District: _____

NO	INDICATORS		VALUES	DATA SOURCE
1	**Background characteristic of the study area**			
2	Name of the district (prove how & why named) : _____			
3	Location	Distance from the regional town		
Distance from the zonal town				
Direction from the regional town				
Direction from the zonal town				
Latitudinal location				
Longitudinal location				
4	History (Prove for any historical aspect): _____			
		Annual Average rainfall		
		Annual Average Temperature		
	Climatic zones	Bereha		
		Kolla		
		Woyna Dega		
		Dega		
		Wurch		
5	**Administrative Organizations**			
	Cluster (Ketena)	Total number of Clusters (Ketena)		
	Kebele	Number of Urban Kebeles		
		Number of Rural Kebeles		
	Boundaries	East		
		West		
		North		
		South		
6	**Demographic and socio-economic status**			
7	Total number of households (HHs)		-	-
		Number of Urban Households		
		Number of Rural Households		
8	Total number of population disaggregated by residence			
		Urban population		
		Rural population		
9	Total number of population disaggregated by Sex			
		Male population		
		Female population		
10	Total number of population disaggregated by age			
	Total number of women in reproductive age groups			

NO	INDICATORS		VALUES	DATA SOURCE
	Total number of pregnant women			
	Total number of surviving infants:			
11	Total Number of Under*five years age children by age			
		Age: 0-24 Months		
		Age: 25-59 Months		
	Main source of economy			
	Household’s average income level per year			
	Infrastructures	# KG Schools		
		# of primary Schools		
		# of Secondary schools		
		# of Health Posts		
		# of Health centers		
		# of Government Hospitals		
		# of private Health Facilities		
		Proportion of health facilities with electricity		
		Proportion of health facilities with water and sanitation facilities (latrine and Hand washing)		
12	**Vital Statistics**			
13	Sex Ratio (Male to Female)			
	Sex	Male:		
		Female:		
14	Dependency ratio:			
	Non-working age population:	Young population (Age 0-14 years)		
		Old population (age 65+ years)		
	Working age population:	15-65 years age population		
15	Crude Birth rate (CBR)			
		Total # of Live Births		
		Mid-Year Population in the same period		
16	Crude Death rate (CDR)			
		Total # Deaths due to all causes in the Year		
		Mid-Year Population in the same period		
17	Still Birth Rate (SBR)			
		Total Number of Live Births		
		Total Number of Still births		
		# of Deaths of Children under 1 Year of age		
18	Maternal Mortality rate (MMR) per 100,000 Live Births			
		# Maternal Deaths due Maternal Causes in the year		
19	**Ten Top leading Causes of Morbidity**			
20	**Ten Top leading Causes of Morbidity in Adults and children**			

NO	INDICATORS	VALUES	DATA SOURCE

Adults

RANK	VALUES	DATA SOURCE
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
All Causes		

Children

RANK	VALUES	DATA SOURCE
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
All Causes		

21	**10 Top Leading Causes of Morbidity in all ages**

Children & Adults · ranks 1–5

RANK	VALUES	DATA SOURCE
1		
2		
3		
4		
5		

Ranks 6–10 and all causes

RANK	VALUES	DATA SOURCE
6		
7		
8		
9		
10		
All Causes		

22	**Health Service Coverages**		
23	**Potential Health Service Coverage**		
	Maternal Health		
24	**Contraceptive acceptance rate**		
	Total new and repeat acceptors		
	Long Acting Family Planning (LAFP) Utilization		
25	**Antenatal Care coverage**		
	Pregnant women who received ANC at least once		
	Pregnant women that received ANC: at least four visits		
26	**Percentage of pregnant & lactating women who were tested for HIV and who know their results (PMTCT)**		
	Pregnant & lactating women who were tested for HIV and who know their results		
	Pregnant & lactating women tested positive for HIV		
	HIV positive pregnant and lactating women who received ART at ANC+L&D+PNC for the first time based on option B+		
	Number of HIV-positive women who get pregnant while on ART and linked to ANC		
	Proportion of pregnant women tested for syphilis		
27	**Delivery & Postnatal Care**		
	Proportion of births attended by skilled health personnel		
	Proportion of births attended by Health Extension Workers		
	Early Postnatal Care Coverage (Within the first 7 days of delivery)		

NO	INDICATORS	VALUES	DATA SOURCE
28	**Number of women receiving comprehensive abortion care services**		
	<18 Years		
	=>18 years		
	Proportion of kebeles declared 'home delivery free'		
29	**Child Health**		
30	**EPI Coverages (Less than 1 year infants)**		
	BCG		
	OPV1		
	OPV3		
	IPV		
	DPT1-HepB1-Hib1		
	DTP3-HepB3-Hib3		
	PCV1		
	PCV3		
	Rota1		
	Rota2		
	Measles		
	Full vaccinated		
	PAB From NNT		
	Pregnant women TT2+		
	Non-pregnant women Pregnant women TT2+		
31	**Nutrition/GMP**		
32	**Percentage of newborns with low birth weight (LBW)**		
	Total # of live births		
	# of Low birth weight (LBW) newborns		
33	**Percentage of underweight Children aged Less than 5 years**		
	# of under 5 Yrs children measured for weight		
	# of under 5 Yrs children measured for weight & recorded as Moderately underweight (MUW)		
	# of under 5 Yrs children measured for weight & recorded as Severely underweight (SUW)		
34	**OTP**		
	Total Number of children screened for malnutrition		
	Number of children screened and have severe acute malnutrition (SAM)		
	Total number of children who enter into severe acute malnutrition treatment (OTP)		
	Total number of children who exit from severe acute malnutrition treatment (OTP)		
	Number of children recovered from SAM		
35	**Vitamin A Supplementation and Deworming**		
	Proportion of children aged 6-59 months who supplemented with vitamin A		
	Proportion of children aged 24-59 months who dewormed with Albendazone		
36	**IMNCI**		
	# of under-five children who visited under five OPD (IMNCI)		
	Proportion of diarrhea among OPD visits in under five children		
	Proportion of Pneumonia among OPD visits in under five children		
	Proportion of Neonatal Asphyxia among live births		
	Proportion of Neonatal sepsis among live births		

NO	INDICATORS	VALUES	DATA SOURCE
37	**Prevention and Control of communicable Diseases**		
38	**TB**		
	Tuberculosis case detection rate (all forms)		
	Number of bacteriologically confirmed New PTB cases detected in the Year		
	Cure rate for bacteriologically confirmed new PTB cases		
	Treatment Success Rate (TSR) among bacteriologically confirmed NEW PTB cases		
	Death rate among all forms of TB cases		
39	**HIV/AIDS**		
	# of individuals who have been tested for HIV and who received their results (HCT)		
	# of Clients receiving HIV test results (at VCT)		
	# of Clients receiving HIV test results (at PITC)		
	# of Clients testing positive for HIV (at VCT & PITC)		
40	**HIV care (Pre-ART and ART)**		
	# of adults and children with HIV infection newly enrolled in Pre ART care		
	# of adults and children with HIV infection newly started on ART		
	Cumulative Number of PLHIV ever started on ART		
	# of adults and children currently on ART		
	# of adults and children with HIV infection newly started on ART		
41	**TB/HIV**		
	# of HIV screening for TB patient		
	# of HIV positive clients diagnosed with active TB		
	# of TB Screening for HIV positive Clients		
	# of TB patients with HIV positive test result		
42	**Leprosy case notification**		
	# of leprosy cases detected		
	Leprosy treatment completion rate		
43	**Sexually transmitted infections : All syndromic Approaches**		
44	**Malaria prevention and control**		
	Malaria positivity rate		
	Total number of slides or RDT performed for malaria diagnosis		
	Number of slides or RDT positive for malaria		
	Proportion of unit structures covered by IRS		
	Facility-based malaria deaths		
45	**Hygiene and Environmental health**		
	Proportion of households access to latrine facilities		
	Number of HHs with improved latrine (Hand washing facility + Slab + ventilation pipe)		
	Number of HHs with unimproved latrine		
	Proportion of households using latrine		
	Proportion of Kebeles declared Open Defecation Free (ODF)		
46	**Community ownership -HDAs**		
47	**Proportion of functional 1 to 5 networks**		
	Functional 1 to 5 networks		
	Expected number of 1 to 5 networks		
48	**Quality of health Services**		
	Outpatient attendance per capita : Number of outpatient visits		
	Admission rate: Number of inpatient admissions		

NO	INDICATORS	VALUES	DATA SOURCE
49	**Bed occupancy rate**		
	Total length of stay (in days)		
	Number of beds		
50	**Average length of stay**		
	Number of inpatient discharges		
51	**Inpatient Mortality Rate**		
	Number of inpatient deaths in the reporting period		
52	**Logistics supply and management**		
	Proportion of availability of essential drugs for Health Post		
	Proportion of availability of essential drugs for Health Center		
53	**Information revolution & Evidence Based Decision Making**		
54	**eHMIS/PHEM implementation**		
	# of HFs having eHMIS/PHEM computers		
	# of HFs implementing eHMIS		
	# of HFs with eMR-MRU computer		
	# of HFs using eMR for registration at MRU		
55	**CHIS implementation**		
	Proportion of CHIS data collection		
	Proportion of CHIS implementation		
56	**Human Resource Information System (HRIS)**		
	Total # of human resource by professional Sub Category		
	# of Health professionals including HEWs		
	Medical Doctors (MDs)		
	Health Officers (Hos)		
	Nurses (BSc)		
	Nurses (Diploma)		
	Midwifery (BSc)		
	Midwifery (Diploma)		
	Medical Laboratory Technologist		
	Medical Laboratory Technician		
	Pharmacist		
	Pharmacy Technician		
	Environmental Health Technician		
	Health Information Technicians (HITs)		
	Health Extension Workers (HEWs)		
	Others (Like Front lines, etc.)		
	# of Supportive Health workers		
	Population to Health Profession Ratio (GP, HO, Nurse, Midwifery, Laboratory)		
57	**Health Budget Administration (Finance)**		
	Total Government Budget Allocated For the District Administration		
	Proportion of Government Budget Allocated For the Health Sector		
	Proportion of Government Budget Utilized		
	Amount disbursed by development partners		
	Main areas of support by development partners		
58	**Health Care Financing (HCF)**		
	Proportion of Health Facilities implementing HCF		

NO	INDICATORS	VALUES	DATA SOURCE
59	**Reform implementation status**		
	Health center reform implementation status		
	Hospital reform implementation status		

CHAPTER 5

Rapid Nutrition Assessment

Rapid nutrition assessment in Yilmana Densa district, northwest Ethiopia, 2018

Tesfahun Taddege GEREMEW[4], Lemma Derseh, Getie Zeru and Meklit Mekonnen

4. Correspondence to : ttaddege@gmail.com↩

Abstracts

Introduction: In Ethiopians, millions of are still under chronic and acute malnutrition that ranks among the top, both in sub-Saharan Africa and the world. A rapid assessment is helpful to inform the prioritization of interventions and allocation of supplies and resources. However, the prevalence of undernutrition among children, the nutrition specific service availability and readiness were poorly understood in the study area. Hence, this rapid assessment was aimed to assess nutrition specific service availability and readiness, undernutrition among children and treatment outcomes.

Method: A facility-based cross-sectional study was conducted in Yilmana Densa district of northwest Ethiopia**.** We randomly selected ten (30%) health posts, five (45%) health centers and the primary hospital in the district. We interviewed staff in the health facilities and analyzed nutrition surveillance data notified during 2010 EFY. Pooled prevalence of undernutrition (underweight and acute malnutrition) was calculated for children from the reports of health facilities. Subgroup analysis and Meta-regression were used to identify variables that could account for the inconsistency of undernutrition prevalence across the reports of health facilities.

Result: The mean nutrition specific service availability and readiness were 67 percent and 72 percent, respectively. The pooled prevalence underweight among children was 10.6% in health posts, 21.6% in health centers including the hospital, and 14.8% in health facilities in general. The pooled prevalence of acute malnutrition was 9.8% in health posts, 16 % in health centers including the hospital, and 12.4% in health facilities in general. In subgroup analysis and meta-regression, the type of health facility and number of children measured for weight were identified as variables that could account for the inconsistency in the prevalence of underweight across the report of health facilities. In addition, the type of health facility was a potential source of heterogeneity for the prevalence of acute malnutrition across the report of health facilities.

Conclusion: The mean nutrition specific service availability and readiness were 67 percent and 72 percent, respectively. The high prevalence of undernutrition found in this assessment highlights the importance of promptly assessing and treating undernutrition in children. In general, the findings in this assessment highlights the need of screening all the targeted children for malnutrition using anthropometric indice both for acute and chronic malnutrition and further investigating the factors associated with variations in the prevalence of undernutrition among health posts and health centers.

Key words: Service availability and readiness, pooled analysis, undernutrition, Children

5.1 Introduction

Malnutrition refers to deficiencies, excesses or imbalances in a person's intake of energy and/or nutrients. The term malnutrition covers two broad groups of conditions. One is 'undernutrition', which includes stunting (low length/height-for-age), acute malnutrition or wasting (low weight-for-length/height or low Mid-upper arm circumference (MUAC) or bilateral pitting oedema), underweight (low weight-for-age) and micronutrient deficiencies or insufficiencies (a lack of important vitamins and minerals). The other is overweight, obesity and diet-related non-communicable diseases (such as heart disease, stroke, diabetes and cancer)[1, 2].

Undernourished children are at increased risk of faltered growth, impaired mental development, and death[3, 4]. Child undernutrition which is highly prevalent in low and middle income countries is crippling global economic growth[5, 6]. Stunting (low height-for-age) is a sign of chronic undernutrition that reflects failure to receive adequate nutrition over a long period. Stunting can also be affected by recurrent and chronic illness. Wasting (low weight-for-height) is a measure of acute undernutrition that represents the failure to receive adequate nutrition currently. Wasting may result from inadequate food intake or from a recent episode of illness that caused weight loss. Underweight (low weight-for-age) is a composite index of height-for-age and weight-for-height that accounts for both acute (wasting) and chronic(stunting) undernutrition, is an indicator of overall undernutrition.[4].

In Ethiopia, malnutrition has been encouragingly decreasing in the last two decades. The efforts made through food security and scaling up of nutrition programs have made tremendous contributions in reducing malnutrition. However, millions of Ethiopians are still under chronic and acute malnutrition that ranks among the top, both in sub-Saharan Africa and the world[7]. According to the 2016 demography and health survey (DHS) report, about 38 percent of children under age 5 years are stunted (short for their age); 10% are wasted (thin for their height); 24% are underweight (thin for their age), and 1% are overweight (heavy for their height), confirmed the public health significance of undernutrition[4]. Amhara region is the most highly affected by child stunting (46%). In the region, the prevalence of child wasting and underweight were 9.8%, and 28.4%, respectively[4].

A community based cross-sectional study in east Belesa district of northwest Ethiopia reported that about 57.7% of the children were stunted and 16 % of the children were wasted[8]. Another community based cross-sectional study in Hadaleala district of Afar region revealed that the prevalence of acute malnutrition was 11.8% ranging from 9.3% to 14.8%[9]. In a recent study at Takusa district of northwest Ethiopia, 19.5% of children were underweight[10].

As indicated in the Sphere project for the humanitarian charter and minimum standards in public health emergency response, the first step in humanitarian response for public health problems is to conduct rapid assessment to identify gaps, and design a prioritized plan of action based on the result. Without a rapid assessment, significant gaps or overlaps in assistance may occur, which not only wastes precious resources at a time of great need, but can also be a cause of further burden to the affected population[11]. A rapid assessment is conducted after the onset of a public health emergency problem as early as possible to locally assess the affected areas and the needs of victims. In addition it can be conducted for any public health problem for the need of prioritization of interventions [11, 12]. A rapid assessment will help inform the prioritization of interventions and allocation of finite supplies and resources. Without the timely availability of reliable and scientifically sound data, public health officials run the risk of making ill-informed decisions that might adversely affect the response effort. Sound information on the supply and quality of health services is necessary for health systems management, monitoring and evaluation [13, 14].

The variation in the prevalence of undernutrition among children and its potential source including the nutrition specific service availability and readiness were poorly understood in the study area. Hence, this rapid assessment may help for generating relevant information on undernutrition among children and response for evidence-based decision making. In addition, it may be helpful in generating baseline evidences for further detail assessment on nutrition specific activities in the area.

5.2 Objectives

1. To describe the nutrition-specific service availability and readiness among health facilities in Yilmana Densa district of Northwest Ethiopia in 2010 EFY.

2. To estimate the prevalence of undernutrition and treatment outcomes among children aged 6 to 59 months in Yilmana Densa district of Northwest Ethiopia in 2010 EFY.

3. To explore potential sources of variation in the prevalence of childhood undernutrition among the reports of health facilities in Yilmana Densa district of Northwest Ethiopia in 2010 EFY.

5.3 Methods

Study design and setting

A facility-based cross-sectional study was conducted from July 10th to 16th 2018.This study was conducted in Yilmana Densa district, the second most populous district in west Gojjam zone, northwest Ethiopia. The district has 36 kebeles (smallest administrative units in Ethiopia), 33 of which are rural. The catchment population of the district was 275,187, of which 35,444 (12.88%) were children aged 6-59 months and 37,271 (13.54%) were children under five years of age. The district has one primary hospital, 11 health centers and 36 health posts (33 rural and 3 urban).

Study participants and Selection

Government health institutions were the study subjects. Randomly selected five health centers (45%), ten health posts (28%) including the hospital were included in this assessment.

Operational definitions

- **Service availability**: The presence of standard health services for a specific health problem [15, 16]. In this case, it was measured by a set of tracer indicators of service for malnutrition i.e., the presence measurement service of child weight, height, age and MUAC including therapeutic and supplementary feeding services for undernourished children.
- **Service readiness:** The state of being ready in health facilities to provide basic health-care interventions relating to a specific health problem [15, 16]. In this case, it was assessed by availability of trained staff on nutrition, and nutrition guidelines, equipment and medicines used for malnutrion screening and management.
- **Underweight:** Children with weight-for-age Z Score less than -2 standard deviations of the WHO child growth standards. Children whose weight-for-age Z-score is below minus three standard deviations (-3 SDs) from the median are considered severely underweight (SUW) and children whose weight-for-age Z Score between -2 and -3 SDs of the WHO child growth standards are considered moderately underweight (MUW) [1, 4, 17].
- **Global Acute malnutrition (GAM)** - A form of undernutrition that provides an aggregate of moderate and severe acute malnutrition. Moderate acute malnutrition (MAM) among children aged 6-59 months refers to MUAC 11cm to <12cm and no edema of both feet, whereas sever acute malnutrition (SAM) among children aged 6-59 months refers to MUAC less than11cm or oedema of both feet [18-20].
- **Therapeutic foods**: Specially formulated foods used in the treatment of severe acute malnutrition which include feeding formulas such as F-75 and F-100, which are therapeutic milks used in the stabilization and rehabilitation phases in an inpatient setting, and ready-to-use therapeutic foods (RTUF) used in the rehabilitation phase, usually in an outpatient setting[21].
- **Outpatient therapeutic programme (OTP)**: Programme offering outpatient care to severely malnourished cases who have appetite and who do not have any medical complications[22].
- **Inpatient therapeutic care (ITC)**: A malnutrition management care for severely malnourished patients admitted to inpatient[22].

Data Source and extraction

The primary data was collected by interviewing staffs from health facilities and the secondary data was extracted from the annual reports of the health facilities using standard checklists adapted from the national nutrition assessment and Ethiopian service availability and readiness assessment tools.

Data management and statistical analysis

The data was summarized into an Excel spreadsheet and analyzed with MS-Excel 2013 and Stata Version 14[23]. Descriptive measures were computed and narrative summary were used. We conducted statistical pooling of results of health facility reports to generate a combined estimates of undernutrition (underweight and acute malnutrition). We assessed the heterogeneity among the prevalence of undernutrition across reports of health facilities using the Higgins's I^2 test statistic [24, 25]. In the presence of heterogeneity, we used a random effects model to estimate the pooled prevalence of undernutrition [26]. Subgroup analysis and meta-regression using random effects model were carried out to explore the potential sources of heterogeneity [24, 27] with a statistical software of Stata version 14. Forest plot was used to present the pooled estimates

Ethical considerations

Legal letters were obtained from West Gojjam zone health department and Yilmana Densa district health office and permission was obtained from the health facilities.

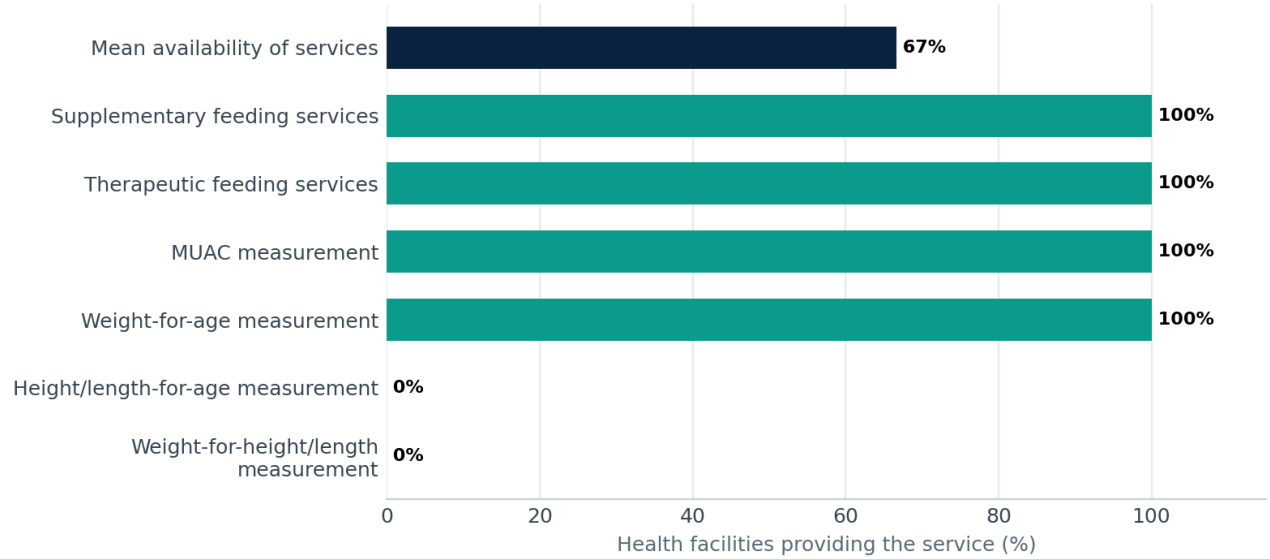
5.4 Results

Characteristics of health facilities in service availability and readiness

Nutrition specific services Availability

Services for malnutrion were deemed available when the providers in the facility screen with each anthropometric indices and manage patients with undernutrition. Hence, the service availability for nutrition screening and management was assessed based on the presence of the 6 services (measurements of weight-for-height/length, weight-for-age, Height/length-for-age and MUAC as well as therapeutic and supplementary feeding services) (Figure 1).

All of health facilities offered weight-for-age and MUAC measurement services and also therapeutic and supplementary feeding services were available in all health facilities. Overall, the mean availability of the six services was 67 percent (Figure 5.1).



Source: Compiled Body of Works in Field Epidemiology, T. T. Geremew (2019)

Figure 5.1. Availability of services for undernutrition among health facilities in Yilmana Densa district, Northwest Ethiopia, 2010 EFY.

Service readiness

Readiness of health facilities to provide malnutrition service was assessed based on the presence of 11 tracer items under the category of trained staff & guidelines and equipment & medicines (Table 5.1).

Child weight scale, tape meter, vitamin A, Albendazone or Mebendazole, RUTF and malnutrition job-aids and/or growth standard charts were available in all health facilities. The availability of at least one trained staff on nutrition was observed in the 94 percent of health facilities. In addition, the availability of the 11 tracer items was 91% at hospital level, 78% at health center level and 80% at health post level. Overall, the availability of the 11 tracer items was 72 percent (Table 5.1)

Table 5.1. Percentage of facilities with basic equipment items available and trained personnel for Nutrition screening and management, Yilmana Densa district, Northwest Ethiopia, 2010 EFY.

Basic equipment and trained personnel	Hospital (N=1)	Health centers (N=5)	Health posts (N=10)	Total (N=16)
Equipment and medicines				
Child weight scale	100.0	100.0	100.0	100.0
Height scale	100.0	80.0	40.0	56.3
MUAC measuring tape	100.0	100.0	100.0	100.0
Albendazone/Mebendazole	100.0	100.0	100.0	100.0
Vitamin A capsule	100.0	100.0	100.0	100.0
RUTF	100.0	100.0	100.0	100.0
F75	100.0	0.0	NA	16.7
F100	0.0	0.0	NA	0.0
Trained staff and guidelines				
Guidelines available for malnutrition	100.0	80.0	0.0	31.3
Malnutrition job-aids/growth standard charts	100.0	100.0	100.0	100.0
At least one trained staff on nutrition	100.0	100.0	80.0	93.8
Overall percentage of availability	90.9	78.2	80.0	72.4

Prevalence of undernutrition (wasting and underweight)

A total of 14,127 (40%) of children aged 6 to 59 months were measured for their weight, of which 12.4% were moderately underweight (MUW) and 4.9% were severely underweight (SUW). The highest prevalence of severely underweight was reported by Kotti health post (10%) among health posts and Addet primary hospital (16.5%) among health centers/hospital. In case of screening for acute malnutrition using Mid-upper arm circumference (MUAC) measurement, a total of 17,082 (48.2%) were measured for MUAC, of which 10.4 % were diagnosed with moderate acute malnutrition (MAM) and 3.5 % with sever acute malnutrition (SAM). The highest percentage of SAM was reported by Debre Mawi HC (5.2%) (Table 5.2).

Table 5.2. Percentage of children with severe acute malnutrion and underweight based on weight-for-age and MUAC measurements, Yilmana Densa, northwest Ethiopia 2018.

Health facilities	Number of children measured for weight	Underweight		Number of children measured for MUAC	Acute malnutrition	
		MUW (%)	SUW (%)		MAM (%)	SAM (%)
Health posts						
Addet zuria HP	975	7.0	2.2	975	9.8	2.1
Tsion HP	449	8.2	2.9	449	5.8	2.7
Kotti HP	576	9.9	2.4	576	6.1	2.6
Kudade HP	359	7.0	3.1	359	5.3	3.9
Danbash HP	496	9.5	4.0	496	5.8	3.4
Sefatera HP	470	6.8	3.8	470	8.9	2.3
Anbesit HP	603	7.5	3.8	603	5.6	4.5
Gube HP	615	6.0	3.1	615	6.8	2.9
Walka HP	579	8.8	2.6	579	7.9	3.3
Senkegna HP	611	7.0	3.1	611	5.4	2.9

Health facilities	Number of children measured for weight	Underweight		Number of children measured for MUAC	Acute malnutrition	
Health centers						
Addet HC	2,713	15.3	5.8	2,713	11.4	4.2
Debre Mewi HC	632	14.7	5.2	632	11.6	5.2
Densa Bata HC	425	16.2	8.0	525	11.2	4.8
Goshey HC	813	13.4	6.0	1,231	12.3	5.1
Walka HC	354	15.8	6.2	577	14.0	4.5
Addet prim. hosp.	3,457	16.5	6.6	5,671	12.3	3.0
Total 14	,127 12.4	4.9 1	7,082	10.4 3	.53	

Pooled estimate of underweight

The overall pooled estimate of underweight prevallenece was 14.9% (95%CI: 11.9-17.8) with a high heterogeneity across health facility reports (I^2=95.9%, P<0.0001). the subgroup analysis by health facility type revealed that the pooled estimate of underweight among children aged 6–59 months was 10.6 percent (95% CI: 9.77-11.50) at health post level with low evidence of heterogeneity across reports (I^2=13.6%; p=0.32), and 21.6% (95%CI: 20.20-22.99) at health center level with low evidence of heterogeneity across reports (I^2=48.0%; P=0.09) (Figure 5.2)

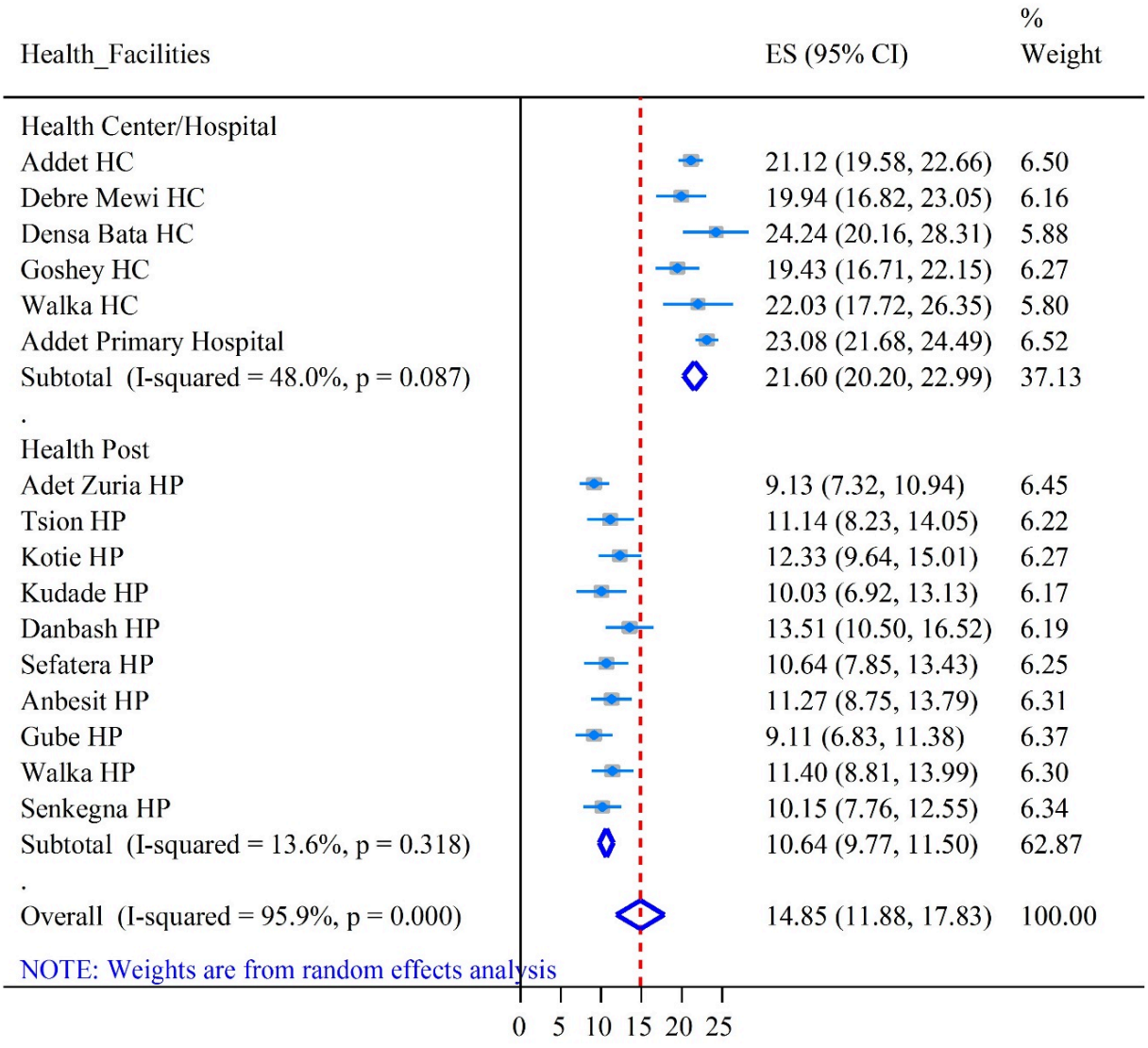


Figure 5.2. Pooled estimates for underweight prevalence by type of health facility among children aged 6-59 months in Yilmana Densa district, Northwest Ethiopia, EFY 2010.

Pooled estimate of global acute malnutrition (GAM)

The overall pooled estimate of GAM among children aged 6–59 months was 12.4% (95% CI: 10.66–14.05) with a high statistical heterogeneity across health facility reports (I^2=89.9%, P<0.0001).The subgroup analysis revealed that the pooled estimate of childhood acute malnutrition was 9.8%(95%CI: 9.02–10.67) at health posts with low evidence of heterogeneity across reports (I^2=12.8%; P=0.33), and 16% (95%CI: 15.19–16.89) at health center level including the hospital with low evidence of heterogeneity across reports (I^2=21.8%; P=0.27). (Figure 5.3)

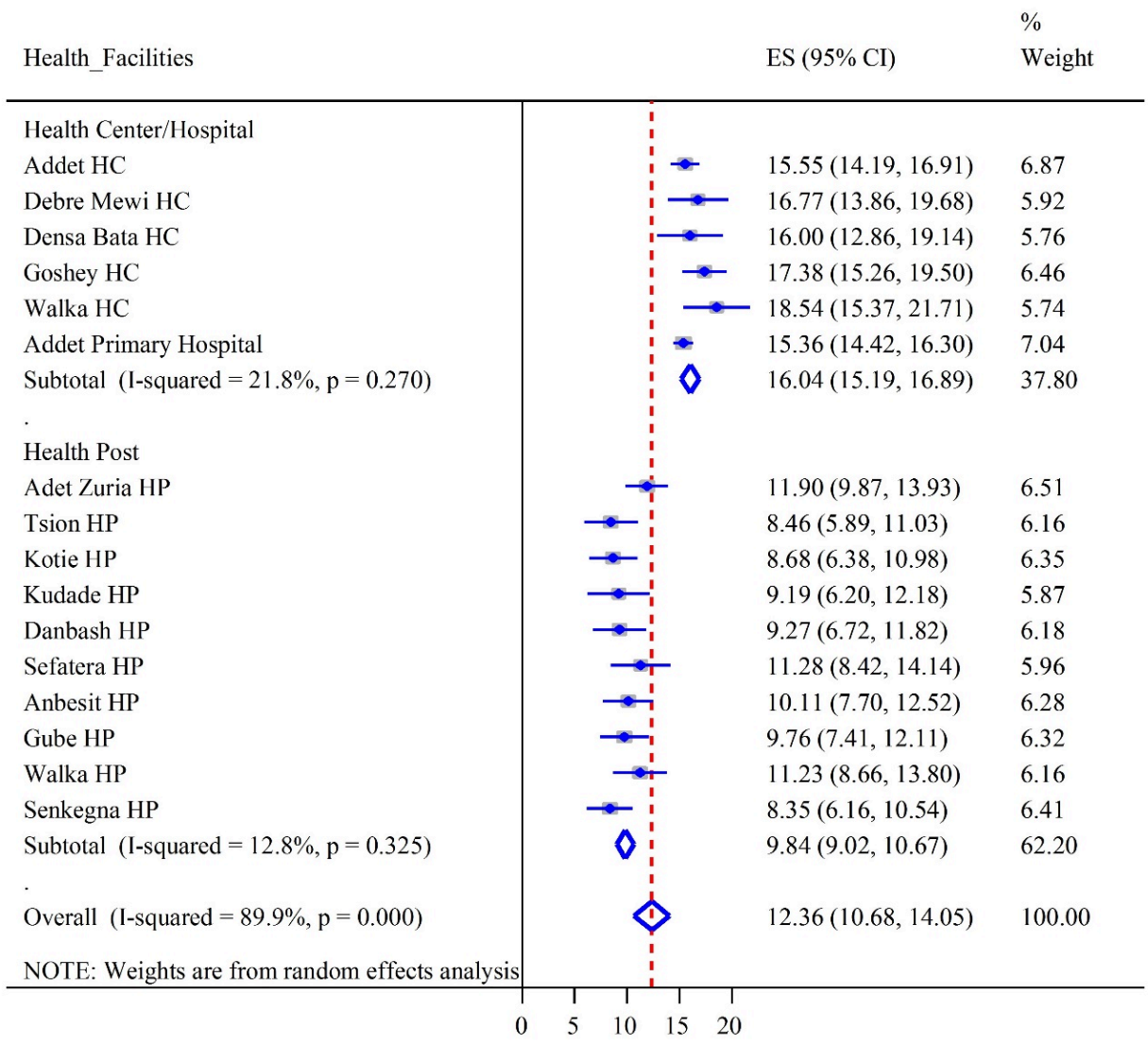


Figure 5.3. Pooled estimates for global acute malnutrition prevalence by type of health facility among children aged 6-59 months in Yilmana Densa district, northwest Ethiopia, EFY 2010.

Meta-regression

In addition to subgroup analysis, Meta regression was conducted to explore the potential sources of variation in the prevalence of underweight and acute malnutrition by considering cluster, type of health facility, and number of children screened for malnutrition. The result of the Meta regression analysis revealed that the type of health facility ($p<0.0001$) and number of children measured for weight ($p=0.03$) were the potential sources of variation in underweight prevalence across the reports of health facilities (Table 5.3).

In case of acute malnutrition prevalence, the type of health facility was identified as a potential source of variation across the reports ($p=0.0001$). However, there was no statistical evidence of cluster and number children measured for MUAC to be the potential sources of inconsistency of acute malnutrion prevalence across reports (Table 5.4).

Table 5.3. Factors associated with the inconsistency of underweight prevalence among children across health facilities in Yilmana Densa district northwest Ethiopia, 2010 EFY.

Variables	Coefficients	P-value
Clusters		
Addet (Constant)	16.18	0.00
Debre Mewi	-2.09	0.66
Densa Bata	-0.20	0.97
Gosheye	-2.94	0.54
Walka	-1.85	0.70
Type of health facility		
Health post (constant)	10.69	0.00
Health center/hospital	10.96	**<0.0001**
Children measured for weight		
Number of children measured for weight	0.003	**0.03**

Table 5.4. Factors associated with inconsistency of prevalence of acute malnutrion among under-five children across health facilities in Yilmana Densa district northwest Ethiopia, 2010 EFY.

Variables	Coefficients	P-value
Clusters		
Addet (Constant)	12.92	0.00
Debre Mewi	-1.41	0.65
Densa Bata	-0.79	0.80
Gosheye	-0.47	0.88
Walka	-0.35	0.91
Type of health facility		
Health Post (Constant)	9.84	0.00
Health Center/Hospital	6.22	**<0.0001**
Children measured for MUAC		
Number of children measured for MUAC	0.001	0.11

Treatment outcome for management of severe acute malnutrition

In all health facilitates, a total of 603 children with severe acute malnutrition (SAM) were enrolled into the therapeutic feeding program (TFP), of which 550 (91.2%) were in the outpatient therapeutic program (OTP) and 53 (8.8%) were in the inpatient therapeutic care. Of the total 603 children enrolled into TFP, 90 percent were recovered, 1.7 percent lost to follow up (LTFU) and 1.7 percent were transferred out. Overall, 93 percent of the children with SAM among those who enrolled in to TFP were exit from the SAM treatment (Table 5.5).

Table 5.5. Treatment outcome for management of severe acute malnutrition in Yilmana Densa district, Northwest Ethiopia, 2010 EFY.

Health Facilities	Number of children enrolled into TFP			Percentage of treatment outcome				Percentage of children exited from SAM treatment
	OTP	ITC	Total	Recovered	LTFU	Transferred	Died	
Addet HC	113	0	113	92.9	1.8	0.9	0.0	95.6
Addet Zuria HP	20	0	20	90.0	0.0	0.0	0.0	90.0
Tsion HP	12	0	12	75.0	16.7	0.0	0.0	91.7
Debre Mewi HC	33	0	33	93.9	0.0	0.0	0.0	93.9
Kotie HP	15	0	15	86.7	0.0	0.0	0.0	86.7
Kudade HP	14	0	14	92.9	0.0	0.0	0.0	92.9
Densa Bata HC	25	0	25	88.0	0.0	4.0	0.0	92.0
Danbash HP	17	0	17	82.4	0.0	0.0	0.0	82.4
Sefatera HP	11	0	11	90.9	0.0	0.0	0.0	90.9
Goshey HC	63	0	63	93.7	0.0	0.0	0.0	93.7
Anbesit HP	27	0	27	96.3	0.0	0.0	0.0	96.3
Gube HP	18	0	18	94.4	0.0	0.0	0.0	94.4
Walka HC	26	0	26	80.8	0.0	3.8	0.0	84.6
Walka HP	19	0	19	94.7	0.0	0.0	0.0	94.7
Senkegna HP	18	0	18	88.9	5.6	0.0	0.0	94.4
Addet Prim. Hospital	119	53	172	87.2	2.9	4.1	0.0	94.2
Total	550	53	603	89.9	1.7	1.7	0.0	93.2

5.5 Discussion

The average availability of service for malnutrion among six tracer items (measurements of weight-for-height/length, weight-for-age, Height/length-for-age and MUAC as well as therapeutic and supplementary feeding services) was 66.7%. None of the health facilities were providing weight-for-height/length, Height/length-for-age measurement service for children. This indicates that that no any system that could capture the stunting among children in the district. Readiness of health facilities to provide malnutrition service was assessed based on the presence of 11 tracer items under the category of trained staff & guidelines and equipment & medicines. Hence, the availability of at least one trained staff on malnutrition diagnosis and treatment was observed in the 94 percent of health facilities. In addition, the availability of the 11 tracer items was 91% at hospital level, 78% at health center level and 80% at health post level. Overall, the availability of the 11 tracer items (readiness) was 72 percent.

According to WHO criteria, the prevalence cut-off values for public health significance for underweight is <10% for Low prevalence, 10-19% for medium prevalence, 20-29% for High prevalence and ≥ 30% for Very high prevalence[28, 29]. Whereas, the prevalence cut-off values for public health significance for wasting (acute malnutrition) is < 5%: Acceptable, 5-9%: Poor, 10-14%: Serious, ≥ 15%: Critical [28, 29]. Hence, the prevalence of underweight (thin for age) (14.9 %) and acute malnutrion (wasting) (12.3 %) in this rapid assessment confirmed a medium and serious public health significance of undernutrition in the study area, respectively.

The prevalence of underweight was high (22%) among children who attended health centers and hospital and medium (11%) among children who attended health posts. However, the prevalence of underweight (thin for their age) (14.9%) in this study was lower than the 2016 Ethiopian demography and health survey report (24%)[4] and the second National Nutrition Program (NNP II) target to decrease the prevalence of under-weigh among under-five children to 17.8 percent in 2018[30]. It is also low as compared with a study finding in Takusa district, Northwest Ethiopia (19.5%)[10]. The test for heterogeneity shows significant inconstancy in the prevalence of under-weight across health facility reports and in the subgroup analysis, the type of health facility (health centers Vs health posts) was a potential source of variation in the prevalence of under-weight across health facility reports. In the Meta regression, the type of health facility (health centers/hospital Vs health posts), number of children measured for weight were associated factors with the inconsistence of underweight prevalence across health facilities' reports.

Proportion of children under five years screened for acute malnutrition is an indicator used for monitoring/identification of nutritional status of children and used to detect acute malnutrition and manage accordingly [17]. However, only nearly half (48.2%) of the children were screened for acute malnutrition using MUAC measurement and 10.5 % of them were diagnosed with moderate acute malnutrition (MAM) and 3.5 % with sever acute malnutrition (SAM). Overall, the prevalence of global acute malnutrition among children aged 6-59 months was found to be 12.4 percent (95% CI; 10.66–14.05). This indicates the risky situation of global acute malnutrition (10% - < 15%) in the district based on the prioritization categories[31]. It also higher than the national prevalence of acute malnutrion (10%)[4], the National Nutrition Program II target to decrease prevalence of acute malnutrition among under 5 children to 6.5 percent in 2018 [30], and a study finding conducted in Bule Hora district, South Ethiopia (10%) [32]. It was also two-fold higher than the prevalence reported by a study conducted in Aleta Chucko and Aleta Wondo districts, Sidama Zone, South Ethiopia(5.6%) [33]. The test for heterogeneity shows significant inconstancy among reports of health facilities and the result of subgroup analysis by health facility type (health centers Vs health posts) shows the type of health facility was a potential source of variation in the prevalence of acute malnutrition. The Meta regression result showed that the type of health facility was a

potential source of inconsistence in the prevalence of acute malnutrition. However, there was no statistical evidence that could show the clusters & number children measured for MUAC were the potential sources of inconsistence in the prevalence of acute malnutrition across health facilities.

Infants and children who are 6–59 months of age and have a mid-upper arm circumference <11 cm or have bilateral edema, should be immediately admitted to a program (OTP or IPC) for the management of severe acute malnutrition[17] to recover from undernutrition. The proportion of children under-five with severe acute malnutrition that recovered from treatment centers (OTP or IPC) was 90 percent. This is higher than the expected cure rate, 75% [17]. A defaulter from a feeding program is an individual who has not attended 3 consecutive visits and it is confirmed that the child is alive for out-patient services. Based on this, the proportion of children under-five with severe acute malnutrition that defaulted or lost to follow up from treatment centers (OTP or IPC) was nearly 2 percent which is lower than the expected default rate (<15%) by the program[17].

5.6 Conclusions

The nutrition specific service availability and readiness at the health facilities was low. The global acute malnutrition (GAM) is a risky situation in the district. Significant variation in the prevalence of underweight and GAM were was observed among the reports of health facilities. The type of health facility and number of children measured for weight were associated factors to account for the inconsistency of undernutrition prevalence across the reports of health facilities. The recovery rate of children under sever acute malnutrition treatment was high. .

5.7 Recommendations

To improve malnutrition detection and management among children, it is better if the district health office and health facilities work more on improving nutrition specific service availability and readiness, particularly on the training of staffs at health posts, availability of guidelines for malnutrition management and height/length measuring scale at health facilities. It is also better to screen all the targeted children for malnutrition using anthropometric indice both for acute and chronic malnutrition. In addition, further study may need to investigate the factors associated with variations in the prevalence of undernutrition among health posts and health centers.

References

1. Guideline: assessing and managing children at primary health-care facilities to prevent overweight and obesity in the context of the double burden of malnutrition. Updates for the Integrated Management of Childhood Illness (IMCI). Geneva: World Health Organization; 2017. Licence: CC BY-NC-SA 3.0 IGO.
2. WHO. What is malnutrition? (www.who.int/features/qa/malnutrition/en/).
3. Central Statistical Agency [Ethiopia] and ICF International. 2012. Ethiopia Demographic and Health Survey 2011. Addis Ababa, Ethiopia and Calverton, Maryland, USA: Central Statistical Agency and ICF International.
4. Central Statistical Agency (CSA) [Ethiopia] and ICF. 2016. Ethiopia Demographic and Health Survey 2016. Addis Ababa, Ethiopia, and Rockville, Maryland, USA: CSA and ICF.
5. Victora CG, Adair L, Fall C, Hallal PC, Martorell R, Richter L, Sachdev HS, Maternal and Child Undernutrition Study Group. Maternal and child undernutrition: consequences for adult health and human capital. Lancet. 2008;371(9609):340–57.
6. Black RE, Victora CG, Walker SP, Zq B, Christian P, de Onis M, et al. Maternal and child undernutrition and overweight in low-income and middleincome countries. Lancet. 2013;382:427–51.
7. Federal Democratic Republic of Ethiopia. National Nutrition Program Multi-sectoral Implementation Guide. June 2016, Addis Ababa, Ethiopia.
8. Fentahun, W., M. Wubshet, and A. Tariku, Undernutrition and associated factors among children aged 6-59 months in East Belesa District, northwest Ethiopia: a community based cross-sectional study. BMC Public Health, 2016. **16**: p. 506.
9. Gizaw, Z., W. Woldu, and B.D. Bitew, Acute malnutrition among children aged 6-59 months of the nomadic population in Hadaleala district, Afar region, northeast Ethiopia. Ital J Pediatr, 2018. **44**(1): p. 21.
10. Nigatu, G., et al., Prevalence and associated factors of underweight among children 6-59 months of age in Takusa district, Northwest Ethiopia. Int J Equity Health, 2018. **17**(1): p. 106.
11. Sphere Handbook (2004), The Sphere Project, Humanitarian Charter and Minimum Standards in Disaster Response, Oxfam Publishing, Oxford www.spherehandbook.org/
12. Maya ARII. Rapid assessment in disasters. JMAJ 56(1): 19–24, 2013.
13. Centers for Disease Control and Prevention (CDC). Disaster Preparedness and Response: Complete Course. Facilitator guide, first edition. Atlanta (GA): CDC; 2014.
14. CDC. Community Assessment for Public Health Emergency Response (CASPER) Toolkit, Second edition. Atlanta (GA); 2012.
15. WHO 2013. Service Availability and Readiness Assessment (SARA) Implementation Guide : An annual monitoring system for service delivery In: WHO (ed.). 20 Avenue Appia, 1211 Geneva 27, Switzerland: World Health Organization.
16. FMOH and EPHI. Ethiopia Service Availability and Readiness Assessment (SARA) 2016. Summary Report. January, 2017.
17. FMOH. Revised Health Management Information System Indicator Reference Guide: Technical Standards: Area 1, August 2017.
18. Federal Democratic Republic of Ethiopia, Ministry of Health: Training Course on the manageent of sever acute mulnutrition; Chart Booklet. Secone Edition, 2013 Addis Ababa, Ethiopia.
19. Updates on the management of severe acute malnutrition in infants and children: guideline. Geneva: World Health Organization; 2013 (apps.who.int/iris/bitstream/10665/95584/1/9789241506328_eng.pdf?ua=1).
20. WHO child growth standards: methods and development. Length/height-for-age, weight-for-age, weight-for-length, weight-for-height and body mass index-for-age. Geneva: World Health Organization; 2006. (www.who.int/nutrition/publications/childgrowthstandards_technical_report_1/en/).
21. Community-based management of severe acute malnutrition. A joint statement by the World Health Organization, the World Food Programme, the United Nations System Standing Committee on Nutrition and the United Nations Children's Fund. Geneva: World Health Organization; 2007. (apps.who.int/iris/bitstream/10665/44295/1/9789280641479_eng.pdf?ua=1&ua=1).
22. FMOH: Guidelines for the management of moderate acute malnutrition, in Ethiopia. Final version September, 2012. Addis Ababa, Ethiopia.
23. StataCorp. 2015. Stata: Release 14. Statistical Software. College Station, TX: StataCorp LP.
24. Sedgwick, P., Meta-analyses: what is heterogeneity? BMJ, 2015. **350**: p. h1435.

25. Higgins, J.P., et al., Measuring inconsistency in meta-analyses. BMJ, 2003. **327**(7414): p. 557-60.

26. DerSimonian R, Laird N. Meta-analysis in clinical trials. Controlled clin trials. 1986;7(3):177–88.

27. Nordmann, A.J., B. Kasenda, and M. Briel, Meta-analyses: what they can and cannot do. Swiss Med Wkly, 2012. **142**: p. w13518.

28. WHO. Physical status: the use and interpretation of anthropometry. Report of a WHO Expert Committee. Technical Report Series No. 854. Geneva, World Health Organization, 1995. whqlibdoc.who.int/trs/WHO_TRS_854.pdf.

29. World Health Organization. Nutrition Landscape Information System country profile indicators interpretation guide. Geneva: World Health Organization; 2010.

30. Federal Democratic Republic of Ethiopia. National Nutrition Program II 2016-2020. June2016, Addis Ababa, Ethiopia.

31. ENCU, DPPC. Emergency Nutrition Intervention Guideline. Addis Ababa, Ethiopia August 2004.

32. Asfaw, M., et al., Prevalence of undernutrition and associated factors among children aged between six to fifty nine months in Bule Hora district, South Ethiopia. BMC Public Health, 2015. **15**: p. 41.

33. ACF International. Maize Livelihood Belt of Aleta Chucko and Aleta Wondo Woredas, Sidama Zone, SNNPR Ethiopia final report 2014. By Jennifer Holden, Action Contre La Faim (ACF): Analyst of Nutrition Causal Analysis. Available at www.ghadvocates.eu/wp-content/uploads/2015/08/Nutrition-Causal-Analysis-Link-NCA-Sidama-Zone-Ethiopia-2014.pdf.

Annex 5.1: Rapid nutrition assessment data extraction checklist

Date of Interview: _____

District: _____, Zone: _____, Region: _____

Section I: Socio-demographic profiles

1. Health facility name _____, Health facility type ____Cluster name _____
2. Catchment population: _____
    a. By sex : Male _____, Female _____
    b. By age: 6-59 months _____, 24-59months _____, < 5 years _____
3. Health facilities
    a. Number of government hospital(s): _____
    b. Number of government health center(s) _____
    c. Number of government health post(s) _____
    d. Number of private clinic(s): _____

Section II: Nutrition specific service availability and readiness assessment questions

4. Check for the presence of the following standard health services for nutrition program (Tracer indicators for service availability). Mark “Yes” if present, otherwise “No”

EQUIPMENT AND MEDICINES	YES	NO
The health facility provides service for child weight measurement	☐	☐
The health facility provides service for child height measurement	☐	☐
The health facility provides service for child MUAC measurement	☐	☐
The health facility provides Therapeutic feeding service for severely malnutrion children	☐	☐
The health facility provides complementary public health interventions for supplementary feeding programs (deworming & Vit. A supplementation)	☐	☐

5. Check for the presence of the following necessary materials and trained staff for malnutrition screening (Tracer indicators for readiness). Mark “Yes” if present, otherwise “No”

SELECTED TRACER INDICATORS	YES	NO
Equipment and medicines	☐	☐
1 Child weight scale	☐	☐
1 Length/height scale	☐	☐
1 MUAC measuring tape	☐	☐
1 Albendazone/Mebendazole	☐	☐

SELECTED TRACER INDICATORS	YES	NO
1 Vitamin A capsule	☐	☐
1 RUTF	☐	☐
1 F75	☐	☐
1 F100	☐	☐
Trained staff and guidelines	☐	☐
1 Guidelines available for malnutrition	☐	☐
1 Malnutrition job-aids/growth standard charts	☐	☐
1 At least one trained staff on nutrition	☐	☐

Section III: Nutrition screening

6. Number of children weighted:_____

7. Number of children with moderately underweight (-2 < Z-score < -3): _____

8. Number of children with severely underweight (Z-score <-3) _____

9. Number children of screened for acute malnutrion (measured for MUAC): _____

10. Number of children with MUAC <11cm or Oedema of both feet :_____

11. MUAC 11cm to <12cm AND no edema of both feet _____

Section IV: Management of sever acute malnutrion and treatment outcomes

12. Number of children enrolled into therapeutic feeding program (TFP)

    a. Outpatient therapeutic program: _____

    b. Inpatient therapeutic program : _____

13. Treatment outcome for management of severe acute malnutrition in children 6-59 months

    a. Number of children recovered ____

    b. Number of children defaulted_____

    c. Number of children transferred_____

    d. Number of children died_____

14. Total number of children who exit from severe acute malnutrition treatment:_____

Section V: Management systems and coordination

15. Is the any assigned focal person for nutrition

16. Does the PHEM weekly report include SAM?

17. Is there a Public Health Emergency preparedness and response plan?

    a. Does it include Nutrition? Observe and comment ____

18. What nutrition screening modality is used in the district?

    a. Routine

    b. CHD

Section VI: Complementary public health interventions for supplementary feeding programs

19. Number of children aged 6-59 months supplemented with Vitamin A:____

20. Number of children aged 24-59months dewormed: _____

PART II

Research and Dissemination

Spatial and multilevel research, scientific manuscripts, conference abstracts and additional outputs.

CHAPTER 6

Research Project

Spatial clustering of measles vaccination and associated factors among children in Ethiopia: Findings from the 2016 demography and health survey

Tesfahun Taddege GEREMEW[5] and Lemma Derseh

5. Correspondence to : ttaddege@gmail.com↩

PUBLISHED VERSION

Geremew TT, Gezie LD, Abejie AN. Geographical variation and associated factors of childhood measles vaccination in Ethiopia: a spatial and multilevel analysis. *BMC Public Health*. 2019;19:1194.

https://doi.org/10.1186/s12889-019-7529-z

Abstract

Background: In Ethiopia, despite considerable improvement of childhood measles vaccination, measles outbreaks have been occurring in most parts of the country. Understanding the neighborhood variation in measles vaccination has been increasingly recognized and crucial for evidence-based decision-making. However, the spatial pattern of first dose of measles-containing vaccine (MCV1) and its predictors are poorly understood in Ethiopia. Hence, this study aimed to explore the spatial pattern and associated factors of childhood MCV1

Methods: A secondary data analysis was conducted using the 2016 Ethiopia demographic and health survey data. A total of 3,722 children nested in 611 enumeration areas were included in the analysis. A purely spatial scan statistics using the discrete Poisson model was employed to detect spatial clusters of childhood MCV1. Multilevel logistic regression models were fitted to identify predictors of childhood MCV1.

Result: The coverage of childhood MCV1 was 54.3% (95% CI = 52.7 - 55.9) in Ethiopia with spatial hetrogeniety (Global Moran's I=0.134, p-value < 0.0001) and a clusterd pattern of areas with low MCV1 coverages. A total of seven statistically significant SaTScan clusters of areas with low MCV1 coverage were detected. The most likely primary SaTScan cluster was detected in Afar region (LLR = 33.79, p< 0.01), secondary cluster in Ethio-Somali region (LLR=26.76, p<0.01), and tertiary cluster in Gambella region (LLR=17.73, P<0.01). Older child age, pentavalent vaccination, higher level maternal education and media exposure were positively associated factors with childhood MCV1, while older maternal age and a problem in distance to a health facility were negatively associated factors.

Conclusion: A clustered pattern of areas with low childhood MCV1 coverage was observed in Ethiopia. Both the individual level characteristics (child's age, first and third doses of pentavalent vaccination, maternal age and education) and community level characteristics (region and distance from health facility) were statistically significant predictors of childhood MCV1. Hence, it is good if the federal ministry of health and other child health programmers give priority of the areas with low MCV1 coverage identified in this study. It is also better to consider the individual and community level determinant factors identified in this study.

Keywords: MCV1, Spatial, Multilevel, Ethiopia

6.1 Introduction

6.1.1 Background

Measles is an acute, highly contagious viral disease caused by the measles virus and transmitted by respiratory droplets[1]**.** It is one of the most infectious human diseases and can cause serious illness, lifelong complications and death, accounting for 114,900 deaths globally in 2014 of which 73,914 (63%) occurred in Africa[2, 3]. Prior to the availability of measles vaccine, measles infected over 90% of children before they reached 15 years of age[4, 5].

The Measles-Rubella Initiative, spearheaded by the American Red Cross, the US Centers for Disease Control and Prevention (CDC), the World Health Organization (WHO), and others, have targeted the measles virus for global elimination. This initiative aimed to reduce annual measles incidence rates(IRs) to less than five cases per million, requiring >90% coverage of at least one dose of Measles-Containing Vaccine (MCV1), recommended at age 9 -12 months, by the end of 2015 and >95% coverage by 2020 in all countries[5]. Improvement in vaccination coverage has decreased measles deaths from over half a million globally in 2000 to 114,900 in 2014, however, the progress has stalled Since 2010[2, 6, 7]. Globally, in 2015, MCV1 coverage had reached 85% and an estimated 134 200 measles deaths had occurred, accounting a 79% decline since 2000[4]. The 2015 vaccination goal was not met and IRs remained relatively unchanged between 2013 and 2014[2]. Throughout sub-Saharan Africa, where the majority of the world's remaining measles burden is found, measles vaccination is predominantly delivered through two activities: MCV1 for children age 9 months through routine immunization activities and MCV2 through supplemental immunization activities (SIAs) to provide MCV to those who were not[8, 9]

Ethiopia has been engaged in expanding immunization services against the six childhood diseases since 1980. The goal of EPI in Ethiopia at the beginning was to increase immunization coverage by 10% annually and achieve universal childhood immunization. In the beginning, the goal of the EPI program was to provide immunization services including measles to all children under two years of age. However, in 1986 the policy has been changed and targeted only under one surviving infants throughout the country[10]. Recently, the MCV1 in Ethiopia shows an improvement, 84% in 2014[11].

6.1.2 Statement of the problem

Measles immunization is an essential component for reducing under-five mortality[12] and the vaccine is recommended for all susceptible children for whom measles vaccination is not contraindicated. Where risk of measles mortality among infants remains high, MCV1 should be administered at 9 months of age[4]. In Ethiopia, an improvement in routine EPI coverage including MCV has been shown due to the combine effect of Reaching Every District (RED) approach, Health extension program and implementation of Enhanced Routine Immunization Activities (ERIAs) and contributed a lot in the reduction of measles-related morbidity and mortality[13]. From 2014 to 2016, the MCV1 coverage was 70%-84% [4, 7, 13-15]. Such improved Measles immunization coverage should have a strong inverse correlation with the prevalence of the measles disease, especially among under-five children, i.e. increasing coverage should be accompanied by decreasing cases of the disease[12].

However, despite this considerable improvement of MCV in Ethiopia, Measles outbreaks continue to occur in most parts of the country and becoming a cause of significant number of under-five morbidity and mortality and determinants are not well understood. As the MCV1 coverage increased from 59% in 2005 to 84% in 2014, the incidence rate of confirmed

measles cases per 100,000 population increased from 0.6 in 2005 to 11.2 in 2014[13]. Ethiopia Contributes 9% to global measles mortality[6, 14]. Among the global estimated 20.8 million infants who did not receive MCV1 through routine immunization services in 2015, approximately 0.7 million (3.4%) were in Ethiopia[16].

Such contradiction may result from heterogeneity of coverage with pockets of missed children, population movements, community resistance, cold-chain failure, poor data collection, and reporting errors, accumulation of unvaccinated children in highly populated areas[5, 13]. Spatial heterogeneity of vaccination coverage can delay disease elimination even in countries with high average nationwide vaccination coverage rates and Substantial heterogeneity in measles vaccination coverage has been demonstrated previously in Sub-Saharan Africa[17, 18], but drivers for this heterogeneity are poorly understood.

6.1.3 Justification

The occurrence of outbreaks is useful to identify gaps in immunization programme performance that may not be evident through monitoring vaccination coverage. One of the weaknesses of immunization programme is heterogeneity of coverage with pockets of missed children[5].

Though the MCV1 rate has increased dramatically over the past 10 years (from 54% in 2000 to 87 in 2016), neighborhood variation can be quite high and this has attracted much attention from public health officials and researchers. In addition, the impact of spatially heterogeneous vaccination has been increasingly recognized in decision-making decisions, resulting in a shift in focus from simply setting country-level targets for coverage, to ensuring uniformly high vaccination levels across countries. Understanding the spatial epidemiology is crucial to take response measures[19]

However, there are limited studies on the spatial epidemiology of MCV1 in Ethiopia and drivers for heterogeneity are poorly understood. Hence, this study will help to explore the spatial distribution of MCV1 and to identify driving factors for spatial heterogeneity of MCV1 among children in Ethiopia. In Addition, this study may contribute information for evidence-based decision making on the intervention for measles immunization, one of the high impact interventions in child health.

6.1.4 Literature review

Spatial Analysis in Public Health

It is a long known basic assumption of epidemiology that diseases are non-randomly distributed in part as a result of geographic clusters of health behaviors, and personal belief exemption[20]. Spatial analysis in Epidemiology roots its foundation in the basic assumptions of descriptive Epidemiology in that it tries to identify the geographical distribution of a health event/disease occurrence, and whether an event/disease distribution is place dependent or whether there is geographical dependence in a health event/disease occurrence between neighboring regions or within the regions. Spatial studies are nowadays becoming an area of interest because of an increased advent of free user-friendly software like SaTScan, GeoDa and the R project[21]. The impact of spatially heterogeneous vaccination has been increasingly recognized in making policy decisions, resulting in a shift in focus from simply setting country-level targets for coverage, to ensuring uniformly high vaccination levels across countries[19]

Some studies were done to see the spatial distributions of immunization coverage around specified localities in sub-Saharan Africa and African Great Lakes region also report on the application of spatial epidemiology to see the distribution of vaccination coverage[22, 23].

Spatial Studies on Immunization

Despite a widespread misinformation about childhood vaccination including measles vaccination, not all geographic areas have the same rates of childhood vaccination[24, 25]. Child vaccination for vaccine-preventable diseases are highly clustered at multiple geographic scales including region, Zone, district, and neighborhood level. This heterogeneity in vaccination coverage has had an immediate health impact on those communities with the lower immunization rates at all spatial scales[26]. As a study on spatial clustering of underimmunization shows, the reasons for clustering are varied[26]. While child vaccination is gaining attention at all levels of public health, there is an urgent need to identify regions with high and low childhood vaccination rates and to intervene in these areas[26].

The impact of spatially heterogeneous childhood vaccination has been increasingly recognized in decision making, resulting in a shift in focus from simply setting country-level targets for coverage, to ensuring uniformly high vaccination coverage across all levels of a country (for example, the strategy of RED). Although a considerable improvement over a country-level focus, a zonal-level or district-level focus may still miss important aspects of geographical heterogeneity in childhood vaccination. By taking averages across administratively-defined areas (that is, Countries or Regions), we may miss zones, districts or enumeration areas of vulnerability that are small or do not reflect national or sub-national administrative boundaries[19].

Measles burden and its vaccination status in Ethiopia

Measles is the commonest vaccine-preventable diseases that occur in Ethiopia, and mothers recognized as a self-limiting common childhood illness of which no medical care is often sought[27]. In 2013, measles incidence was 7.2 cases per 100,000 populations. A total of 243 measles outbreaks were confirmed in 2013 compared to 146 in 2012 with a total of 192 affected districts in 2013 compared to 125 in 2012[13]. Outbreaks of vaccine-preventable diseases drain immunization resources and highlight inadequacies in both routine and supplemental immunization services. In 2015, large measles outbreaks occurred in Africa, posing major challenges to immunization programmes in affected countries. Learning to maintain immunization services will be key to building more resilient health systems that can withstand the strains of outbreaks and emerging diseases[6].

Even though the coverage of MCV1 has shown an improvement from 81% in 2010 to 84% in 2014, Measles outbreak continues to occur, Measles IR per 100,000 population was 7.7 in 2010 and 11.1 in 2014[13]. The Ethiopia Demography and Health surveys have shown progress in measles vaccination coverage. The percentage of children age 12-23 months who received first dose measles vaccination increased from 21% in 2000 to 35% in 2005, 56% in 2011, and then decreased to 54% in 2016. The proportion of children age 12-23 months with no vaccination decreased from 25% in 2005 to 16% in 2016[28-31]. The recent 2016 Ethiopia Demography and Health Survey final report shows that the percentage of children age 12-35 months who received first dose measles vaccine at any time before the survey, according to evidences from a vaccination card, health facility, or the mother’s report, is 54% at national level which varies across regions with lowest (30%) in Afar region and highest (93%) in Addis Ababa[28].

Determinant factors of measles vaccination

Different studies have reported on the possible socio-demographic and related factors that are associated with low measles vaccination status. Based on this, we reviewed literatures on the possible factors associated with childhood measles vaccination with categories of maternal and/or maternal related factors, child-related factors and possible opportunities during pregnancy and delivery for childhood vaccination.

Maternal/Paternal related factors:

Mothers' and fathers' background characteristics play a vital role in child health interventions including child vaccination. As a study done in Ethiopia in 2015 showed, children whose mother with higher educational status, better quantile, media exposure are more likely to receive childhood vaccination including measles vaccination[32]. This study also reported that children whose father with higher educational status are more likely to receive childhood vaccination including measles vaccination [32]. In contrary to this, Children in the family with the number of children aged less than 5 years in the household was negatively associated with childhood immunization[32].

Another study on Extended Program on Immunization (EPI) coverage in selected Ethiopian zones revealed that wealth index is a determinant factor for complete vaccination including measles vaccination in all zones. However, determinant factors varied across zones, in Yem Zone, for example, parity and maternal age were independent predictors for complete vaccination including measles vaccination. Likewise, maternal education, marital status and wealth quantile were the determinant factors for complete vaccination including first dose measles vaccination in Zone 3. This study also showed that wealth quantile, maternal education, maternal age and living area (Zone) were significant determinants of measles vaccination dropout among children who received the first dose of Pentavalent vaccine[33]. The recent 2016 EDHS final report revealed that nearly three fourth of Children age 12-23 months in urban areas and nearly half of children in urban areas received first dose measles vaccine[28]

Children's characteristics

Child character is also play a vital role in childhood vaccination. Age of the children can determine the childhood vaccination and delaying in childhood vaccination which extends the period of vulnerability of children to vaccine preventable diseases[34]. Parents' sex preference for their children can also determine children's health care visit including visits for childhood vaccination services. As indicated in the 2016 Ethiopia demographic and health survey descriptive report, the percentage of female children is relatively higher than the percentage of male children[28]. In addition, the report describes that the first dose measles vaccination coverage among children age 12-23 declines from 59% to 46% as the birth order of children increases form first order births to for sixth or higher order births[28].

Opportunities for child vaccination

Childhood vaccination including first dose measles vaccination can be enhanced by different opportunities. Health service accessibility is one of the opportunities that can create opportunities for childhood vaccination. A study done in sub-Saharan Africa reported that clustering of low measles vaccination coverage was more likely in populations with limited access to health care[35]. Distance from health facilities is an indicator of health service accessibility which can in turn determines the first dose measles vaccination among children. A nearest health facility may create an opportunity for a child to be vaccinated for measles.

In addition, DTP1-HepB1-Hib1 (Pentavalent first dose) vaccination coverage indicates availability of access to and initial use of immunization services. It is a good indicator of health system performance and access to the beneficiary[12]. DTP3-HepB3-Hib3 (Pentavalent third dose) vaccination coverage indicates the continuity of use by parents or care takers, client satisfaction with services, and capability of the system to deliver a series of vaccinations. It is a good indicator of health system performance and utilization by the beneficiary. Children who have received DTP1-HepB1-Hib1 (Pentavalent first dose) and DTP3-HepB3-Hib3 (Pentavalent third dose) vaccination are likely to receive first dose measles vaccination[12].

6.1.5 Conceptual framework

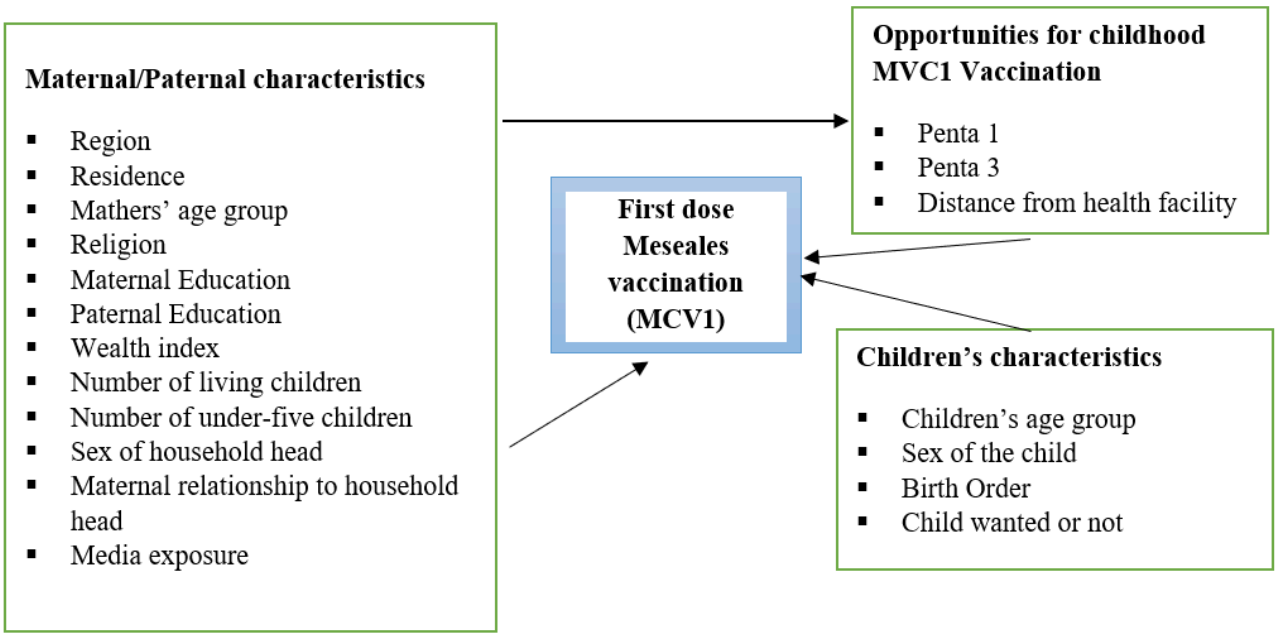


Figure 6.1. Conceptual Framework for determinants of first dose Measles vaccination among children in Ethiopia[36]

(Adapted from Adebiyi. F, 2013)

6.2 Objectives

GENERAL OBJECTIVE

★ To assess the spatial pattern and determinant of first dose measles vaccination (MCV1) among children in Ethiopia.

SPECIFIC OBJECTIVES

1. To explore the spatial pattern of first dose measles vaccination among children
2. To identify factors associated with the first dose measles vaccination among children

6.3 Methods

Study design and Setting

An in-depth analysis of the 2016 Ethiopia demographic and health survey (EDHS) data was conducted. EDHS is a national and subnational representative household survey which is conducted every five year. Ethiopia is second most populous country, located in the horn of Africa, between 3 and 15 degrees north latitude and 33 and 48 degrees east longitude. Administratively, Ethiopia is divided into 11 geographical regions and each region is sub-divided into zones, each zone into districts. The districts in-further are sub-divided into kebeles (the smallest administrative units). During the 2007 population and housing census of Ethiopia, each kebele was sub-divided again into enumeration areas (EAs), which were used as a primary sampling unit in this demographic and health survey.

The health system of Ethiopia, which focuses on a preventive health care, is organized into three health tire system: Primary health care unit (PHCU), general hospital and specialized hospital. Routine child immunization is primarily provided at PHCUs through static and outreach sites[37] and the availability of routine childhood vaccination services in Ethiopia is 80 percent[38].

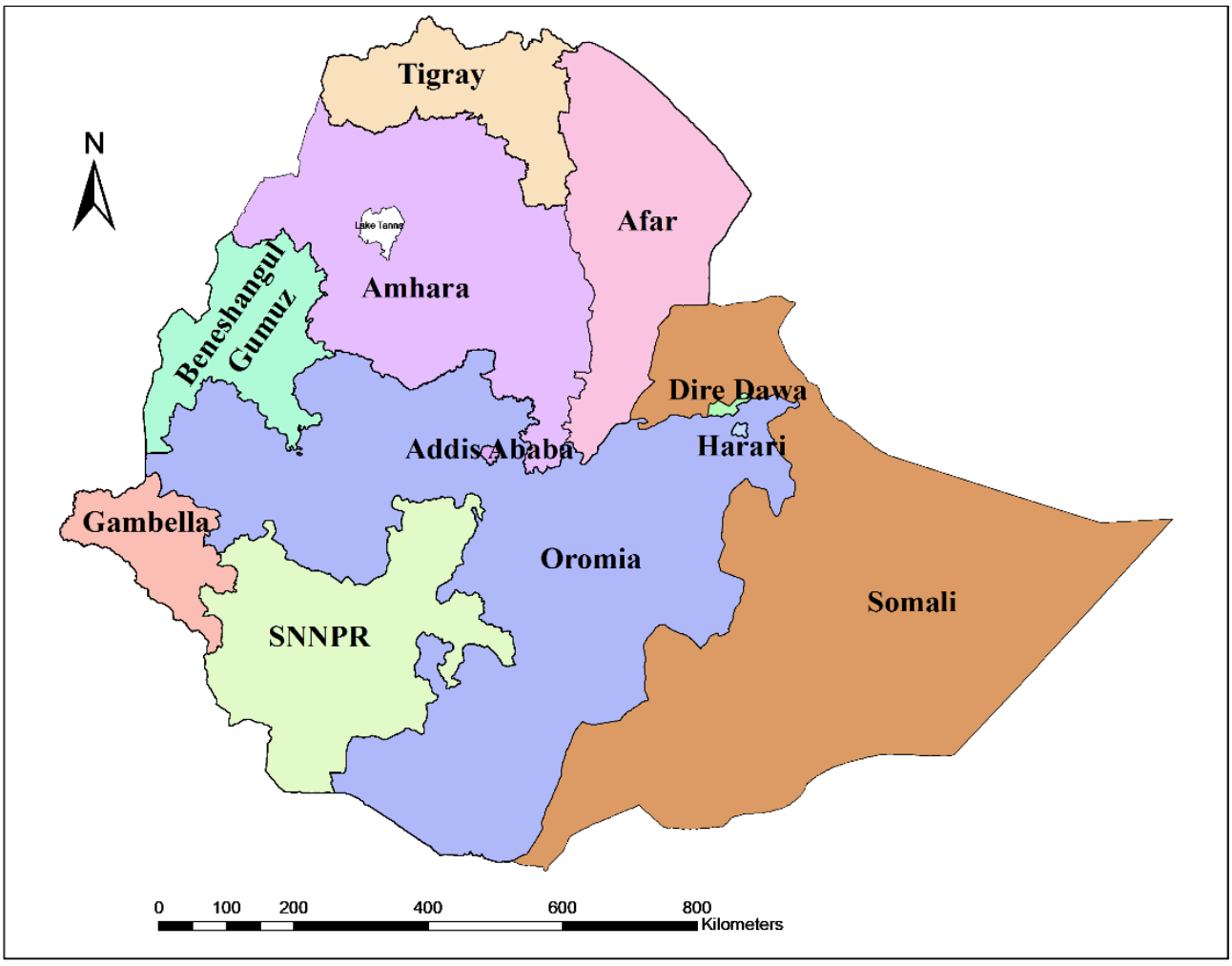


Figure 6.2. Regional states of federal democratic republic of Ethiopia[39].

Population

Source population

- All children aged 12-35 months in Ethiopia

Study population

- Children age 12-35 months in the selected Enumeration Areas (EAs) of Ethiopia

Eligibility criteria

Children aged 12-35 months were eligible and children whose geographical coordinates were not available at the global positioning system (GPS) were excluded from this study

Sampling technique and sample size

The 2016 EDHS used a stratified, two-stage cluster sampling design using the enumeration areas (EAs) of the 2007 Ethiopian population and housing census (PHC) as the primary sampling unit and households as the secondary sampling unit. In the first stage, a total of 645 EAs (202 in urban areas and 443 in rural areas) were selected with probability proportional to EAs size (PPS) from the complete list of 84,915 EAs created for the 2007 PHC sampling frame. In the second stage of selection, a fixed number of 28 households per cluster were selected with an equal probability systematic selection from the newly created household listing. Finally, a total of 16,650 households were successfully interviewed, yielding a response rate of 98%. A total of 3,722 children aged 12–35 months were included in this analysis. The detailed methodology has been published in the 2016 EDHS final report[28].

Data source and extraction

The 2016 Ethiopia demographic and health survey datasets, which are publicly available to all registered users, were requested and downloaded from MeasureDHS program website (https://www.dhsprogram.com/data/dataset_admin). After reviewing and understanding the details of the EDHS data structure and dataset types, we selected the recommended dataset type for childhood vaccination record in the last three years before the survey. Data on childhood MCV1 and potential independent variables (individual and community level characteristics) were extracted accordingly.

Study Variables

Dependent variables

The response variable of this study was MCV1 vaccination status of the last child aged 12–35 months. The survey collected this data from written vaccination records (including the infant immunization card and other health cards), mothers' verbal reports and health facility records. If a child had received MCV1, it was classified as "Yes", otherwise "No".

Independent Variables

Table 6.1 depicted the potential predictor variables included in this study along with their categories at two levels: Individual and community levels. Individual-level (level I) variables include socio-demographic and economic characteristics, while community-level (level II) variables include the common characteristics of study subjects based on the enumeration areas or clusters within the 2016 EDHS data.

Table 6.1. Independent variables and categorization

Variables	Categories
1. Individual level factors	
Child Characteristics	
Age of child (months)	(1) 12-23; (2) 24-35
Sex of the child	(1) Male; (2) Female
Birth order	(1) 1; (2) 2-3; (3) 4-5; (4) 6+
Acceptance of the child	(1) Wanted then, (2) Wanted later & (3) Wanted no more
DPT1-HepB1-Hib1	(0) Not vaccinated; (1) Vaccinated
DPT3-HepB3-Hib3	0) Not vaccinated; (1) Vaccinated
Maternal/Paternal characteristics	
Mother's age (years)	(1)15-19; (2) 20-34; (2) 35-49
Religion	(1)Orthodox; (2)Muslim; (3)Protestant; (4) Others
Mother's education	(1)No education; (2)Primary; (3)Secondary; (4)Higher
Father's education	1)No education; (2)Primary; (3)Secondary; (4)Higher
Wealth index	(1) Poorest; (2)Poorer; (3) Rich; (4) Richer ; (5)Richest
Sex of household head	(1) Male; (2) Female
Number of living children	(1) 1; (2) 2-3; (3) 4-5; (4) 6+
Head of household	(1) Male; (2) Female
Mother's relation to the head of the household	1. Wife; (2) head; (3)Daughter; (4) Others
Regular media exposure	1. No; (1) Yes
2. Community level factors	
Residence	(1) Urban & (2) Rural
Region	(1)Addis Ababa; (2)Tigray; (3)Afar; (4)Amhara; (5) Oromiya; (6) Somali; (7) Benishangul-Gumuz; (8) SNNP; (9) Gambella; (10) Harari & (11) Dire Dawa
Distance to health facility	1. A big problem, (2) Not a big problem

Operational definitions

- **Clustering**: An unusual aggregation of children with the same MCV1 status (Vaccinated or unvaccinated) in a specific geographical area[21, 40].

- **Measles-containing Vaccines**: Vaccines containing antigens of measles only (M) or a combination of measles with rubella (MR), mumps (MM, MMR) and varicella (MMRV) vaccines[41].

- **Children who received MCV1:** Children who received a one dose measles-containing vaccine for the first time at any time before the survey (based on the evidence from vaccination cards, health facility records, or the mother's report) [28].

- **Mass media exposure:** Respondents were asked how often they read a newspaper, listened to the radio, or watched television and responded at least once a week are considered to be regularly exposed to that form of media[28].

Data management and analysis

After data extraction, further recoding of variables was done to better suit with other studies for comparison. We used SaTScan 9.6 and ArcGIS 10.3 for spatial analysis and Stata Version 14 for non-spatial statistical analysis.

Sample weights were applied to compensate for the unequal probability of selection between the strata that has been geographically defined. The detailed explanation of the weighting procedure can be found in the methodology of the EDHS final report[28].

Spatial analysis: The aggregated MCV1 count and proportion data was joined to the geographic coordinates based on each EA unique identification code. The global spatial autocorrelation was assessed with ArcGIS version 10.3 using the Global Moran's I statistic (Moran's I) to evaluate whether the pattern expressed is clustered, dispersed, or random across the study areas[42]. When p-value indicates statistical significance, a positive Moran's I index value indicates tendency toward clustering while a negative Moran's I index value indicates tendency toward dispersion[42].

In the presence of positive global spatial autocorrelation, we employed Kulldorff's method of purely spatial scan statistic using the discrete Poisson model to detect the local spatial clusters of areas with high or low rates. SaTScan 9.6 software was used for the cluster detection [43]. "SaTScan^{TM} is a trademark of Martin Kulldorff. The SaTScan^{TM} software was developed under the joint auspices of (i) Martin Kulldorff, (ii) the National Cancer Institute, and (iii) Farzad Mostashari of the New York City Department of Health and Mental Hygiene[21, 43, 44]". It uses a circular window which moves systematically throughout the study area to identify significant SaTScan clustering of children who received MCV1. We used a maximum of 10 % (to avoid the detection of large clusters) of the population at risk for the Spatial cluster size and the analysis was done using standard Monte Carlo hypothesis testing with 999 Monte Carlo replicates. A cluster is statistically significant when its log likelihood ratio (LLR) is greater than the Standard Monte Carlo critical value for 0.05 significance level and p-value <0.05[21, 43].

Multilevel logistic regression analysis: Considering the hierarchical nature of the data in which 3,722 children were nested within 611 EAs, we fitted multilevel logistic regression models to identify predictors of childhood MCV1 at two levels: individual and community levels. Four models were fitted with the first model (model I) without any explanatory variable (empty model) to detect the existence of possible contextual effect, the second model (model II) adjusted for individual-level variables, the third model (Model III) adjusted for community-level variables and the fourth model (Model IV) adjusted for both the individual- and community-level variables. The goodness of fit of the model was assessed based on the deviance information criteria (DIC) and Akaike's Information Criterion (AIC) and the fourth model with the smallest value of the information criterion, was selected as the final better fit model for the data[45]. For measures of association (fixed effect), adjusted odds ratio with 95% confidence intervals was used to declare statistical significance.

For measures of variation (random effects), Intra-class correlation coefficient (ICC), median odds ratio (MOR) and proportional change in variance (PCV) statistics were computed. ICC was used to explain cluster variation, the variation between individuals within the same cluster and it was calculated using the formula:

$$\mathrm{ICC} = \frac{V_A}{V_A + \frac{\pi^2}{3}} = \frac{V_A}{V_A + 3.29}$$

where V_A is the estimated variance of clusters[46]. The proportional change in variance (PCV) measures the total variation attributed by individual level factors and community level factors in the multilevel regression model and it was calculated as

$$\mathrm{PCV} = \frac{V_A - V_B}{V_A}$$

Where V_A = variance of the initial model, and V_B = variance of the model with more terms [46]. The MOR is the median odds ratio between the individual of higher propensity and the individual of lower propensity when comparing two individuals from two different randomly chosen clusters. It is a measure of unexplained cluster heterogeneity, the variation between clusters (the second-level variation) by comparing two persons from two randomly chosen, different clusters and it was computed using the formula:

$$\mathrm{MOR} = \exp\left(\sqrt{2 \times V_A} \times 0.6745\right) \approx \exp\left(0.95\sqrt{V_A}\right)$$

where V_A is the cluster level variance [46, 47]. The measure is always greater than or equal to 1. If the MOR is 1, there is no variation between clusters (no second level variation) [48].

Ethical Consideration

The data of the 2016 Ethiopia demographic and health survey was used for this study with a permission from the Measure DHS program after being registered and submiting a request with briefly stated objctives of the study. Ethical approval was obtained from the Institutional Review Board of the Institute of Public Health of the University of Gondar and the ICF International Institutional Review Board. The data has been used only for this registered research and it could not be passed on to other researchers. The shapefiles for Ethiopia's administrative boundaries was obtained from the openAFRICA website [https://africaopendata.org/dataset/ethiopia-shapefiles][39]. The detaile of the ethical issues has been published in the 2016 Ethiopia DHS final report, which can be accessed at: http://www.dhsprogram.com/publications[28].

6.4 Results

About 54.3% (95% CI = 50.3 - 58.3) of children aged 12 - 35 months were received MCV1 in Ethiopia. Nearly half of the children (51 %) were within the age group of 12–23 months with the mean age of 22.95 (SD ± 7.04) months. Nearly 70% and 49% of the children had received first and third dose of pentavalent, respectively. Over 63 percent of mothers of the children had no education, about one-fourth of them were in the poorest wealth quintile and 81 percent had no regular media exposure (Table 6.2).

Table 6.2. Percentage of children age 12-35 months who received the first dose Measles vaccine at any time before the survey, according to background characteristics, Ethiopia, 2016.

Background Characteristics	Children aged 12-35 months		Children aged 12-35 months who received MCV1	
	N	%	N	%
Child age (months) (mean=22.95 (SD ± 7.04)				
12 - 23	1976	51.0	1069	54.1
24 - 35	1901	49.0	1036	54.5
Sex of child				
Male	1935	49.9	1056	54.6
Female	1941	50.1	1049	54
Birth Order				
1	753	19.4	451	59.9
2 - 3	1161	29.9	674	58.1
4 - 5	900	23.2	475	52.8
6+	1063	27.4	505	47.5
Wanted child				
Wanted then	2832	73.1	1491	52.7
Wanted later	702	18.1	430	61.3
Wanted no more	343	8.8	183	53.5
DPT1-HepB1-Hib1 (Pentavalent first dose) vaccine				
No	1158	29.9	121	10.4
Yes	2718	70.1	1984	73
DPT3-HepB3-Hib3 (Pentavalent third dose) vaccine				
No	1983	51.2	505	25.5
Yes	1893	48.8	1599	84.5
Maternal Age (Years)				
15 -19	137	3.5	77	56.2
20 - 34	2850	73.5	1587	55.7
35 - 49	889	22.9	441	49.6
Religion				
Orthodox	1349	34.8	887	65.8
Muslim	1530	39.5	617	40.4
Protestant	861	22.2	529	61.4
Others*	136	3.5	71	52.2
Mother's educational				

Background Characteristics	Children aged 12-35 months		Children aged 12-35 months who received MCV1	
No education	2474	63.8	1188	48
Primary	1100	28.4	669	60.8
Secondary & Higher	302	7.8	248	82.1
Husband/partner's education (N=3655)				
No education	1746	47.8	820	47
Primary	1462	40.0	838	57.3
Secondary & Higher	447	12.2	319	71.4
Mother's occupation				
Did not work	2071	53.4	1024	49.4
Non-Professional	1731	44.7	1025	59.2
Professional	74	1.9	56	75.7
Father's occupation (N=3655)				
Did not work	275	7.5	111	40.2
Nonprofessional	620	17.0	303	48.9
Professional	2760	75.5	1564	56.7
Maternal relation to the household dead				
Wife	3091	79.7	1670	54
Head	406	10.5	224	55.2
Daughter	211	5.4	122	57.8
Others	169	4.4	89	52.7
Sex of the household head				
Male	3314	85.5	1792	54.1
Female	562	14.5	312	55.5
Number of living children				0
1	684	17.6	413	60.4
2 - 3	1281	33.0	759	59.3
4 - 5	1022	26.4	529	51.8
6+	890	23.0	404	45.4
Maternal health care decision making (N=3655)				
Jointly with her husband	2439	66.7	1333	54.7
Husband/partner alone	727	19.9	351	48.3
By herself alone	479	13.1	286	59.6
Other	10	0.3	8	80
Wealth index				
Poorest	965	24.9	412	42.7
Poorer	860	22.2	445	51.7
Middle	808	20.8	419	51.9
Richer	676	17.4	393	58.1
Richest	567	14.6	436	76.9
Mass Media Exposure				
Not regular	3154	81.4	1601	50.8
Regular	722	18.6	504	69.7
Total	3,876 10	0 21 05	54.3	

The percentage of children who received first dose measles vaccine (MCV1) varied across the regions of the country with the highest (94 percent) in Addis Ababa and lowest (36percent) in Somali region. The distance to a health facility was a big problem in the three fifths (60 percent) of respondents (Table 6.3).

Table 6.3. Percentage of children age 12-35 months who received the first dose Measles vaccine at any time before the survey, according to community level background characteristics, Ethiopia DHS 2016.

Background Characteristics	Total children aged 12-35 months		Children aged 12-35 months who received MCV1	
	Frequency	Percent	Frequency	Percent
Region				
Addis Ababa	95	2.5	90	93.8
Tigray	279	7.2	227	81.1
Afar	42	1.1	12	28.6
Amhara	718	18.5	434	60.4
Oromia	1712	44.2	722	42.1
Somali	132	3.4	47	35.6
Benishangul Gumuz	43	1.1	32	76.2
SNNPR	818	21.1	517	63.1
Gambella	10	0.2	6	60
Harari	10	0.2	5	50
Dire Dawa	17	0.4	14	82.4
Residence				
Urban	433	11.2	337	77.8
Rural	3443	88.8	1768	51.4
Distance to health facility				
Big Problem	2332	60.2	1114	47.8
Not a big problem	1545	39.8	991	64.1
Total	3876	100.0	2105	54.3

Spatial pattern childhood MCV1

In the global spatial autocorrelation analysis based on feature locations and attribute values, a clustering pattern of childhood MCV1 was observed across the study areas (Global Moran's I = 0.134, p-value < 0.0001) (Figure 6.3).

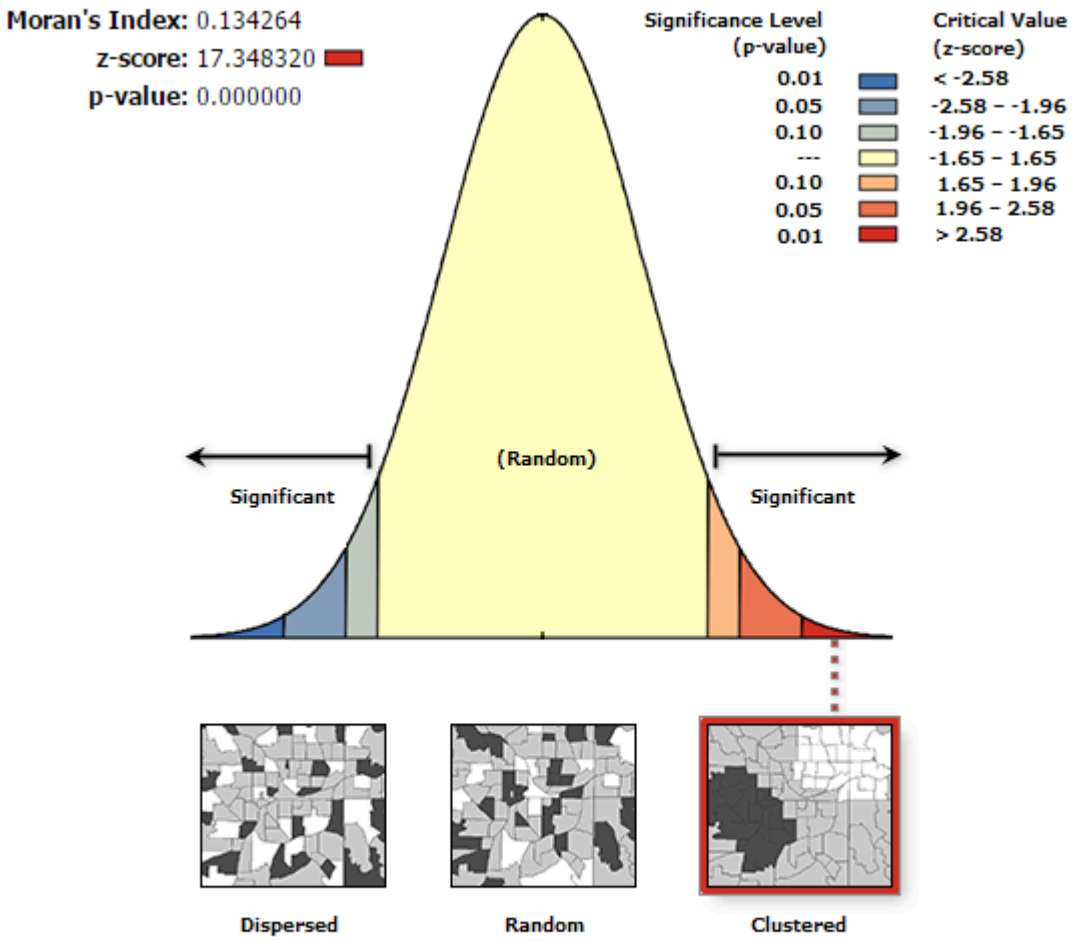


Figure 6.3. The global spatial autocorrelation based on feature locations and attribute values (MCV1) across the study areas in Ethiopia, 2016.

The SaTScan spatial analysis detected a total of seven statistically significant SaTScan clusters areas with low childhood MCV1 coverage, which means that the prevalence of MCV1 is lower inside the SaTScan circular window compared to outside the SaTScan window. The most likely primary SaTScan cluster of areas with low coverage of MCV1 was detected in Afar region (LLR = 33.79, p< 0.01), specifically in Zone 1, Zone 2 and Zone 4 administrative zones and the most likely secondary Spatial SaTScan cluster (LLR=26.76, p<0.01) in Eastern Ethiopia, specifically in Afder, Shabelle, Korah, Doolo, Nogob, Jarar, Fatan administrative zones of Ethio-Somali region.

The 3rd and 4th most likely clusters of areas with low rates of MCV1 were detected in Nuer & Agnuak Administrative zones of Gambella region (LLR=17.73, P<0.01) and Zone 2 administration of Afar region (LLR=14.54, P<0.01), respectively. In addition, the 5th most likely SaTScan cluster (LLR=14.49, P<0.01) was detected in Illuababur & Jimma Administrative zones of Oromia Region, while the 6th most likely SaTScan cluster (LLR=14.18, P<0.01) in Zone 3 and

Zone 4 of Afar region, North Shewa of Amhara region, West and East Hararge of Oromia Region and Siti zone of Somali region. The 7th most likely SaTScan cluster (LLR=12.27, P<0.01) was detected in East Hararge Zone of Oromia region and Fagan Zone of Ethio-Somali region (Table 6.4 and Figure 6.4).

Table 6.4. The most likely clusters from a purely spatial scan statistic (discrete Poisson model) of children received MCV1 in Ethiopia, 2016.

Clusters	Regional location (zones)	Radius (km)	LLR*	C.V	P-Value
Primary cluster	Afar Region (Zone 1, Zone 2 & Zone 3)	265.1	33.79	9.85	0.001
Secondary cluster	Ethio-Somali region (Afder, Shabelle, Korah, Doolo, Nogob, Jarar, Fatan)	339.6	26.76	9.83	0.001
3rd cluster	Gambella region (Nuer & Agnuak)	91.7	17.73	9.45	0.001
4th cluster	Afar region (Zone 2)	8.9	14.54	9.28	0.001
5th cluster	Oromia region (Illuababur & Jimma)	78.7	14.49	9.45	0.002
6th cluster	Afar region (Zone 3 & 4), Amhara region (North Shewa), Oromia Region (West and East Hararge) and Ethio-Somali Region (Siti)	139.1	14.18	9.40	0.001
7th cluster	Oromia region (East Hararge Zone), and Ethio-Somali region (Fatan Zone)	39.8	12.27	9.69	0.005

*Log likelihood ratio, **Standard Monte Carlo Critical Value for 0.05 significance level.

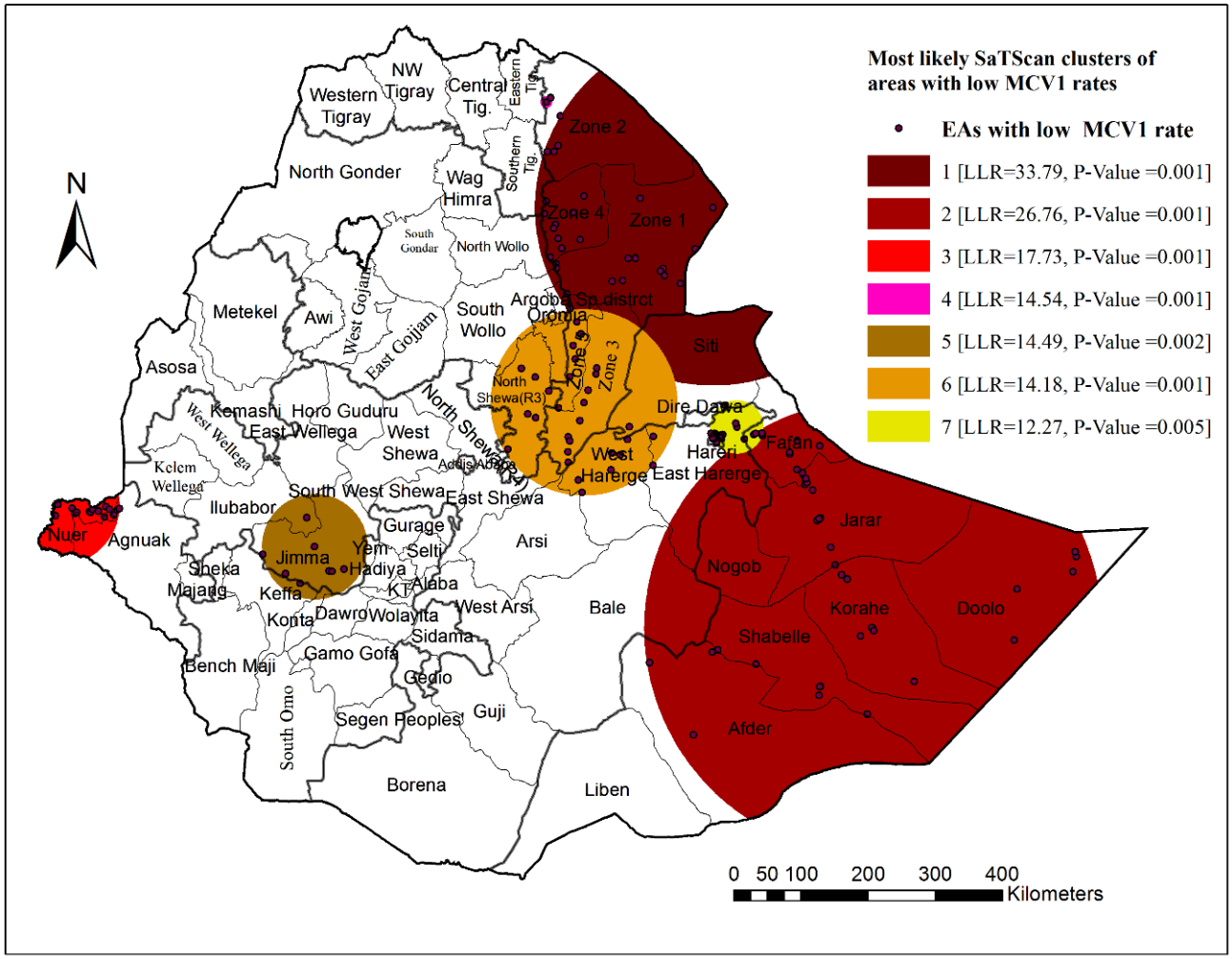


Figure 6.4. The spatial clustering of areas with low childhood MCV1 coverage in Ethiopia, 2016.

Factors associated with the childhood MCV1 in Ethiopia: Multilevel Analysis

The results of multilevel logistic regression models for individual and community level factors are displayed in Table 5. In the final full model (model IV) where all individual and community level factors are included, child's age; child's Penta 1 & Penta 3 and Vaccination status; maternal age; education status of mother; low household wealth index; region; and distance from health facility were factors that were significantly associated with childhood first dose measles vaccination.

Individual level factors: Children aged 24–35 months old were 1.53 times (AOR = 1.53; 95 % CI: 1.25-1.88) more likely to be vaccinated for MCV1 than children aged 12-23 months old. Children who have received first dose of pentavalent vaccine (AOR = 9.09; 95 % CI: 6.86-12.03) and the third dose of pentavalent vaccine (AOR = 7.12; 95 % CI: 5.51-9.18) were more likely to receive first dose of measles vaccine. Children whose mothers' age 20-34 years were 55 % (AOR = 0.45; 95 % CI: 0.27–0.76) less likely to receive MCV1 compared to those children whose mothers age 15-19 years. In addition, children with maternal age 35-45 years were 55 % (AOR = 0.45; 95 % CI: 0.25–0.82) less likely to receive MCV1 compared to those children whose mothers age 15-19 years. The odds of MVC1 vaccination were increased by 62 % (AOR = 1.62; 95 % CI: 1.03–2.55) in children with secondary & higher maternal education compared to children whose mothers had no education (Table 5).

Community level factors: The odds of childhood MCV1 vaccination in Tigray (AOR =0.32; 95 % CI: 0.13-0.81), Oromia (AOR 0.24; 95 % CI 0.10-0.58), Ethio-Somali (AOR =0.39; 95 % CI: 0.16-0.95), Gambella (AOR=0.39; 95%CI: 0.16-0.99) and Harari (AOR=0.27; 95%CI: 0.11-0.66) regions were lower compared to Addis Ababa. The odds of MVC1 vaccination were increased by 31 % (AOR = 1.31; 95 % CI: 1.12-1.61) in children living in areas where distance to health facility is not a big problem compared to children living in areas where distance to health facility is a big problem (Table 5).

Table 6.5. Multilevel logistic regression analysis of individual and community level factors with childhood MCV1 among 12–23-month old children in Ethiopia, 2016.

Variables	Model II AOR (95% CI)	Model III AOR (95% CI)	Model IV AOR (95% CI)
Individual level factors			
Child characteristics			
Child's age (months)			
12 - 23	1.00	-	1.00
24 - 35	1.53 (1.2-1.88)	-	1.53 (1.25-1.88)
Sex of child			
Male	1.00	-	1.00
Female	1.11 (0.92-1.35)	-	1.10 (0.91-1.34)
Birth Order			
1	1.00	-	1.00
2 - 3	1.01 (0.62-1.63)	-	1.00 (0.61-1.63)
4 - 5	1.03 (0.55-1.93)	-	1.03 (0.55-1.92)
6+	0.86 (0.41-1.81)	-	0.87 (0.42-1.83)
Child Wanted			
Wanted then	1.00	-	1.00
Wanted later	0.93 (0.70-1.23)	-	0.94 (0.71-1.25)
Wanted no more	1.52 (0.98-2.37)	-	1.58 (1.00-2.47)
DPT1-HepB1-Hib1 (Pentavalent first dose) vaccine			
Not vaccinated	1.00	-	1.00
Vaccinated	9.23 (6.98-12.22)	-	9.09 (6.86-12.03)
DPT3-HepB3-Hib3 (Pentavalent third dose) vaccine			
Not vaccinated	1.00	-	1.00
Vaccinated	8.15 (6.33-10.48)	-	7.12 (5.51-9.18)
Maternal Characteristics			
Maternal age (in Years)			
15 - 19	1.00	-	1.00
20 - 34	0.46 (0.28-0.78)	-	0.45 (0.27-0.76)
35 - 49	0.49 (0.27-0.90)	-	0.45 (0.25-0.82)
Religion			

Variables	Model II AOR (95% CI)	Model III AOR (95% CI)	Model IV AOR (95% CI)
Orthodox Christian	1.00	-	1.00
Muslim	0.71 (0.54-0.94)	-	0.95 (0.65-1.38)
Protestant	1.04 (0.74-1.46)	-	1.32 (0.85-2.03)
Others*	1.05 (0.54-2.03)	-	1.32 (0.66-2.62)
Maternal Education			
No education	1.00	-	1.00
Primary	1.00 (0.78-1.29)	-	0.99 (0.77-1.28)
Secondary & Higher	1.71 (1.10-2.66)	-	1.62 (1.03-2.55)
Mother's occupation			
Did not work	1.00	-	1.00
Non-Professional	1.07 (0.86-1.33)	-	1.04 (0.83-1.29)
Professional	0.97 (0.40-2.35)	-	0.93 (0.38-2.25)
Mather's relation to the household head			
Wife	1.00	-	1.00
Head	0.72 (0.37-1.42)	-	0.70 (0.35-1.38)
Daughter	1.39 (0.83-2.32)	-	1.35 (0.81-2.26)
Others	1.08 (0.61-1.91)	-	1.05 (0.58-1.87)
Sex of the household head			
Male	1.00	-	1.00
Female	1.54 (0.83-2.87)	-	1.57 (0.84-2.93)
Number of living children	1.00		
1	1.00	-	1.00
2 - 3	1.21 (0.74-1.97)	-	1.21 (0.74-1.98)
4 - 5	1.26 (0.66-2.40)	-	1.29 (0.68-2.47)
6+	1.16 (0.55-2.48)	-	1.18 (0.55-2.53)
Wealth index			
Poorest	1.00	-	1.00
Poorer	1.12 (0.83-1.52)	-	1.16 (0.85-1.59)
Middle	0.94 (0.68-1.30)	-	0.99 (0.71-1.40)
Richer	1.06 (0.75-1.50)	-	1.18 (0.83-1.70)
Richest	1.35 (0.93-1.98)	-	1.34 (0.82- 2.20)
Exposure to mass media			
No regular exposure	1.00	-	1.00
Regular exposure	1.33 (0.98-1.80)	-	1.32 (0.97-1.79)
Community level factors			
Region			
Addis Ababa	-	1.00	1.00
Tigray	-	0.72 (0.30-1.73)	0.65 (0.27-1.56)
Afar	-	0.04 (0.02-0.09)	0.32 (0.13-0.81)

Variables	Model II AOR (95% CI)	Model III AOR (95% CI)	Model IV AOR (95% CI)
Amhara	-	0.25 (0.10-0.58)	0.46 (0.19-1.11)
Oromia	-	0.12 (0.05-0.27)	0.24 (0.10-0.58)
Somali	-	0.05 (0.02-0.13)	0.39 (0.16-0.95)
Benishangul	-	0.60 (0.25-1.47)	0.63 (0.25-1.56)
SNNPR	-	0.32 (0.14-0.75)	0.42 (0.17-1.01)
Gambella	-	0.15 (0.06-0.36)	0.39 (0.16-0.99)
Harari	-	0.11 (0.04-0.25)	0.27 (0.11-0.66)
Dire Dawa	-	0.63 (0.25-1.60)	0.69 (0.27-1.79)
Residence			
Urban	-	1.00	1.00
Rural	-	0.30 (0.21-0.44)	1.03 (0.64-1.65)
Distance to Health facility			
Big problem	-	1.00	1.00
Not a big problem	-	1.42 (1.17-1.72)	1.31 (1.12-1.61)

*Catholic, traditional and other unclassified
**Daughter-in-low, sister, other relative, no relation

Measures of variation (Random-effects)

Table 6 depicted multilevel logistic regression analysis results for odds of childhood MCV1 Vaccination. The null model (model I) revealed statistically significant variation in MCV1 coverages across communities [$\tau = 2.68$, $p< 0.001$] and the ICC showed that 45% of the variability in the odds of receiving MCV1 was due to community-level factors.

After the model was adjusted for individual-level characteristics, the variation in the odds of a child receiving MCV1 remained statistically significant [$\tau=0.56$, $p<0.001$] across the communities and about 15% of the variance in childhood MCV1 was attributed community-level factors. Similarly, about 79% of the variation in the odds of children who received MCV1 between communities was attributed to the individual factors adjusted in the model (Model 2).

After adjusting the community-level characteristics, model 3 found a slightly increased variance of a child being vaccinated for MCV1 [$\tau = 1.12$, $p< 0.001$] across the communities, as compared to the variance reported in model 2. Model 3 further identified that about 25% of the variability in the odds of a child being MCV1 vaccinated was due to community-level characteristics (ICC = 25.3%). Similarly, about 58% of the variability in the odds of children receiving MCV1 between communities could be explained by the community-level characteristics included in Model 3 (PCV = 58.3%).

Finally, after the statistical adjustment of the individual and community-level characteristics simultaneously, Model 4 depicted significant variability across the communities with regards to the odds of a child receiving MCV1[$\tau = 0.48$; $p<0.001$]. About 13% of the variability in the odds of a child being MCV1 vaccinated was due to the community-level factors (ICC = 12.7%). Similarly, the PCV revealed that about 82% of the variance in the odds of MCV1 vaccination (PCV = 82.1%) across communities was due to the simultaneous effect of both individual and community-level characteristics adjusted in Model 4.

Results from the MOR also confirmed evidence of community contextual phenomena shaping the odds of being a ‘’MCV1 Vaccinated child’‘. The high MOR (4.73) in Model 1 between children with a higher and lower propensity of being’‘vaccinated’’ in a community suggests that the community heterogeneity is substantial. Including all factors reduced the unexplained heterogeneity between communities to an MOR of 2.95.

Table 6.6. Measures of variation in childhood MCV1 in Ethiopia, 2016.

Measures of variation	Model 1[a]	Model 2[b]	Model 3[c]	Model 4[d]
Community level				
Variance (SE)	2.676(0.091)*	0.556(0.085)*	1.117 (0.074)*	0.479(0.086)*
PCV (%)	Reference	79.2	58.3	82.1
ICC (%)	44.9	14.5	25.3	12.7
MOR[e]	4.73	2.03	2.73	1.93
Model fit statistics				
DIC (-2log likelihood)	4,462	2,984	4,142	2,953
AIC	4,466	3,050	4,170	3,043

*P-value <0.001

SE: Standard Error; PCV: Proportional Change in Variance; ICC: Intraclass Correlation Coefficient; MOR: Median Odds Ratio; DIC: Deviance Information Criterion, AIC: Akaike’s Information Criterion

[a]Model 1 is the empty model, a baseline model without any explanatory variable

[b]Model 2 is adjusted for individual-level factors

[c]Model 3 is adjusted for community-level factors

[d]Model 4 is final model adjusted for both individual and community-level factors

[e]Increased risk (in median) that one would have if moving to a neighborhood/cluster with a higher risk

6.5 Discussion

In this study, a total of 3,722 children aged 12-35 months nested within 611 clusters were included from the 2016 EDHS data. The prevalence of childhood MCV1 was found to be 54.3 percent (95% CI = 50.3 - 58.3) which was low compared to the recent DHS reports of most low and middle income countries such as Egypt, 96%[49], Kenya, 87%[50], Rwanda, 95%[51], Ghana, 89%[52], Zimbabwe, 82%[53] and Uganda, 80%[54]. It was also low as compared to a local survey finding in the selected zones of Ethiopia by JSI-l10k , 80%[33], the national report in 2015, 87%[55], the 2016 national target for childhood MCV1, 91%[55] and the recommended herd immunity threshold, 95%[9, 16]. This childhood MCV1 coverage below the recommended herd immunity threshold (95%) may indicate that not also the country but also the regions with high levels of childhood MCV1 coverage may still be at considerable risk for measles outbreaks.

While vaccine exemptions are gaining attention at all levels of public health, there is an urgent need to identify regions or local areas exhibiting high exemption rates and concomitant low immunization coverage, and to intervene in these areas. Spatial heterogeneity in vaccination coverage has an immediate health impact on communities with the lower immunization rates at all spatial scales[56].

In the global spatial autocorrelation analysis, a clustering pattern of childhood MCV1 across the study areas was observed (Global Moran's I = 0.134, p-value<0.0001). This indicates that approximately the same coverages of childhood MCV1 were aggregated in specific areas. Hence, the SaTScan spatial analysis detected seven statistically significant most likely SaTScan clusters of areas with low MCV1 coverage. The most likely primary SaTScan cluster of areas with low MCV1 coverage was detected in Afar region; specifically in Zone 1, Zone 2 and Zone 4 administrative zones (LLR = 33.79, p< 0.01), and the most likely secondary SaTScan cluster in Ethio-Somali region, specifically in Afder, Shabelle, Korah, Doolo, Nogob, Jarar and Fatan administrative zones (LLR=26.76, p<0.01). The 3rd and 4th most likely clusters of areas with low MCV1 coverage were detected in Gambella region; specifically in Nuer and Agnuak Administrative zones (LLR=17.73, P<0.01) and Afar region, particularly in Zone 2 administrative zone (LLR=14.54, P<0.01), respectively. In addition, the 5th most likely SaTScan cluster (LLR=14.49, P<0.01) was detected in Illuababur & Jimma Administrative zones of Oromia Region, while the 6th most likely SaTScan cluster (LLR=14.18, P<0.01) in zone 3 and zone 4 administrative zones of Afar region, North Shewa zone of Amhara region, West and East Hararge zones of Oromia region and Siti zone of Ethio-Somali region. The last (7th) most likely SaTScan cluster (LLR=12.27, P<0.01) was detected in East Hararge zone of Oromia region and Fatan zone of Ethio-Somali region. This local clustering of low childhood MCV1 coverages indicates that children who lived in the above mentioned geographical locations had a low probability of receiving MCV1 compared with those who lived outside the SaTScan clusters. This may be due to differences in health service accessibility and community background characteristics. This considerable spatial heterogeneities and geographic clustering of low MCV1 coverage areas found in our analysis suggest that regions with high levels of sub-national MCV1 coverage may still be at considerable risk for measles outbreaks[23].

The multilevel regression analysis in this study confirmed that the variation in the childhood MCV1 in Ethiopia was attributed to both individual and community level factors. This finding is supported by evidences from multilevel analyses done for the determinants of childhood full immunization in Ethiopia[32] and Democratic Republic of Congo[57]. In the final model (model IV), individual and community-level factors accounted for about 82.1 % of the variation observed for childhood MCV1.

We found that children aged 24-35 months were more likely to receive MCV1 compared to children aged 12-23 months (AOR=1.53; 95 % CI: 1.25-1.88). This could be due to late initiation of childhood vaccination and Vaccine hesitancy for early ages. This indicates delaying in childhood vaccination which extends the period of vulnerability of children to

vaccine preventable diseases[34]. We also found that children who have received the first dose of pentavalent vaccine (AOR = 9.09; 95 % CI: 6.86-12.03) and third dose of pentavalent vaccine (AOR = 7.12; 95 % CI: 5.51-9.18) were more likely to receive MCV1. This may be due to differences in childhood vaccination service availability and utilization among communities. It could also be justified as; women whose children lacked the first and third doses of pentavalent vaccination may miss the opportunities for information on the importance of childhood measles vaccination. Even though, there were no variables included in this analysis to directly measure the availability of vaccination services in the communities, pentavalent first dose vaccination coverage is a good proxy indicator for the availability of access to and initial use of childhood vaccination services[12]. Hence, a high proportion of pentavalent first dose coverage may indicate the availability of childhood vaccination services in the communities. In addition, the third dose pentavalent vaccination coverage indicates the continuity of use by parents or care takers, client satisfaction with services, and capability of the system to deliver a series of vaccinations[12]. Hence, a high proportion of pentavalent third dose vaccination may indicate better childhood vaccination service utilization and the health system performance in relation to vaccination services.

Children from mothers with secondary or higher education had higher odds of being vaccinated for MCV1 (AOR = 1.62; 95 % CI: 1.03–2.55). Similar findings were reported in studies on childhood immunization from Ethiopia [32], Democratic Republic of Congo[57], Nigeria [58, 59], Kenya[60] and in India[61]. Education was identified as a strong predictor of childhood immunization in other several studies [62-65]. Educated mothers are generally more aware of the importance of available health and immunization services, have better communication skills, and tend to better utilize available services[66, 67].

In regard to community characteristics, geographic region of the study participants was found to be statistically significant explanatory variable of childhood MCV1. Children from Tigray, Oromia, Ethio-Somali, Gambella and Harari regions were less likely to receive MCV1 than children from Addis Ababa. This may be due to the differences in religion, culture, population size, geographic nature and level of development in the nine regional states and two city administrative councils found in Ethiopia. This could be linked with differences in vaccine supply, availability of health care providers and accessibility of health facilities. Hence, these regional divergences tend to affect the range of child MCV1 vaccination across the country. Vaccine hesitancy or exemption can lead to relatively large outbreaks even within regions with high vaccination coverage [17, 68-70]. Subpopulations with low vaccination coverage across country are a particular concern for measles elimination because these transient populations are often not covered by national immunization programs

Unlike what has been documented elsewhere[71-74], this study did not found significant association between childhood MCV1 and maternal place of residence. This may be due to the fact that the small sample size they used and the time in the study was conducted by itself may be a factor. Nevertheless, the finding is consistent with the findings of studies conducted in central Ethiopia by Etana et al. (2012) and Nigeria by Aremu et al. (2010), in which both studies found non-significant association between likelihood of childhood MCV1 and place of residence[62, 75] and this may need further investigation.

Finally, this study revealed that children who lived in areas where distance from health facility is not a big problem were more likely to receive childhood MCV1 which is consistent with a study finding in sub-Saharan Africa [35].

The national and subnational representativeness of the data with a large sample size, as well as a community-based study with a high response rate are strengths of this study to give high statistical power. Another important strength is the use of multilevel analysis to identify other factors beyond individual-level factors that could not be addressed by using ordinary logistic regression analysis. However, this study is not free from limitation and should be mentioned. Due to the retrospective nature of the data, and the maternal verbal reports for childhood vaccination, recall bias might have been introduced.

6.6 Conclusion

A clustered pattern of areas with low childhood MCV1 coverage was observed in Ethiopia. Statistically significant local clusters of areas with low childhood MCV1 were detected in Ethio-Somali, Afar, Gambella, and Oromia regions of the country. Both the individual level characteristics (child's age, first and third doses of pentavalent vaccination, maternal age and education) and community level characteristics (geographic region and distance from health facility) were statistically significant predictors of childhood MCV1

6.7 Recommendation

The considerable spatial heterogeneities and geographic clustering of low MCV1 vaccine coverage areas found in our analysis suggest that the country and regions with high levels of sub-national coverage may still be at considerable risk for measles outbreaks. Hence, it is good if the federal ministry of health and other concerned child health programmers give priority of the areas with low MCV1 coverage identified in this study. It is also better to consider the individual and community level determinant factors identified in this study.

References

1. The Ethiopian Health and Nutrition Research Institute (EHNRI): Guideline on measles surviellance and outbreak management, 3rd Edition. Addis Ababa, Ethiopia, January 2012.
2. Perry RT, et al., Progress toward regional measles elimination - Worldwide, 2000-2014. MMWR. Morb. Mortal. Wkly. Rep. United States; 2015;64:1246–51.
3. World Health Organization (2016). Immunization Coverage: Fact sheet No. 286. 2016 . Available at: www.who.int/mediacentre/factsheets/fs286/en/ (accessed on 24th January 2018).
4. WHO: Summary of the WHO position on Measles Vaccine- April 2017. Available at: www.who.int/entity/immunization/policy/position_papers/WHO_PP_measles_vaccine_summary_2017.pdf. (Accessed on 26th February 2018).
5. World Health Organization (2012): Global measles and rubella strategic plan : 2012-2020.
6. Strategic Advisory Group of Experts on Immunization (SAGE): 2016 Midterm Review Report of Global Vaccine Action Plan.
7. WHO and UNICEF Estimates for National Immunization Coverage (WUENIC), 2010-2015.
8. WHO (2017): Summary of WHO Position Papers - Recommendations for Routine Immunization. Available at: www.who.int/immunization/policy/Immunization_routine_table1.pdf?ua=1 (Accessed on 26th February 2018).
9. WHO | Measles Surveillance Data. (2016). Available at: www.who.int/immunization/monitoring_surveillance/burden/vpd/surveillance_type/active/measles_monthlydata/en/ (Accessed on 26th February 2018).
10. Lidetu S, Okubazgi G (1993) Childhood diseases and immunization. The ecology of health and disease in Ethiopia, Westview Press, USA.
11. Ethiopia National Expanded Programme on Immunization (2015). comprehensive multiyear plan, 2016-2020; Federal Ministry of Health, Addis Ababa.
12. Federal Ministry of Health. Revised Health Management Information System Indicator Reference Guide; Technical Standards: Area 1; Addis Ababa Ethiopia, Augest 2017.
13. Akalu HB (2015) Review on Measles Situation in Ethiopia; Past and Present. J Trop Dis 4: 193. doi:10.4172/2329-891X.1000193.
14. WHO, Department of Evidence, Information and Research estimates for child causes of death, 2000-2015 (updated 5 February 2016).
15. World Health Organization (2017): Global Health Observatory (GHO) data>Measles-containingvaccine first-dose (MCV1) immunization coverage among 1-year-olds. Available at: www.who.int/gho/immunization/measles/en/ [Accessed 26th February 2018].
16. WHO|WER Index 2016: Weekly Epidemiological Record, 11 November 2016, vol. 91, 45 (pp. 525–536). Available at: www.who.int/wer/2016/wer9145/en/.
17. Zipprich J, et al., Measles outbreak–California, December 2014-February 2015. MMWR. Morb.Mortal. Wkly. Rep. United States. 2015. 64: p. 153–154.
18. Suijkerbuijk A, et al., Economic costs of measles outbreak in the Netherlands, 2013-2014. Emerg Infect Dis United States. 2015. 21: p. 2067–2069.
19. WHO | Reaching Every District (RED) approach: A way to improve immunization performance. 2011. Available at: www.who.int/bulletin/volumes/86/3/07-042127/en/ (accessed on 26th February 2018).
20. CDC: Principles of Epidemiology in Public Health Practice , An Introduction to Applied Epidemiology and Biostatistics. Third Edition, 2012.
21. Kulldorff M.(2018). SaTScanTM User Guide for Version 9.6. March, 2018. www.satscan.org/.
22. Brownwright, T.K., Z.M. Dodson, and W.G. van Panhuis, Spatial clustering of measles vaccination coverage among children in sub-Saharan Africa. BMC Public Health, 2017. 17(1): p. 957.
23. Takahashi, S., et al., The geography of measles vaccination in the African Great Lakes region. Nat Commun, 2017. 8: p. 15585.
24. Parker Fiebelkorn A, Redd SB, Gallagher K, Rota PA, Rota J, Bellini W, et al. Measles in the United States during the postelimination era. J Infect Dis 2010; 202:1520-8; PMID:20929352; Available at: dx.doi.org/10.1086/656914.

25. May T, Silverman RD. 'Clustering of exemptions' as a collective action threat to herd immunity. Vaccine 2003; 21:1048-51; PMID:12559778; Available at: dx.doi.org/10.1016/S0264-410X(02)00627-8.

26. Ernst, K.C. and E.T. Jacobs, Implications of philosophical and personal belief exemptions on reemergence of vaccine-preventable disease: the role of spatial clustering in under-vaccination. Hum Vaccin Immunother, 2012. 8(6): p. 838-41.

27. Berhane Y, Bekele A, Tesfaye F (2000) Immunization (EPI) in Ethiopia: acceptance, coverage, and sustainability. Ethiop Med J 1:1-60.

28. Central Statistical Agency (CSA) [Ethiopia] and ICF. 2016. Ethiopia Demographic and Health Survey 2016. Addis Ababa, Ethiopia, and Rockville, Maryland, USA: CSA and ICF.

29. Central Statistical Agency [Ethiopia] and ICF International. 2012. Ethiopia Demographic and Health Survey 2011. Addis Ababa, Ethiopia and Calverton, Maryland, USA: Central Statistical Agency and ICF International.

30. Central Statistical Agency [Ethiopia] and ORC Macro. 2006. Ethiopia Demographic and Health Survey 2005. Addis Ababa, Ethiopia and Calverton, Maryland, USA: Central Statistical Agency and ORC Macro.

31. Central Statistical Authority [Ethiopia] and ORC Macro. 2001. Ethiopia Demographic and Health Survey 2000. Addis Ababa, Ethiopia and Calverton, Maryland, USA: Central Statistical Authority and ORC Macro.

32. Abadura, S.A., et al., Individual and community level determinants of childhood full immunization in Ethiopia: a multilevel analysis. BMC Public Health, 2015. **15**: p. 972.

33. JSI-L10K: Extended Program on Immunization (EPI) coverage in selected Ethiopian zones: A baseline survey for L10K's Routine Immunization Improvement Initiative. Addis Ababa, Ethiopia. 2015.

34. Simone Périnet, Marilou Kiely, Gaston De Serres & Nicolas L. Gilbert (2018): Delayed measles vaccination of toddlers in Canada: Associated socio-demographic factors and parental knowledge, attitudes and beliefs, Human Vaccines & Immunotherapeutics, DOI: 10.1080/21645515.2017.1412899.

35. Brownwright, T.K., Z.M. Dodson, and W.G. van Panhuis, Spatial clustering of measles vaccination coverage among children in sub-Saharan Africa. BMC Public Health, 2017. **17**(1): p. 957.

36. Adebiyi F: Determinants of Full Child Immunization Among 12-23 Months Old In Nigeria. Unpublished Article, 2013: 24-39.

37. Federal Minstry of Health (FMOH): Ethiopian Growth & Transformation Plan, 2010 - 2015.

38. Ethiopian Public Health Institute (EPHI) : Ethiopia Services Availability and Readiness Assessment 2016: Summary Report. Addis Ababa, Ethiopia, January, 2017.

39. Alelign, A., Z. Tekeste, and B. Petros, Prevalence of malaria in Woreta town, Amhara region, Northwest Ethiopia over eight years. BMC Public Health, 2018. **18**(1): p. 990.

40. K., A., W. D., and S. H., Spatial distribution of under immunization among children 12-23 months old in butajira, southern ethiopia, 2016. Unpublished Article.

41. Federal Ministery of Health (FMOH): Immunization in Practice National Guideline for Health Workers, Modules for health workers, 2014 Update.

42. Anselin L, Getis A. Spatial statistical analysis and geographic information systems. Ann Reg Sci. 1992;26(1):19–33.

43. Kulldorff M. and Information Management Services, Inc. SaTScan™ v9.6: Software for the spatial and space-time scan statistics. www.satscan.org, 2018.

44. Bernoulli, Discrete Poisson and Continuous Poisson Models: Kulldorff M. A spatial scan statistic. Communications in Statistics: Theory and Methods, 26:1481-1496, 1997. www.satscan.org/papers/k-cstm1997.pdf.

45. Akaike, H. 1974. A new look at the statistical model identification. IEEE transactions on Automatic Control 19: 716-723.

46. Merlo, J., et al., A brief conceptual tutorial of multilevel analysis in social epidemiology: using measures of clustering in multilevel logistic regression to investigate contextual phenomena. J Epidemiol Community Health, 2006. **60**(4): p. 290-7.

47. Austin PC, Stryhn H, Leckie G, Merlo J. Measures of clustering and heterogeneity in multilevel Poisson regression analyses of rates/count data. Statistics in Medicine. 2018;37:572–589. doi.org/10.1002/sim.7532.

48. Halonen, J.I., et al., Quantifying neighbourhood socioeconomic effects in clustering of behaviour-related risk factors: a multilevel analysis. PLoS One, 2012. **7**(3): p. e32937.

49. Ministry of Health and Population/Egypt, El-Zanaty and Associates/Egypt, and ICF International. 2015. Egypt Demographic and Health Survey 2014. Cairo, Egypt: Ministry of Health and Population and ICF International. .

50. Kenya National Bureau of Statistics, Ministry of Health/Kenya, National AIDS Control Council/Kenya, Kenya Medical Research Institute, National Council for Population and Development/Kenya, and ICF International. 2015. Kenya Demographic and Health Survey 2014. Rockville, MD, USA: Kenya National Bureau of Statistics, Ministry of Health/Kenya, National AIDS Control Council/Kenya, Kenya Medical Research Institute, National Council for Population and Development/Kenya, and ICF International. .

51. National Institute of Statistics of Rwanda (NISR) [Rwanda], Ministry of Health (MOH) [Rwanda], and ICF International. 2015. Rwanda Demographic and Health Survey 2014-15. Rockville, Maryland, USA: NISR, MOH, and ICF International.

52. Ghana Statistical Service - GSS, Ghana Health Service - GHS, and ICF International. 2015. Ghana Demographic and Health Survey 2014. Rockville, Maryland, USA: GSS, GHS, and ICF International. .

53. Zimbabwe National Statistics Agency and ICF International. 2016. Zimbabwe Demographic and Health Survey 2015: Final Report. Rockville, Maryland, USA: Zimbabwe National Statistics Agency (ZIMSTAT) and ICF International. .

54. Uganda Bureau of Statistics - UBOS and ICF. 2018. Uganda Demographic and Health Survey 2016. Kampala, Uganda and Rockville, Maryland, USA: UBOS and ICF. .

55. Federal Democratic Republic of Ethiopia, Ministry of Health: Health Sector Transformation Plan-HSTP, July 2015-June 2020 (EFY 2008-2012), May 2015.

56. Kacey Ernst & Elizabeth T. Jacobs (2012) Implications of philosophical and personal belief exemptions on re-emergence of vaccine-preventable disease: The role of spatial clustering in under-vaccination, Human Vaccines & Immunotherapeutics, 8:6, 838-841, DOI: 10.4161/hv.19743.

57. Acharya, P., et al., Individual- and community-level determinants of child immunization in the Democratic Republic of Congo: A multilevel analysis. PLoS One, 2018. **13**(8): p. e0202742.

58. Olorunsaiye, C.Z. and H. Degge, Variations in the Uptake of Routine Immunization in Nigeria: Examining Determinants of Inequitable Access. Global Health Communication, 2016. **2**(1): p. 19-29.

59. Fatiregun, A.A. and A.O. Okoro, Maternal determinants of complete child immunization among children aged 12-23 months in a southern district of Nigeria. Vaccine, 2012. **30**(4): p. 730-6.

60. Maina, L.C., S. Karanja, and J. Kombich, Immunization coverage and its determinants among children aged 12 - 23 months in a peri-urban area of Kenya. Pan Afr Med J, 2013. **14**: p. 3.

61. Phukan, R.K., M.P. Barman, and J. Mahanta, Factors associated with immunization coverage of children in Assam, India: over the first year of life. J Trop Pediatr, 2009. **55**(4): p. 249-52.

62. Etana, B. and W. Deressa, Factors associated with complete immunization coverage in children aged 12-23 months in Ambo Woreda, Central Ethiopia. BMC Public Health, 2012. **12**: p. 566.

63. Kumar, D., A. Aggarwal, and S. Gomber, Immunization status of children admitted to a tertiary-care hospital of north India: reasons for partial immunization or non-immunization. J Health Popul Nutr, 2010. **28**(3): p. 300-4.

64. Shrivastwa, N., et al., Predictors of Vaccination in India for Children Aged 12-36 Months. Am J Prev Med, 2015. **49**(6 Suppl 4): p. S435-44.

65. Wiysonge, C.S., et al., Individual and contextual factors associated with low childhood immunisation coverage in sub-Saharan Africa: a multilevel analysis. PLoS One, 2012. **7**(5): p. e37905.

66. Govindasamy P, Ramesh B. Maternal education and the utilization of maternal and child health services in India. 1997.

67. Vikram K, Vanneman R, Desai S. Linkages between maternal education and childhood immunization in India. Social science & medicine. 2012; 75(2):331–9.

68. Centers for Disease Control and Prevention (CDC). Notes from the field: Measles outbreak among members of a religious community—Brooklyn, New York, March–June 2013. Morb. Mortal. Wkly. Rep. 2013, 62, 752–753.

69. Phadke, V.K.; Bednarczyk, R.A.; Salmon, D.A.; Omer, S.B. Association between vaccine refusal and vaccine-preventable diseases in the United States: A review of measles and pertussis. JAMA 2016, 315, 1149–1158.

70. Gastanaduy, P.A.; Budd, J.; Fisher, N.; Redd, S.B.; Fletcher, J.; Miller, J.; McFadden, D.J., 3rd; Rota, J.; Rota, P.A.; Hickman, C.; et al. A measles outbreak in an underimmunized amish community in Ohio. N. Engl. J. Med. 2016, 375, 1343–1354.

71. Ibnouf A, Vanden B, Maarse J. Factors influencing immunization coverage among children under five years of age in Khartoum State, Sudan: University of Maastricht. S Afr Fam Pract. 2007;49(8):14.

72. Roy, SG. “Risk F actors for Childhood Immunization Incompletion in Ethiopia” (2010). Public Health Theses. Georgia State University, Paper 90. Available from: digitalarchive.gsu.edu/iph_theses/90.

73. Kidane T, Yigzaw A, Sahilemariam Y, Bulto T, Mengistu H, Malay T, et al. National EPI coverage survey report in Ethiopia, 2006. Ethiop J Health Dev. 2008;22(2):148–57.

74. Mathew JL. Inequality in childhood immunization in India: a systematic review. Indian Pediatr. 2012;49:203–23.

75. Aremu O, Lawoko S, Dalal K. Childhood vitamin A capsule supplementation coverage in Nigeria: a multilevel analysis of geographic and socioeconomic inequities. The Scientific World Journal: TSW Child Health. 2010.

CHAPTER 7

Scientific Manuscripts

Geographical variation in childhood measles vaccination and associated factors in Ethiopia: A spatial and multilevel analysis[6]

6. Submitted to BMC Public Health↩

PUBLISHED VERSION

Geremew TT, Gezie LD, Abejie AN. Geographical variation and associated factors of childhood measles vaccination in Ethiopia: a spatial and multilevel analysis. *BMC Public Health*. 2019;19:1194.

https://doi.org/10.1186/s12889-019-7529-z

Tesfahun Taddege GEREMEW, Lemma Derseh

Abstract

Background: In Ethiopia, despite considerable improvement of measles vaccination, measles outbreaks have been occurring in most parts of the country. Understanding the neighborhood variation in measles vaccination has been increasingly recognized, and is crucial for evidence-based decision-making. However, the spatial pattern of first dose of measles-containing vaccine (MCV1) and its predictors are poorly understood. Hence, this study aimed to explore the spatial pattern and associated factors of childhood MCV1 coverage

Methods: An in-depth analysis the 2016 Ethiopia demographic and health survey data was conducted and a total of 3,722 children nested in 611 enumeration areas were included in the analysis. Poisson-based purely spatial scan statistics was employed to explore the spatial patterns of childhood MCV1. Multilevel logistic regression models were fitted to identify factors associated with childhood MCV1.

Result: Spatial hetrogeniety of childhood MCV1 was observed (Global Moran's I=0.134, p-value < 0.0001) and seven statistically significant SaTScan clusters of areas with low MCV1 coverage were detected. The most likely primary SaTScan cluster was detected in Afar region, secondary cluster in Ethio-Somali region, and tertiary cluster in Gambella region. In the final model of the multilevel analysis, individual and community level factors accounted for 82% of the variance in the odds of MCV1 vaccination. Older child age (AOR=1.53; 95%CI: 1.25-1.88), pentavalent vaccination first dose (AOR=9.09; 95%CI: 6.86-12.03) and third dose (AOR=7.12; 95%CI: 5.51-9.18, secondary and above maternal education (AOR=1.62; 95%CI: 1.03–2.55) and media exposure were the factors that increased the odds of MCV1 vaccination at the individual level. Children with older maternal age had lower odds of receiving MCV1. Living in Tigray, Oromia, Ethio-Somali, Gambella and Harari regions were factors associated with lower odds of MCV1 from the community-level factors. Children far from health facilities had higher odds of receiving MCV1 (AOR=1.31, 95%CI=1.12-1.61).

Conclusion: A clustered pattern of areas with low childhood MCV1 coverage was observed in Ethiopia. Both individual and community level factors were significant predictors of childhood MCV1. Hence, it is good to give priority for the areas with low childhood MCV1 coverage, and to consider the identified factors for vaccination interventions.

Keywords: Children, MCV1, Spatial, Multilevel, Ethiopia

7.1 Background

Measles is an acute, highly contagious viral disease caused by the measles virus and transmitted by respiratory droplets[1]**.** It is one of the most infectious human diseases and can cause serious illness, lifelong complications and death, accounting for 114,900 deaths globally in 2014 of which 73,914 (63%) occurred in Africa[2, 3]. Prior to the availability of measles vaccine, measles infected over 90% of children before they reached 15 years of age[4, 5]. World Health Organization (WHO) has targeted the measles virus for global elimination to reduce annual measles incidence rates(IRs) to less than five cases per million, requiring >90% coverage of at least one dose of Measles-Containing Vaccine (MCV1), recommended at age 9 -12 months, by the end of 2015 and >95% coverage by 2020 in all countries[5]. Evidences showed that improvement in vaccination coverage has decreased measles deaths from over half a million globally in 2000 to 114,900 in 2014[2, 6, 7]. Globally, in 2015, MCV1 coverage had reached 85% and an estimated 134 200 measles deaths had occurred, accounting a 79% decline since 2000[4]. The 2015 vaccination goal was not met and IRs remained relatively unchanged between 2013 and 2014[2].

Ethiopia contributed 9% to global measles mortality[6, 8] and among the global estimated 20.8 million infants who did not receive MCV1 through routine immunization services in 2015, approximately 0.7 million (3.4%) were in Ethiopia[9].

Different reports from Ethiopia stated the improvement childhood immunization coverage including MCV1 through the combine effect of Reaching Every District (RED) approach, Health extension program and implementation of Enhanced Routine Immunization Activities (ERIAs)[7, 10-12]. Measles immunization coverage should have a strong inverse correlation with the prevalence of the measles disease, especially among under-five children-i.e. increasing coverage should be accompanied by decreasing cases of the disease[13]. However, despite considerable improvement of childhood MCV1 in Ethiopia, measles outbreaks continue to occur in most parts of the country. The occurrence of outbreaks is useful to identify gaps in immunization programme performance that may not be evident through monitoring vaccination coverage. One of the weaknesses of immunization programme is heterogeneity of coverage with pockets of missed children[5].

As the MCV1 coverage increased from 59% in 2005 to 84% in 2014[4, 7, 8, 11, 14], the incidence rate of confirmed measles cases per 100,000 population increased from 0.6 in 2005 to 11.2 in 2014[11]. Such contradiction may be resulted from heterogeneity of MCV1 coverage with pockets of missed children or accumulation of unvaccinated children in highly populated areas and community resistance[5, 11]. Spatial heterogeneity of vaccination coverage can delay disease elimination even in countries with high average nationwide vaccination coverages[15, 16]. Understanding the neighborhood variation in measles vaccination has been increasingly recognized and crucial for evidence-based decision-making. However, the spatial pattern of childhood MCV1 and its predictors are poorly understood in Ethiopia. Hence, this study aimed to explore the spatial pattern of childhood MCV1 and its predictors in Ethiopia.

7.2 Methods

Study design and setting

An in-depth analysis of the 2016 Ethiopia demographic and health survey (EDHS) data was conducted. EDHS is a national and subnational representative household survey which is conducted every five year. Ethiopia is the oldest, independent and second populous country, located in the horn of Africa, between 3 and 15 degrees north latitude and 33 and 48 degrees east longitude. Administratively, Ethiopia is divided into nine regional states and two city administrations and then into zone administrations. The zone administrations are sub-divided into districts and the district in-further sub-divided into kebeles (the smallest administrative units). During the 2007 census, each kebele was sub-divided into census enumeration areas (EAs), which were used as a primary sampling unit in the demographic and health survey.

The health system of Ethiopia, which focuses on a preventive health care, is organized into three health tire system: Primary health care unit (PHCU), general hospital and specialized hospital. Routine child immunization is primarily provided at PHCUs through static and outreach sites[17] and the availability of routine childhood vaccination services in Ethiopia is 80 percent[18].

Study population and eligibility

Children age 12-35 months in the selected enumeration areas (EAs) of Ethiopia were the study population. All children age 12-35 months were included and children whose geographical locations were not available at the global positioning system (GPS) were excluded from this study.

Sampling technique and sample size

The 2016 EDHS used a stratified, two-stage cluster sampling design using the enumeration areas (EAs) of the 2007 Ethiopian population and housing census (PHC) as the primary sampling unit and households as the secondary sampling unit. In the first stage, a total of 645 EAs (202 in urban areas and 443 in rural areas) were selected with probability proportional to EAs size (PPS) from the complete list of 84,915 EAs created for the 2007 PHC sampling frame. In the second stage of selection, a fixed number of 28 households per cluster were selected with an equal probability systematic selection from the newly created household listing. Finally, a total of 16,650 households were successfully interviewed, yielding a response rate of 98%. A total of 3,722 children aged 12–35 months were included in this analysis. The detailed methodology has been published in the 2016 EDHS final report[19].

Data source and extraction

The 2016 EDHS datasets, which are publicly available to all registered users, were requested and downloaded from MeasureDHS program website (https://www.dhsprogram.com/data/dataset_admin). After reviewing and understanding the details of the DHS data structure and dataset types, we selected the recommended dataset type for childhood vaccination record in the last three years before the survey. Data on childhood MCV1 and potential independent variables (individual and community level characteristics) were extracted accordingly.

Study Variables

Dependent Variables

The outcome variable of this study was the MCV1 vaccination status of the last child aged 12–35 months. In the 2016 EDHS, this was obtained in three ways: from written vaccination records (including the infant immunization card and other health cards), mothers' verbal reports and health facility records. It was categorized as "Yes" if the child had received MCV1 and "No" otherwise.

Independent Variables

The individual and community-level variables included in the study as explanatory variables are shown in Table 1, along with the coding. Individual-level variables include socio-demographic and economic characteristics (level one). Community-level variables describe the cluster of the communities living in the same geographical living environment (level two). These two hierarchical levels were used to create a multilevel analysis for this study. Communities were based on sharing a common primary sampling unit (enumeration area or cluster) within the 2016 EDHS data.

Table 7.1. Independent variables and categorization

Variables	Categories
3. Individual Level Factors	
Child characteristics	
Age of child (months)	(1) 12-23; (2) 24-35
Sex of the child	(1) Male; (2) Female
Birth order	(1) 1; (2) 2-3; (3) 4-5; (4) 6+
Acceptance of the child	(1) Wanted then, (2) Wanted later & (3) Wanted no more
DPT1-HepB1-Hib1	(0) Not vaccinated; (1) Vaccinated
DPT3-HepB3-Hib3	0) Not vaccinated; (1) Vaccinated
Maternal/Paternal characteristics	
Mother's age (years)	(1)15-19; (2) 20-34; (2) 35-49
Religion	(1)Orthodox; (2)Muslim; (3)Protestant; (4) Others
Mother's education	(1)No education; (2)Primary; (3)Secondary; (4)Higher
Father's education	1)No education; (2)Primary; (3)Secondary; (4)Higher
Wealth index	(1) Poorest; (2)Poorer; (3) Rich; (4) Richer ; (5)Richest
Sex of household head	(1) Male; (2) Female
Number of living children	(1) 1; (2) 2-3; (3) 4-5; (4) 6+
Head of household	(1) Male; (2) Female
Mother's relation to the head of the household	2. Wife; (2) head; (3)Daughter; (4) Others
Regular media exposure	2. No; (1) Yes
4. Community Level Factors	
Residence	(1) Urban; (2) Rural

Variables	Categories
Region	(1)Addis Ababa; (2)Tigray; (3)Afar; (4)Amhara; (5) Oromiya; (6) Somali; (7) Benishangul-Gumuz; (8) SNNP; (9) Gambella; (10) Harari ; (11) Dire Dawa
Distance to health facility	2. A big problem, (2) Not a big problem

Operational definitions

Clustering: An unusual aggregation of children with the same MCV1 status (vaccinated or unvaccinated) in a specific geographical area[20, 21].

Measles-containing vaccines(MCV): Vaccines containing antigens of measles only (M) or a combination of measles with rubella (MR), mumps (MM, MMR) and varicella (MMRV) vaccines[22].

Children who received MCV1: Children who received a one dose measles-containing vaccine for the first time at any time before the survey (based on the evidence from vaccination cards, health facility records, or the mother's report)[19].

Mass media exposure: Respondents were asked how often they read a newspaper, listened to the radio, or watched television and responded at least once a week are considered to be regularly exposed to that form of media[19].

Data management and statistical analysis

After data extraction, further recoding of variables was done to better suit with other studies for comparison. We used SaTScan 9.6 and ArcGIS 10.3 for spatial analysis and Stata version 14 for non-spatial statistical analysis.

Sample weights were applied to compensate for the unequal probability of selection between the strata that has been geographically defined. The detailed explanation of the weighting procedure can be found in the methodology of the EDHS final report[19].

Spatial Analysis: The aggregated MCV1 count and proportion data was joined to the geographic coordinates based on each EA unique identification code. To evaluate whether the pattern expressed is clustered, dispersed, or random across the study areas, global spatial autocorrelation (GSA) was assessed with ArcGIS version 10.3 using the Global Moran's I statistic (Moran's I) [23]. When p-value indicates statistical significance, a positive Moran's I index value indicates tendency toward clustering while a negative Moran's I index value indicates tendency toward dispersion[23].

In the presence of positive GSA, we employed Kulldorff's method of purely spatial scan statistic using the discrete Poisson model to detect the local spatial clusters of areas with high or low childhood MCV1 coverages. SaTScan 9.6 software was used for the cluster detection [24]. "SaTScan™ is a trademark of Martin Kulldorff. The SaTScan™ software was developed under the joint auspices of (i) Martin Kulldorff, (ii) the National Cancer Institute, and (iii) Farzad Mostashari of the New York City Department of Health and Mental Hygiene[21, 24, 25]". It uses a circular window which moves systematically throughout the study area to identify significant SaTScan clustering of children who received MCV1. We used a maximum of 10 % (to avoid the detection of large clusters) of the population at risk for the spatial cluster size and the analysis was done using standard Monte Carlo hypothesis testing with 999 Monte Carlo replicates. A cluster is statistically significant when its log likelihood ratio (LLR) is greater than the Standard Monte Carlo critical value (C.V) for 0.05 significance level and p-value <0.05[21, 24].

Multilevel logistic regression Analysis: Considering the hierarchical nature of the data in which 3,722 children were nested within 611 EAs, we fitted multilevel logistic regression models to identify predictors of childhood MCV1 at two levels: individual and community levels. Four models were fitted with the first model (model I) without any explanatory variable (empty model) to detect the existence of possible contextual effect, the second model (model II) adjusted for individual-level variables, the third model (Model III) adjusted for community-level variables and the fourth model (Model IV) adjusted for both the individual- and community-level variables. The goodness of fit of the model was assessed based on the deviance information criteria (DIC) and Akaike's Information Criterion (AIC) and the fourth model with the smallest value of the information criterion, was selected as the final better fit model for the data[26]. Adjusted odds ratio with 95% confidence interval was used to declare statistical significance in the measures of association (Fixed effect).

Intraclass correlation coefficient (ICC), median odds ratio (MOR) and proportional change in variance (PCV) statistics were calculated for measures of variation (random effects). ICC was used to explain cluster variation, the variation between individuals within the same cluster and it was calculated using the formula:

$$\mathrm{ICC} = \frac{V_A}{V_A + \frac{\pi^2}{3}} = \frac{V_A}{V_A + 3.29}$$

where V_A is the estimated variance of clusters, which has been described elsewhere[27]. The MOR is the median odds ratio between the individual of higher propensity and the individual of lower propensity when comparing two individuals from two different randomly chosen clusters. It is a measure of unexplained cluster heterogeneity, the variation between clusters (the second-level variation) by comparing two persons from two randomly chosen, different clusters and it was computed using the formula:

$$\mathrm{MOR} = \exp\left(\sqrt{2 \times V_A} \times 0.6745\right) \approx \exp\left(0.95\sqrt{V_A}\right)$$

where V_A is the cluster level variance [27, 28]. The measure is always greater than or equal to 1. If the MOR is 1, there is no variation between clusters (no second level variation) [29]. The proportional change in variance (PCV) measures the total variation attributed by individual level factors and community level factors in the multilevel regression model. It was calculated as

$$\mathrm{PCV} = \frac{V_A - V_B}{V_A}$$

Where V_A = variance of the initial model, and V_B = variance of the model with more terms [27]

Ethical Consideration

The data of the 2016 Ethiopia demographic and health survey was used for this study with a permission from the Measure DHS program after being registered and submiting a request with briefly stated objctives of the study. Ethical approval was obtained from the Institutional Review Board of the Institute of Public Health of the University of Gondar and the ICF International Institutional Review Board. The data has been used only for this registered research and it could not be passed on to other researchers. The shapefiles for Ethiopia's administrative boundaries was obtained from the openAFRICA website [https://africaopendata.org/dataset/ethiopia-shapefiles][30]. The detaile of the ethical issues has been published in the 2016 EDHS final report, which can be accessed at: http://www.dhsprogram.com/publications [19].

7.3 Results

Characteristics of study participants and MCV1 coverage

Nearly half of the children (51 %) were within the age group of 12–23 months with the mean age of 23 months (SD ± 7). Nearly 70% of the children had received pentavalent first dose (DPT1-HepB1-Hib1) vaccine and 49% of children had received third dose of pentavalent (DPT3-HepB3-Hib3) vaccine. Over 63 percent of mothers of the children had no education, about one-fourth of them were in the poorest wealth quintile and 81 percent had no regular media exposure. The overall prevallenece of childhood MCV1 in Ethiopia was 54.3% (95% CI = 52.7 - 55.9) with variation across the regions of the country, highest in Addis Ababa, 94% and lowest in Ethio-Somali region, 36%. The distance to a health facility was a big problem in the three-fifth (60%) of respondents (Table 7.2).

Table 7.2. Percentage of children age 12-35 months who received MCV1 at any time before the survey (weighted) by background characteristics, Ethiopia, 2016.

Background Characteristics	Total number Children		Children who received MCV1	
	N	%	N	%
Individual level characteristics				
Child Age (in Months) (Mean=22.95 (SD ± 7.04)				
12 - 23	1976	51.0	1069	54.1
24 - 35	1901	49.0	1036	54.5
Sex of child				
Male	1935	49.9	1056	54.6
Female	1941	50.1	1049	54
Birth Order				
1	753	19.4	451	59.9
2 - 3	1161	29.9	674	58.1
4 - 5	900	23.2	475	52.8
6+	1063	27.4	505	47.5
Wanted child				
Wanted then	2832	73.1	1491	52.7
Wanted later	702	18.1	430	61.3
Wanted no more	343	8.8	183	53.5
DPT1-HepB1-Hib1 vaccine				
Not vaccinated	1158	29.9	121	10.4
Vaccinated	2718	70.1	1984	73
DPT3-HepB3-Hib3 vaccine				
Not vaccinated	1983	51.2	505	25.5
Vaccinated	1893	48.8	1599	84.5
Maternal age (years)				
15 -19	137	3.5	77	56.2
20 - 34	2850	73.5	1587	55.7
35 - 49	889	22.9	441	49.6
Religion				

Background Characteristics	Total number Children		Children who received MCV1	
Orthodox	1349	34.8	887	65.8
Muslim	1530	39.5	617	40.4
Protestant	861	22.2	529	61.4
Others*	136	3.5	71	52.2
Mother's educational				
No education	2474	63.8	1188	48
Primary	1100	28.4	669	60.8
Secondary & Higher	302	7.8	248	82.1
Husband/partner's education (N=3655)				
No education	1746	47.8	820	47
Primary	1462	40.0	838	57.3
Secondary & Higher	447	12.2	319	71.4
Mother's occupation				
Did not work	2071	53.4	1024	49.4
Non-Professional	1731	44.7	1025	59.2
Professional	74	1.9	56	75.7
Father's occupation (N=3655)				
Did not work	275	7.5	111	40.2
Nonprofessional	620	17.0	303	48.9
Professional	2760	75.5	1564	56.7
Maternal relation to the household dead				
Wife	3091	79.7	1670	54
Head	406	10.5	224	55.2
Daughter	211	5.4	122	57.8
Others	169	4.4	89	52.7
Sex of the household head				
Male	3314	85.5	1792	54.1
Female	562	14.5	312	55.5
Number of living children				0
1	684	17.6	413	60.4
2 - 3	1281	33.0	759	59.3
4 - 5	1022	26.4	529	51.8
6+	890	23.0	404	45.4
Maternal health care decision making (N=3655)				
Jointly with her husband	2439	66.7	1333	54.7
Husband/partner alone	727	19.9	351	48.3
By herself alone	479	13.1	286	59.6
Others	10	0.3	8	80
Wealth index				
Poorest	965	24.9	412	42.7
Poorer	860	22.2	445	51.7
Middle	808	20.8	419	51.9
Richer	676	17.4	393	58.1

Background Characteristics	Total number Children		Children who received MCV1	
Richest	567	14.6	436	76.9
Mass Media Exposure				
Not regular	3154	81.4	1601	50.8
Regular	722	18.6	504	69.7
Community level characteristics				
Region				
Addis Ababa	95	2.5	90	93.8
Tigray	279	7.2	227	81.1
Afar	42	1.1	12	28.6
Amhara	718	18.5	434	60.4
Oromia	1712	44.2	722	42.1
Somali	132	3.4	47	35.6
Benishangul Gumuz	43	1.1	32	76.2
SNNPR	818	21.1	517	63.1
Gambella	10	0.2	6	60
Harari	10	0.2	5	50
Dire Dawa	17	0.4	14	82.4
Residence				
Urban	433	11.2	337	77.8
Rural	3443	88.8	1768	51.4
Distance to health facility				
Big Problem	2332	60.2	1114	47.8
Not a big problem	1545	39.8	991	64.1
Total	3876	100	2105	54.3

*Catholic, traditional and other unclassified
**Daughter-in-low, sister, other relative, no relation

Spatial pattern of childhood MCV1

The global spatial autocorrelation analysis based on feature locations and attribute values revealed a clustering pattern of childhood MCV1 across the study areas (Global Moran’s I = 0.134, p-value < 0.0001) (Figure 7.1).

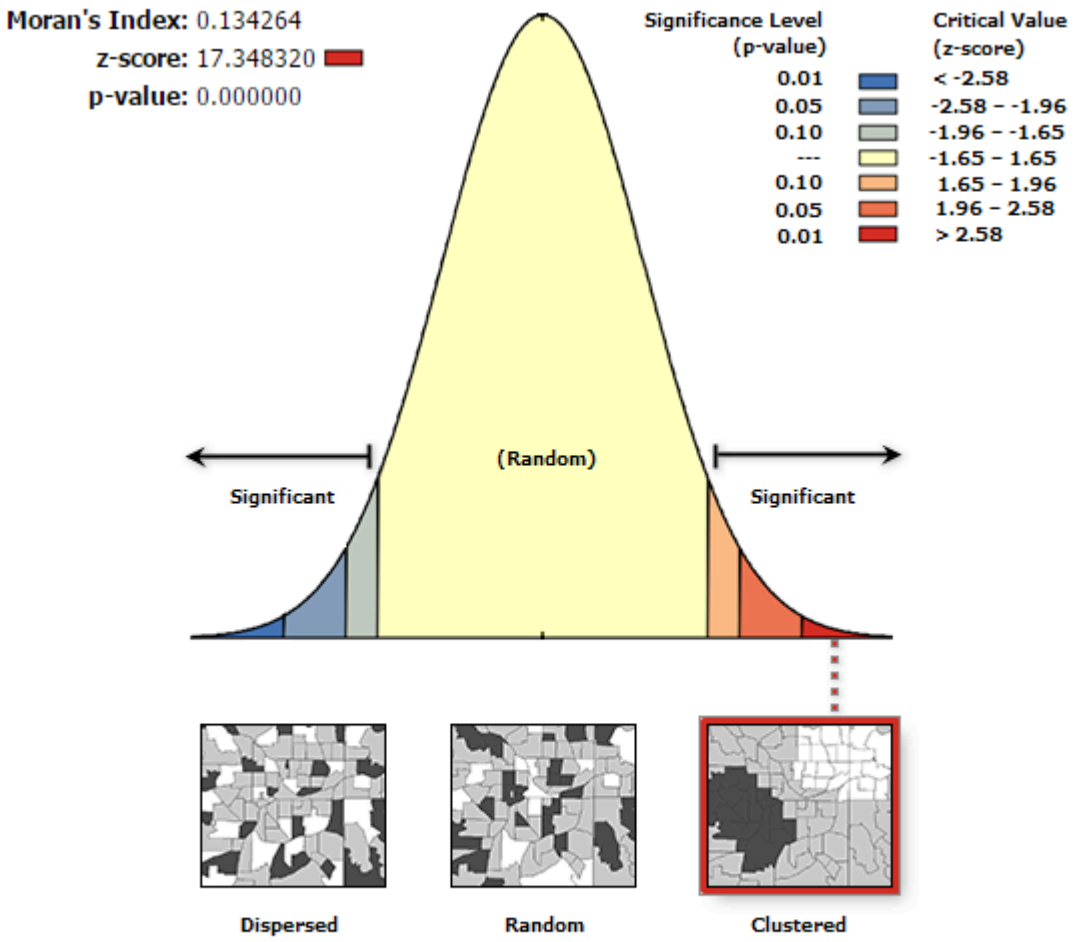


Figure 7.1. The global spatial autocorrelation based on feature locations and attribute values (MCV1) across the study areas in Ethiopia, 2016.

The SaTScan spatial analysis detected a total of seven statistically significant SaTScan clusters areas with low childhood MCV1 coverage, which means that the prevalence of MCV1 is lower inside the SaTScan circular window compared to outside the SaTScan window. The most likely primary SaTScan cluster of areas with low coverage of MCV1 was detected in Afar region (LLR = 33.79, p< 0.01), specifically in Zone 1, Zone 2 and Zone 4 administrative zones and the most likely secondary Spatial SaTScan cluster (LLR=26.76, p<0.01) in Eastern Ethiopia, specifically in Afder, Shabelle, Korah, Doolo, Nogob, Jarar, Fatan administrative zones of Ethio-Somali region.

The 3rd and 4th most likely clusters of areas with low rates of MCV1 were detected in Nuer & Agnuak Administrative zones of Gambella region (LLR=17.73, P<0.01) and Zone 2 administration of Afar region (LLR=14.54, P<0.01), respectively. In addition, the 5th most likely SaTScan cluster (LLR=14.49, P<0.01) was detected in Illuababur & Jimma Administrative zones of Oromia Region, while the 6th most likely SaTScan cluster (LLR=14.18, P<0.01) in Zone 3 and

Zone 4 of Afar region, North Shewa of Amhara region, West and East Hararge of Oromia Region and Siti zone of Somali region. The 7th most likely SaTScan cluster (LLR=12.27, P<0.01) was detected in East Hararge Zone of Oromia region and Fagan Zone of Ethio-Somali region (Table 7.3 & Figure 7.2).

Table 7.3. The most likely clusters from a purely spatial scan statistic (discrete Poisson model) of children received MCV1 in Ethiopia, 2016.

Clusters	Location (Administrative zones)	Radius (km)	LLR*	C.V	P-Value
Primary cluster	Afar Region (Zone 1, Zone 2 & Zone 3)	265.1	33.79	9.85	0.001
Secondary cluster	Ethio-Somali region (Afder, Shabelle, Korah, Doolo, Nogob, Jarar, Fatan)	339.6	26.76	9.83	0.001
3rd cluster	Gambella region (Nuer & Agnuak)	91.7	17.73	9.45	0.001
4th cluster	Afar region (Zone 2)	8.9	14.54	9.28	0.001
5th cluster	Oromia region (Illuababur & Jimma)	78.7	14.49	9.45	0.002
6th cluster	Afar region (Zone 3 & 4), Amhara region (North Shewa), Oromia Region (West and East Hararge) and Ethio-Somali Region (Siti)	139.1	14.18	9.40	0.001
7th cluster	Oromia region (East Hararge Zone), and Ethio-Somali region (Fatan Zone)	39.8	12.27	9.69	0.005

*Log likelihood ratio, **Standard Monte Carlo Critical Value for 0.05 significance level.

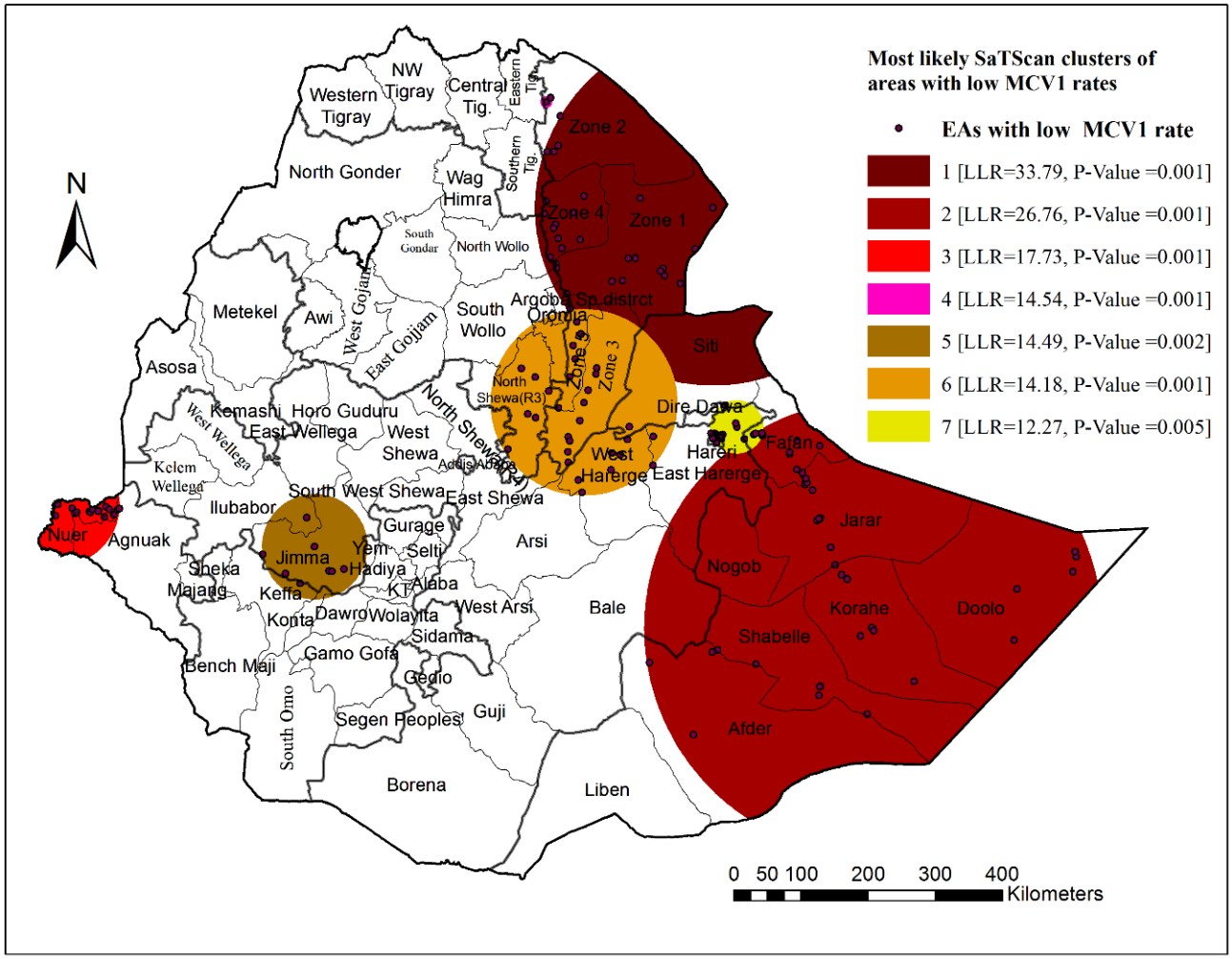


Figure 7.2. The spatial clustering of childhood MCV1 in Ethiopia, 2016.

Factors associated with the childhood MCV1 in Ethiopia: Multilevel Analysis

The results of multilevel logistic regression models for individual and community level factors are displayed in Table 5. In the final full model (model IV) where all individual and community level factors are included, child's age; child's Penta 1 & Penta 3 and Vaccination status; maternal age; education status of mother; low household wealth index; region; and distance from health facility were factors that were significantly associated with childhood first dose measles vaccination.

Individual level factors: Children aged 24–35 months old were 1.53 times (AOR = 1.53; 95 % CI: 1.25-1.88) more likely to be vaccinated for MCV1 than children aged 12-23 months old. Children who have received first dose pentavalent vaccine (DPT1-HepB1-Hib1) (AOR = 9.09; 95 % CI: 6.86-12.03) and third dose pentavalent vaccine (DPT3-HepB3-Hib3) (AOR = 7.12; 95 % CI: 5.51-9.18) were more likely to receive first dose of measles vaccine. Children whose mothers' age 20-34 years were 55 % (AOR = 0.45; 95 % CI: 0.27–0.76) less likely to receive MCV1 compared to those children whose mothers age 15-19 years. In addition, children with maternal age 35-45 years were 55 % (AOR = 0.45; 95 % CI: 0.25–0.82) less likely to receive MCV1 compared to those children whose mothers age 15-19 years. The odds of MVC1 vaccination were increased by 62 % (AOR = 1.62; 95 % CI: 1.03–2.55) in children with secondary and higher maternal education compared to children whose mothers had no education (Table 7.4).

Community level factors: The odds of childhood MCV1 vaccination in Tigray (AOR =0.32; 95 % CI: 0.13-0.81), Oromia (AOR 0.24; 95 % CI 0.10-0.58), Ethio-Somali (AOR =0.39; 95 % CI: 0.16-0.95), Gambella (AOR=0.39; 95%CI: 0.16-0.99) and Harari (AOR=0.27; 95%CI: 0.11-0.66) regions were lower compared to Addis Ababa. The odds of MVC1 vaccination were increased by 31 % (AOR = 1.31; 95 % CI: 1.12-1.61) in children living in areas where distance to health facility is not a big problem compared to children living in areas where distance to health facility is a big problem (Table 7.4).

Table 7.4. Multilevel logistic regression analysis of individual and community level factors associated with childhood MCV1 in Ethiopia, 2016.

Variables	Model II AOR (95% CI)	Model III AOR (95% CI)	Model IV AOR (95% CI)
Individual level factors			
Child characteristics			
Child's age (months)			
12 - 23	1.00	-	1.00
24 - 35	1.53 (1.2-1.88)	-	1.53 (1.25-1.88)
Sex of child			
Male	1.00	-	1.00
Female	1.11 (0.92-1.35)	-	1.10 (0.91-1.34)
Birth Order			
1	1.00	-	1.00
2 - 3	1.01 (0.62-1.63)	-	1.00 (0.61-1.63)
4 - 5	1.03 (0.55-1.93)	-	1.03 (0.55-1.92)
6+	0.86 (0.41-1.81)	-	0.87 (0.42-1.83)
Child Wanted			
Wanted then	1.00	-	1.00
Wanted later	0.93 (0.70-1.23)	-	0.94 (0.71-1.25)
Wanted no more	1.52 (0.98-2.37)	-	1.58 (1.00-2.47)
DPT1-HepB1-Hib1 (Pentavalent first dose) vaccine			
Not vaccinated	1.00	-	1.00
Vaccinated	9.23 (6.98-12.22)	-	9.09 (6.86-12.03)
DPT3-HepB3-Hib3 (Pentavalent third dose) vaccine			
Not vaccinated	1.00	-	1.00
Vaccinated	8.15 (6.33-10.48)	-	7.12 (5.51-9.18)
Maternal Characteristics			
Maternal age (in years)			
15 - 19	1.00	-	1.00
20 - 34	0.46 (0.28-0.78)	-	0.45 (0.27-0.76)
35 - 49	0.49 (0.27-0.90)	-	0.45 (0.25-0.82)
Religion			

Variables	Model II AOR (95% CI)	Model III AOR (95% CI)	Model IV AOR (95% CI)
Orthodox Christian	1.00	-	1.00
Muslim	0.71 (0.54-0.94)	-	0.95 (0.65-1.38)
Protestant	1.04 (0.74-1.46)	-	1.32 (0.85-2.03)
Others*	1.05 (0.54-2.03)	-	1.32 (0.66-2.62)
Maternal Education			
No education	1.00	-	1.00
Primary	1.00 (0.78-1.29)	-	0.99 (0.77-1.28)
Secondary and higher	1.71 (1.10-2.66)	-	1.62 (1.03-2.55)
Mother's occupation			
Did not work	1.00	-	1.00
Non-professional	1.07 (0.86-1.33)	-	1.04 (0.83-1.29)
Professional	0.97 (0.40-2.35)	-	0.93 (0.38-2.25)
Mather's relation to the household head			
Wife	1.00	-	1.00
Head	0.72 (0.37-1.42)	-	0.70 (0.35-1.38)
Daughter	1.39 (0.83-2.32)	-	1.35 (0.81-2.26)
Others	1.08 (0.61-1.91)	-	1.05 (0.58-1.87)
Sex of the household head			
Male	1.00	-	1.00
Female	1.54 (0.83-2.87)	-	1.57 (0.84-2.93)
Number of living children	1.00		
1	1.00	-	1.00
2 - 3	1.21 (0.74-1.97)	-	1.21 (0.74-1.98)
4 - 5	1.26 (0.66-2.40)	-	1.29 (0.68-2.47)
6+	1.16 (0.55-2.48)	-	1.18 (0.55-2.53)
Wealth index			
Poorest	1.00	-	1.00
Poorer	1.12 (0.83-1.52)	-	1.16 (0.85-1.59)
Middle	0.94 (0.68-1.30)	-	0.99 (0.71-1.40)
Richer	1.06 (0.75-1.50)	-	1.18 (0.83-1.70)
Richest	1.35 (0.93-1.98)	-	1.34 (0.82- 2.20)
Regular exposure to mass media			
No	1.00	-	1.00
Yes	1.33 (0.98-1.80)	-	1.32 (0.97-1.79)
Community level factors			
Region			
Addis Ababa	-	1.00	1.00
Tigray	-	0.72 (0.30-1.73)	0.65 (0.27-1.56)
Afar	-	0.04 (0.02-0.09)	0.32 (0.13-0.81)

Variables	Model II AOR (95% CI)	Model III AOR (95% CI)	Model IV AOR (95% CI)
Amhara	-	0.25 (0.10-0.58)	0.46 (0.19-1.11)
Oromia	-	0.12 (0.05-0.27)	0.24 (0.10-0.58)
Somali	-	0.05 (0.02-0.13)	0.39 (0.16-0.95)
Benishangul	-	0.60 (0.25-1.47)	0.63 (0.25-1.56)
SNNPR	-	0.32 (0.14-0.75)	0.42 (0.17-1.01)
Gambella	-	0.15 (0.06-0.36)	0.39 (0.16-0.99)
Harari	-	0.11 (0.04-0.25)	0.27 (0.11-0.66)
Dire Dawa	-	0.63 (0.25-1.60)	0.69 (0.27-1.79)
Residence			
Urban	-	1.00	1.00
Rural	-	0.30 (0.21-0.44)	1.03 (0.64-1.65)
Distance to Health facility			
Big problem	-	1.00	1.00
Not a big problem	-	1.42 (1.17-1.72)	1.31 (1.12-1.61)

*Catholic, traditional and other unclassified
**Daughter-in-low, sister, other relative, no relation

Measures of variation (random-effects)

Table 7.5 depicted multilevel logistic regression analysis results for odds of childhood MCV1 Vaccination. The null model (Model I) revealed significant variability on MCV1 coverage across communities [variance = 2.68, p< 0.001]. Similarly, the ICC showed that 45% of the variability in the odds of receiving MCV1 was due to community-level factors.

After adjusting for individual-level characteristics, the variation in the odds of a child receiving MCV1 remained statistically significant [variance =0.56, p<0.001], across the communities. At the same time, about 15% of the variance in MCV1 among the children was because of community-level factors. Similarly, about 79% of the variation in the odds of children who received MCV1 between communities was attributed to the individual factors adjusted in the model (Model 2).

After adjusting the community-level characteristics, model 3 found a slightly increased variance of a child being vaccinated for MCV1 [variance = 1.12, p< 0.001] across the communities, as compared to the variance reported in model 2. Model 3 further identified that about 25% of the variability in the odds of a child being MCV1 vaccinated was due to community-level characteristics (ICC = 25.3%). Similarly, about 58% of the variability in the odds of children receiving MCV1 between communities could be explained by the community-level characteristics included in Model 3 (PCV = 58.3%).

Finally, after the statistical adjustment of the individual and community-level characteristics simultaneously, model 4 depicted significant variability across the communities with regards to the odds of a child receiving MCV1[variance = 0.48; p<0.001]. About 13% of the variability in the odds of a child being fully vaccinated was due to the community-level factors (ICC = 12.7%). Similarly, the PCV revealed that about 82% of the variance in the odds of MCV1 vaccination (PCV = 82.1%) across communities was due to the simultaneous effect of both individual and community-level characteristics adjusted in Model 4. Results from the MOR also confirmed evidence of community contextual phenomena

shaping the odds of being a ‘’MCV1 Vaccinated child’‘. The high MOR (4.73) in Model 1 between children with a higher and lower propensity of being’‘vaccinated’’ in a community suggests that the community heterogeneity is substantial. Including all factors reduced the unexplained heterogeneity between communities to an MOR of 2.95 (Table 7.5).

Table 7.5. Model fit statistics and measures of variation (random intercept models) in childhood MCV1 in Ethiopia, 2016.

Measures of variation	Model 1[a]	Model 2[b]	Model 3[c]	Model 4[d]
Community level				
Variance (SE)	2.676(0.091)*	0.556(0.085)*	1.117 (0.074)*	0.479(0.086)*
PCV (%)	Reference	79.2	58.3	82.1
ICC (%)	44.9	14.5	25.3	12.7
MOR[e]	4.73	2.03	2.73	1.93
Model fit statistics				
DIC (-2log likelihood)	4,462	2,984	4,142	2,953
AIC	4,466	3,050	4,170	3,043

*P-value <0.001
SE: Standard Error; PCV: Proportional Change in Variance; ICC: Intraclass Correlation Coefficient; MOR: Median Odds Ratio; DIC: Deviance Information Criterion, AIC: Akaike’s Information Criterion
[a]Model 1 is the empty model, a baseline model without any explanatory variable
[b]Model 2 is adjusted for individual-level factors
[c]Model 3 is adjusted for community-level factors
[d]Model 4 is final model adjusted for both individual and community-level factors
[e]Increased risk (in median) that one would have if moving to a neighborhood/cluster with a higher risk

7.4 Discussion

In this study, a total of 3,722 children aged 12-35 months nested within 611 clusters were included from the 2016 EDHS data. The prevalence of childhood MCV1 was found to be 54.3 percent (95% CI = 52.7 - 55.9) which was low compared to the recent DHS reports of most low and middle income countries such as Egypt, 96%[31], Kenya, 87%[32], Rwanda, 95%[33], Ghana, 89%[34], Zimbabwe, 82%[35] and Uganda, 80%[36]. It was also low as compared to a local survey finding in the selected zones of Ethiopia by JSI-l10k , 80%[37], the national report in 2015, 87%[38], the 2016 national target for childhood MCV1, 91%[38] and the recommended herd immunity threshold, 95%[9, 39]. This childhood MCV1 coverage below the recommended herd immunity threshold (95%) may indicate that not also the country but also the regions with high levels of childhood MCV1 coverage may still be at considerable risk for measles outbreaks.

While vaccine exemptions are gaining attention at all levels of public health, there is an urgent need to identify regions or local areas exhibiting high exemption rates and concomitant low immunization coverage, and to intervene in these areas. Spatial heterogeneity in vaccination coverage has an immediate health impact on communities with the lower immunization rates at all spatial scales[40].

In the global spatial autocorrelation analysis, a clustering pattern of childhood MCV1 across the study areas was observed (Global Moran's I = 0.134, p-value<0.0001). This indicates that approximately the same coverages of childhood MCV1 were aggregated in specific areas. Hence, the SaTScan spatial analysis detected seven statistically significant most likely SaTScan clusters of areas with low MCV1 coverage. The most likely primary SaTScan cluster of areas with low MCV1 coverage was detected in Afar region; specifically in Zone 1, Zone 2 and Zone 4 administrative zones (LLR = 33.79, p< 0.01), and the most likely secondary SaTScan cluster in Ethio-Somali region, specifically in Afder, Shabelle, Korah, Doolo, Nogob, Jarar and Fatan administrative zones (LLR=26.76, p<0.01). The 3rd and 4th most likely clusters of areas with low MCV1 coverage were detected in Gambella region; specifically in Nuer and Agnuak Administrative zones (LLR=17.73, P<0.01) and Afar region, particularly in Zone 2 administrative zone (LLR=14.54, P<0.01), respectively. In addition, the 5th most likely SaTScan cluster (LLR=14.49, P<0.01) was detected in Illuababur & Jimma Administrative zones of Oromia Region, while the 6th most likely SaTScan cluster (LLR=14.18, P<0.01) in zone 3 and zone 4 administrative zones of Afar region, North Shewa zone of Amhara region, West and East Hararge zones of Oromia region and Siti zone of Ethio-Somali region. The last (7th) most likely SaTScan cluster (LLR=12.27, P<0.01) was detected in East Hararge zone of Oromia region and Fatan zone of Ethio-Somali region. This local clustering of low childhood MCV1 coverages indicates that children who lived in the above mentioned geographical locations had a low probability of receiving MCV1 compared with those who lived outside the SaTScan clusters. This may be due to differences in health service accessibility and community background characteristics. This considerable spatial heterogeneities and geographic clustering of low MCV1 coverage areas found in our analysis suggest that regions with high levels of sub-national MCV1 coverage may still be at considerable risk for measles outbreaks[41].

The multilevel regression analysis in this study confirmed that the variation in the childhood MCV1 in Ethiopia was attributed to both individual and community level factors. This finding is supported by evidences from multilevel analyses done for the determinants of childhood full immunization in Ethiopia[42] and Democratic Republic of Congo[43]. In the final model (model IV), individual and community-level factors accounted for about 82.1 % of the variation observed for childhood MCV1.

We found that children aged 24-35 months were more likely to receive MCV1 compared to children aged 12-23 months (AOR=1.53; 95 % CI: 1.25-1.88). This could be due to late initiation of childhood vaccination and Vaccine hesitancy for early ages. This indicates delaying in childhood vaccination which extends the period of vulnerability of children to

vaccine preventable diseases[44]. We also found that children who have received the first dose of pentavalent vaccine (AOR = 9.09; 95 % CI: 6.86-12.03) and third dose of pentavalent vaccine (AOR = 7.12; 95 % CI: 5.51-9.18) were more likely to receive MCV1. This may be due to differences in childhood vaccination service availability and utilization among communities. It could also be justified as; women whose children lacked the first and third doses of pentavalent vaccination may miss the opportunities for information on the importance of childhood measles vaccination. Even though, there were no variables included in this analysis to directly measure the availability of vaccination services in the communities, pentavalent first dose vaccination coverage is a good proxy indicator for the availability of access to and initial use of childhood vaccination services[13]. Hence, a high proportion of pentavalent first dose coverage may indicate the availability of childhood vaccination services in the communities. In addition, the third dose pentavalent vaccination coverage indicates the continuity of use by parents or care takers, client satisfaction with services, and capability of the system to deliver a series of vaccinations[13]. Hence, a high proportion of pentavalent third dose vaccination may indicate better childhood vaccination service utilization and the health system performance in relation to vaccination services.

Children from mothers with secondary or higher education had higher odds of being vaccinated for MCV1 (AOR = 1.62; 95 % CI: 1.03–2.55). Similar findings were reported in studies on childhood immunization from Ethiopia [42], Democratic Republic of Congo[43], Nigeria [45, 46], Kenya[47] and in India[48]. Education was identified as a strong predictor of childhood immunization in other several studies [49-52]. It could be explained by educated mothers are expected to be more aware of the importance of the available health services, have better communication skills, and tend to better utilize available services[53, 54].

In regard to community characteristics, geographic region of the study participants was found to be statistically significant explanatory variable of childhood MCV1. Children from Tigray, Oromia, Ethio-Somali, Gambella and Harari regions were less likely to receive MCV1 than children from Addis Ababa. This may be due to the differences in religion, culture, population size, geographic nature and level of development in the nine regional states and two city administrative councils found in Ethiopia. This could be linked with differences in vaccine supply, availability of health care providers and accessibility of health facilities. Hence, these regional divergences tend to affect the range of child MCV1 vaccination across the country. Vaccine hesitancy or exemption can lead to relatively large outbreaks even within regions with high vaccination coverage [15, 55-57]. Subpopulations with low vaccination coverage across country are a particular concern for measles elimination because these transient populations are often not covered by national immunization programs

Unlike what has been documented elsewhere[58-61], this study did not found significant association between childhood MCV1 and maternal place of residence. This may be due to the fact that the small sample size they used and the time in the study was conducted by itself may be a factor. Nevertheless, the finding is consistent with the findings of studies conducted in central Ethiopia by Etana et al. (2012) and Nigeria by Aremu et al. (2010), in which both studies found non-significant association between likelihood of childhood MCV1 and place of residence[49, 62] and this may need further investigation.

Finally, this study revealed that children who lived in areas where distance from health facility is not a big problem were more likely to receive childhood MCV1 which is consistent with a study finding in sub-Saharan Africa [63].

The national and subnational representativeness of the data with a large sample size, as well as a community-based study with a high response rate are strengths of this study to give high statistical power. Another important strength is the use of multilevel analysis to identify other factors beyond individual-level factors that could not be addressed by using ordinary logistic regression analysis. The use of combination of method (spatial and regression statistics) to identify priority areas

and predictors can also be mentioned as the strength of this study. However, this study is not free from limitation and should be mentioned. Due to the retrospective nature of the data, and the maternal verbal reports for childhood vaccination, recall bias might have been introduced.

7.5 Conclusion

A clustered pattern of areas with low childhood MCV1 coverage was observed in Ethiopia. Statistically significant local clusters of areas with low childhood MCV1 were detected in Ethio-Somali, Afar, Gambella, and Oromia regions of the country. Both the individual level characteristics (child's age, first and third doses of pentavalent vaccination, maternal age and education) and community level characteristics (geographic region and distance from health facility) were statistically significant predictors of childhood MCV1. Hence, it is good if the federal ministry of health and other concerned child health programmers give priority of the areas with low MCV1 coverage identified in this study. It is also better to consider the individual and community level determinant factors identified in this study.

Abbreviations

AIC: Akaike's information criterion; AOR: adjusted odds ratio; CI: confidence intervals; COR: Crude odd ratio; DIC: Deviance information criterion; DPT-HepB-Hib: a combination vaccine containing diphtheria, pertussis, tetanus toxoid, hepatitis B and Haemophilus influenza type b; EAs: Enumeration areas; EDHS: Ethiopian demographic health survey; GIS: geographic information system; ICC: intra-cluster correlation; IR: Incidence rate; LLR: Log likelihood ratio; MCV1: First dose of measles-containing vaccine MOR: median odds ratio; PCV: proportional change in variance; SD: standard deviation; WHO: world health organization

Declarations

Competing interest

The authors declare that they have no competing interest

Acknowledgments

The authors would like to thank the MeasureDHS program for their permission to download and use the 2016 Ethiopia demographic and Health survey datasets.

Ethical approval

Ethical approval was obtained from the Institutional Review Board of the Institute of Public Health of the University of Gondar and the ICF International Institutional Review Board. The data has been used only for this registered research and it could not be passed on to other researchers.

Authors' contributions

TT conceived the study, conducted the statistical analysis and interpretation, and drafted the manuscript. LD and AN assisted the data analysis and interpretation, reviewed the draft of consecutive manuscript and did the final editing. All of the authors read and approved the final version of the manuscript.

Consent for publication

Not applicable

Funding

The authors have funding support from University of Gondar.

Author details

[1]Ethiopian Field Epidemiology and Laboratory Training Program, Department of Epidemiology and Biostatistics, Institute of Public Health, College of Medicine and Health Sciences, University of Gondar, P.O. Box 169, Gondar, Ethiopia.

[2]Department of Reproductive Health, School of Public Health, College of Medicine and Health Sciences, Bahir Dar University, P.O. Box 79, Bahir Dar, Ethiopia

[3]Department of Epidemiology and Biostatistics, Institute of Public Health, College of Medicine and Health Sciences, University of Gondar, P.O. Box 169, Gondar, Ethiopia

[4]Department of Human Nutrition and Food Sciences, College of Health Sciences, Debre Markos University, P.O. Box 269, Debre Markos, Ethiopia.

[5]PhD student at the academic center of excellence for human nutrition, food science and post-harvest technology, Hawassa University, Ethiopia

References

1. The Ethiopian Health and Nutrition Research Institute (EHNRI): Guideline on measles surviellance and outbreak management, 3rd Edition. Addis Ababa, Ethiopia, January 2012.
2. Perry RT, et al., Progress toward regional measles elimination - Worldwide, 2000-2014. MMWR. Morb. Mortal. Wkly. Rep. United States; 2015;64:1246–51.
3. World Health Organization (2016). Immunization Coverage: Fact sheet No. 286. 2016 . Available at: www.who.int/mediacentre/factsheets/fs286/en/ (accessed on 24th January 2018).
4. WHO: Summary of the WHO position on Measles Vaccine- April 2017. Available at: www.who.int/entity/immunization/policy/position_papers/WHO_PP_measles_vaccine_summary_2017.pdf. (Accessed on 26th February 2018).
5. World Health Organization (2012): Global measles and rubella strategic plan : 2012-2020.
6. Strategic Advisory Group of Experts on Immunization (SAGE): 2016 Midterm Review Report of Global Vaccine Action Plan.
7. WHO and UNICEF Estimates for National Immunization Coverage (WUENIC), 2010-2015.
8. WHO, Department of Evidence, Information and Research estimates for child causes of death, 2000-2015 (updated 5 February 2016).
9. WHO|WER Index 2016: Weekly Epidemiological Record, 11 November 2016, vol. 91, 45 (pp. 525–536). Available at: www.who.int/wer/2016/wer9145/en/.
10. Merlo, J., et al., A brief conceptual tutorial of multilevel analysis in social epidemiology: linking the statistical concept of clustering to the idea of contextual phenomenon. J Epidemiol Community Health, 2005. **59**(6): p. 443-9.
11. Akalu HB (2015) Review on Measles Situation in Ethiopia; Past and Present. J Trop Dis 4: 193. doi:10.4172/2329-891X.1000193.
12. Ethiopia National Expanded Programme on Immunization (2015). comprehensive multiyear plan, 2016-2020; Federal Ministry of Health, Addis Ababa.
13. Federal Ministry of Health. Revised Health Management Information System Indicator Reference Guide; Technical Standards: Area 1; Addis Ababa Ethiopia, Augest 2017.
14. World Health Organization (2017): Global Health Observatory (GHO) data>Measles-containingvaccine first-dose (MCV1) immunization coverage among 1-year-olds. Available at: www.who.int/gho/immunization/measles/en/ [Accessed 26th February 2018].
15. Zipprich J, et al., Measles outbreak–California, December 2014-February 2015. MMWR. Morb.Mortal. Wkly. Rep. United States. 2015. 64: p. 153–154.
16. Suijkerbuijk A, et al., Economic costs of measles outbreak in the Netherlands, 2013-2014. Emerg Infect Dis United States. 2015. 21: p. 2067–2069.
17. Federal Minstry of Health (FMOH): Ethiopian Growth & Transformation Plan, 2010 - 2015.
18. Ethiopian Public Health Institute (EPHI) : Ethiopia Services Availability and Readiness Assessment 2016: Summary Report. Addis Ababa, Ethiopia, January, 2017.
19. Alemu, M.B., et al., Utilization of insecticide treated bed net and associated factors among households of Kola Diba town, North Gondar, Amhara region, Ethiopia. BMC Res Notes, 2018. **11**(1): p. 575.
20. K., A., W. D., and S. H., Spatial distribution of under immunization among children 12-23 months old in butajira, southern ethiopia, 2016. Unpublished Article.
21. Kulldorff M.(2018). SaTScanTM User Guide for Version 9.6. March, 2018. www.satscan.org/.
22. Federal Ministery of Health (FMOH): Immunization in Practice National Guideline for Health Workers, Modules for health workers, 2014 Update.
23. Anselin L, Getis A. Spatial statistical analysis and geographic information systems. Ann Reg Sci. 1992;26(1):19–33.
24. Kulldorff M. and Information Management Services, Inc. SaTScan™ v9.6: Software for the spatial and space-time scan statistics. www.satscan.org, 2018.
25. Victora, C.G., L. Adair, and C. Fall, Maternal and child undernutrition: consequences for adult health and human capital. Lancet, 2008. **371**.

26. Akaike, H. 1974. A new look at the statistical model identification. IEEE transactions on Automatic Control 19: 716-723.

27. Merlo, J., et al., A brief conceptual tutorial of multilevel analysis in social epidemiology: using measures of clustering in multilevel logistic regression to investigate contextual phenomena. J Epidemiol Community Health, 2006. **60**(4): p. 290-7.

28. Austin PC, Stryhn H, Leckie G, Merlo J. Measures of clustering and heterogeneity in multilevel Poisson regression analyses of rates/count data. Statistics in Medicine. 2018;37:572–589. doi.org/10.1002/sim.7532.

29. Halonen, J.I., et al., Quantifying neighbourhood socioeconomic effects in clustering of behaviour-related risk factors: a multilevel analysis. PLoS One, 2012. **7**(3): p. e32937.

30. Alelign, A., Z. Tekeste, and B. Petros, Prevalence of malaria in Woreta town, Amhara region, Northwest Ethiopia over eight years. BMC Public Health, 2018. **18**(1): p. 990.

31. Ministry of Health and Population/Egypt, El-Zanaty and Associates/Egypt, and ICF International. 2015. Egypt Demographic and Health Survey 2014. Cairo, Egypt: Ministry of Health and Population and ICF International. .

32. Kenya National Bureau of Statistics, Ministry of Health/Kenya, National AIDS Control Council/Kenya, Kenya Medical Research Institute, National Council for Population and Development/Kenya, and ICF International. 2015. Kenya Demographic and Health Survey 2014. Rockville, MD, USA: Kenya National Bureau of Statistics, Ministry of Health/Kenya, National AIDS Control Council/Kenya, Kenya Medical Research Institute, National Council for Population and Development/Kenya, and ICF International. .

33. National Institute of Statistics of Rwanda (NISR) [Rwanda], Ministry of Health (MOH) [Rwanda], and ICF International. 2015. Rwanda Demographic and Health Survey 2014-15. Rockville, Maryland, USA: NISR, MOH, and ICF International.

34. Ghana Statistical Service - GSS, Ghana Health Service - GHS, and ICF International. 2015. Ghana Demographic and Health Survey 2014. Rockville, Maryland, USA: GSS, GHS, and ICF International. .

35. Zimbabwe National Statistics Agency and ICF International. 2016. Zimbabwe Demographic and Health Survey 2015: Final Report. Rockville, Maryland, USA: Zimbabwe National Statistics Agency (ZIMSTAT) and ICF International. .

36. Uganda Bureau of Statistics - UBOS and ICF. 2018. Uganda Demographic and Health Survey 2016. Kampala, Uganda and Rockville, Maryland, USA: UBOS and ICF. .

37. JSI-L10K: Extended Program on Immunization (EPI) coverage in selected Ethiopian zones: A baseline survey for L10K's Routine Immunization Improvement Initiative. Addis Ababa, Ethiopia. 2015.

38. Federal Democratic Republic of Ethiopia, Ministry of Health: Health Sector Transformation Plan-HSTP, July 2015-June 2020 (EFY 2008-2012), May 2015.

39. WHO | Measles Surveillance Data. (2016). Available at: www.who.int/immunization/monitoring_surveillance/burden/vpd/surveillance_type/active/measles_monthlydata/en/ (Accessed on 26th February 2018).

40. Kacey Ernst & Elizabeth T. Jacobs (2012) Implications of philosophical and personal belief exemptions on re-emergence of vaccine-preventable disease: The role of spatial clustering in under-vaccination, Human Vaccines & Immunotherapeutics, 8:6, 838-841, DOI: 10.4161/hv.19743.

41. Takahashi, S., et al., The geography of measles vaccination in the African Great Lakes region. Nat Commun, 2017. 8: p. 15585.

42. Abadura, S.A., et al., *Individual and community level determinants of childhood full immunization in Ethiopia: a multilevel analysis.* BMC Public Health, 2015. **15**: p. 972.

43. Acharya, P., et al., *Individual- and community-level determinants of child immunization in the Democratic Republic of Congo: A multilevel analysis.* PLoS One, 2018. **13**(8): p. e0202742.

44. *Simone Périnet, Marilou Kiely, Gaston De Serres & Nicolas L. Gilbert (2018): Delayed measles vaccination of toddlers in Canada: Associated socio-demographic factors and parental knowledge, attitudes and beliefs, Human Vaccines & Immunotherapeutics, DOI: 10.1080/21645515.2017.1412899.*

45. Olorunsaiye, C.Z. and H. Degge, *Variations in the Uptake of Routine Immunization in Nigeria: Examining Determinants of Inequitable Access.* Global Health Communication, 2016. **2**(1): p. 19-29.

46. Fatiregun, A.A. and A.O. Okoro, *Maternal determinants of complete child immunization among children aged 12-23 months in a southern district of Nigeria.* Vaccine, 2012. **30**(4): p. 730-6.

47. Maina, L.C., S. Karanja, and J. Kombich, *Immunization coverage and its determinants among children aged 12 - 23 months in a peri-urban area of Kenya.* Pan Afr Med J, 2013. **14**: p. 3.

48. Phukan, R.K., M.P. Barman, and J. Mahanta, *Factors associated with immunization coverage of children in Assam, India: over the first year of life.* J Trop Pediatr, 2009. **55**(4): p. 249-52.

49. Etana, B. and W. Deressa, *Factors associated with complete immunization coverage in children aged 12-23 months in Ambo Woreda, Central Ethiopia.* BMC Public Health, 2012. **12**: p. 566.

50. Kumar, D., A. Aggarwal, and S. Gomber, *Immunization status of children admitted to a tertiary-care hospital of north India: reasons for partial immunization or non-immunization.* J Health Popul Nutr, 2010. **28**(3): p. 300-4.

51. Shrivastwa, N., et al., *Predictors of Vaccination in India for Children Aged 12-36 Months.* Am J Prev Med, 2015. **49**(6 Suppl 4): p. S435-44.

52. Wiysonge, C.S., et al., *Individual and contextual factors associated with low childhood immunisation coverage in sub-Saharan Africa: a multilevel analysis.* PLoS One, 2012. **7**(5): p. e37905.

53. *Govindasamy P, Ramesh B. Maternal education and the utilization of maternal and child health services in India. 1997.*

54. *Vikram K, Vanneman R, Desai S. Linkages between maternal education and childhood immunization in India. Social science & medicine. 2012; 75(2):331–9.*

55. *Centers for Disease Control and Prevention (CDC). Notes from the field: Measles outbreak among members of a religious community—Brooklyn, New York, March–June 2013. Morb. Mortal. Wkly. Rep. 2013, 62, 752–753.*

56. *Phadke, V.K.; Bednarczyk, R.A.; Salmon, D.A.; Omer, S.B. Association between vaccine refusal and vaccine-preventable diseases in the United States: A review of measles and pertussis. JAMA 2016, 315, 1149–1158.*

57. *Gastanaduy, P.A.; Budd, J.; Fisher, N.; Redd, S.B.; Fletcher, J.; Miller, J.; McFadden, D.J., 3rd; Rota, J.; Rota, P.A.; Hickman, C.; et al. A measles outbreak in an underimmunized amish community in Ohio. N. Engl. J. Med. 2016, 375, 1343–1354.*

58. *Ibnouf A, Vanden B, Maarse J. Factors influencing immunization coverage among children under five years of age in Khartoum State, Sudan: University of Maastricht. S Afr Fam Pract. 2007;49(8):14.*

59. *Roy, SG. "Risk F actors for Childhood Immunization Incompletion in Ethiopia" (2010). Public Health Theses. Georgia State University, Paper 90. Available from: digitalarchive.gsu.edu/iph_theses/90.*

60. *Kidane T, Yigzaw A, Sahilemariam Y, Bulto T, Mengistu H, Malay T, et al. National EPI coverage survey report in Ethiopia, 2006. Ethiop J Health Dev. 2008;22(2):148–57.*

61. *Mathew JL. Inequality in childhood immunization in India: a systematic review. Indian Pediatr. 2012;49:203–23.*

62. *Aremu O, Lawoko S, Dalal K. Childhood vitamin A capsule supplementation coverage in Nigeria: a multilevel analysis of geographic and socioeconomic inequities. The Scientific World Journal: TSW Child Health &.*

63. Brownwright, T.K., Z.M. Dodson, and W.G. van Panhuis, *Spatial clustering of measles vaccination coverage among children in sub-Saharan Africa.* BMC Public Health, 2017. **17**(1): p. 957.

CHAPTER 8

Conference Abstracts

Spatial clustering and determinants of women's postnatal health checkup in Ethiopia: Results from the 2016 demographic and health survey[7]

7. Presented at the 1st Annual scientific conference of Amhara Public Health Institute, Bahir Dar; the 2nd EFELTP annual scientific conference, Addis Ababa; the 1st national scientific conferences of Hawassa University, Hawassa.↩

PUBLISHED VERSION

Sisay MM, Geremew TT, Demlie YW, Alem AT, Beyene DK, Melak MF, et al. Spatial patterns and determinants of postnatal care use in Ethiopia: findings from the 2016 demographic and health survey. *BMJ Open*. 2019;9(6):e025066.

https://doi.org/10.1136/bmjopen-2018-025066

Tesfahun Taddege GEREMEW*, Yeshambel W., Asaye T., Desalew K., Malede M., Kassahun A., Tadesse A

*Correspondence: ttaddege@gmail.com

Abstract

Introduction: Postnatal care is one of the high impact maternal health interventions to reduce maternal mortality, the major public health problem in most resource-limited settings, especially in Sub-Saharan Africa. In Ethiopia, the estimated maternal mortality ratio was 412 deaths per 100,000 live births in 2016. However, studies on spatial epidemiology of women's postnatal care are limited and the driving factors are poorly understood. Hence, this study aimed to explore geographical patterns and predictors of women's postnatal health checkup after discharge or delivery at home in Ethiopia.

Methods: A secondary data analysis was conducted using the 2016 Ethiopia demographic and health survey data, which was collected with a stratified two stage cluster sampling method. A total of 7, 193 women who gave live birth in last five years preceding the survey were included in this analysis. We employed poison-based purely spatial scan statistics to detect spatial clusters of women's health checkup after discharge or delivery at home and a multilevel logistic regression models were used to identify associated factors.

Results: Only 6.3% of women in Ethiopia received postnatal health checkup after discharge or delivery at home. The SaTScan spatial analysis identified a total of six statistically significant SaTScan clusters for women's health checkup after discharge or delivery at home, of which, three were clusters of areas with low prevalence of postnatal care. The most likely SaTScan clusters of areas with low rates of women's postnatal health checkup were detected in Southwestern Ethiopia (LLR = 18.07, p < 0.0001), Southeast Ethiopia (LLR=14.29, P < 0.001) and Eastern Ethiopia (LLR=10.18, P=0.024). In the final model of multilevel logistic regression analysis**,** individual and community-level factors accounted for about 58.2 % of the variation observed in women's health checkup after discharge or delivery at home. Higher educational level [AOR= 2.06; 95%CI: 1.31-3.26], richest wealth quantile [AOR = 1.71; 95%CI: 1.07-2.71] and antenatal care service [AOR=2.35; 95%CI: 1.68-3.29] were individual level factors associated with higher odds women's postnatal health checkup**,** whereas, living out of Tigray region was a community-level factor associated with lower odds of women's postnatal health checkup.

Conclusion: The spatial distribution of women's health checkup after discharge or delivery at home was not random in Ethiopia. Both individual level factors (educational status, wealth index and antenatal care) and a community-level factor (region) were statistically significant determinants of women's postnatal health checkup. Hence**,** it is good if the FMOH and other maternal health programmers give priority for the areas with low prevalence of women's postnatal health checkup (coldspots) identified in this study. It is also better to work more on promoting antenatal care services and women education.

Key words: Postnatal care, Spatial, Multilevel Ethiopia

Acute watery diarrhea outbreak investigation and response at Abune-hara holly water in Bahir Dar Zuriya district, Northwest Ethiopia, 2017[8]

8. Presented at the 2nd EFELTP annual scientific conference, EPHI, Addis Ababa↩

Tesfahun Taddege GEREMEW[1*], Lemma Derseh, Meklit Mekonnen and Getie Zeru

*Correspondence: ttaddege@gmail.com

Abstract

Introduction: Globally, Acute Watery Diarrhea represents an estimated burden of 1.4 - 4.3 million patients and it has remained as an important public health problem in developing countries. Recently, AWD has become a public health problem in Ethiopia. Since the beginning of 2017 to 7 July 2017, a total of 38,230 cases with 792 deaths (CFR 2.1%) were reported. However, there limited studies on the history of exposures for AWD patients. Hence, the purpose of this study was to identify the risk factors of AWD outbreak to guide responses activities.

Methods: A case-control study was conducted from May 24 to June 17, 2017 at Abune-Hara Holly Water in Bahir-Dar Zuria District, Northwest Ethiopia. A total of 141 cases and 282 controls were included. The data was collected through Computer Assisted Personal Interviewing (CAPI) using android device loaded with Epi-Info^{TM} templet and analyzed using SPSS V_{22}. We carried out multivariable logistic regression to identify associated factors and odds ratio with 95 % CI was used to declare statistical significance.

Result: A total of 141 epidemiologically linked AWD cases and three deaths (CFR 2.1%) were recorded. The median age was 35 years [IQR 26-50]. Close contact history [OR=5.81, 95%CI=2.85-11.85], water interruption than any other before [OR=4.30, 95%CI =2.09-8.85], living in overcrowded room [OR=3.85, 95%CI=1.48-10.05], hand washing after toilet [OR=0.15, 95%CI=0.08-0.28] and before eating [OR=0.10, 95%CI =0.02-0.41], storing water in narrow mouthed plastic container [OR=0.42, 95% CI =0.23-0.78] and eating food immiadetely after cocked [OR = 0.21, 95%CI =0.12-0.38] were statistically significant associated factors.

Conclusion: The AWD case fatality rate was high**.** Close contact history, living in overcrowded room, Water supply interruption were the possible risk factors. The study result suggests a need to give more attention for Case management, adequate water supply and water storage in narrow-mouthed container, feeding practices, building additional rooms and increasing the awareness of hygienic practices for the Holly Water users.

Key words: AWD, Outbreak, Abune-Hara Holly Water, Northwest Ethiopia

Geographical variations in childhood measles vaccination and associated factors in Ethiopia: a spatial and multilevel analysis

PUBLISHED VERSION

Geremew TT, Gezie LD, Abejie AN. Geographical variation and associated factors of childhood measles vaccination in Ethiopia: a spatial and multilevel analysis. *BMC Public Health*. 2019;19:1194.

https://doi.org/10.1186/s12889-019-7529-z

Tesfahun Taddege GEREMEW*, Lemma Derseh

*Correspondence: ttaddege@gmail.com

Abstract

Background: In Ethiopia, despite considerable improvement of measles vaccination, measles outbreaks have been occurring in most parts of the country. Understanding the neighborhood variation in measles vaccination has been increasingly recognized, and is crucial for evidence-based decision-making. However, the spatial pattern of first dose of measles-containing vaccine (MCV1) and its predictors are poorly understood. Hence, this study aimed to explore the spatial pattern and associated factors of childhood MCV1 coverage

Methods: An in-depth analysis the 2016 Ethiopia demographic and health survey data was conducted and a total of 3,722 children nested in 611 enumeration areas were included in the analysis. Poisson-based purely spatial scan statistics was employed to explore the spatial patterns of childhood MCV1. Multilevel logistic regression models were fitted to identify factors associated with childhood MCV1.

Result: Spatial hetrogeniety of childhood MCV1 was observed (Global Moran's I=0.134, p-value < 0.0001) and seven statistically significant SaTScan clusters of areas with low MCV1 coverage were detected. The most likely primary SaTScan cluster was detected in Afar region, secondary cluster in Ethio-Somali region, and tertiary cluster in Gambella region. In the final model of the multilevel analysis, individual and community level factors accounted for 82% of the variance in the odds of MCV1 vaccination. Older child age (AOR=1.53; 95%CI: 1.25-1.88), pentavalent vaccination first dose (AOR=9.09; 95%CI: 6.86-12.03) and third dose (AOR=7.12; 95%CI: 5.51-9.18, secondary and above maternal education (AOR=1.62; 95%CI: 1.03–2.55) and media exposure were the factors that increased the odds of MCV1 vaccination at the individual level. Children with older maternal age had lower odds of receiving MCV1. Living in Tigray, Oromia, Ethio-Somali, Gambella and Harari regions were factors associated with lower odds of MCV1 from the community-level factors. Children far from health facilities had higher odds of receiving MCV1 (AOR=1.31, 95%CI=1.12-1.61).

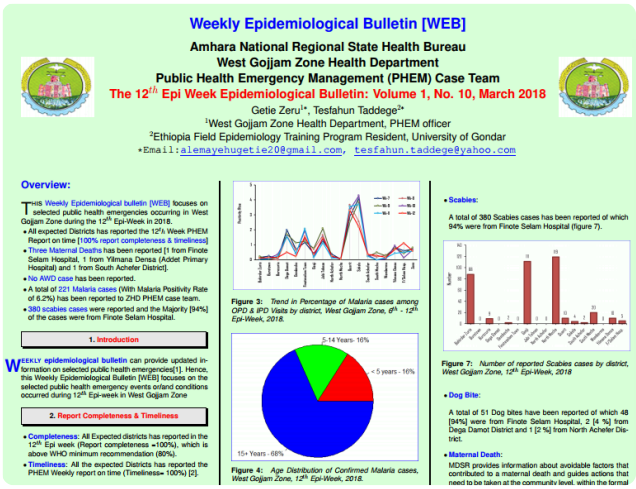
Conclusion: A clustered pattern of areas with low childhood MCV1 coverage was observed in Ethiopia. Both individual and community level factors were significant predictors of childhood MCV1. Hence, it is good to give priority for the areas with low childhood MCV1 coverage, and to consider the identified factors for vaccination interventions.

Keywords: Children, MCV1, Spatial, Multilevel, Ethiopia

CHAPTER 9

Additional Outputs

9.1 Sample weekly bulletins



Weekly Epidemiological Bulletin [WEB]

Amhara National Regional State Health Bureau
West Gojjam Zone Health Department
Public Health Emergency Management (PHEM) Case Team
The 12th Epi Week Epidemiological Bulletin: Volume 1, No. 10, March 2018

Getie Zeru[1]*, Tesfahun Taddege[2]*
[1]West Gojjam Zone Health Department, PHEM officer
[2]Ethiopia Field Epidemiology Training Program Resident, University of Gondar
*Email: alemayehugetie20@gmail.com, tesfahun.taddege@yahoo.com

Overview:

This Weekly Epidemiological bulletin [WEB] focuses on selected public health emergencies occurring in West Gojjam Zone during the 12th Epi-Week in 2018.

- All expected Districts has reported the 12th/ Week PHEM Report on time [100% report completeness & timeliness]
- Three Maternal Deaths has been reported [1 from Finote Selam Hospital, 1 from Yilmana Densa (Addet Primary Hospital) and 1 from South Achefer District].
- No AWD case has been reported.
- A total of 221 Malaria cases (With Malaria Positivity Rate of 6.2%) has been reported to ZHD PHEM case team.
- 380 scabies cases were reported and the Majority [94%] of the cases were from Finote Selam Hospital.

1. Introduction

WEEKLY epidemiological bulletin can provide updated information on selected public health emergencies[1]. Hence, this Weekly Epidemiological Bulletin [WEB] focuses on the selected public health emergency events or/and conditions occurred during 12th Epi-week in West Gojjam Zone

2. Report Completeness & Timeliness

- Completeness: All Expected districts has reported in the 12th Epi week (Report completeness =100%), which is above WHO minimum recommendation (80%).
- Timeliness: All the expected Districts has reported the PHEM Weekly report on time (Timeliness= 100%) [2].

Figure 3: *Trend in Percentage of Malaria cases among OPD & IPD Visits by district, West Gojjam Zone, 6th - 12th Epi-Week, 2018.*

Figure 4: *Age Distribution of Confirmed Malaria cases, West Gojjam Zone, 12th Epi-Week, 2018.*

- Scabies:

A total of 380 Scabies cases has been reported of which 94% were from Finote Selam Hospital (figure 7).

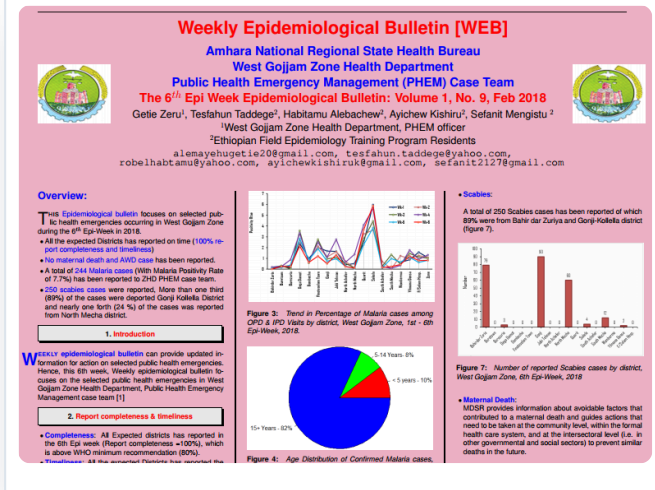
Figure 7: *Number of reported Scabies cases by district, West Gojjam Zone, 12th Epi-Week, 2018*

- Dog Bite:

A total of 51 Dog bites have been reported of which 48 [94%] were from Finote Selam Hospital, 2 [4 %] from Dega Damot District and 1 [2 %] from North Achefer District.

- Maternal Death:

MDSR provides information about avoidable factors that contributed to a maternal death and guides actions that need to be taken at the community level, within the formal



Weekly Epidemiological Bulletin [WEB]

Amhara National Regional State Health Bureau
West Gojjam Zone Health Department
Public Health Emergency Management (PHEM) Case Team
The 6th/ Epi Week Epidemiological Bulletin: Volume 1, No. 9, Feb 2018

Getie Zeru[1], Tesfahun Taddege[2], Habitamu Alebachew[2], Ayichew Kishiru[2], Sefanit Mengistu [2]
[1]West Gojjam Zone Health Department, PHEM officer
[2]Ethiopian Field Epidemiology Training Program Residents
alemayehugetie20@gmail.com, tesfahun.taddege@yahoo.com, robelhabtamu@yahoo.com, ayichewkishiruk@gmail.com, sefanit2127@gmail.com

Overview:

This Epidemiological bulletin focuses on selected public health emergencies occurring in West Gojjam Zone during the 6th Epi-Week in 2018.

- All the expected Districts has reported on time (100% report completeness and timeliness)
- No maternal death and AWD case has been reported.
- A total of 244 Malaria cases (With Malaria Positivity Rate of 7.7%) has been reported to ZHD PHEM case team.
- 250 scabies cases were reported, More than one third (89%) of the cases were deported Gonji Kollella District and nearly one forth (24 %) of the cases was reported from North Mecha district.

1. Introduction

WEEKLY epidemiological bulletin can provide updated information for action on selected public health emergencies. Hence, this 6th week, Weekly epidemiological bulletin focuses on the selected public health emergencies in West Gojjam Zone Health Department, Public Health Emergency Management case team [1]

2. Report completeness & timeliness

- Completeness: All Expected districts has reported in the 6th Epi week (Report completeness =100%), which is above WHO minimum recommendation (80%).

Figure 3: *Trend in Percentage of Malaria cases among OPD & IPD Visits by district, West Gojjam Zone, 1st - 6th Epi-Week, 2018.*

Figure 4: *Age Distribution of Confirmed Malaria cases,*

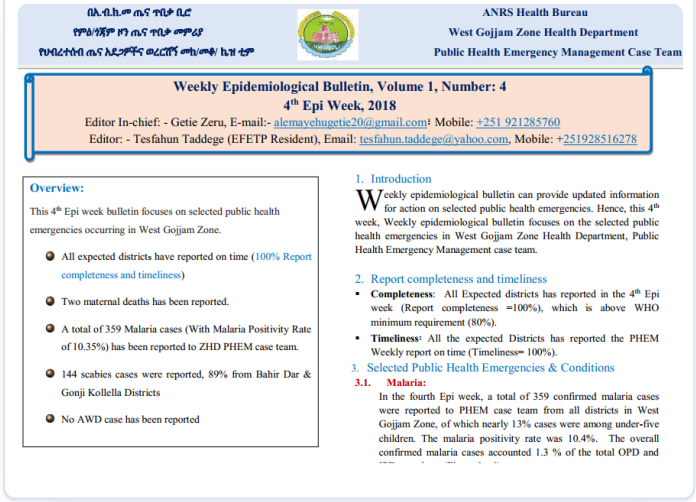
- Scabies:

A total of 250 Scabies cases has been reported of which 89% were from Bahir dar Zuriya and Gonji-Kollella district (figure 7).

Figure 7: *Number of reported Scabies cases by district, West Gojjam Zone, 6th Epi-Week, 2018*

- Maternal Death:

MDSR provides information about avoidable factors that contributed to a maternal death and guides actions that need to be taken at the community level, within the formal health care system, and at the intersectoral level (i.e. in other governmental and social sectors) to prevent similar deaths in the future.



በኢ.ፌ.ዲ.ሪ.መ ጤና ጥቢቃ ቢሮ
የምዕራ/ጎጃም ዞን ጤና ጥቢቃ መምሪያ
የህብረተሰብ ጤና አደጋዎችና ወረርሽኝ መከ/መቆ/ ኬዝ ቲም

ANRS Health Bureau
West Gojjam Zone Health Department
Public Health Emergency Management Case Team

Weekly Epidemiological Bulletin, Volume 1, Number: 4
4th Epi Week, 2018

Editor In-chief: - Getie Zeru, E-mail:- alemayehugetie20@gmail.com; Mobile: +251 921285760
Editor: - Tesfahun Taddege (EFETP Resident), Email: tesfahun.taddege@yahoo.com, Mobile: +251928516278

Overview:

This 4th Epi week bulletin focuses on selected public health emergencies occurring in West Gojjam Zone.

- All expected districts have reported on time (100% Report completeness and timeliness)
- Two maternal deaths has been reported.
- A total of 359 Malaria cases (With Malaria Positivity Rate of 10.35%) has been reported to ZHD PHEM case team.
- 144 scabies cases were reported, 89% from Bahir Dar & Gonji Kollella Districts
- No AWD case has been reported

1. Introduction

Weekly epidemiological bulletin can provide updated information for action on selected public health emergencies. Hence, this 4th week, Weekly epidemiological bulletin focuses on the selected public health emergencies in West Gojjam Zone Health Department, Public Health Emergency Management case team.

2. Report completeness and timeliness

- **Completeness**: All Expected districts has reported in the 4th Epi week (Report completeness =100%), which is above WHO minimum requirement (80%).
- **Timeliness**: All the expected Districts has reported the PHEM Weekly report on time (Timeliness= 100%).

3. Selected Public Health Emergencies & Conditions

3.1. **Malaria:**

In the fourth Epi week, a total of 359 confirmed malaria cases were reported to PHEM case team from all districts in West Gojjam Zone, of which nearly 13% cases were among under-five children. The malaria positivity rate was 10.4%. The overall confirmed malaria cases accounted 1.3 % of the total OPD and

9.2 Sample poster for scientific presentation[9]

9. Presented at the 13th TRAC 2018 annual scientific conference on tuberculosis, Addis Ababa, March 21-23, 2018 and at the 2nd EFELTP annual scientific conference↩

A FIVE YEAR TUBERCULOSIS SURVEILLANCE DATA ANALYSIS, WEST GOJJAM ZONE, NORTHWEST ETHIOPIA [2012/3–2016/7]

Tesfahun Taddege[1*], Lemma D[2]., Giete Z[3]., Meklit M[4].

[1]University of Gondar, EFELTP Resident, [2]Univesity of Gondar, [3]West Gojjam ZHD, [4]Amhara RHB

*Email:tesfahun.taddege@yahoo.com, Mobile:+251928516278

1. Introduction

TUBERCULOSIS is a global concern for both developing and developed countries. In 2015, there were an estimated 1.4 million TB deaths among HIV negative people, about 84% occurred in the WHO African Region[1].

Ethiopia is one of the TB HBCs[1,2]. Tuberculosis data analysis is essential for TB Program evaluation to generate evidence for decision making. Hence, this analysis aimed to describe the TB case notification & treatment outcome in West Gojjam Zone.

2. Methods

- Study setting
  - West Gojjam Zone, Northwest Ethiopia
  - Has a catchment popn of 2,560,131, 7 Hospitals [1 private & 6 gov't], 103 HCs [81% TB diagnostic] & 374 HPs [68% provide DOTs Service], 154 private clinics[4% TB diagnostic].
- Design & Period:
  - A descriptive Cross-Sectional Study
  - From July 2012/3 to June 2016/7.
- Data Source, Extraction and analysis:
  - eHMIS database & TB Annual Reports.
  - A total of 15,682 all forms of TB included
  - Analyzed with MS-EXcel 2013 & ArcGIS 10.3

3. Results

TB case Notification:

- A 15,682 patients with all forms of new and relapse TB were notified.
- 3118 (19.9%) cases were New SS+ PTB
- In 2016/17 TB CDR was 54.8% which is Consistent with TB CDR in Amhara region,52%[3] but Lower than the national targe[4] [Figure 1]

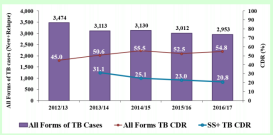


Figure 1: *Trend in number of all forms of TB Cases, all forms TB CDR and SS+ PTB CDR, West Gojjam Zone , 2012/13-2016/17*

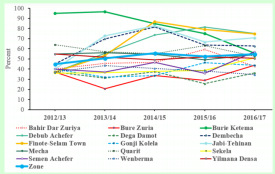


Figure 2: *Trend in TB Case Detection Rate by district , West Gojjam zone (2012/3-2016/7)*

TB Incidence:

TB IR (per 100,000 popn) was 114, below the 2016/7 HSTP target, 191 cases per 100 000 population[4] [Figure 3]

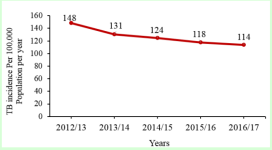


Figure 3: *Trend of TB incidence rate per 100,000 population in West Gojjam Zone, Northwest Ethiopia, 2012/3 - 2016/7*

Table 1: Percentage distribution of all forms of new and relapse TB Cases by background characteristics, West Gojjam Zone, Northwest Ethiopia, 2014/15-2016/17

Background Characteristics	Years		
	2014/15	2015/16	2016/17
Sex			
Females	25.1	24.8	25.4
Males	30.4	27.7	29.4
Age Group (in years)			
0-14	5.3	5.3	5.5
15-24	13.4	12.1	13.7
25-34	13.2	13.2	12.5
35- 44	9.4	9.3	9.3
45-54	7.5	6.2	6.9
55-64	3.6	3.8	3.5
65+	2.1	1.8	2.2
Unknown/ND	0.7	0.9	0.9
TB category			
PTB (New)	24.6	20.5	20.9
EPTB (New)	28.8	30.0	31.9
Relapse Cases	2.1	1.9	1.8
TB Case Detection Rate	**55.5**	**52.5**	**54.8**
Total number of New & Relapse Cases	**3126**	**3012**	**2953**

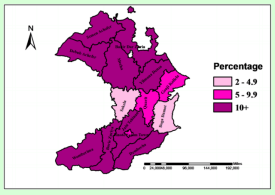


Figure 5: *Percentage of under 15 years children among all forms of TB patients by district, West Gojjam Zone, Northwest Ethiopia, 2016/17*

TB Treatment Outcomes:

- In 2016/7, the TB TSR was 94.6%, higher than the WHO target, 85 % and 2016 Global TB report, 89% [1](Figure 6).

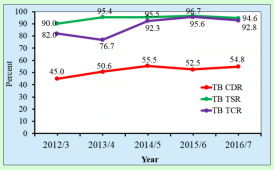


Figure 6: *Trend in TB CDR, TB treatment success rate and TB Cure rate, West Gojjam Zone, Northwest Ethiopia (2012/13 - 2016/17)*

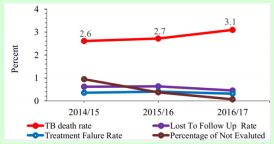


Figure 7: *Trend in unsuccessful treatment outcome (Rates), West Gojjam Zone (2014/15-2016/17)*

TB Screening among HIV Positive Clients:

Table 2: Percentage of HIV positive Clients Screened For TB and HIV positive rate among HIV Positive clients screened for TB, West Gojjam Zone, Northwest Ethiopia (2014/15 - 2016/17)

Years	Number HIV positive clients who received HIV care	HIV Positive Clients Screened For TB		HIV positive clients Screened for TB and confirmed with active TB	
		Number	Percent	Number	Percent
2014/15	22,666	21,703	95.8	64	0.29
2015/16	28,365	28,187	99.4	71	0.25
2016/17	29,038	29,037	100.0	54	0.19
Total	**80,069**	**78,927**	**98.6**	**189**	**0.24**

Drug Resistance TB:

- Over the Five years, a total of 35 patients with DR-TB [20% RR, 80% MDR] were notified.
- The mean age was 30 years [SD + 12.7] and two-third were males
- 80 % were Smear +Ve PTB and 71 % Were non-reactive for HIV.
- 6% treatment Completed, 26% cured, 6 % died, 11% LTFU and 46% on treatment
- In 2016/7, the MDR TB accounted 3.1% from Re-treatments and 0.14% from all forms

4. Conclusion

- TB CDR in WestGojjam Zone was low.
- TB Death rate is increasing from time to time
- The 2016/7 MDR TB cases accounted 3.1 % of re-treatment and 0.14 % all form TB cases
- It is better that West Gojjam ZHD to give emphasis for TB case detection and further investigation is needed to identify the reasons of for Low TB CDR and increasing TB Death Rates

Acknowledgment

- Mentors: Mr. Lemma D., Mrs. Meklit M. and Mr. Getie Z.
- West Gojjam Zone Health Department HMIS and TB Officers, EFELTP, University of Gondar

References

[1] WHO: Global TB Report 2016

[2] Zenebe, Y., et al., Profile of tuberculosis and its response to anti-TB drugs among tuberculosis patients treated under the TB control Programme at Felege-Hiwot Referral Hospital, Ethiopia. BMC Public Health, 2016. 16: p. 688

[3] FMOH: Health & health related indicator EFY 2007

[4] FMOH: HSTP - 2015/16-2019/2020.

The 2nd EFELTP Annual Scientific Conference, June 25–28, 2018, Addis Ababa, Ethiopia

9.3 Publication

PUBLISHED VERSION

Negesse A, Getaneh T, Temesgen H, Taddege T, Jara D, Abebaw Z. Prevalence of anemia and its associated factors in human immuno deficiency virus infected adult individuals in Ethiopia: a systematic review and meta-analysis. *BMC Hematology*. 2018;18:32.

https://doi.org/10.1186/s12878-018-0127-y

Negesse et al. BMC Hematology (2018) 18:32
https://doi.org/10.1186/s12878-018-0127-y

BMC Hematology

RESEARCH ARTICLE Open Access

Prevalence of anemia and its associated factors in human immuno deficiency virus infected adult individuals in Ethiopia. A systematic review and meta-analysis

Ayenew Negesse[1*], Temesgen Getaneh[2], Habtamu Temesgen[1], Tesfahun Taddege[3], Dube Jara[4,5] and Zeleke Abebaw[6]

Abstract

Background: Anemia is a common hematologic disorder among human Immunodeficiency virus (HIV) infected adult Individuals. However, there is no concrete scientific evidence established at national level in Ethiopia. Hence, this review gave special emphasis on Ethiopian HIV infected adult individuals to estimate pooled prevalence of anemia and its associated factors at national level.

Methods: Studies were retrieved through search engines in PUBMED/Medline, Cochrane Library, and the web of science, Google and Google scholar following the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P). Joanna Briggs Institute Meta-Analysis of Statistical Assessment and Review Instrument (JBI-MAStARI) was used for critical appraisal of the included studies. Random effects meta-analysis was used to estimate the pooled prevalence of anemia and associated factors at 95% Confidence interval with its respective odds ratio (OR). Meta regression was also carried out to identify the factors. Moreover, Sub-group analysis, begs and egger test followed by trim-and-fill analysis were employed to assess heterogeneity and publication bias respectively.

Result: A total of 532 articles were identified through searching of which 20 studies were included in the final review with a total sample size of 8079 HIV infected adult individuals. The pooled prevalence of anemia was 31.00% (95% CI: 23.94, 38.02). Cluster of Differentiation 4 (CD4) count <= 200 cells/μl with OR = 3.01 (95% CI: 1.87, 4.84), World Health Organization (WHO) clinical stage III&IV with OR = 2.5 (95% CI: 1.29, 4.84), opportunistic infections (OIs) with OR = 1.76 (95% CI: 1.07, 2.89) and body mass index (BMI) < 18.5 kg/M^2 with OR = 1.55 ((95% CI: 1. 28, 1.88) were the associated factors.

Conclusion: This review demonstrates high prevalence of anemia among HIV infected adults. Low CD4 count, WHO clinical stage III&IV, OIs and low level of BMI were found to have significant association with the occurrence of anemia. Therefore, the responsible stockholders including anti retro viral treatment (ART) clinics should strengthen the system and procedures for the early diagnosis of opportunistic infection and screening of underlying problems. There should be also early screening for OIs and under nutrition with strict and frequent monitoring of HIV infected individuals CD4 count.

Keywords: Anemia, Adult individuals, Pooled prevalence, Ethiopia, HIV/AIDS



Compiled Body of Works in Field Epidemiology

Field residency outputs of the Ethiopian Field Epidemiology and Laboratory Training Program, University of Gondar: outbreak investigation, surveillance, health profiling, rapid assessment, research and dissemination in northwest Ethiopia.

AUTHOR

Tesfahun Taddege GEREMEW

CONTACT

ttaddege@gmail.com

PORTFOLIO

tesfahuntaddege.github.io

GET ONLINE VERSION →

© 2019 Tesfahun Taddege GEREMEW · University of Gondar